



AEC Environment Fundamentals

From Site Acquisition to Closeout,
and the Software and AI That Run It

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Technology information in Part 3 is current as of September 2026.

Introduction

Most people in AEC know one part of a project well. An architect knows design but has never read a pro forma. A superintendent knows the jobsite but was never in the room when the land was bought. A developer can underwrite a deal but has never written an RFI.

The people building software for this industry often know even less, because they learned the one workflow their product serves.

This course is for people who know one part and want to understand the rest. It follows a building from the first look at a site through design, construction, and turnover, and at each step it covers who does the work, what documents they produce, what software they use, and where AI is changing the job.

About the Author

I'm Brooks Mitchell. I don't need to give my whole resume here, but I started my post-college career as a Field Manager in residential construction in Orlando, FL. I spent my days driving from site to site, talking with superintendents, coordinating trades for installation of windows, doors, trusses, trim, and stairs, and sweating my ass off.

As of the time of writing this, I help architects and real estate folks implement AI software into their workflows.

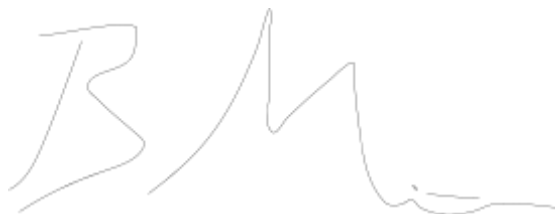
In between, I did a few things, like software engineering for oil and gas accounting, and IoT software and hardware.

I had a smattering of knowledge from construction, architecture, and operations, and wanted a more complete and thorough understanding of how everything fit together and flowed. So I wrote this for myself.

In writing this I did utilize several LLMs. I also drew upon my own experience and spent hundreds of hours talking to AEC professionals of all kinds about their industries and AI.

I've learned an immeasurable amount from writing this, and I hope you gain just as much from reading it.

For any feedback, questions, or criticisms, please reach out to me on LinkedIn:
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A handwritten signature in grey ink, consisting of the letters 'B' and 'M' followed by a stylized, flowing line that ends in a small horizontal stroke.

- Brooks Mitchell

How This Is Organized

The material has three parts that follow the life of a project in order, followed by a summary.

Part 1: Real Estate Development

How sites are analyzed and bought.

Part 1 follows a deal through five decision gates, from the first look at a market to closing on the land and financing. At each gate the money at risk goes up and walking away gets more expensive.

It covers market and site selection, zoning and entitlements, physical due diligence, feasibility and conceptual budgeting, the capital stack, and the purchase transaction, and it ends with the handoff to the design team.

Part 2: Design and Construction

The design and build process.

Part 2 follows one reference project, a 200-unit garden-style apartment community, from the day the developer closes on the land to the day the property manager receives the keys.

It covers the owner's program and delivery method, design as a sequence of commitments, cost control during design, the construction document set, bidding and buyout, preconstruction planning, construction, and closeout.

Part 3: Technology and AI

The software behind each phase, and where AI is changing the work.

Part 3 walks the same lifecycle a third time and covers the software used at each phase: the market leaders, their main competitors, and where AI is already in use.

It ends with the topics that cut across every phase: data standards and interoperability, liability and professional licensure, the effect on AEC jobs, and how firms evaluate and adopt new technology.

Synthesis: Lifecycle Recap

A single module that runs one project through all three parts in sixteen steps. For each step it summarizes who is involved, what they produce, which tools they use, what AI does today, and what information gets lost in the handoff to the next step.

Let's Get Started

A building passes through many hands before anyone moves in. A developer finds and buys the site, a design team turns the deal's assumptions into drawings and specifications, a construction team prices, plans, and builds it, and an operator runs it for decades. Each group knows its own step and hands a compressed version of what it knows to the next.

This course follows through every one of those steps, from site selection to turnover, and at each step explains the work, the documents, the software, and what AI is and isn't changing. It is written for people who know one part of AEC and want to understand the rest.

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Part 1 - Real Estate: How Sites Are Analyzed and Bought

Introduction

Every project arrives at the design team pre-loaded with constraints.

The site is already chosen. The program is already fixed - this many units, this much rentable area, this parking ratio. The budget already exists and is not a target but a commitment someone made to someone else. The schedule has a date in it that seems arbitrary and is not. The delivery method has been selected. Sometimes the massing has been approved by a city council in a public hearing, and the building cannot get materially larger or smaller without starting that process over.

Nobody explains where any of it came from.

If you have worked in design or construction, you have experienced the downstream effects of this. The owner who will not consider a better mechanical system. The three-week schedule fight that seems disproportionate to anything happening in the field. The change that gets rejected without discussion because it "affects rentable area." The renovation that stalls because something in an existing building triggered a code requirement nobody priced. Each of these feels, from the inside, like unreasonableness or bad management.

None of it is. All of it is the consequence of decisions made months or years earlier, by a group of people with their own logic, their own incentives, and their own sequence - a process that concluded before the first line was drawn.

Part 1 is that process.

What This Part Covers

The work that happens between "there is a piece of land" and "there is a project."

Specifically: how a site is found and screened, how demand for a use is established, how zoning and title and soil and existing conditions are investigated, how a concept is priced before it is designed, where the money comes from and what it demands in return, how the property is actually put under contract and bought, and what gets handed to the design team at the end of it.

This is the phase variously called predevelopment, acquisition, or feasibility. It has no CSI format, no phase deliverables, no standard contract, and no certification. It is nonetheless where the majority of a project's cost and risk is determined, because it is where the irreversible decisions are made.

Who This Is For

Someone with real experience in one part of the built environment who wants a working model of the rest of it.

The assumption throughout is that you are competent and under-exposed, not a beginner. You may have run residential projects and never seen a commercial capital stack. You may know construction documents cold and have never read a title commitment. You may work with architects and developers daily and have never watched the moment a deal goes hard.

No prior finance background is assumed.

How It's Organized

A development deal advances through a series of decision points where the money becomes progressively harder and walking away becomes progressively more expensive. That structure - not a topic list - is the spine of Part 1.

Gate	The question	Modules
Gate 0	Is this worth looking at?	1, 2
Gate 1	Do we tie it up?	7
Gate 2	Do we go hard?	3, 4A, 4B, 5
Gate 3	Can we build what we underwrote?	3
Gate 4	Do we close and start?	6, 8

Gate 4 is where "Part 2 - The Design and Construction Process" begins. The last page of Part 1 is the first page of the design process.

Module 1 - The Development Ecosystem and Deal Lifecycle. The cast, the sequence, and the gates. Who does what, who owes what to whom, and why "the Owner" in your contract documents is usually six parties in a trench coat.

Module 2 - Market and Site Selection. Whether demand exists. Supply, absorption, comps, and the highest and best use test that governs every decision after it.

Module 3 - Legal and Regulatory Due Diligence. What you are permitted to build. Zoning, entitlement, discretionary approval, easements, and private restrictions. Where most deals die.

Module 4A - Physical Due Diligence: Vacant Land. What the dirt will tell you. Survey, geotechnical, environmental, utilities, access, stormwater.

Module 4B - Physical Due Diligence: Existing Buildings. Condition, hazardous materials, and the existing-building code triggers that determine whether a renovation or conversion is viable at all.

Module 5 - Concept Feasibility and Cost. Where a real estate deal becomes an architecture problem. Test fits, efficiency, conceptual estimating, and the site development costs that are the most common source of underwriting error.

Module 6 - The Capital Stack. Whose money, on what terms, and how the lender's requirements reach into design and construction long before anyone meets the lender.

Module 7 - Site Control and the Transaction. The mechanics of buying it. Letters of intent, purchase agreements, earnest money, title, options, ground leases, closing.

Module 8 - The Handoff to Design. What gets transmitted, what gets lost, what was already promised to a city on your behalf, and how the delivery method decision gets made.

How Each Module Works

Every module follows the same structure so you can read it once for orientation and return to it as reference:

- **Concepts** - the working model
- **Artifacts** - the documents actually produced, and who produces them
- **Decision gate** - what this work is for and when it must be finished
- **Failure modes** - how this goes wrong in practice
- **What happened next** - the two running deals at this stage
- **Optional: try this** - something you can do with public data if you want to

The Two Deals

Two composite example projects run through all nine modules.

Kipling Row is a 3.2-acre infill parcel with a vacant church on it, on a suburban arterial, where a merchant developer wants a rezoning to build 34 for-sale townhomes. Its risk is concentrated in a political approval.

Six Corners is a 78,000 SF office building from 1984, 40 percent occupied, priced far below replacement cost, that a value-add sponsor intends to convert to residential. Its risk is concentrated in the existing structure, and the conversion thesis does not survive.

They are carried deliberately. Residential and commercial, ground-up and existing, entitlement risk and physical risk. Where they behave the same way - title, soil, survey - the lesson is that the process is universal. Where they diverge - demand analysis, cost structure, capital sources - the divergence is the lesson.

What Is Not Covered

- Property management and operations.
- Brokerage practice and licensure.
- Residential mortgage lending.
- Public-sector development, which is a genuine specialization with its own procurement statutes and acquisition powers; it is flagged where it materially differs and otherwise left alone.
- Institutional portfolio strategy.
- Anything approaching investment advice.

Nothing here is legal, financial, or investment advice, and the two deals are composites rather than real transactions.

A Note on Terminology

Development and construction share a vocabulary and disagree about what several words mean. "Program," "feasibility study," "due diligence," "owner," and "closing" all mean one thing in this Part and something different in Part 2. These collisions are mapped explicitly at the end of Module 1 and reconciled in a combined glossary in Module 8. If a term reads strangely, check there first - it is usually a collision rather than an error.

1.1 - The Development Ecosystem and Deal Lifecycle

In construction documentation, there is a party called "the Owner." Contracts name it, drawings reference it, the specification's Division 01 assigns it duties. It appears in the documents as a single actor with a single set of interests.

It almost never is one.

By the time an architect signs an agreement with "the Owner," that Owner is typically a limited liability company formed sixty days earlier, capitalized by three to six separate parties who negotiated against each other for months, holding a piece of land it bought under conditions imposed by a lender who has not yet funded, to build a program that was fixed by a market study and a zoning approval before the architect was hired.

This module establishes the cast and the sequence: who those parties are, what each of them wants, the order in which they commit money, and the points at which the deal can still be abandoned. Every module that follows is one of those parties doing their work at one of those points.

Section 1 - The Deal Team: Roles and Responsibilities

1.1 The Developer / Sponsor

The developer is the party that assembles the deal. In the construction documents they become "the Owner." In the capital structure they are usually called the sponsor or the general partner (GP).

Key responsibilities:

- Sources the site and negotiates control of it
- Forms the investment thesis: what gets built here, for whom, at what cost, sold or held for how long
- Underwrites the deal (builds and defends the pro forma)
- Raises the equity and negotiates the debt
- Hires and directs every consultant, including the design team
- Carries the pre-development spend at risk, before any outside capital commits
- Signs the guarantees the lender requires (completion, carve-out, and sometimes repayment)

- Makes the go/no-go call at each decision gate

The developer is the only party in the ecosystem whose exposure is asymmetric in both directions. Consultants get paid whether or not the deal closes. Lenders get repaid before equity sees anything. The developer eats the pre-development spend if the deal dies and takes the disproportionate upside if it works. This asymmetry explains nearly every behavior you will observe from developers that seems irrational from a design or construction perspective: the urgency, the reluctance to spend on drawings before entitlement, the willingness to litigate over three weeks of schedule.

What the developer is NOT:

- A fiduciary to the design team, the contractor, or the community
- Necessarily the party with the money (most developers deploy far more of other people's capital than their own)
- Necessarily the long-term owner (see 1.2)

1.2 Owner Types

The single most useful predictor of how an owner will behave during design and construction is what they intend to do with the building when it is finished.

Merchant developer. Builds to sell, usually at or shortly after stabilization. Time to exit is the dominant variable. Will trade long-term operating cost for first cost almost every time, because they will not be paying the operating cost. This is the owner most likely to value-engineer the envelope and the mechanical system, and the one most likely to push a schedule that the construction team considers unrealistic.

Long-term hold / owner-operator. Builds to own and operate for ten to thirty years. Willing to spend on durability, maintainability, and energy performance because they hold the operating expense. More receptive to life-cycle cost arguments, more likely to have standards and prototypes, more likely to have an in-house facilities voice at the table during design.

Owner-occupier. A corporation, hospital system, university, or religious institution building for its own use. The building is not the business - it is infrastructure for the business. These owners frequently lack development expertise and hire an owner's representative (1.8) to supply it. They are also the owners most likely to change the program mid-design, because the business needs changed.

Institutional owner / REIT / fund. Deploys pooled capital against a defined strategy and a defined hold period. Highly process-driven. Investment committee approval gates are real and calendared, and a deal that misses a committee date waits weeks. These owners bring standardized documents, standardized reporting, and low tolerance for deviation.

Public owner. Municipalities, school districts, transit agencies, state and federal agencies. Acquires land differently (including by dedication, surplus disposition, and eminent domain) and procures differently - by statute rather than by negotiation. Qualifications-based selection for

design services under the Brooks Act or its state equivalents, competitive sealed bidding above statutory thresholds, public advertisement, prevailing wage. Public development is a genuine specialization and is out of scope for this material; it is flagged here and again in Module 8 so you recognize it when you see it, not so you can practice it.

1.3 The Capital

Money in a development deal is stacked, and position in the stack determines both cost and control. The mechanics are Module 6; the parties are here.

Common equity (LP and GP). The GP is the developer. The limited partners (LPs) are the passive investors - high-net-worth individuals, family offices, funds, syndicated groups. The GP typically contributes a small minority of the equity and earns a disproportionate share of the profits above a return threshold. LPs supply most of the cash and take last position in the payout order. LP agreements grant "major decision" rights: changes to the budget above a threshold, changes to the business plan, sale, refinance. Those rights are why a developer sometimes cannot approve a change order without a phone call.

Preferred equity and mezzanine debt. Fills the gap between senior debt and common equity when the senior lender will not lend enough. More expensive than senior debt, cheaper than equity, and structurally more aggressive - it typically carries rights to take over the deal on default.

Senior construction lender. Usually a bank. Provides the largest single source of capital, funds by draw against work in place, and holds first lien on the property. The lender's requirements shape the project directly: a fixed budget, often a guaranteed maximum price, retainage terms, an interest reserve, an approved contractor, an approved architect, and an inspecting consultant who visits monthly and signs off before each draw funds. The lender is not on the design team and never attends a design meeting, and yet the lender's underwriting is often the reason a design decision cannot be reopened.

Permanent lender / takeout. Retires the construction loan once the building stabilizes. Cares about the income stream, not the construction process.

Public and incentive capital. Tax increment financing, PACE, tax abatements, opportunity zone equity, low-income housing tax credits, historic rehabilitation tax credits, state and utility grants. Every one of these attaches compliance conditions that reach into design and construction - design review, wage requirements, income restrictions, energy targets, historic standards. Free money is never free; it is money with a specification attached.

1.4 The Brokerage Layer

Land broker. Represents the seller in nearly all cases. Produces the offering memorandum (OM), runs the marketing process, sources buyers. The offering memorandum (OM) is a marketing document, not an underwriting document. Read it for facts (acreage, parcel numbers, asking price, seller motivation) and discard the conclusions.

Investment sales broker. Same function for improved, income-producing property. Produces an OM with a rent roll, trailing financials, and a seller's pro forma. The seller's pro forma is an argument, not an analysis.

Leasing broker / tenant rep. Matters at two points: providing the market rent and concession reality that underpins the underwriting, and later leasing the finished building. A good leasing broker consulted during feasibility will tell you what the market will not accept - unit mix, ceiling heights, dock counts, bay depths - while it is still free to change.

1.5 The Diligence Consultants

These are the parties who tell the developer what is actually true about the site. They are covered in depth in Modules 3, 4A, and 4B. Introduced here by role:

Consultant	Produces	Answers
Surveyor	ALTA/NSPS survey, topographic survey	Where are the boundaries, easements, and contours
Geotechnical engineer	Geotechnical report	What will the soil support and what will foundations cost
Environmental consultant	Phase I / Phase II ESA, wetlands delineation	Is the site contaminated or jurisdictionally restricted
Civil engineer	Yield study, utility and grading concepts	How much can physically fit and what does the site work cost
Land use attorney	Zoning opinion, entitlement strategy	What am I legally permitted to build
Entitlement consultant	Approval strategy, community outreach	How do I get permission and how long will it take
Market study firm	Market and feasibility study	Is there demand at the rent I need
Appraiser	Appraisal (as-is, as-complete, as-stabilized)	What is it worth, per the lender

Property condition consultant	PCA (ASTM E2018)	What is broken and what will it cost to fix
Traffic engineer	Traffic impact study	What off-site improvements will be required of me

Two notes that matter in practice.

The appraiser is engaged by the lender, not the borrower, and federal rules require that independence. The developer pays for the appraisal and does not direct it. A low appraisal reduces the loan proceeds and can kill a deal that otherwise works.

The market study firm and the appraiser answer different questions and are frequently confused. The market study asks whether demand exists at the assumed rent. The appraisal asks what the asset is worth. A project can appraise well and still fail to lease.

1.6 The Transaction Professionals

Real estate counsel. Drafts and negotiates the letter of intent, the purchase and sale agreement, the joint venture agreement, and the loan documents. In several states, an attorney rather than a title company runs the closing.

Title company / escrow agent. Searches title, issues the title commitment, holds the earnest money in escrow, and issues the title insurance policy at closing. Neutral to both parties.

Entity formation counsel. Forms the single-purpose entity that will actually own the property.

That last item deserves emphasis, because it is the practical bridge to Part 2. The entity that signs the AIA agreement with the architect and the contract with the general contractor is usually a single-purpose LLC formed for this one project. It holds one asset. It has no operating history and, prior to loan closing, frequently no meaningful capital. This is intentional - it isolates liability and satisfies the lender's requirement for a bankruptcy-remote borrower. It also means that the "Owner" named in the contract documents may have no assets other than the project itself, which is precisely why lien rights, payment bonds, and personal or corporate guarantees exist.

1.7 The Entitlement and Approval Layer

This is a different set of public actors than the ones you meet in construction, and confusing the two is one of the most common errors made by people entering development from the design or build side.

Planning department staff. Reviews land use applications for conformance with the zoning ordinance and comprehensive plan. Makes recommendations. Usually professional and predictable.

Planning commission / zoning board. Appointed body. Holds public hearings. Recommends or decides, depending on the application type and jurisdiction.

City council / county commission. Elected body. Final authority on rezonings, development agreements, and most discretionary approvals. Political, and responsive to constituents who show up.

Neighborhood associations and organized opposition. Not a formal authority, and frequently decisive anyway.

Special districts and external agencies. State DOT for access to a state road, the water and sewer district for capacity, the Army Corps for wetlands, the state historic preservation office for historic properties, the utility for interconnection.

The distinction that matters:

The planning department decides **whether you may build it**. The building department decides **whether what you drew is code-compliant**. These are separate reviews, by separate staff, against separate bodies of law, at separate times. Entitlement risk is generally binary and political. Building code risk is generally technical and solvable. Confusing the two leads people to assume that a code consultant can fix an entitlement problem, which is not a thing that happens.

Module 3 develops this; Part 2 Module 7 handles the code side.

1.8 Early Design and Construction Input

The design and construction team you know from Part 2 does exist during this phase, in a reduced and unglamorous form.

Architect (feasibility / test fit role). Produces massing studies, yield studies, and test fits to establish whether the program physically fits and at what efficiency. Frequently unpaid or paid at a nominal rate against a promise of the full commission if the deal closes. Sometimes required to produce near-schematic-level drawings for an entitlement submittal, months before the land is under firm control.

Civil engineer (concept role). Often more influential than the architect at this stage, because on many sites the binding constraints are grading, stormwater, access, and utilities rather than the building.

General contractor / estimator (preconstruction role). Provides conceptual pricing. Also frequently unpaid, on the same speculative basis.

Owner's representative. A hired project management function for owners without one in-house. Common with owner-occupiers and institutions. The owner's rep is the party that translates between the owner's business language and the design and construction team's technical language.

Recognize the incentive structure here. A meaningful portion of pre-development professional work is performed at risk, in exchange for positioning on the eventual commission. It is why feasibility-stage drawings are thin, why estimates carry wide contingencies, and why the developer's request for "just a quick massing" is not the small favor it is presented as.

1.9 Fiduciaries, Vendors, and Counterparties

The most useful way to hold the entire cast in your head is by the nature of the obligation each party owes.

Fiduciaries owe a duty of loyalty and must act in the other party's interest.

- The GP owes fiduciary duties to the LPs
- The broker owes fiduciary duties to whoever engaged them, and only to them
- Attorneys owe them to their client
- Licensed design professionals owe a professional standard of care to their client and a duty to public safety, which are not the same duty and occasionally conflict

Vendors owe performance of a defined scope for a defined fee. They do not owe you the answer you want.

- Every diligence consultant in 1.5
- Environmental and geotechnical consultants deserve specific mention: their reports are relied upon by lenders and are written defensively for that reason

Counterparties are negotiating against you and owe you nothing beyond the contract and the applicable law.

- The seller
- The lender
- The tenant
- The contractor, once a price is agreed

Regulators owe a duty to the public, not to you.

- Every party in 1.7, plus the building official and fire marshal in Part 2

Newcomers to development make the same category error repeatedly: they treat a vendor's report as advice, a counterparty's pro forma as analysis, or a regulator's informal comment as an approval. None of those hold.

Section 2 - Product Types and Why They Underwrite Differently

A development deal is a bet on an income stream. The shape of that income stream differs enough by product type that the same site can be a good deal for one use and an impossible one for another. This is the underlying logic of the highest and best use test in Module 2.

Product	Revenue unit	Typical lease term	Re-prices	Dominant risk
Multifamily	Rent per unit per month	12 months	Annually	New supply, interest rates
Build-to-rent / SFR	Rent per home per month	12 months	Annually	Scattered-site operations, supply
Industrial / logistics	Rent per SF per year, NNN	3–10 years	On rollover	Tenant credit, location, obsolescence of clear height
Retail	Rent per/SF plus percentage rent	5–10 years, anchors longer	On rollover	Anchor loss, co-tenancy clauses, traffic
Office	Rent per SF, gross or NNN	5–10 years	On rollover	Obsolescence, TI and leasing cost, structural demand shift
Hospitality	Revenue per available room, nightly	One night	Daily	Operating volatility, brand, capital intensity
Self-storage	Rent per unit per month	Month to month	Monthly	Local supply saturation
Senior housing	Rent plus care revenue	Varies	Varies	Operator quality, licensure, staffing

Data center	Rent per kW of critical load	10–15 years	On rollover	Power availability, interconnection queue
Raw land	None	-	-	Pure carry cost and entitlement risk

Three implications worth internalizing now:

Lease duration governs how fast the asset can respond to inflation. Multifamily and self-storage reprice within a year. A twelve-year industrial lease with fixed escalations does not. This is why the same interest rate move affects two product types very differently.

Tenant improvement (TI) and leasing cost varies enormously. Industrial shells lease with modest tenant improvement allowances. Office space can require allowances that consume years of rent. A pro forma that omits TI and leasing commissions is not an underwriting.

Hospitality and senior housing are operating businesses wearing real estate costumes. They are valued on business performance, staffed daily, and carry risk that has nothing to do with the building.

2.1 Speculative, Build-to-Suit, and Owner-Occupied

Speculative. Built without a committed tenant or buyer. Highest risk, highest return, and the version most exposed to timing. The lender will require more equity and often a leasing threshold before releasing the takeout.

Build-to-suit. A tenant commits to a lease before construction begins. Financing is straightforward because the income stream is contracted. The tenant's requirements drive the program and often the schedule, and the penalty for late delivery is contractual and specific.

Owner-occupied. No leasing risk because the owner is the user. Financing looks more like corporate credit than real estate credit.

The design and construction implications diverge immediately. A speculative building is designed to be leasable to an unknown user, which means flexible, generic, and finished to a shell-plus-allowance standard. A build-to-suit is designed to one user's specification, with their standards and their acceptance criteria. If you have only worked on one of these, the other will feel like a different industry.

Section 3 - The Deal Lifecycle and the Five Gates

3.1 The Gates

A development deal advances through a series of decision points at which the money becomes progressively harder and the option to walk away becomes progressively more expensive. Everything in Part 1 is organized around these.

Gate	Question	What commits	Reversibility
Gate 0	Is this worth looking at?	Staff time	Total
Gate 1	Do we tie it up?	Refundable deposit, initial legal	Nearly total
Gate 2	Do we go hard?	Deposit becomes non-refundable; diligence spend is sunk	Losing the deposit
Gate 3	Can we build what we underwrote?	Entitlement spend, design at risk, carry	Very expensive
Gate 4	Do we close and start?	Land payment, loan closing, guarantees	Effectively none

Gate 0 - Screening. A parcel comes across the desk. The developer spends hours, not dollars: assessor records, GIS, zoning map, a rough rent and cost check. Most deals die here and should. (Module 2)

Gate 1 - Site control. A letter of intent is negotiated, then a purchase and sale agreement is signed with a refundable earnest money deposit and a defined due diligence period. The developer now has the exclusive right to study the site. This is cheap and it is the whole point of the structure - it buys the option to investigate before committing. (Module 7)

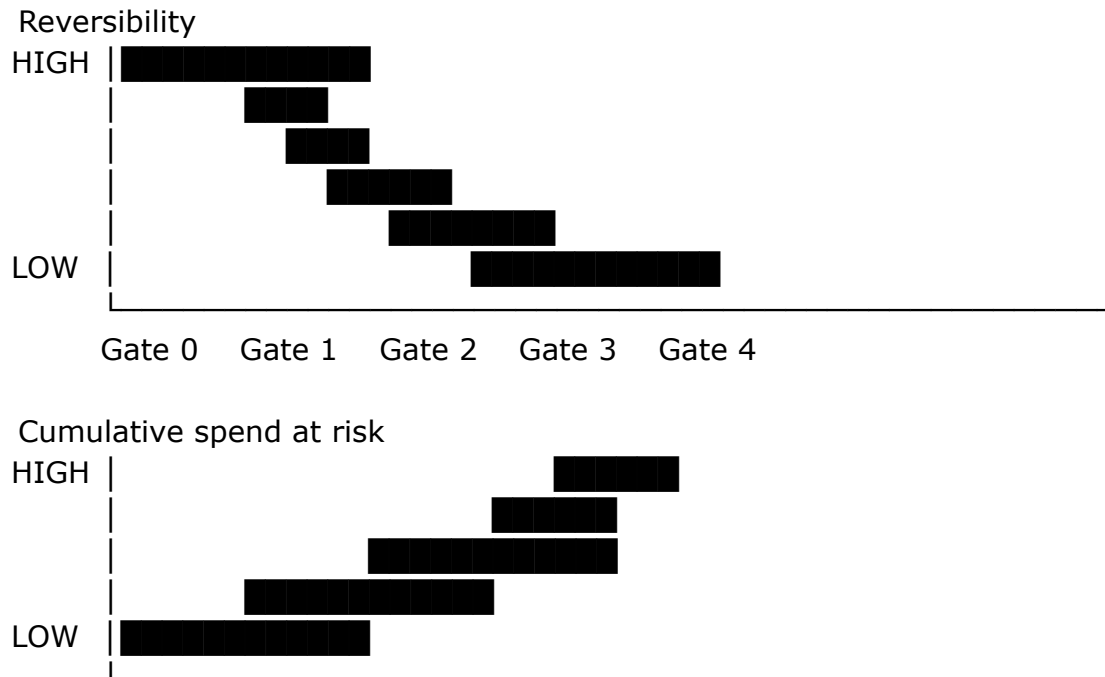
Gate 2 - Going hard. The due diligence period expires. The earnest money becomes non-refundable. Every consultant report from Modules 3, 4A, and 4B must be in hand and resolved before this date, or the developer is guessing with real money. Extensions are commonly negotiated and usually cost additional non-refundable deposit. This is the single most consequential date in the deal.

Gate 3 - Entitlement. The discretionary approval is granted or denied. Binary, political, and frequently slower than any schedule contemplates. Sophisticated developers try to place this before Gate 2 by negotiating an entitlement contingency; sellers resist, and price accordingly. (Module 3)

Gate 4 - Closing and notice to proceed. Land is purchased, the construction loan closes, equity funds, guarantees are signed, and the general contractor is authorized to begin. Part 2 begins here. (Modules 6, 7, 8)

3.2 The Money Hardening Curve

The reason the gates matter is that spend and reversibility move in opposite directions.



The crossing point is Gate 2. Before it, information is cheap and commitment is low. After it, commitment is high and new information is a problem rather than a decision input. Nearly every catastrophic development outcome traces back to information that arrived after Gate 2 and should have arrived before it.

3.3 Who Is in the Room, and When

Gate 0 — Developer, land broker

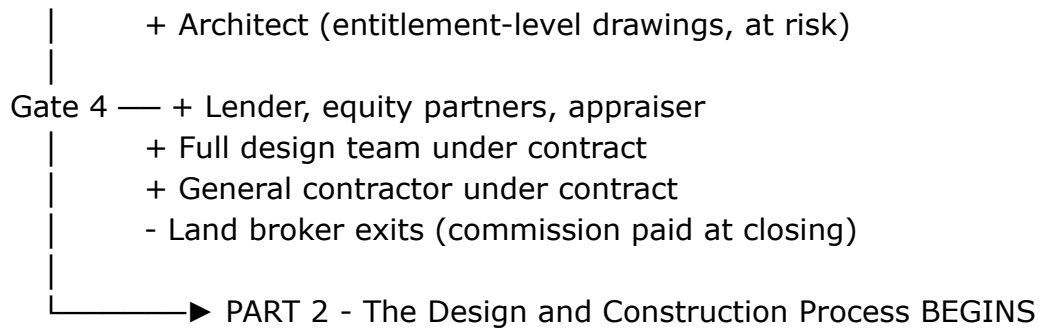
Gate 1 — + Real estate counsel, title company
 + Land use attorney (zoning opinion)

DUE DILIGENCE PERIOD

+ Surveyor, geotech, environmental
 + Civil engineer, market study, PCA (if improved)
 + Architect (test fit), GC (conceptual pricing)

Gate 2 — Deposit goes hard. Diligence consultants largely exit.

Gate 3 — + Entitlement consultant, public agencies



Note the handoff pattern. The parties who know the most about the site - the geotechnical engineer, the environmental consultant, the surveyor - do their work and largely leave before the design team arrives. Their reports are the design team's only inheritance. Whether those reports are transmitted completely, and whether anyone reconciles them against the design, is the subject of Module 8.

Section 4 - Vocabulary Collisions

Development and construction share a vocabulary and disagree about what several words mean. Fix these now.

Term	In development	In construction
Feasibility study	Does the deal produce an acceptable return? A financial analysis.	Sometimes used for pre-design investigation of physical or code viability.
Program	Unit mix, rentable area targets, parking count - the revenue-generating configuration.	The owner's project requirements: spaces, adjacencies, performance criteria.
Due diligence	A contractually defined period with a hard expiration date, after which money is at risk.	Generic care and investigation. No fixed date.
Development	The whole enterprise of creating the asset.	Sometimes the site work only.
Owner	A single-purpose entity answerable to LPs, a lender, and a guarantor.	A single party with decision authority in the contract documents.
Project manager	The developer's employee running the deal.	The GC's office-side manager, or the architect's PM. Three different jobs, one title.
Closing	The purchase transaction. Occurs before construction.	Closeout. Occurs after construction.
Contingency	An underwriting reserve, watched by the lender and the equity.	A budget line within the construction estimate.

Hard costs

Bricks and sticks:
construction contract
value.

Not standard CDT usage; the
estimate is organized by
UniFormat or MasterFormat
instead.

Approval

An entitlement granted by
a legislative or
quasi-judicial body.

A permit or a submittal review
action.

The one to watch is **program**. When a developer says the program is fixed, they mean the revenue model is fixed and cannot move without re-underwriting the deal. When an architect hears the program is fixed, they hear that the space list is settled and the design can proceed. Those are compatible statements right up until someone proposes a change that alters rentable area, at which point they are not.

Section 5 - Artifacts Produced in This Phase

Documents you should be able to recognize by the end of Part 1:

Artifact	Prepared by	Purpose
Offering memorandum	Seller's broker	Market the property
Screening memo	Developer	Gate 0 go/no-go
Market study	Market analyst	Demand support
Zoning report / opinion	Land use attorney	What may be built
ALTA/NSPS survey	Surveyor	Boundaries, easements
Geotechnical report	Geotech engineer	Foundation basis
Phase I ESA	Environmental consultant	Contamination screen
Property condition assessment	PCA consultant	Existing building condition
Test fit / yield study	Architect, civil	What physically fits
Development budget	Developer	Total cost
Pro forma	Developer	Return analysis
Sources and uses	Developer	Capital structure
Appraisal	Appraiser, engaged by lender	Value for loan sizing
Letter of intent	Buyer, via counsel	Non-binding deal terms
Purchase and sale agreement	Counsel	Binding acquisition terms
Title commitment	Title company	Ownership and encumbrances

Loan term sheet

Lender

Debt terms

Owner's project
requirements

Owner / owner's rep

Design brief

Section 6 - The Two Deals

Two projects run through every module in Part 1. Both are composites, and both are built from ordinary situations rather than dramatic ones.

Deal A - Kipling Row (residential)

A 3.2-acre parcel on a suburban arterial. Current improvement is a closed church building of roughly 12,000 square feet, vacant two years, plus a large surface parking lot. The congregation has merged with another and the property is listed by a local land broker at \$1.95 million.

Current zoning is single-family residential, which permits about 11 detached lots after right-of-way and stormwater. The developer's thesis is a rezoning to a planned unit development supporting 34 attached townhomes for sale, which is roughly the density required to make the land price work.

- **Owner type:** merchant developer, three-person shop, build-to-sell
- **Product:** for-sale attached residential
- **Delivery risk concentration:** entitlement (Gate 3)
- **Capital:** local high-net-worth equity, regional bank construction loan

Deal B - Six Corners (commercial)

A 78,000 square foot two-story suburban office building, completed 1984, on 6.1 acres with surface parking. Occupancy is 40 percent across three tenants, all on leases with under three years remaining. Asking price is \$4.4 million, well below replacement cost, and the seller is an out-of-state fund winding down.

The buyer is a small value-add sponsor. The initial thesis is an office-to-residential conversion. That thesis will not survive Module 4B.

- **Owner type:** value-add sponsor, five-year hold
- **Product:** to be determined by diligence - the point of the example
- **Delivery risk concentration:** existing conditions and code triggers (Gate 2)
- **Capital:** small fund equity, debt fund bridge loan

What Happened at Gate 0

Kipling Row. Two hours of work. County assessor confirmed acreage and ownership. The zoning map confirmed single-family, and the comprehensive plan future land use map showed the corridor designated for medium-density residential - meaning a rezoning would be consistent with adopted policy, which is the difference between a hard rezoning and a plausible one. Three recent townhome closings within a mile supported a base price around \$415,000. A rough check at 34 units suggested the land price was defensible if, and only if, the rezoning succeeded.

Advanced to Gate 1, with the entire thesis resting on a political approval nobody had yet asked for.

Six Corners. Ninety minutes. The price worked out to roughly \$56 per square foot against a replacement cost above \$200, which is the number that draws every value-add buyer to every vacant office building in America. Zoning was a commercial district that permitted multifamily by right, which appeared to remove entitlement risk entirely and made the conversion thesis look clean. The sponsor noted the 1984 vintage and the 40 percent occupancy as diligence items and moved on. **Advanced to Gate 1**, with two facts unexamined: the depth of the floor plate, and what a change of occupancy would trigger.

Both deals passed Gate 0. Only one of them passed for the right reasons.

Failure Modes

Reading the offering memorandum as underwriting. The OM is written by the seller's agent to produce a sale. Its rent assumptions, its stabilized occupancy, and its "value-add opportunity" narrative are advocacy. Extract the parcel number and the asking price and rebuild everything else.

Treating a vendor's report as advice. A Phase I ESA tells you whether recognized environmental conditions exist. It does not tell you whether to buy the site. Consultants answer the question in their scope, defensively, because lenders rely on their reports.

Confusing entitlement risk with code risk. These are different regulators, different law, different timing, and different fixes. Entitlement risk is often binary and cannot be engineered away.

Mistaking the developer for the money. The developer directs the project and does not unilaterally control it. LP major-decision rights and lender approval rights are real constraints that appear, from the design side, as unexplained delay.

Assuming the design team inherits what diligence learned. The geotech, the survey, and the environmental report are produced before the architect is engaged, by consultants who are gone by the time design begins. Transmission is a deliberate act, and when it does not happen, the consequences appear during construction.

Underestimating the pre-development burn. The spend between Gate 1 and Gate 4 is entirely at the developer's risk and can reach several hundred thousand dollars on a modest deal. That fact drives the urgency, the thin drawings, and the schedule pressure that the design and construction team will experience later and interpret as unreasonableness.

Optional: Try This

Pick a parcel you can see from your window or drive past regularly. Pull it up on your county assessor's site and your municipal GIS. In under thirty minutes, answer: who owns it, what is it zoned, what does the comprehensive plan say it should become, and what is the assessed value per acre relative to a sale nearby. That is a Gate 0 screen. Most of them end in "no," which is the correct answer and the reason the gate exists.

Next - Module 2: Market and Site Selection. Whether demand exists, how highest and best use is actually tested, and where the numbers behind a Gate 0 screen come from.

1.2 - Market and Site Selection: Is There Demand?

Most people enter development through a site. Someone knows a piece of land, or inherits a building, or drives past a vacant lot every morning and starts imagining what could go there. The site comes first and the use is reverse-engineered to fit it.

Professionals do it the other way around. They establish that unmet demand exists for a specific product at a specific price point in a specific geography, and then go looking for sites that can serve it.

A site is not an opportunity. A site plus demonstrated demand plus a legal path to build is an opportunity. This module covers the demand half. Module 3 covers the legal path.

The work here is deliberately cheap. Everything in this module can be done from public data and a subscription you may already have, in hours rather than weeks, before a dollar of earnest money or a single consultant invoice. That is the point of Gate 0: eliminate the deals that cannot work before you start paying to find out.

Section 1 - Geography: Market, Submarket, Trade Area

Demand is local, and "local" means something different for every product type. Getting the geography wrong invalidates everything downstream, because you will be comparing your project to the wrong competition.

Market. The metropolitan statistical area (MSA). Denver-Aurora-Lakewood. Phoenix-Mesa-Chandler. This is the unit at which employment, migration, and most published economic data are reported. It is almost always too large to make a development decision on. A metro can have 3 percent vacancy in aggregate and 19 percent vacancy in the four blocks around your site.

Submarket. A subdivision of the metro used by the data providers and understood by brokers - "Southeast Suburban," "North I-25 Corridor," "Central Business District." Submarket boundaries are conventions, not law, and different providers draw them differently. This is the level at which supply and vacancy data becomes meaningful.

Trade area. The actual geography from which your customers will come. This is the one you have to define yourself, and it is product-specific:

Product	Trade area is defined by	Typical shape
Multifamily	Commute tolerance to employment nodes, school attendance boundaries	3–5 mile radius, distorted by highways and barriers
For-sale residential	School district, price band, commute	Follows school boundaries closely
Neighborhood retail	Drive time	5–10 minute drive time polygon
Regional retail	Drive time	20–30 minute drive time polygon
Industrial / logistics	Labor shed plus distance to distribution nodes	30–45 minute labor drive time; highway interchange proximity
Self-storage	Drive time, very tight	1–3 mile radius, or 5 minutes
Senior housing	Adult children's residence as much as the resident's	5–10 miles, income-qualified

Note that a trade area is a polygon, not a circle. Rivers, interstates, rail lines, and mountains cut trade areas in half. A radius drawn across a freeway with no crossing for two miles is fiction. GIS tools will produce drive-time polygons; use those rather than radii wherever the geography is not flat and gridded.

Barrier analysis. The most valuable five minutes in site selection is identifying what prevents a competitor from building next door. Barriers can be physical (no developable land remaining), regulatory (a moratorium, a restrictive zoning district, an infrastructure concurrency requirement), or economic (land basis too high to support new construction). A submarket with no barriers to entry will have its rent growth competed away by new supply, which is exactly the point at which your building is leasing up.

Section 2 - Demand Drivers

Demand for real estate is derived demand. Nobody wants an apartment; they want to live somewhere near a job they have. Every demand analysis therefore traces back to some underlying economic activity.

2.1 Employment

The primary driver for nearly every product type.

- **Total employment and growth rate.** Reported by BLS at the metro level. The trend over five to ten years matters more than any single year.
- **Employment by sector.** A metro that added 40,000 jobs in warehousing is a different market than one that added 40,000 jobs in professional services. The first drives industrial and workforce housing; the second drives office and higher-end multifamily.
- **Concentration risk.** A metro where one employer or one sector is 30 percent of the base is one corporate decision away from a different market. Energy towns, single-mill towns, and government towns all exhibit this.
- **Major employers and announced expansions.** Announcements are not jobs. Track them, discount them heavily until construction starts.
- **Wage levels by sector.** Determines what rent or price the demand can actually support, which is the link between employment data and your pro forma.

2.2 Population and Migration

- **Population growth,** decomposed into natural increase (births minus deaths) and net migration. Migration-driven growth is more volatile and more responsive to economic conditions; natural increase is slow and predictable.
- **Domestic in- and out-migration,** available at the county level from IRS Statistics of Income data, which also reports the aggregate income of migrating households. Who is arriving matters as much as how many.
- **Age cohorts.** The 20–34 cohort drives rental demand. The 30–45 cohort drives for-sale demand. The 75-plus cohort drives senior housing, on a roughly fifteen-year lag from the 60-plus cohort you can already count.

2.3 Household Formation

The most direct driver of housing demand, and the one most often skipped because it requires a small amount of arithmetic rather than a single lookup.

Population growth alone does not create housing demand - households do. A metro adding 20,000 people at an average household size of 2.4 needs roughly 8,300 additional housing units, before accounting for obsolescence, second homes, and vacancy. Household size is itself moving: declining household size creates housing demand even in a metro with flat population.

Then split the demand by tenure. Ownership rate in the relevant cohort, applied to new households, gives you the for-sale versus rental split. That split is sensitive to mortgage rates and price-to-income ratios, which is why a rate move changes rental demand without changing population at all.

2.4 Income Distribution

Not median income. Distribution.

A project's target rent or price implies a household income, conventionally at a 30 percent rent-to-income ratio for rental and roughly a 3x to 4x price-to-income ratio for ownership. Run that backwards from your target rent to get the required income, then count how many households in the trade area actually have it. That count is the qualified demand pool.

Capture rate is the fraction of that pool you need. If your trade area contains 6,000 qualified renter households, annual turnover is 40 percent, and you need to lease 200 units in a year, you need to capture 8 percent of the households moving in a year. Whether that is reasonable depends on what else is available to them, which is Section 3. A required capture rate above roughly 10 to 15 percent should make you uncomfortable; above 20 percent, the deal is asking the market to behave unusually.

For income-restricted and affordable products, the arithmetic runs against Area Median Income as published by HUD, and the qualifying bands are defined rather than chosen.

2.5 Traffic and Access

Primarily a retail and commercial driver, published by state DOTs as annual average daily traffic (AADT) counts by segment.

- Raw counts matter less than counts combined with the ability to access the site: median cuts, signalization, turn lanes, right-in/right-out constraints.
- Corner sites at signalized intersections carry a real premium for a reason.
- Traffic counts trend. A segment with declining counts over a decade indicates a corridor losing relevance.
- For residential, high traffic counts are a liability rather than an asset, except at the corridor edge.

2.6 Logistics Geography

For industrial and distribution, the relevant drivers are almost entirely locational.

- Distance and drive time to interstate interchanges, intermodal facilities, ports, and airports
- Population reachable within a one-day and two-day truck drive
- Labor availability within a 30- to 45-minute commute, at warehouse wage levels
- Utility capacity, particularly power - which for modern distribution and any data-adjacent use has become the binding constraint rather than land

Section 3 - Supply

Demand analysis without supply analysis is optimism. The question is never "is there demand" - there is always some demand. The question is whether demand exceeds what is already being built to serve it.

3.1 The Vocabulary

Inventory. Total existing competitive stock in the submarket, in units or square feet.

Vacancy rate. Share of inventory unoccupied. Compare to the submarket's long-run average, not to zero. Every product type has a structural vacancy rate below which it does not go; for multifamily this is typically 4 to 6 percent, representing normal turnover.

Net absorption. Change in occupied space over a period. This is the demand measurement that matters, because it counts actual leases rather than sentiment. Negative absorption means tenants are leaving faster than arriving.

Deliveries. Completed and placed in service during the period.

Pipeline. Everything not yet delivered, which should always be reported in stages rather than as one number (see 3.2).

Months of supply. Pipeline divided by monthly net absorption. A submarket with 1,800 units in the pipeline absorbing 90 units a month has 20 months of supply. This single ratio is the most useful supply metric there is, and it is the one most frequently omitted from offering memoranda.

Concessions. Free rent, reduced deposits, buyer incentives, tenant improvement allowances above market. Concessions are how a market softens before asking rents move, which makes them a leading indicator.

Asking rent versus effective rent. Effective rent nets out concessions and is the only number worth underwriting. A building advertising \$2,100 with two months free on a twelve-month lease is achieving roughly \$1,750. Sellers quote asking rent. Always.

Shadow supply. Competitive stock that does not appear in any pipeline: single-family homes offered for rent, condos rented by owners, sublease space, accessory dwellings. In some submarkets this exceeds the formal pipeline.

3.2 Reading a Pipeline Honestly

A pipeline reported as a single number is useless, because the stages have wildly different probabilities of completion.

Stage	Probability of delivering	Treat as
Announced / rumored	Low	Note it, count it at zero
Entitlement application filed	Modest	Count partially, with a long timeline
Entitled, not permitted	Meaningful	Count at a discount
Permitted	High	Count fully
Under construction	Very high	Count fully, with a known delivery date

Three tests to apply to anything in the pipeline:

1. **Does it actually compete?** A luxury high-rise and a garden-style workforce product in the same submarket are not the same supply. Compare product type, price point, and unit mix.
2. **When does it deliver relative to my lease-up?** A project delivering eighteen months before you is a competitor during your lease-up. One delivering three years after you is the next owner's problem.
3. **Can I verify it independently?** Building permit data from the jurisdiction is the most reliable pipeline source there is, because a permit means someone paid a fee. Press releases and broker pipeline reports include projects that will never happen.

The discipline that separates good underwriting from bad is counting supply you would rather not count. Every developer is inclined to discount competing projects and to believe their own will deliver. Assume the reverse.

Section 4 - Comparables and Adjustment Logic

Comparables AKA Comps establish what the market is actually paying. All three types answer the same structural question - what would a buyer or tenant pay for something like this - and all three require adjustment, because nothing is exactly like anything else.

4.1 The Three Types

Rent comps. What competing properties achieve, in effective rent per unit or per square foot. Establishes the revenue line of your pro forma. Source: shopping the competition directly, plus

commercial data providers. Calling competing properties as a prospective tenant remains the single most reliable method and nobody does enough of it.

Sales comps. What similar improved properties have traded for, expressed as price per unit, price per square foot, and implied capitalization rate. Establishes your exit value. Source: county recorder for the transaction, data providers for the terms.

Land comps. What comparable sites have traded for. This is the hardest of the three because land is the least standardized asset there is. Express land comps in the denominator that actually drives value:

- **Per acre or per square foot** - useful only for like-zoned, like-condition sites
- **Per entitled unit ("per door")** - the meaningful metric for residential land
- **Per buildable square foot** - the meaningful metric where FAR governs

A four-acre site zoned for 12 units and a four-acre site zoned for 90 units are not comparable on a per-acre basis and are roughly comparable on a per-door basis. This is why entitlement creates value: it changes the denominator.

4.2 The Adjustment Sequence

Appraisal practice applies adjustments in a specific order, because some are multiplicative on the others. The sequence is worth learning even if you never do it formally, because it identifies what you must adjust for:

1. **Property rights conveyed** - fee simple versus leased fee versus leasehold
2. **Financing terms** - seller financing below market inflates the observed price
3. **Conditions of sale** - foreclosure, estate sale, related-party transfer, a 1031 buyer under a deadline paying above market
4. **Expenditures immediately after purchase** - a site sold with \$400,000 of required demolition has an effective price higher than its recorded price
5. **Market conditions (time)** - adjust older comps forward or back for market movement; this is where most errors occur
6. **Location**
7. **Physical characteristics** - size, condition, vintage, amenity, topography, frontage

Steps 1 through 5 come first and in order. Steps 6 and 7 are applied to the result.

Two practical rules. First, fewer and closer comps beat more and distant ones; three genuinely comparable transactions are worth more than twelve adjusted heavily. Second, if your adjustments total more than about 25 percent gross, the comp is not comparable and you are constructing the answer you want.

Section 5 - Highest and Best Use

This is the conceptual anchor of Part 1, and the reason its modules are ordered the way they are.

Highest and best use is a formal test from appraisal practice, defined as the reasonably probable use of a property that is legally permissible, physically possible, financially feasible, and maximally productive. The four tests are applied **in that order**, and the order is the entire point: a use that fails an earlier test is eliminated regardless of how well it performs on a later one.

Test	The question	Where Part 1 covers it
1. Legally permissible	May I build it here? Zoning, deed restrictions, easements, private covenants, environmental regulation.	Module 3
2. Physically possible	Will the site or structure support it? Size, shape, soil, topography, utilities, existing building capacity.	Modules 4A, 4B
3. Financially feasible	Does it produce a positive return? Revenue from Module 2, cost from Module 5, capital from Module 6.	Modules 2, 5, 6
4. Maximally productive	Among the uses that survive tests 1 through 3, which produces the greatest value?	The whole exercise

Read that table again. It is the table of contents for Part 1.

5.1 As Vacant and As Improved

The test is applied twice to any property with an existing building, and the comparison of the two answers is what determines whether the building survives.

Highest and best use as vacant asks: if this site were empty, what would be built on it? This establishes land value.

Highest and best use as improved asks: given the building that exists, what is the most productive thing to do with it - continue the current use, renovate, convert, or demolish?

The existing structure contributes value only if the improved use is worth more than land value minus demolition cost. When the best use as vacant is worth more than the best use as

improved, the building is worth less than nothing and is a demolition line item. This is an uncomfortable conclusion that is reached correctly all the time, and reached incorrectly even more often, because buyers of underperforming buildings tend to assume the structure is an asset because they paid for it.

5.2 Interim Use

A use is not required to be permanent. A site whose highest and best use is a mid-rise in eight years, when the light rail line opens, may have an interim highest and best use as surface parking or self-storage today. Land banking is the explicit version of this strategy. Carry cost - taxes, insurance, debt service on the land - is what makes or breaks it.

5.3 Why This Matters to People Coming From Design or Construction

Highest and best use is the reason the program arrives at the design team fixed. It was not a preference. It was the output of a four-part elimination against legal, physical, and financial constraints, and the surviving use is usually the only one that worked. When a developer says the unit count cannot change, this is generally what is behind it.

Section 6 - Data Sources and Reliability Tiers

Every number you will use has a provenance, a lag, and a bias. Sort them before you use them.

Tier 1 - Primary and authoritative

Direct from a government body of record. Low bias, variable freshness, free.

Source	Provides	Lag
County assessor	Ownership, parcel size, assessed value, improvement data, tax history	Days to months
County recorder / clerk	Deeds, recorded sale prices (in disclosure states), liens, easements, plats	Days
Municipal / county GIS	Parcel geometry, zoning overlay, floodplain, utilities, aerials, drive-time tools	Varies

Building department permit data	The most reliable supply pipeline available	Days to weeks
Census / ACS	Population, households, household size, tenure, income distribution, age cohorts	1–2 years; ACS is a rolling estimate with margins of error
BLS	Employment by sector, wages, unemployment	1–3 months
IRS SOI migration	County-to-county migration flows with income	~2 years
HUD	Area Median Income, fair market rents	Annual
FEMA	Flood zones	Varies
State DOT	Traffic counts, planned improvements	1–2 years

The assessor and the GIS layer together will answer most Gate 0 questions in under half an hour and cost nothing. Note that roughly a dozen states are non-disclosure states where sale prices are not public record, which materially changes how comps are sourced.

Tier 2 - Commercial aggregators

Aggregated, normalized, and substantially more convenient. Reliable in dense markets, thin in secondary and rural ones, and always worth spot-checking against Tier 1.

CoStar, Yardi Matrix, RealPage, CBRE and JLL research, Placer.ai for foot traffic, Dodge and ConstructConnect for the construction pipeline, MLS for residential comps, Zonda and John Burns for housing.

These data providers will be explored in depth in Part 3 - Tech and AI.

Two caveats that matter. These providers define submarket boundaries themselves, so numbers from two providers will not reconcile. And their coverage is a function of how many properties report to them, which means a "submarket vacancy rate" in a thin market can rest on a handful of properties.

Tier 3 - Interested parties

Use for facts, never for conclusions.

The offering memorandum. The seller's pro forma. Chamber of commerce and economic development materials. Developer press releases. Broker market reports, which are produced by firms whose revenue depends on transaction volume.

None of this is dishonest. All of it is advocacy. The parcel number in an OM is a fact. The projected rent growth is a position.

6.1 Reconciliation

When two sources disagree, and they will, prefer the one closest to a recorded transaction or a paid fee. A recorded deed beats a reported sale. A pulled permit beats an announced project. A signed lease beats an asking rent.

Section 7 - Site Selection Criteria by Product Type

Once demand is established, screening sites becomes a checklist exercise. The criteria differ sharply by product, which is why a site that is wrong for one use is often right for another.

Multifamily

- 3 to 10 acres for garden or wrap; under 2 acres for mid-rise
- Proximity to employment nodes and transit; commute under 30 minutes to major employment
- Retail and grocery within convenient distance
- School quality, which affects rent even for renters without children
- Zoning permitting the required density, or a plausible path to it
- Site geometry allowing efficient parking; irregular sites kill efficiency
- Not directly adjacent to incompatible uses

For-sale residential

- School district is frequently the single dominant variable
- Lot yield after right-of-way, stormwater detention, and open space requirements - typically 20 to 35 percent of gross acreage is lost
- Topography, because grading cost scales quickly
- Absorption pace in the price band; a subdivision that takes six years to sell out has a carry problem
- Availability of comparable recent closings to support appraisals for buyers' mortgages

Industrial / logistics

- Interchange access, typically within 3 to 5 miles
- Site depth adequate for truck court and trailer parking; a shallow site cannot serve a modern building regardless of acreage

- Rectangular geometry, minimal grade
- Power capacity and, increasingly, the utility's interconnection timeline
- Labor availability at target wage within a 30- to 45-minute commute
- Zoning permitting the use plus outdoor storage, and truck routing that works

Retail

- Traffic counts and, more importantly, ingress and egress
- Visibility and signage rights
- Co-tenancy: what anchors the trade area and whether it is healthy
- Trade area demographics matched to the concept
- Parking ratio required by both code and the tenant's own standards

Self-storage

- Tight trade area with rooftop count and household density
- Visibility from an arterial
- Existing competitive supply within 3 miles, which is the whole analysis
- Low land basis, because revenue per acre is modest

7.1 The Screening Matrix

For any product, the criteria become a weighted matrix so that sites can be compared rather than argued about. Assign weights by importance, score each site, and compare. The value of the exercise is not the total score - it is that the weights force you to state in advance what matters, before you have fallen in love with a particular site.

Include knockout criteria that override the score entirely. A site in a floodway, a site with no sewer access and no path to it, a site with a required capture rate above 20 percent: these fail regardless of how well they score elsewhere.

Section 8 - The Gate 0 Screen

The deliverable for this module is one page. It should take under two hours to produce and it should end in "no" most of the time.

Site facts. Parcel number, acreage, ownership, current improvements, asking price, price per acre and per buildable unit.

Demand. Trade area definition, qualified household count, required capture rate, primary demand driver in one sentence.

Supply. Competitive inventory, current vacancy against the long-run average, pipeline by stage, months of supply.

Pricing. Three rent or sale comps with effective, not asking, figures. Implied revenue for the proposed program.

Preliminary highest and best use. The use being proposed, and which of the four tests is most at risk.

Knockouts. Anything disqualifying, and anything that must be true for the deal to work.

Recommendation. Advance to Gate 1, pass, or revisit under stated conditions.

The last line is the one that matters. A screen that does not end in a decision is a research exercise.

Artifacts

Artifact	Prepared by	Purpose
Screening memo	Developer	The Gate 0 decision
Site screening matrix	Developer	Compare candidate sites against stated criteria
Market study	Independent market analyst	Third-party demand support, often required by lenders and equity
Broker opinion of value	Broker	Informal valuation; free, non-certified, and interested
Demographic and trade area report	Data provider or GIS	Household counts, income distribution, drive-time geography
Rent comp survey	Developer, leasing broker	Effective rent support for the pro forma

A note on the market study. It is a formal third-party document, ordered when a lender or equity partner requires independent demand support, and it typically costs several thousand dollars and takes several weeks. It comes after Gate 1, not at Gate 0. The screening work in this module is what determines whether it is worth ordering one.

Failure Modes

Analyzing the market instead of the trade area. Metro-level statistics will support almost any project. The relevant question is always what is happening within the polygon your residents or customers will actually come from.

Using asking rents. Concessions are how markets soften first. Underwriting asking rent in a market with two months free overstates revenue by roughly 15 percent, which is usually the entire margin.

Counting the pipeline optimistically. Discounting competitors while believing your own project will deliver on schedule is the standard cognitive error of the industry. Permit data is the corrective.

Ignoring shadow supply. In markets with substantial single-family rental or condo-rental activity, formal multifamily inventory is not the competitive set.

Reasoning from replacement cost. "It's selling at a third of replacement cost" is a statement about construction costs, not about demand. If nobody wants the use, the building will trade below replacement cost forever, and correctly so.

Skipping highest and best use as vacant. Buyers of existing buildings routinely assume the structure has value because they paid for it. The test is whether the improved use beats land value net of demolition, and sometimes it does not.

Required capture rates that nobody sanity-checks. The arithmetic runs in five minutes and disqualifies more bad deals than any other single check in this module.

Falling in love before screening. The screening matrix exists to make criteria explicit before a preferred site distorts them. Weights assigned after the fact are not criteria; they are rationalization.

What Happened Next

Kipling Row

The demand work was straightforward and mostly favorable.

The trade area was defined as the school attendance boundary plus a three-mile radius, cut on the west by a rail line with no crossing for 1.8 miles - which removed about a third of the nominal radius. Within the remaining polygon, the primary driver was move-up and move-down buyers being priced out of detached product: detached median in the trade area had reached \$612,000, while attached product was closing in the low \$400s. Household formation in the

30–45 cohort supported roughly 340 for-sale units of annual demand in the price band across the submarket.

Three townhome closings within a mile, adjusted for time and unit size, supported a base price of \$415,000 and a blended average of \$437,000 across the proposed mix.

The supply check was the useful part. County permit data showed a 60-unit attached project permitted two miles north, delivering in approximately eighteen months - meaning it would be in active sales throughout Kipling Row's own sell-out. Combined pipeline against a submarket absorption pace of roughly 9 attached units per month produced about 13 months of supply, which is tight but tolerable. At 34 units, Kipling Row's required capture was under 10 percent of annual trade area demand in the band. Acceptable.

Highest and best use, as vacant: detached single-family, which is what the site legally permits, yields about 11 lots after right-of-way and detention. At the \$1.95 million asking price that is roughly \$177,000 per lot on land, against finished lot values that do not support it. Detached is legally permissible, physically possible, and **not financially feasible at this price.**

Attached at 34 units is financially feasible, physically possible, and **not legally permissible.**

That is the deal in one sentence. The entire thesis is the conversion of a use from impermissible to permissible, and every dollar of value creation lives in Module 3. **Advanced to Gate 1** with the understanding that the market work, however favorable, is not the risk.

Six Corners

The demand work here produced three findings, two of which the sponsor expected and one of which did not register until much later.

Office is not coming back in this submarket. Vacancy at 24.1 percent against a long-run average near 12. Six consecutive quarters of negative net absorption. Effective rents down 18 percent from peak once concessions - averaging over a year free on five-year terms - were netted out. Roughly 30 percent of the submarket's vacant space was sublease. The conclusion that the existing use is failing was correct, obvious, and confirmed everything the sponsor already believed.

Multifamily demand in the trade area is real. Vacancy at 5.2 percent, positive absorption, effective rents supporting roughly \$1.95 per square foot on the unit sizes a conversion would produce. A conversion, on market grounds alone, works. This is the finding that kept the sponsor moving forward.

Industrial is the tightest product in the submarket, and nobody was looking at it.

Vacancy at 3.4 percent. Small-bay flex and light industrial under 20,000 square feet had essentially no availability, with rents up 31 percent over three years. The site sits 2.3 miles from an interstate interchange with adequate depth for a truck court. This finding sat in the screening memo as a single line and was not acted on for another four months.

The land comp work produced the number that should have been the headline. Industrial-zoned land in the submarket was trading around \$400,000 per acre. At 6.1 acres, the dirt alone is worth roughly \$2.4 million against a \$4.4 million asking price - meaning the sponsor was paying approximately \$2.0 million, or \$25 per square foot, for the improvements.

Highest and best use as vacant: something worth at least \$2.4 million in land. Highest and best use as improved: unknown, and entirely dependent on whether the 1984 structure can be reused for anything.

Advanced to Gate 1 with the market case for a conversion appearing strong, and with the actual question - whether \$25 per square foot is a bargain or an overpayment for a building that cannot be converted - deferred to physical due diligence.

Optional: Try This

Take the parcel you screened after Module 1 and build the one-page Gate 0 screen from Section 8.

Everything required is free. County assessor for the parcel facts. Municipal GIS for the zoning, floodplain, and a drive-time polygon. Census QuickFacts or ACS table B25118 for household counts and income distribution in the tract or place. The building department's permit search for the pipeline. Three comps from the MLS, the recorder, or a listing site.

Work backwards from a target rent or price to a required income, count the qualified households, and compute your required capture rate. Then look up how many competing units are permitted within two miles.

Most of the time the answer is no. Write the "no" down with its reason. A file of screened-and-rejected sites with stated reasons is how developers build market knowledge, and it is the closest thing to reps that this phase offers.

Next - Module 3: Legal and Regulatory Due Diligence.

1.3 - Legal and Regulatory Due Diligence: What Are You Allowed to Build?

Building code risk is technical. It has a correct answer. A qualified professional can determine it in advance, an alternate means and methods request exists for the edge cases, and a competent design team resolves nearly all of it before permit. When it goes wrong, it goes wrong expensively but solvably: redesign, resubmit, pay for the delay.

Entitlement risk is none of those things. It is political, it is frequently discretionary, it is decided in a public room by people who are accountable to voters rather than to a code, and it is often binary. There is no alternate means and methods request for a city council that votes no. The deal does not get more expensive. The deal ends, and the pre-development spend is gone.

This is the first test of highest and best use - legally permissible - and it is applied first for a reason. A use that fails here cannot be rescued by excellent design, aggressive value engineering, or a better capital structure.

The analysis in this module happens during due diligence, before Gate 2. The approval itself, when one is required, lands at Gate 3. The gap between those two points is where developers lose money.

Section 1 - Reading a Zoning Ordinance

A zoning ordinance has two components that must be read together: **the map**, which assigns every parcel to a district, and **the text**, which states what may be done in each district. Neither is useful alone. The map tells you the district; the text tells you what the district means.

Every ordinance is organized differently, but all of them regulate the same four things.

1.1 Use

What activities are permitted. Ordinances express this as a use table with columns for each district and rows for each use, coded as:

- **P - Permitted by right.** Allowed, subject to meeting the dimensional standards.
- **C - Conditional / special use.** Allowed only with a discretionary approval, after a hearing, with conditions attached.
- **A - Accessory.** Allowed only as subordinate to a principal use.
- **Blank or X - Prohibited.**

Two traps. First, use definitions are frequently narrower than ordinary English: "multifamily dwelling," "townhouse," "rowhouse," and "two-family dwelling" can be four distinct uses with four different treatments. Second, an ordinance may permit a use and then attach **use-specific standards** buried in a separate article - minimum unit sizes, separation requirements from other like uses, limits on hours, additional buffering. Finding the P in the table is the beginning of the analysis, not the end.

1.2 Density and Intensity

How much of the use is allowed.

- **Density** - dwelling units per acre, for residential. Sometimes expressed gross (of the whole site) and sometimes net (excluding right-of-way, steep slopes, floodplain, or detention). The difference is routinely 20 to 30 percent of yield.
- **Floor area ratio (FAR)** - total building floor area divided by lot area. An FAR of 2.0 on a 40,000 SF lot permits 80,000 SF of building, in any configuration the other rules allow. Check the definition of floor area: whether parking, mechanical space, and below-grade area count varies by jurisdiction and changes the answer materially.
- **Lot coverage** - the share of the lot that may be covered by building footprint.
- **Impervious surface coverage** - the share that may be covered by anything that sheds water, including pavement. This constrains parking as much as building and is frequently the binding constraint on suburban sites.

1.3 Bulk and Placement

Where the building may sit and how large it may be.

- **Setbacks** - front, side, rear, and often a separate street-side setback on corner lots. Build-to lines in urban and form-based districts do the opposite, requiring the building to reach the line.
- **Height** - the sleeper item, because how height is measured varies enormously. Some ordinances measure to the roof deck, some to the midpoint of a pitched roof, some to the highest point. Some cap stories in addition to feet. The reference elevation may be existing grade, finished grade, average grade, or a defined grade plane. On a sloping site these produce different answers, and the difference is sometimes a full story.
- **Building separation** on multi-building sites
- **Stepbacks and daylight planes** adjacent to lower-intensity districts
- **Open space and landscaping** - minimum percentages, sometimes with usable-versus-total distinctions, buffer yards, tree preservation and replacement obligations

1.4 Parking and Loading

- **Minimum parking ratios**, by use. Expressed per unit, per bedroom, or per thousand square feet.
- **Maximum ratios**, increasingly common in transit-served districts.

- Dimensional standards: stall size, aisle width, drive aisle geometry, accessible stall count and location.
- **Bicycle parking, loading berths, queuing** for drive-through uses.
- Reductions available for shared parking, transit proximity, or parking studies - usually discretionary.

Parking minimums deserve special attention because they are the constraint most likely to defeat a project that otherwise complies. A site can have the density, the FAR, and the height and still be undevelopable at that intensity because the required parking will not physically fit at surface, and the economics of structured parking are a different deal entirely. This is covered in Module 5.

1.5 The Definitions Article

Read it. Every consequential term - lot area, building height, grade, story, floor area, dwelling unit, family - is defined, and the definitions control. A very large share of zoning analysis errors come from applying the ordinary meaning of a word that the ordinance defines differently.

1.6 The Binding Constraint

Run every dimensional standard and see which one caps the project first. It is rarely the one people assume.

A site zoned for 24 units per acre may yield 18 because the setbacks and required open space leave insufficient buildable area. A site with generous FAR may cap at half of it because the height limit and the required floor plate cannot reach it. A site with adequate density may fail on impervious coverage once parking and drives are counted.

The answer is never "the zoning allows 24 units per acre, therefore we get 24 units per acre." The answer is a test fit, which is Module 5. The purpose of this module is to establish which rules apply.

Section 2 - Beyond the Base District

Base zoning is rarely the whole picture.

Overlay districts. Layered on top of base zoning, adding requirements without replacing the base. Common overlays: historic, floodplain, airport height hazard, corridor design, hillside, transit-oriented, groundwater protection, view protection. An overlay can be more restrictive, or it can grant additional entitlement in exchange for compliance. You must check for them specifically; they often appear on a separate GIS layer that is not displayed by default.

Planned Unit Development (PUD) and planned districts. A negotiated, site-specific zoning district in which the standards are written for that project. This is the primary vehicle for projects

that do not fit any standard district. The developer proposes the standards; the jurisdiction approves them; the approved plan becomes the zoning for that parcel. PUDs offer flexibility at the cost of discretion - everything is negotiable, which means everything is subject to negotiation, including things that would have been by right in a standard district. PUD approval is legislative and carries full political risk.

Form-based codes. Regulate physical form - building placement, frontage type, height in stories, facade composition - rather than use. Common in downtown and corridor redevelopment. Generally more permissive on use and more prescriptive on appearance. They tend to make entitlement more predictable and design less flexible, which is a trade many developers will take.

Specific plans, area plans, and master plans. Adopted sub-area plans that may carry regulatory weight or may be advisory. Determine which. A project consistent with an adopted plan has a substantially easier political path; a project inconsistent with one will be confronted with it at the hearing.

Inclusionary zoning and density bonuses. Requirements to provide affordable units, or incentives offering additional density, height, or reduced parking in exchange for them. Occasionally these are the only path to a feasible density, which makes an affordability requirement an economic input rather than a constraint.

The comprehensive plan. Not regulation, but it matters more than its legal status suggests. The future land use map expresses the jurisdiction's adopted intention for each area. A rezoning consistent with the comprehensive plan is a request to do what the city already said it wanted. A rezoning inconsistent with it is a request to override adopted policy, in public, against staff recommendation. The difference in probability is enormous, and checking it takes four minutes.

Section 3 - By-Right Versus Discretionary

This is the distinction that determines the risk profile of the entire deal.

By-right (ministerial). The project complies with all applicable standards. The jurisdiction verifies compliance and must approve. No hearing, no discretion, no political exposure. Timeline measured in weeks to a few months. Cost measured in review fees.

Discretionary. The project requires an approval that a decision-making body may grant or deny. Timeline measured in months to years. Outcome is not assured at any price.

The spectrum, from least to most discretionary:

Approval type	Decided by	Standard	Political exposure
Building permit / ministerial site plan	Staff	Objective compliance	None
Administrative site plan review	Staff	Mostly objective, some judgment	Low
Variance	Board of adjustment	Hardship, usually strict	Low to moderate
Conditional use / special use permit	Planning commission or board	Enumerated criteria, quasi-judicial	Moderate
Special exception	Board of adjustment	Enumerated criteria	Moderate
Subdivision / plat approval	Commission, sometimes council	Mostly objective	Low to moderate
Site plan requiring commission approval	Planning commission	Mixed	Moderate
Rezoning / map amendment	Elected body	Legislative discretion	High
PUD approval	Elected body	Legislative discretion	High
Text amendment	Elected body	Legislative discretion	High
Development agreement	Elected body	Negotiated	High

3.1 The Instruments

Variance. Relief from a dimensional standard - a setback, a height, a coverage requirement. The legal standard in most states is a hardship arising from a physical peculiarity of the property that was not self-created, and that standard is applied seriously in some jurisdictions and loosely in others. Critically, **a use variance is unavailable in many states**: you cannot get a variance to do something the district does not permit. Variances are for dimensions, not for uses. Underwriting a deal on the expectation of a use variance is, in much of the country, underwriting on something that legally cannot be granted.

Conditional use permit / special use permit. Approval for a use the ordinance contemplates in the district but requires review of. The criteria are enumerated in the ordinance - compatibility, traffic, buffering, hours - and the body applies them quasi-judicially, meaning it is supposed to decide on the record rather than on preference. It attaches conditions, and those conditions become binding obligations that will appear in your construction documents.

Rezoning / map amendment. Changing the district assigned to the parcel. A legislative act by the elected body. There is no entitlement to a rezoning and no legal standard that compels approval. A council may deny a rezoning that staff recommended, that complies with the comprehensive plan, and that meets every technical criterion, for no stated reason beyond the testimony of the room.

Text amendment. Changing the ordinance itself. Slower, broader, and occasionally the cleanest path when the problem is a rule that no longer fits anyone's intent.

Development agreement. A negotiated contract between the developer and the jurisdiction, addressing phasing, infrastructure, exactions, and - critically - **vesting**, which is the developer's protection against changes in the rules after approval. More on this in 4.4.

3.2 Political Risk Is Not Schedule Risk

The most common and most expensive mistake in this entire module is treating an entitlement as a long lead time rather than as a coin flip.

Schedule risk has a cost: carry, extension payments, delayed revenue. It is priced. Binary risk has a different shape entirely: a probability-weighted outcome where one branch is total loss of pre-development spend and the option on the site. A deal that requires a rezoning is not a deal that takes eight months longer. It is a deal that works or does not exist, resolved by a vote.

Things that drive that vote, none of which appear in the ordinance:

- **Organized neighborhood opposition.** The reliable predictor is not the merits; it is how many people show up. Traffic, schools, and "neighborhood character" are the standard grounds, and density is the standard target.
- **Protest petitions.** In many states, owners of a threshold percentage of adjacent property can file a protest that raises the approval requirement to a supermajority. A handful of neighbors can convert a 4-3 approval into a denial.

- **Election timing.** A rezoning heard six weeks before a municipal election is heard by different people than one heard six weeks after.
- **Moratoria.** Jurisdictions can and do suspend acceptance of applications, sometimes for a year or more, sometimes specifically in response to a project.
- **Staff recommendation.** Not binding, and highly influential. Losing staff is not fatal, but it converts a probable approval into a contested one.
- **Precedent anxiety.** Bodies deny applications they otherwise like because of what approving it would mean for the next twelve.

3.3 Managing It

The standard toolkit, applied before Gate 2 wherever possible:

- **Pre-application meetings** with planning staff. Free, informal, non-binding, and the single highest-value hour in the entitlement process. Staff will tell you what will be contested.
- **An entitlement contingency in the purchase agreement**, allowing termination if the approval is not obtained. Sellers resist it and price it, but it moves binary risk off the buyer.
- **Community outreach ahead of the hearing**, which does not convert opponents but does reduce the number of people who show up surprised.
- **Local land use counsel**, whose value is largely knowledge of how the specific body has behaved.
- **A fallback plan** that works by right, even at lower returns. A deal with a by-right floor is a fundamentally different risk than one without.

Section 4 - Land Division and Infrastructure Obligations

Zoning says what may be built. A second body of regulation governs dividing land and paying for the public infrastructure the project consumes.

4.1 Subdivision and Platting

Required whenever land is divided into lots, and often whenever lot lines are adjusted at all.

- **Preliminary plat** - the design-level approval. Lot layout, streets, easements, drainage, open space. Usually requires commission and sometimes council approval.
- **Final plat** - the recordable instrument, approved after construction drawings for public improvements are approved. Recording the plat is what legally creates the lots.
- **Dedication** - conveyance to the public of right-of-way, easements, and sometimes park land or sites for public facilities.

- **Improvement agreement and security** - the developer agrees to construct public improvements and posts a bond or letter of credit guaranteeing completion. The security is released after acceptance and a warranty period, which is typically a year or two after construction ends.

The consequential number here is **yield loss**. Gross acreage minus right-of-way, detention, open space requirements, buffers, and unbuildable area. Twenty to thirty-five percent is normal on residential land, and considerably more on constrained sites. A pro forma built on gross acreage is wrong before it starts.

4.2 Impact Fees and Exactions

Impact fees are schedule-based charges assessed to fund system-wide infrastructure: roads, water, sewer, parks, schools, fire. They are calculated by formula, published, and generally not negotiable. They can be very large - five figures per residential unit in some jurisdictions - and they are typically due at building permit, which makes them a construction-period cash requirement rather than a land cost.

Exactions are project-specific requirements to dedicate land or construct improvements: a turn lane, a traffic signal, an off-site sewer extension, a trail connection. These are negotiated and can be substantial. Off-site improvements in particular are a classic underwriting miss, because they are identified during traffic and utility review, which happens after the deal is already moving.

Constitutional limits apply. Under the *Nollan* and *Dolan* line of cases, an exaction must have an essential nexus to a legitimate government interest and must be roughly proportional to the project's impact. This is a real constraint and also an expensive one to enforce; most developers negotiate rather than litigate.

Concurrency and adequate public facilities requirements. Some jurisdictions prohibit approval unless infrastructure capacity exists to serve the project - road level of service, school capacity, water and sewer capacity. Where these exist, they can halt otherwise-compliant development entirely, or shift the cost of the capacity expansion to the developer.

4.3 Annexation

A site outside municipal limits may require annexation before urban services are available. Annexation is legislative, negotiated, and typically bundled with a zoning designation and a development agreement in a single proceeding. It adds time and adds a second political body to the process.

4.4 Vesting

The right to complete a project under the rules in effect when it was approved, rather than the rules as they later change. Vesting rules vary enormously by state, and the trigger may be approval of a plat, issuance of a permit, substantial construction, or a development agreement.

This matters when entitlement and construction are separated by years, which they often are. An unvested approval in a jurisdiction that changes its ordinance is worth less than it appears.

Section 5 - Private Restrictions

Everything to this point is public regulation. There is a parallel system of private restrictions that runs with the land, is enforceable by private parties, and is invisible on every zoning map.

These appear in **Schedule B, Part II of the title commitment** - the exceptions from coverage. The title company is telling you what it will not insure against, which is a list of everything encumbering the property. Read every exception document. Not the summary; the recorded instruments themselves.

Easements. A right held by another party to use part of your property. Utility, access, drainage, ingress-egress, conservation, aviation (aircraft overflight). The practical question is always the same: where is it, how wide is it, and what does it prohibit. A twenty-foot utility easement crossing the middle of the buildable area can eliminate a building. Easements can sometimes be relocated or vacated, which requires the consent of the holder and their cooperation on their timeline.

Covenants, conditions and restrictions (CC&Rs). Recorded restrictions typically created when a larger tract was subdivided. They can control use, architecture, materials, setbacks, signage, fencing, and maintenance obligations, and they are frequently more restrictive than zoning. An architectural review committee under CC&Rs may have approval rights over your design and may be composed of your competitors.

Deed restrictions. Restrictions imposed by a prior grantor, often permanent. Common examples: a former retail owner prohibiting competing uses, a church prohibiting certain uses on land it sold, an environmental covenant restricting groundwater use or requiring a cap to remain undisturbed.

Reciprocal easement agreements (REAs). Common on multi-parcel commercial developments. Govern shared access, shared parking, common area maintenance, and - often - permitted and prohibited uses across all parcels. An REA can prohibit a use that zoning permits, and amending it requires the consent of every party to it. Those parties have no obligation to be reasonable and every incentive to charge for their signature.

Mineral rights and severed estates. In much of the western United States, surface and mineral estates are commonly severed, and the mineral estate is generally dominant, meaning the mineral owner has a right of reasonable surface access. In active basins this is a live development issue requiring surface use agreements or lease of the minerals. In Colorado specifically, statutory notice requirements apply to residential development over severed minerals. Water rights are severed and separately owned as well.

Historic designation. Distinguish carefully: **local landmark designation** carries regulatory control over alteration and demolition, enforced by a preservation commission. **National Register listing** is largely honorific for privately funded projects and carries review obligations only where there is a federal nexus. Local designation is the one that can prevent you from doing things. Both matter for tax credit eligibility.

Encroachments and prescriptive rights. Revealed by the survey rather than the title search, which is why the two must be reconciled. A neighbor's fence, driveway, or building over the line, or a long-used informal path, can mature into a legal right.

5.1 Curing Exceptions

Schedule B items are not necessarily deal-killers. The standard responses, in order of preference:

1. **Release or termination** by the holder, recorded
2. **Relocation** to a non-interfering position, at the developer's cost
3. **Subordination** to the lender's lien
4. **Partial release** covering only the development area
5. **Title endorsement** insuring over the risk, which does not remove the restriction but shifts the loss
6. **Accept and design around it**

Each of these requires someone else's cooperation and therefore takes time you must have inside the due diligence period. This is the practical reason that due diligence periods on complicated sites need to be 90 to 120 days rather than 30.

Section 6 - Environmental and Overlay Regulation

A third parallel system, running on its own schedule, with its own reviewers.

Federal environmental review (NEPA) applies where there is a federal nexus: federal funding, a federal permit, or federal land. Most private development has no federal nexus and no NEPA obligation. When it does - a Corps permit, federal financing, a highway access approval - the review can add a year or more.

State analogs. Several states impose environmental review on private projects without any federal involvement. California's CEQA is the most consequential, and in that state environmental review is often the principal entitlement risk rather than a side process. Washington's SEPA and New York's SEQRA operate similarly. If you work in one of these states, this is a specialization in itself.

Wetlands and waters. Federal jurisdiction under Clean Water Act Section 404, administered by the Army Corps. A **delineation** identifies wetlands on the site; a **jurisdictional determination**

establishes whether they are federally regulated. The scope of federal jurisdiction has moved significantly in recent years and continues to be litigated, and several states impose their own broader wetlands regulation regardless. Impacts require a permit and typically mitigation. Timelines run from months for a nationwide permit to well over a year for an individual permit.

Floodplain. FEMA flood insurance rate maps define zones. The critical distinction is between the **floodway**, where development is effectively prohibited because it would obstruct flow, and the **flood fringe**, where development is permitted subject to elevation requirements. Maps can be amended through the CLOMR/LOMR process where the mapping is inaccurate or where fill changes conditions, on a timeline of many months. Flood zone status also drives insurance cost, which flows through to operating expenses and therefore to value.

Endangered species. Presence of listed species or designated critical habitat triggers consultation and can require a habitat conservation plan and incidental take permit. Long timelines.

Historic and archaeological review. Section 106 review applies where there is a federal nexus. State and local processes may apply otherwise. Discovery of archaeological resources during construction stops work, which is a schedule risk rather than an entitlement risk but belongs on the same register.

Coastal. Coastal zone management programs impose an additional permitting layer, and in some states the coastal authority is more powerful than the local jurisdiction.

Airport height hazard. Proximity to an airport triggers FAA notice requirements for structures penetrating defined surfaces, plus local overlay restrictions on both height and noise-sensitive uses. Includes construction cranes, which is a genuine and frequently forgotten item.

The pattern to internalize: each of these is a separate permit, from a separate agency, on a separate timeline, and they do not run sequentially. The entitlement schedule has to account for all of them in parallel, and the longest one governs.

Section 7 - Where Zoning Ends and Building Code Begins

This is the handshake to Part 2, and it is the single most valuable distinction in this module for a reader with an AEC background.

	Land use / zoning	Building code
Legal source	Local ordinance, adopted under state enabling legislation	Model code (IBC, IRC, IEBC) adopted by state or local government with amendments
Question answered	May this use, at this intensity, exist here?	Is this building safe and compliant as designed and built?
Reviewer	Planning staff, commissions, elected officials	Plans examiner, building official, fire marshal
Nature of the standard	Often discretionary, sometimes political	Technical and objective
Public participation	Hearings, notice, testimony, opposition	None
When it happens	Before design is complete; often before land is purchased	After design is substantially complete
Typical outcome of failure	Denial. The project does not exist.	Correction comments. The project is redesigned.
Appeal path	Elected body, then courts on the record	Board of appeals, alternate means and methods request
Who de-risks it	Land use attorney, entitlement consultant	Architect, engineers, code consultant
Covered in	Part 1 Module 3	Part 2 Module 7

Three consequences follow.

They fail at different times. Zoning fails at Gate 3, before you own the land. Code fails at plan review, after you own it, after design is paid for, and often after the construction loan has closed. Both are expensive; only one is survivable by redesign.

They are not substitutes. A code consultant cannot fix an entitlement problem and a land use attorney cannot fix a code problem. This sounds obvious and is violated constantly by people who understand one system deeply and assume the other works the same way.

Conditions of approval migrate into the construction documents. This is the mechanism by which Module 3 reaches into Part 2 (construction). When a commission grants a conditional use permit with eighteen conditions - a masonry percentage on the facade, a six-foot fence along the north property line, a restriction on delivery hours, a landscape buffer with specified caliper trees, a requirement to fund half a traffic signal - those conditions are now binding obligations on the project. They must be transmitted to the design team, incorporated into the drawings and specifications, and verified at closeout. The building official will not enforce them, because they are not code. Planning will, at certificate of occupancy.

Conditions of approval that never made it into the documents are one of the more common causes of a project sitting complete and unable to obtain a CO. Module 8 covers the transmittal.

Section 8 - The Entitlement Schedule

The artifact that converts all of this into a decision input.

An entitlement schedule lists every required approval, the body that grants it, the submittal requirements, the calendar, and the dependencies. It is built backwards from the date the project must start construction.

Rough durations, which vary enormously and should always be verified with staff:

Approval	Typical elapsed time
Pre-application meeting	2–4 weeks to schedule
Administrative site plan	2–5 months
Conditional use permit	3–6 months
Variance	2–4 months
Rezoning or PUD	6–18 months

Rezoning plus annexation	9–24 months
Preliminary plat	3–6 months
Final plat and improvement agreement	2–5 months after preliminary
Corps individual permit	12–24 months
Floodplain map amendment (CLOMR/LOMR)	6–18 months
Building permit	1–6 months, after all of the above

Four scheduling realities that consistently surprise people:

Bodies meet monthly. Missing a submittal deadline by a day costs a month, not a day. Agendas fill and items get bumped. Summer and December recesses are real.

Resubmittal loops are the norm. A first submittal that is not substantially complete draws comments and restarts the review clock. Two to three rounds is typical.

Continuances happen. A commission may continue an item to the next meeting for more information, or to allow further neighborhood discussion. Two continuances add two to three months.

Approvals expire. A conditional use permit or preliminary plat that is not acted on within a stated period lapses. Extensions are usually available and usually require going back before the body.

8.1 Presenting It Honestly

The schedule is not the analysis. A schedule communicates duration; entitlement risk is mostly probability. The screening memo needs both:

- What approvals are required, and which are discretionary
- For each discretionary approval, an honest probability and the basis for it - comprehensive plan consistency, staff posture, the body's recent record on similar applications, whether organized opposition exists
- What the site is worth if the approval is denied, which is the downside
- Whether a by-right fallback exists
- Total pre-development spend at risk before the vote

That last figure, set against the probability, is the actual decision. Everything else in this module is input to it.

Artifacts

Artifact	Prepared by	Purpose
Zoning report / zoning analysis	Land use attorney or zoning report vendor	Current district, permitted use, dimensional standards, compliance of existing improvements
Zoning opinion letter	Land use attorney	A reasoned legal opinion, relied upon by lenders
Entitlement strategy memo	Land use attorney or entitlement consultant	Required approvals, path, probability, timeline
Entitlement schedule	Developer with consultants	Calendar and dependencies
Title commitment , Schedule B	Title company	Private encumbrances; the checklist for Section 5
Exception documents	Title company	The actual recorded instruments, which must be read
Pre-application meeting notes	Developer	Staff's informal position, which is not binding and is highly predictive
Conditions of approval	Approving body	Binding obligations carried into design and construction
Site plan / entitlement drawings	Architect and civil engineer	The application exhibit; frequently at or near schematic level, produced at risk

One note on the zoning report. Commercially available zoning reports are inexpensive and useful for confirming current classification and whether existing improvements are conforming. They are not a legal opinion and they do not assess entitlement probability. Lenders on complex deals will require an attorney's opinion letter.

Failure Modes

Underwriting a density the site cannot legally carry. The most expensive error in Part 1. A pro forma built on units per acre, without a test fit against setbacks, height, open space, impervious coverage, and parking, will overstate yield on most sites. The gap is typically 10 to 30 percent, which is usually the entire profit.

Treating entitlement risk as schedule risk. It is binary. A rezoning does not make the deal slower; it makes the deal conditional. Price it as a probability with a total-loss branch, not as carry cost.

Assuming a variance can fix a use problem. In most states a use variance is unavailable, and even where available the hardship standard is real. Variances are for dimensions.

Reading Schedule B summaries instead of the recorded documents. The summary says "easement recorded at Book 4412, Page 88." It does not say the easement is thirty feet wide and runs diagonally through the buildable area. Someone has to pull the document and plot it on the survey.

Missing an overlay. Overlays frequently live on a GIS layer that is off by default, and they can impose height limits, design review, or use restrictions that the base district does not.

Ignoring private restrictions because zoning permits the use. CC&Rs, deed restrictions, and REAs are enforceable by private parties who are not obligated to be reasonable and who may be your direct competitors.

Confusing the comprehensive plan with the zoning ordinance. The plan expresses intent and is not directly enforceable; the ordinance is the law. But plan consistency is the best available predictor of rezoning outcomes, so checking only one of them is a mistake in either direction.

Letting the due diligence period expire with exceptions uncured. Releases, relocations, and amendments all require another party's cooperation. Thirty days is not enough time on any site with real title complexity.

Forgetting that conditions of approval are contract terms. They bind the project, they are enforced at certificate of occupancy by a department that does not attend construction meetings, and they must be transmitted to the design team in writing.

What Happened Next

Kipling Row

This module is the entire deal.

The zoning report confirmed R-1 single-family, minimum 7,000 square foot lots, 35-foot height, 25-foot front setback. Townhomes are not a permitted use in R-1 at any density. There is no dimensional variance path to a use that is not allowed, and the jurisdiction does not grant use variances.

The only paths were a rezoning to the R-3 attached residential district or a PUD written for the project. Counsel recommended the PUD, on the grounds that R-3 as written would permit 44 units and would draw heavier opposition than a PUD capped at 34 with negotiated buffering and architectural commitments. The PUD also allowed the developer to offer conditions voluntarily, which is a meaningful posture at a hearing.

The favorable facts: the comprehensive plan's future land use map designates the corridor **medium-density residential**, meaning the request is consistent with adopted policy rather than a departure from it. The site fronts an arterial and abuts the rail corridor on one side, which weakens the standard compatibility argument. Planning staff, at the pre-application meeting, indicated informal support and identified two issues they expected to be contested: the proposed access point and the transition to the detached homes on the east property line.

The unfavorable facts: the two council members representing the district had both voted against a similar attached project eighteen months earlier. A neighborhood association existed, was active, and had organized against that project. A municipal election fell within the likely hearing window. And the state's protest petition statute meant that owners of 20 percent of the adjacent frontage could force a three-quarters supermajority - and the eastern boundary was six detached homes, any two of which could trigger it.

The title work produced two Schedule B items requiring attention: a ten-foot utility easement along the arterial frontage, which was compatible with the plan, and a 1961 deed restriction from the original grantor prohibiting "commercial or industrial purposes," which counsel opined did not restrict residential use. A restrictive covenant endorsement was requested from the title company.

Entitlement schedule: 4 months to a complete PUD submittal including entitlement-level drawings prepared at risk, 3 to 5 months of staff review and resubmittal, planning commission, then council, with a 30-day appeal period after. Ten to fourteen months to a vested approval, assuming no continuances. Preliminary plat could run concurrently with the later stages. Pre-development spend before the council vote: approximately \$290,000, of which \$185,000 was non-refundable deposit and consultant work already committed.

Counsel's honest probability: **60 to 65 percent**. Plan consistency and staff support on one side; two hostile council members, an organized association, an election, and a live protest petition risk on the other.

The by-right fallback - 11 detached lots - does not support the purchase price. There is no floor.

The deal advanced to Gate 1 with an entitlement contingency in the purchase agreement, which the seller granted in exchange for a higher price and non-refundable deposit installments at 60-day intervals. Whether that was the right trade is Module 7.

Six Corners

The zoning analysis took four days and came back clean, which is what made the private restrictions so surprising.

The site is zoned C-2 general commercial. Multifamily is permitted **by right** at up to 30 units per acre, which on 6.1 acres would permit 183 units - far more than a 78,000 square foot building could ever produce. Light industrial and flex are also permitted by right in C-2 in this jurisdiction, subject to a limitation on outdoor storage. Height limit 45 feet; the existing building is 28. The existing improvements are fully conforming.

No rezoning. No conditional use. No hearing. Administrative site plan review only, estimated at three to four months, with a building permit behind it. Entitlement risk on the public side was, for practical purposes, zero - which is precisely the profile that lets a value-add sponsor move quickly and is exactly why the sponsor had not yet slowed down.

Then the title commitment arrived, with fourteen Schedule B exceptions. Twelve were routine. Two were not.

Exception 9 was a **reciprocal easement agreement recorded in 1986**, when the property was developed as the middle parcel of a three-building office park. The REA governs shared access from the signalized entrance, cross-parking, and common area maintenance across all three parcels. It also contains a use restriction: the parcels are restricted to **office and office-supporting uses**, with residential and manufacturing expressly prohibited. The REA runs until 2036 and may be amended only with the written consent of all three parcel owners.

Zoning permits multifamily by right. The REA prohibits it. The private restriction controls.

The two adjacent parcels are owned by an owner-occupant medical group and a family partnership that has held its building since 1991. Either can refuse consent, for any reason or none. Counsel's assessment was that consent was obtainable and would be priced - the medical group in particular had an interest in the parking arrangement and would likely trade consent for a parking covenant.

Exception 11 was a **severed mineral estate**, with the minerals reserved by a 1958 grantor and never reunited with the surface. No active leasing in the immediate area, but the reservation is dominant and would require a surface use agreement or a subordination for any financing. Routine in this state, and another item requiring a third party's signature inside the diligence period.

Two consequences the sponsor did not register at the time. First, the REA amendment turned a by-right project into a negotiation with two unrelated parties, which is a discretionary approval in everything but name - with the difference that the approving parties have a commercial interest

and will charge rather than hold a hearing. Second, the REA's prohibited-use list includes manufacturing but not warehousing, distribution, or general commercial service uses.

That second fact mattered four months later.

Advanced to Gate 2 preparation with clean public entitlement, one significant private restriction requiring third-party consent, and a mineral reservation requiring a surface agreement. The diligence period was extended by 30 days specifically to pursue the REA amendment.

Optional: Try This

Return to the parcel you screened in Module 2 and answer four questions from public sources.

What district is it in, and what does the ordinance permit there by right? Pull the actual use table, not a summary.

What does the comprehensive plan's future land use map designate for it? If the plan and the zoning disagree, you have found a parcel where a rezoning is plausible - which is the most common source of value creation in land.

What overlays apply? Turn on every GIS layer, not the default set.

Run the dimensional standards against the site and find the binding constraint. It will usually be parking, open space, or impervious coverage rather than density.

Then, if you want the real version of this exercise: read the minutes or watch the recording of the last three rezoning hearings in that jurisdiction. Nothing in an ordinance will teach you as much about your actual approval probability as watching the body that decides it.

Next - Module 4A: Physical Due Diligence, Vacant Land. Test two of highest and best use. What the survey, the soil, and the utilities will support, and why the dirt is where budgets are really made.

1.4A - Physical Due Diligence: Vacant Land

Module 3 asked whether you are allowed to build. This module asks whether the site will let you.

It is the second test of highest and best use - physically possible - and it is the module where the budget is actually made. Not estimated, made. The cost of a project is determined far more by soil conditions, utility capacity, grading, and stormwater than by any decision the design team will make later, and all of it is knowable before you own the land.

There is a specific and unforgiving deadline attached. Everything in this module must resolve before **Gate 2**, when the earnest money goes non-refundable. Before that date, a bad geotechnical report is information and the correct response is to renegotiate or walk. After it, the same report is a loss. The distinction between a disciplined developer and an unlucky one is almost entirely a matter of which side of Gate 2 the bad news arrives on.

One structural note that matters more than it appears. The consultants in this module - the surveyor, the geotechnical engineer, the environmental consultant - do their work, produce a report, and leave. They are gone months before an architect is engaged. Their reports are the design team's entire inheritance from this phase, and whether those reports are transmitted completely, read carefully, and reconciled against the design is the subject of Module 8. A surprising number of construction-phase disasters are documented in a geotechnical report that someone filed and nobody opened.

Section 1 - The Survey

1.1 Survey Types

Four different products, routinely confused:

Boundary survey. Establishes property lines from the legal description and found monuments. The minimum.

Topographic survey. Existing elevations, contours, visible improvements, trees, and surface utility structures. What the civil engineer needs to design grading and drainage.

ALTA/NSPS Land Title Survey. The commercial standard, prepared to a set of minimum detail requirements jointly adopted by the American Land Title Association and the National Society of Professional Surveyors. Its defining characteristic is that it **reconciles the survey against the title commitment** - every Schedule B exception that is capable of being shown is plotted on the drawing. It is what lenders and title companies require on commercial transactions, and it is the one that answers the questions you actually have.

As-built survey. Documents constructed conditions. A closeout deliverable, not a diligence one.

1.2 What an ALTA Survey Shows

- Boundary lines, bearings and distances, and the monuments found or set
- Gross and net land area
- **Every plottable Schedule B exception**, keyed to its recording reference
- Existing improvements and their dimensions
- **Encroachments**, in both directions
- Access to a public right-of-way, and whether it is direct or by easement
- Rights-of-way, and whether the dedicated width matches the constructed width
- Flood zone designation per the current FIRM
- A surveyor's certification running to named parties

1.3 Table A: The Optional Items

The base ALTA standard covers boundary and title reconciliation. The additional content is ordered from a schedule of optional items known as Table A, and what you order determines what you get. Ordering an ALTA survey without specifying Table A items is a common and expensive error - the survey arrives technically correct and missing the information you needed.

The items that matter most for development, by name rather than number (the numbering is revised periodically, most recently in the 2021 standards, so specify by description and confirm with the surveyor):

- **Contours and vertical relief**, which is what turns the ALTA into a usable topographic survey
- **Zoning classification and setback lines**, typically based on a zoning report the client supplies
- **Evidence of underground utilities**, which has an important limitation discussed below
- **Parking counts and striping**
- **Names of adjoining owners**
- **Evidence of recent earth moving, construction, or dumping**
- **Off-site easements benefiting the property**
- **Proposed changes to street rights-of-way**
- **Wetland delineation markers**, where a delineation has been performed

The underground utility item deserves a warning. It is generally satisfied by plotting observed surface evidence and utility records provided by the companies, not by locating anything. Utility records are frequently wrong. If the precise location of a line matters - and on an infill site it usually does - you order a private subsurface utility engineering locate, which is a separate scope at a separate cost.

1.4 How to Read One

Reconcile it to Schedule B yourself. Take the exception list from Module 3 and check each item against the survey. Some will be plotted. Some will be noted as "blanket in nature, not plottable," which means an easement exists somewhere on the property without a defined location - a meaningful problem that is easy to skim past.

Plot the buildable area. Boundary, minus easements, minus setbacks, minus required buffers, minus floodplain, minus anything unbuildable. What remains is the real site. It is always smaller than the parcel.

Check access. Does the property physically touch a public right-of-way with a usable frontage, or does access depend on an easement across someone else's land? If it is the latter, read the easement: width, permitted uses, maintenance obligations, whether it permits the traffic your project will generate.

Look for gaps, gores, and overlaps. Adjoining legal descriptions that do not close against each other create strips of disputed or unowned land. Old, subdivided, and repeatedly conveyed parcels are where these live.

Check the certification. It runs to named parties. The developer, the lender, and the title company all need to be named, and adding a party after the fact requires the surveyor's agreement and a fee.

Compare surveyed acreage to the tax record. They frequently disagree. The survey controls, and a 2 percent discrepancy on a per-acre purchase price is a real number.

Section 2 - Topography and Earthwork

The most under-analyzed cost in development, because it is invisible until someone runs the numbers.

The earthwork balance. A site is "balanced" when the cut required to create the building pads and drainage equals the fill needed elsewhere. An unbalanced site requires either exporting material or importing it. Both are priced per cubic yard, both include trucking, and trucking cost is a function of haul distance - meaning identical sites in different locations have different earthwork costs. On a constrained urban site, export can run several times the unit cost of on-site movement.

Unsuitable material. Excavated material is not automatically usable as structural fill. Organics, debris, and high-plasticity clays may have to be exported and replaced with imported structural fill, which converts a balanced site into a double haul.

Slope. Parking has maximum slope requirements - accessible stalls and access aisles are limited to a maximum slope in all directions, which on a sloping site forces terracing. Drive aisles have

practical and code limits. Building pads must be flat. The steeper the site, the more the design is driven by grading rather than by planning.

Retaining walls. The cost cliff. A site that needs a few short walls is a normal site. A site needing tall walls, and particularly walls requiring engineered systems, guardrails, drainage, and global stability analysis, is a different budget. Retaining wall cost scales with the square footage of exposed face, which scales with the square of the height.

Rock. Shallow bedrock changes excavation from moving dirt to breaking rock - ripping, hoe-ramming, or blasting. Unit costs increase by a large multiple, and blasting near existing structures brings vibration monitoring, preconstruction surveys, and neighbor claims.

Buried debris and undocumented fill. Common on previously developed sites and on rural land that hosted informal dumping. Neither its extent nor its content is known until excavation, which makes it a contingency item rather than a line item.

Section 3 - The Geotechnical Report

3.1 Scope and What It Costs

A geotechnical investigation drills borings across the site, samples and tests the soil, measures groundwater, and issues recommendations for foundations, floor slabs, pavements, and earthwork.

Boring count and depth scale with project size and structure type. A small commercial building might warrant four to eight borings; a large site with multiple structures, considerably more. Cost for a straightforward project typically runs in the low tens of thousands of dollars, and the elapsed time - scheduling a rig, drilling, lab testing, report preparation - is generally three to six weeks. This makes it one of the long poles in a due diligence period and one of the first things to order.

3.2 What the Report Contains

- **Boring logs.** Soil profile by depth, sample descriptions, blow counts from standard penetration testing, groundwater encountered during and after drilling
- **Laboratory results.** Moisture content, Atterberg limits, gradation, consolidation, unconfined compression, and - critically in expansive soil regions - **swell testing**
- **Subsurface profile and interpretation**
- **Foundation recommendations,** with allowable bearing pressure and estimated settlement
- **Floor slab recommendations**
- **Seismic site class,** which feeds directly into structural design under ASCE 7 and IBC Chapter 16
- **Lateral earth pressures** for below-grade walls

- **Pavement sections** for the parking and drives
- **Earthwork recommendations:** subgrade preparation, suitable fill, compaction, moisture conditioning
- **Groundwater and dewatering considerations**

3.3 The Problem Conditions

Condition	What it does	Where it is common
Expansive clay and claystone	Swells when wetted, heaves slabs and foundations	Front Range, Texas, Oklahoma, upper Midwest
Collapsible soils	Settle abruptly when wetted	Arid West, loess regions
Soft or organic soils, peat	Low bearing capacity, long-term settlement	Former wetlands, river valleys, coastal
Undocumented fill	Unknown density and content; usually must be removed or improved	Any previously developed site
High groundwater	Dewatering, buoyancy, below-grade waterproofing, no basements	Coastal, river-adjacent, high water table regions
Shallow bedrock	Excavation cost; also good bearing	Mountain West, New England, Appalachia
Karst	Sinkhole risk, voids	Florida, Missouri, Kentucky, Pennsylvania
Liquefiable soils	Loss of strength under seismic loading	Seismic zones with saturated sands
Corrosive or sulfate soils	Concrete and pipe material requirements	Arid West

3.4 The Remediation Ladder

The useful mental model is that remediation options form a ladder, and where you land on it is one of the larger single determinants of project cost. Ordered from least to most expensive:

1. **Standard subgrade preparation** - scarify, moisture condition, recompact. Baseline.
2. **Moisture conditioning and control of expansive soils** - wetting to target moisture and maintaining it, plus drainage and landscape controls. Modest cost, requires ongoing discipline.
3. **Over-excavation and replacement** - remove problem soil to a depth and replace with engineered fill. Cost scales directly with depth and area.
4. **Chemical stabilization** - lime or cement treatment of expansive or weak subgrade.
5. **Surcharge and preload** - place fill, wait for settlement, remove. Cheap in dollars, expensive in schedule; months of waiting.
6. **Ground improvement** - stone columns, rammed aggregate piers, deep soil mixing. Significantly more expensive, faster than preload.
7. **Structural floor systems** - structural slabs over void form, so the floor does not bear on expansive soil. Standard practice in some markets, and a real per-square-foot premium.
8. **Deep foundations** - drilled piers or caissons, augercast piles, driven piles. The most expensive answer and often the only one.
9. **Permanent dewatering** - ongoing operational cost forever, not just a construction cost.

The spread between the bottom and the top of this ladder on a residential project is routinely \$10,000 to \$20,000 per unit. On a large commercial structure it can be seven figures. This is why the geotechnical report is ordered early and why its findings are a renegotiation trigger rather than a design detail.

3.5 Reliance

A geotechnical report is addressed to the party that commissioned it and is legally relied upon by that party. Lenders, equity partners, and the eventual design team will each need a **reliance letter** extending the consultant's professional obligation to them. These are routine and inexpensive, and they must be requested. A report inherited from a seller, addressed to the seller, with no reliance extended to you, has limited value.

Section 4 - Environmental: The Phase I ESA

4.1 What It Actually Is

A Phase I Environmental Site Assessment, performed to ASTM E1527.

A Phase I is a records and reconnaissance investigation performed to satisfy the **All Appropriate Inquiries** rule, which is the prerequisite for the innocent landowner, contiguous

property owner, and bona fide prospective purchaser defenses to liability under CERCLA. Its purpose is to preserve your defense against liability for contamination you did not cause. It involves **no sampling of any kind**.

This distinction has a practical consequence: a "clean" Phase I means no evidence of a problem was found in the records or on a walkthrough, not that the site is clean.

4.2 Components

- **Regulatory database review** - federal, state, tribal, and local environmental records for the site and surrounding properties within defined search radii
- **Historical research** - aerial photographs, fire insurance maps, city directories, topographic maps, title and land use records, typically back to first developed use or 1940
- **Site reconnaissance** - a walkthrough documenting current conditions, storage, staining, drums, transformers, wells, and evidence of dumping
- **Interviews** - with owners, occupants, and local officials
- **Report and opinion** by an environmental professional

4.3 The Findings Vocabulary

Precision matters here, because the terms have defined meanings and are routinely misused.

Recognized Environmental Condition (REC). The presence or likely presence of hazardous substances or petroleum products in, on, or at the property due to a release, a likely release, or a material threat of a release. This is the finding that matters.

Historical REC (HREC). A past release that has been addressed to the satisfaction of the regulatory authority, with no current controls required. Resolved.

Controlled REC (CREC). A past release addressed to regulatory satisfaction **with controls remaining in place** - a cap that must not be disturbed, a deed restriction, a groundwater use prohibition, a vapor mitigation system. Not resolved; managed. A CREC constrains what you can do with the site, which makes it a development issue rather than a historical footnote.

De minimis condition. Present but not a threat and not subject to enforcement. Not a REC.

Business environmental risk. Issues outside the REC framework that a buyer may nonetheless care about.

Vapor migration is within the scope of the REC analysis under current standards, and it matters disproportionately for residential development. Regulatory screening levels for vapor intrusion are substantially more stringent for residential occupancy than for commercial or industrial occupancy. The same soil gas concentration can be acceptable under a warehouse and unacceptable under an apartment. This is one of the mechanisms by which a site's contamination history dictates its highest and best use.

Non-scope items. A standard Phase I does not address asbestos, lead-based paint, lead in drinking water, radon, mold, wetlands, or endangered species. These are added by scope or ordered separately.

Shelf life. Certain components go stale after 180 days, and the report as a whole after one year. A seller-provided Phase I from two years ago does not preserve your defense.

4.4 Site Histories That Produce Findings

- **Gas stations and fueling** - underground storage tanks, the single most common finding
- **Dry cleaners** - chlorinated solvents, particularly PCE, which migrates readily in groundwater and produces vapor intrusion problems at long distances from the source. A dry cleaner upgradient of your site is a problem even if it never touched your parcel.
- **Auto repair, machine shops, metal plating** - solvents, metals, waste oil
- **Rail corridors and yards** - herbicides, creosote, petroleum, metals
- **Agricultural land** - pesticide and herbicide residues; historic orchards are known for arsenic and lead
- **Historic fill** - urban sites filled with whatever was available at the time
- **Heating oil tanks** - extremely common on older institutional and residential properties and frequently undocumented

4.5 Phase II and What Follows

A Phase II Environmental Site Assessment is invasive: soil borings, groundwater monitoring wells, soil gas sampling, laboratory analysis. It is scoped to investigate the specific RECs the Phase I identified. Cost ranges from a few tens of thousands for a limited scope to well into six figures. Elapsed time is typically four to ten weeks including laboratory turnaround, which for a 60-day due diligence period is most of the clock - and the Phase II cannot start until the Phase I is done.

This sequencing is the reason environmental work is ordered on day one of diligence, and the reason a due diligence extension is often required on any site with a history.

If contamination is confirmed, the paths forward are:

- **Remediation to closure**, with a no further action determination from the state
- **A state voluntary cleanup program**, which provides a defined process and a liability release at completion
- **Engineering and institutional controls** - caps, vapor barriers and sub-slab depressurization, deed restrictions, groundwater use prohibitions - which leave contamination in place and manage exposure
- **Environmental insurance**, covering cost overrun and unknown conditions
- **Seller indemnity or escrow holdback**, which is only as good as the seller's continued existence
- **Price adjustment** reflecting the cost and the residual risk
- **Walking away**

Two practical notes. Lenders are conservative about contaminated collateral because of their own liability exposure, and an unresolved Phase II can make a deal unfinanceable regardless of the developer's own risk tolerance. And a sub-slab depressurization system with a deed restriction is a permanent operational obligation that must be disclosed to every future buyer and tenant.

Section 5 - Natural Resource Constraints

Wetlands. A delineation identifies wetland boundaries on the ground using soil, hydrology, and vegetation indicators. A jurisdictional determination from the Corps establishes which of them are federally regulated; some states regulate more broadly regardless. Impacts require a permit, typically with mitigation through a mitigation bank or on-site creation. Timelines range from a few months for a nationwide permit to well over a year for an individual permit.

Threatened and endangered species. Habitat assessment, and where suitable habitat is present, protocol-level surveys.

Cultural and archaeological resources. Records review, and where indicated, a pedestrian survey or shovel testing.

5.1 The Seasonality Problem

This section contains the most underappreciated scheduling risk in physical due diligence: **many of these investigations can only be performed during specific times of year.**

Wetland delineations require the growing season for vegetation identification. Protocol surveys for many species have defined windows tied to breeding, nesting, or emergence - migratory birds, bats, prairie dog associates, various amphibians. Miss the window and the next opportunity is a year away.

A 60-day due diligence period beginning in November on a site with potential wetlands or listed-species habitat cannot complete the required investigations, no matter how much money is spent. This must be identified at Gate 1, when the due diligence period is being negotiated, not discovered at day 40. Where the timing cannot work, the responses are a longer or seasonally-defined diligence period, a closing contingent on the study result, or a price reflecting the unresolved risk.

Section 6 - Utilities: Availability Versus Capacity

The most common framing error in the entire module. "There's sewer in the street" answers the wrong question. The questions are whether the system has capacity for your demand, whether the utility will commit to serving you, what it will cost to connect, and when.

Water. Main size, location, and static pressure. **Fire flow**, verified by a hydrant flow test - inadequate fire flow can require an on-site tank and pump, or an off-site main upsize, either of which is a substantial cost. Tap and system development fees, which can be significant per unit. In much of the western United States, **water rights dedication or cash-in-lieu payment** is a condition of service and a real per-unit cost that surprises developers from other regions.

Sanitary sewer. Main size, depth, and location, and whether the site can be served by gravity. A site that cannot reach gravity sewer needs a lift station, which is capital cost plus permanent operating and maintenance obligation, and which some jurisdictions refuse to accept for public ownership. **Downstream capacity** is the critical question: an adequately sized main at your frontage may discharge into a downstream segment already at capacity, and the fix - an off-site upsize - may be your obligation. Sewer moratoria exist and halt development entirely.

Electrical power. This has moved from a routine item to a primary schedule driver, and it deserves the most attention.

- Available service capacity at the site, and whether the local distribution system can support the project's connected load
- Whether a new or upgraded transformer, switchgear, feeder, or substation capacity is required
- **Equipment lead times**, which for transformers and switchgear have run from many months to well over a year and remain extended
- The utility's own design and construction timeline, which is theirs and not yours
- Cost allocation: line extension charges, contributions in aid of construction, and refund or reimbursement provisions if others later connect
- For large loads, the interconnection study process, which is a multi-month exercise of its own

The practical consequence is that power has become a critical-path item that must be initiated during due diligence rather than during design. A project that identifies its transformer requirement at construction document stage may find that the equipment arrives after the building is otherwise complete.

Natural gas. Availability, main capacity, and cost of extension. Note that a growing number of jurisdictions restrict or prohibit gas service in new construction, which is a code and policy question with direct design consequences.

Telecommunications and fiber. Routine for most product types, and a primary site selection criterion for anything data-related.

6.1 Will-Serve Letters

A will-serve or capacity availability letter is the utility's written statement that it can and will serve the project at a stated demand. Lenders require them. Read the conditions carefully: they typically expire, are contingent on the applicant paying for required improvements, and reserve

capacity only for a stated period. A will-serve letter is a commitment with conditions, not a guarantee.

6.2 Off-Site Improvements

The costs that appear late and belong to nobody in the original budget: main extensions, upsizing, road widening, turn lanes, signals, sidewalk and curb along the frontage, utility undergrounding.

Partial relief mechanisms exist and are worth pursuing - **latecomer or recovery agreements** reimbursing you when subsequent developers connect to infrastructure you built, **reimbursement districts** and **metropolitan or special districts** that finance infrastructure through subsequent tax revenue, and **development agreements** allocating cost. All of these take time to negotiate and all of them belong to Module 3's entitlement process.

Section 7 - Access, Traffic, and Stormwater

Access. Where the driveways can go is not the developer's choice. Access spacing standards govern distance from intersections and from other driveways; medians determine whether an access is full-movement or right-in/right-out. Jurisdiction matters - access to a state highway requires a permit from the state DOT, which is a separate agency with separate standards and a separate timeline from the city reviewing everything else.

Access location is a frequent source of conflict between the site plan and the reviewing agencies, and it is not a minor one: moving an entrance can reorganize an entire site layout.

Traffic impact study. Required above a trip-generation threshold. Trip generation is estimated using standard ITE rates by land use, distributed onto the network, and evaluated against level-of-service standards at affected intersections. Where the project degrades service below the standard, mitigation is required and generally funded by the developer: turn lanes, signal installation or modification, acceleration lanes, sometimes a proportional contribution to a larger improvement. A signal is a large number, and a developer-funded signal appearing in a Module 3 condition of approval is an event that has ended deals.

Stormwater. Two distinct obligations that are often conflated:

- **Quantity control** - detention or retention so that post-development peak discharge does not exceed pre-development rates, usually for multiple design storms
- **Quality control** - treatment of runoff before discharge, through water quality capture volume, bioretention, or other best management practices, under MS4 permit requirements

The land consumed is the underwriting issue. Detention and water quality facilities routinely occupy 5 to 15 percent of a site, and that land is removed from the yield calculation.

Underground detention exists, is far more expensive, and is used when land is more valuable than the systems that replace it. Regional detention facilities, where available, let a developer pay a fee instead of building on site, which is nearly always the better answer.

Additionally, construction disturbance above one acre triggers a **NPDES construction general permit** and a stormwater pollution prevention plan, which is a construction-phase compliance obligation with real inspection and penalty exposure.

A note that applies to redevelopment specifically: sites developed before modern stormwater regulation frequently have no detention at all. Redevelopment triggers current standards, meaning a fully paved 1980s site must find room for detention it never had. That land comes out of the yield.

Section 8 - Sequencing and Cost

The reason due diligence periods are negotiated in Module 7 rather than assumed.

Study	Typical cost	Elapsed time	Order it
ALTA survey with Table A items	\$5k–\$25k	3–6 weeks	Day 1
Geotechnical investigation	\$10k–\$50k	3–6 weeks	Day 1
Phase I ESA	\$2.5k–\$6k	2–4 weeks	Day 1
Phase II ESA	\$20k–\$150k+	4–10 weeks	Only after Phase I
Wetland delineation	\$5k–\$20k	2–6 weeks, seasonal	Day 1 if indicated
Species survey	\$5k–\$40k	Seasonal windows	Immediately, or accept a year
Traffic impact study	\$8k–\$30k	4–8 weeks	Week 1–2
Utility research and will-serve	\$0–\$10k	4–12 weeks	Day 1

Subsurface utility locate	\$5k–\$20k	2–4 weeks	As needed
Conceptual civil / yield study	\$10k–\$40k	3–6 weeks	Week 1

Figures vary widely by market, site size, and complexity; treat them as relative magnitudes rather than quotes.

Three sequencing rules:

Order the long poles on day one. Survey, geotech, Phase I, and utility research all start immediately. Nothing is gained by staging them and a great deal is lost.

Phase II cannot start until Phase I finishes. This dependency alone consumes most of a 60-day period. Any site with a plausible history needs 90 to 120 days or a pre-negotiated extension.

Seasonal items govern everything. If a required survey has a window, the window determines the schedule regardless of what the purchase agreement says.

Total physical due diligence spend on a modest site runs \$50,000 to \$150,000, and considerably more where a Phase II or extensive civil work is involved. All of it is at risk. All of it is spent before Gate 2, which is the entire design of the process: you spend six figures to avoid spending seven.

Section 9 - Converting Findings Into the Deal

Every finding in this module resolves into one of four outcomes. Making this explicit is what separates due diligence as a process from due diligence as a pile of reports.

A budget line. A known cost, priced, added to the development budget. Drilled piers, an off-site sewer extension, a turn lane, imported structural fill. This flows directly into Module 5.

A yield reduction. Land consumed by detention, wetlands, easements, or grading, reducing the unit count or building area the site supports. This flows into the pro forma as reduced revenue, which is usually worse than an equivalent cost.

A schedule item. Transformer lead time, a Corps permit, a seasonal survey window, a utility's construction queue. Schedule is cost, through carry and delayed revenue.

A risk to price, insure, or refuse. Unresolved contamination, a CREC with permanent controls, undocumented fill of unknown extent. Priced through a discount, transferred through insurance or indemnity, or declined.

The findings then drive one of three decisions before Gate 2: **accept** and go hard, **renegotiate** price or terms against the findings, or **terminate** and recover the deposit. The third is the whole point of the structure, and the discipline to use it is what the due diligence period is for.

Artifacts

Artifact	Prepared by	Purpose
ALTA/NSPS land title survey	Licensed surveyor	Boundary, easements plotted, encroachments, access, area
Topographic survey	Surveyor	Existing grades for civil design
Geotechnical report	Geotechnical engineer	Foundation, slab, pavement, and earthwork basis
Phase I ESA	Environmental professional	CERCLA liability defense; identification of RECs
Phase II ESA	Environmental professional	Confirmation and extent of suspected contamination
Wetland delineation and JD	Environmental consultant / Corps	Jurisdictional boundaries and permit path
Utility will-serve letters	Each utility	Written commitment of capacity
Fire flow test report	Fire department or engineer	Available fire flow
Traffic impact study	Traffic engineer	Required off-site mitigation
Conceptual grading and utility plan	Civil engineer	Earthwork quantities, detention sizing, yield

Failure Modes

Ordering an ALTA without specifying Table A items. The survey is correct and useless. Contours, zoning, utilities, and off-site easements are all optional and all must be requested.

Accepting seller-provided reports at face value. They are addressed to the seller, may be stale, and were scoped to answer the seller's questions. Reliance letters and, where the report is old, new work.

Reading the geotechnical recommendations without reading the boring logs. The recommendations are the consultant's conservative answer. The logs are the data, and they show variability across the site that a summary recommendation flattens.

Treating "utilities are available" as an answer. Availability is not capacity, capacity is not committed capacity, and committed capacity is not free.

Discovering power constraints during design. Transformer and switchgear lead times now exceed the design phase on many projects. This is a Gate 2 item.

Missing a seasonal window. No amount of money accelerates a growing-season delineation. Identify seasonality at Gate 1 when the diligence period is being negotiated.

Underestimating land consumed by detention. Five to fifteen percent of the site, removed from yield, on top of easements, setbacks, and buffers. On redevelopment sites with no existing detention this is a new requirement entirely.

Sequencing Phase I and Phase II inside a 60-day period. Arithmetically impossible on most sites. Negotiate the period against the investigation plan, not against convention.

Letting findings sit as reports rather than resolving them into the budget. A file of consultant PDFs is not due diligence. The output is a revised development budget, a revised yield, and a decision.

What Happened Next

Kipling Row

Physical diligence ran concurrently with the PUD application and produced \$740,000 of budget movement on a deal with roughly \$2 million of projected profit.

Survey. Actual acreage came back at 3.14 rather than the 3.2 in the listing, a minor adjustment against a per-acre price. More consequentially, the survey documented a gravel drive along the north property line used by the adjacent parcel for at least fifteen years, with no recorded easement. Counsel flagged a potential prescriptive easement claim. Resolved by negotiating a recorded access easement in a defined location that did not conflict with the site plan, in exchange for the developer paving it - \$38,000 and three weeks.

Topography. Fourteen feet of fall across the site toward the rail corridor. Favorable for gravity drainage, unfavorable for cost: the layout required terracing and approximately 340 linear feet of retaining wall averaging five feet of exposed face. \$145,000.

Geotechnical. Eight borings to 25 feet. Claystone bedrock at 8 to 14 feet with measured swell potential in the 3 to 5 percent range, which is unremarkable for the Front Range and unremarkable in its consequences. The recommendation was drilled piers with structural floor systems over void form for the townhome foundations, rather than slab-on-grade with moisture conditioning.

The delta was approximately \$11,500 per unit, or \$391,000 across 34 units. This single finding was the largest number in the module, it was entirely predictable from the regional geology, and it had not been in the original underwriting.

Environmental. The Phase I found no on-site RECs from the church use, but historical fire insurance maps showed a 1,000-gallon heating oil tank serving the original 1948 structure, with no record of removal. A limited Phase II located the tank under what is now the parking area, confirmed it was in place and had leaked. Removal, excavation of impacted soil, confirmation sampling, and a state no-further-action determination: \$61,000 and seven weeks, which required a 30-day extension of the due diligence period.

Utilities. Water and sewer adequate in the arterial. Fire flow tested acceptable. Cash-in-lieu of water rights dedication at \$3,200 per unit - \$108,800, a line the developer had never encountered on prior projects in an adjacent jurisdiction that handles it differently. Power and gas routine.

Access and stormwater. Access spacing standards required the entrance approximately 180 feet east of where the PUD site plan had placed it, which was the issue staff had flagged at the pre-application meeting in Module 3. The revised layout absorbed it without losing units. Detention and water quality consumed 0.34 acres - about 11 percent of the site - which had been assumed at 8 percent. Combined with the easement resolution, net yield held at 34 units, with no margin left for further reduction.

Where it landed. Budget up approximately \$740,000. Yield unchanged. Neither finding was disqualifying and all of them were knowable in advance, which is exactly the argument the developer took into the price renegotiation in Module 7.

And all of it remains contingent on a council vote that has not yet happened.

Six Corners

The vacant-land track still applies to an improved site, and here it produced the finding that should have ended the conversion thesis four months before anything else did.

Survey. The ALTA plotted the 1986 REA easements, confirming the shared access from the signalized entrance and the cross-parking arrangement. It also established the parking count at 312 stalls against an REA obligation to maintain not fewer than 290 available for shared use across the three parcels. The practical effect is that only about 22 stalls' worth of the parking field can be built on without an REA amendment - which stacks on top of the use-restriction amendment already required from Module 3. Any redevelopment of the surface lot is a negotiation with the same two neighbors.

Environmental. The Phase I identified an off-site REC. A dry cleaner had operated in a strip center two parcels upgradient from approximately 1979 to 2004, with a documented PCE release, an open state file, and a groundwater plume mapped in the direction of the subject property.

A Phase II with soil gas sampling was ordered. PCE in soil gas beneath the building came back **above residential vapor intrusion screening levels and below commercial and industrial screening levels.**

That single result means: residential occupancy requires vapor mitigation - a sub-slab depressurization system, ongoing monitoring, a permanent operational obligation, disclosure to every future resident, and a lender and insurer who will each have opinions. Commercial or light industrial occupancy requires none of it.

The conversion thesis was now carrying a mitigation cost, a permanent operating obligation, and a financing complication that the industrial alternative did not have. The sponsor recorded it as a cost line and kept going.

Power. Existing service was 800 amps at 277/480, appropriate for a 1984 office building. A residential conversion at any reasonable unit count would require a substantially larger service and a new transformer. The utility quoted a 52-week lead time on the transformer and would not begin its design process without a load letter. A light industrial or flex conversion with modest process load fits within or near the existing service.

Stormwater. The site was built in 1984 with no detention, under regulations that did not require it. Redevelopment triggers current standards. Retrofit detention would consume roughly 0.4 acres of the parking field - the same parking field constrained by the REA.

Geotechnical. A limited investigation for any new foundations found competent bearing and no significant issues. The only good news in the module.

Where it landed. Three independent findings, each of which favored an industrial or commercial reuse over a residential conversion: vapor intrusion screening levels, electrical service capacity, and an REA that permits commercial service uses while prohibiting residential outright.

The sponsor was still underwriting apartments.

Module 4B is where that stops.

Optional: Try This

Find a published geotechnical report - many are public record through state DOT project files, university capital projects, and municipal bid document archives - and read the boring logs before reading the recommendations. See if you can predict the foundation recommendation from the logs alone. It is the single fastest way to develop intuition for how soil becomes cost.

Then, for the parcel you have been screening: pull the FEMA flood map, the county soil survey, and the utility service area maps. Ask the one question that matters. Not whether service exists, but who you would have to call to find out whether it has capacity, and how long they would take to answer.

Next - Module 4B: Physical Due Diligence, Existing Buildings. Condition, hazardous materials, and the existing-building code triggers that determine whether a reuse is viable at all. The longest module in Part 1, and the one where Six Corners is decided.

1.4B - Physical Due Diligence: Existing Buildings and Infrastructure

Every argument for buying an existing building sounds the same. It is trading below replacement cost. The bones are good. The location cannot be reproduced. Somebody already paid for the structure, the envelope, and the entitlements, and you are buying all of it at a discount.

Sometimes that is true. More often, the building is trading below replacement cost because the market has correctly concluded that it is not worth replacing, and the discount is not an opportunity but an accurate price.

The difference between those two cases is not determined by the purchase price. It is determined by a set of physical and regulatory facts that are fully knowable before Gate 2 and that most buyers do not investigate until after. Three of them do most of the work:

Condition tells you what you must spend to keep the building operating as it is.

The code cliff tells you what the building code will require the moment you change it, which is frequently a larger number than the renovation itself.

Geometry tells you whether the building can physically become the thing you want it to be, and geometry does not respond to money.

This is the second test of highest and best use applied to an improved property, and it is the test that resolves Module 2's other question: whether highest and best use as improved beats highest and best use as vacant. When it does not, the building is a demolition line item regardless of what you paid for it.

A note on the schedule before anything else. **Existing-building diligence needs more time than vacant-land diligence, and the reason is dependency chains.** A hazardous materials survey cannot be scoped until the renovation concept exists. A code analysis cannot be performed until the alteration level is known, which depends on the design concept, which depends on the structural and geometric findings. Estoppel certificates require tenant cooperation on the tenants' timeline. Sixty days is not enough. Ninety to one hundred twenty is the working minimum, and that is a Gate 1 negotiation, which is why Module 7 spends time on due diligence period length.

Section 1 - The Property Condition Assessment

1.1 What It Is

A Property Condition Assessment performed to ASTM E2018 is a walkthrough-level evaluation of a building's physical systems by an assessor, resulting in a Property Condition Report (PCR). It is the existing-building analogue of the Phase I: a records-and-observation exercise, not a testing program.

Scope typically includes:

- Site: paving, drainage, retaining structures, landscaping, lighting
- Structural frame and foundation, by observation
- Building envelope: roof, walls, windows, doors, sealants
- Interior elements and finishes
- Mechanical, electrical, plumbing, and fire protection systems
- Vertical transportation
- Accessibility, at a screening level only
- Life safety and code, at a screening level only
- Document review: prior reports, permits, maintenance records, warranties
- Interviews with building staff

Scope explicitly excludes: testing of any system, destructive investigation, opening of concealed spaces, hazardous materials, environmental conditions, structural calculations, and any determination of code compliance. Each of those is a separate scope, and the significant ones are separate sections of this module.

1.2 Remaining Useful Life

The central concept. Each system is assigned an expected useful life, an effective age, and a remaining useful life, and the report projects when replacement will be required.

Typical expected useful lives, which vary widely with climate, maintenance, and quality:

System	Expected useful life
Built-up or modified bitumen roof	15–25 years
Single-ply membrane roof	15–25 years
Asphalt paving	15–25 years, with resurfacing at intervals
Exterior sealants	10–20 years

Aluminum windows and curtain wall	30–40 years, glazing and seals shorter
Packaged rooftop HVAC units	15–20 years
Chillers	20–25 years
Boilers	25–35 years
Air handlers	20–30 years
Electrical distribution	30–40 years
Domestic water and waste piping	40–75 years, material dependent
Hydraulic elevator	20–30 years to modernization
Traction elevator	25–35 years to modernization

The number that matters is not the age but the remaining life relative to your hold period. A system with eight years left is a non-event for a five-year hold and a certainty for a fifteen-year hold.

1.3 Immediate Repairs Versus Replacement Reserves

The PCR divides costs into two categories that are used completely differently.

Immediate repairs are deficiencies requiring attention now: life safety issues, code violations, active water intrusion, systems past failure. These are a cash requirement at or shortly after closing, and they are funded either from acquisition capital or from a lender-held escrow.

Replacement reserves are the projected cost of replacing systems over the analysis period, usually ten or twelve years, expressed as an annual per-square-foot or per-unit figure. These are an operating expense line, reducing net operating income and therefore value.

1.4 How a PCA Feeds an Acquisition Budget Differently Than a Hold Underwriting

This is the distinction that most often gets lost, and it changes what the same report means.

For a hold underwriting - buy it, operate it, sell it later - the PCA's output is a reserve figure. You are keeping the building as it is. Immediate repairs are a capital line; replacement reserves are an expense line that depresses NOI and, at a given cap rate, value. The lender will size a replacement reserve escrow. The analysis is straightforward.

For an acquisition preceding a repositioning - buy it, change it, re-lease it - the PCA is a different document entirely, and reading it as a reserve schedule is a serious error.

Three reasons:

Systems you are replacing anyway have no remaining life value. If the renovation gut-replaces the HVAC, the fact that the existing units have four years left is irrelevant. Do not carry it as a cost and do not carry it as an asset.

Systems you intend to keep must be evaluated against the new use, not the old one. An electrical service adequate for an office is not adequate for apartments. A roof with ten years of life over an office has ten years of life over a warehouse and considerably less over a use that adds rooftop equipment.

The PCA screens code and accessibility; it does not analyze them. The PCR will note that restrooms appear non-compliant. It will not tell you that your renovation triggers an accessible path of travel obligation. That analysis is Section 3, it is the largest number in most repositioning budgets, and it is not in the PCA.

The practical consequence: **on a repositioning deal, the PCA is scoped alongside a renovation concept, not before one.** A PCA ordered on day one of diligence with no concept behind it will produce a competent report answering the wrong question.

Section 2 - Hazardous Materials

Out of scope for both the Phase I and the PCA. Ordered separately, and on renovation deals, ordered early, because abatement is one of the few costs in development that cannot be value-engineered.

2.1 Asbestos

The dominant hazardous materials issue in renovation.

The common belief that asbestos was banned in 1978 is wrong and expensive. What happened in the late 1970s was the prohibition of specific product categories, principally sprayed-on friable insulation and fireproofing. **Many non-friable asbestos-containing products remained in production into the 1980s, and some asbestos-containing materials remain legal today.** Vinyl floor tile and mastic, roofing felts and coatings, joint compound, cementitious board, gaskets, and duct wrap all persisted well past the date most people assume.

The practical rule: **any building constructed before roughly 1990 should be assumed to contain asbestos until surveyed**, and in most jurisdictions a survey is legally required before demolition or renovation regardless of vintage, under NESHAP and state regulations.

Two survey levels. A **pre-renovation or pre-demolition survey** is invasive: it samples suspect materials in the areas to be disturbed, including inside walls and above ceilings. A **management survey** for an operating building is less invasive. The pre-renovation survey is the one a repositioning deal needs, it requires access to the space, and it requires knowing which areas will be disturbed - another dependency on having a concept.

Cost consequences: abatement is performed by licensed contractors under containment, with air monitoring, regulated disposal, and notification periods before work begins. It is schedule-critical because it must generally precede other trades and cannot overlap with them. It is one of the few scopes where the price is what it is.

2.2 Lead

Lead-based paint was banned for residential use in 1978. It was not banned for commercial and industrial coatings, and structural steel primers containing lead are common in buildings of all vintages.

The regulatory consequence that surprises people: OSHA's lead in construction standard applies to worker exposure from disturbing lead-containing coatings **at any concentration**, not only above a residential threshold. Welding, cutting, grinding, or torching coated steel triggers exposure controls, monitoring, and disposal requirements. Residential renovation adds the EPA Renovation, Repair and Painting rule for pre-1978 buildings.

2.3 PCBs

Polychlorinated biphenyls were used in fluorescent light ballasts, transformers, hydraulic equipment, and - the one that catches people - **building sealants and caulking**, primarily in construction between roughly 1950 and 1979.

Ballasts are the routine item, handled as regulated waste during a lighting replacement. Sealants are the serious one: PCB-containing caulk at window perimeters and expansion joints can contaminate adjacent masonry and soil, and remediation obligations under TSCA are substantial. Any envelope or window replacement on a building of that era warrants sealant sampling before a price is committed.

2.4 Mold and Moisture

Not federally regulated in the way the preceding materials are, and a real cost and liability nonetheless. Partially vacant buildings with deferred maintenance are the classic case: an aging roof over unoccupied, unconditioned space produces moisture problems nobody reported because nobody was there. Remediation cost depends entirely on extent, which is not knowable from a walkthrough. Where a building has a roof past its life and low occupancy, assume moisture until shown otherwise.

2.5 Other Items

Underground storage tanks at older industrial and institutional properties, covered in 4A. **Mercury** in thermostats, switches, and some lamps. **Refrigerants**, where phased-out refrigerants drive equipment replacement independent of remaining life. **Radiological sources** in certain medical and industrial equipment. **Silica** as a construction-phase exposure issue rather than a material present in the building.

2.6 Why Abatement Cannot Be Value-Engineered

Worth stating plainly, because it is the reason this section belongs in an underwriting module rather than a construction one.

Most construction costs respond to design changes. A cladding system can be downgraded, a mechanical system simplified, a finish package reduced. Abatement responds to none of that. The material is present in a quantity, the law requires it be removed under a defined procedure by licensed personnel with monitoring and regulated disposal, and the only variable is how much you disturb. A renovation concept that reduces the disturbed area reduces abatement cost; nothing else does.

When a repositioning budget comes in high and the team begins value engineering, abatement is a fixed floor. Underwrite it as one.

Section 3 - The Code Cliff

This is the most important section in Part 1 for a reader with an AEC background, and it is where adaptive reuse deals die.

They do not die at plan review. By plan review the land is bought, the design is paid for, and the loan has closed. They die at plan review only in the sense that a patient dies at the hospital. The cause of death was administered months earlier, at underwriting, when nobody priced the trigger.

The remainder of this section describes the framework of the International Existing Building Code. **Verify everything against the edition your jurisdiction has adopted and its local amendments**, which are extensive in this area - some jurisdictions adopt the IEBC, some use the existing-building provisions of the IBC, some maintain their own code, and retroactive requirements frequently arrive through the fire code or a local ordinance rather than the IEBC at all. The framework is stable; the specifics are not.

3.1 Legal Nonconforming Status, and How It Is Lost

An existing building is generally permitted to remain in its existing condition even where that condition does not meet current code. This is the foundational principle of existing-building regulation and it is what makes the existing stock legal.

It is also conditional. Nonconforming status is lost or curtailed by:

- **Alteration**, in proportion to the extent of the work
- **Change of occupancy classification**, which is the big one
- **Discontinuance of the use** for a defined period, under the zoning ordinance rather than the building code - typically six months to two years, after which the nonconforming *use* right lapses and the property reverts to what the district permits
- **Damage exceeding a threshold percentage of value**, after which reconstruction must comply
- **Retroactive requirements** adopted by ordinance, which apply regardless of whether any work is done

Two of these deserve emphasis for a buyer of a vacant building. **Discontinuance** means a building vacant for two years may have lost the right to resume its prior nonconforming use - a zoning question, checked in Module 3, that has nothing to do with the building code. And **retroactive requirements** - high-rise sprinkler ordinances, existing-building fire alarm mandates, seismic retrofit ordinances in certain jurisdictions, elevator safety upgrades - apply on a schedule set by the ordinance and are inherited by whoever owns the building when the deadline arrives.

3.2 The Three Compliance Methods

The IEBC offers three routes, and the choice among them is a real design and cost decision made early.

Prescriptive Compliance Method. A relatively simple path in which alterations must comply with the IBC for new construction, with defined exceptions for existing conditions. Straightforward to administer, generally the most demanding, and best suited to limited work.

Work Area Compliance Method. The route most repositioning projects take, and the one worth understanding in detail. Requirements scale with the extent and nature of the work, organized through the concept of a **work area** and three alteration levels. This method comprises the chapters on repairs, the three alteration levels, change of occupancy, additions, historic buildings, and relocated buildings.

Performance Compliance Method. A scoring system evaluating the building across a set of safety parameters - height, area, compartmentation, egress, fire protection, and others - and requiring the composite score to meet a threshold. It permits trade-offs: exceeding requirements in one area to offset a deficiency in another. Valuable on buildings that cannot meet prescriptive requirements in a specific respect but are demonstrably safe overall. It requires an experienced code consultant and a building official willing to engage with it.

The decision matters at underwriting because the methods produce different costs on the same project, and a deal that fails under one may work under another.

3.3 The Work Area Concept

Work area means the portions of the building affected by the alterations, as shown on the construction documents. It excludes areas where only like-for-like replacement occurs.

This definition has a consequence that drives design: **the work area is drawn by the design team, and where it is drawn determines which requirements attach.** Percentage thresholds are computed against it. A renovation concept that keeps the work area below a threshold has a materially different cost than one that crosses it, and the difference can exceed the cost of the additional work.

This is legitimate and it is also the most common place where a project's scope is shaped by code thresholds rather than by program. It is worth knowing before you underwrite, because it means the renovation scope and the code cost are not independent variables.

3.4 The Three Alteration Levels

Level 1 - Replacement in kind. Removal and replacement, or covering, of existing materials, elements, equipment, or fixtures with new materials serving the same purpose. New carpet, new roof membrane, replacement equipment of the same type. Requirements are limited: the replacement materials must meet current requirements for those materials.

Level 2 - Reconfiguration. Any reconfiguration of space, addition or elimination of a door or window, reconfiguration or extension of a system, or installation of additional equipment. Level 2 includes everything required at Level 1, plus requirements addressing the work area for means of egress, fire protection, accessibility, structure, and energy.

This is the level nearly every real tenant improvement and most renovations hit, because moving one wall or adding one door is a reconfiguration. Within Level 2, additional requirements attach where the work area exceeds a percentage - commonly 50 percent - of the floor area, with automatic sprinkler protection being the most consequential of them.

Level 3 - Extensive work. Where the work area exceeds 50 percent of the aggregate area of the building. Level 3 includes Levels 1 and 2, and adds requirements applying to the **entire building** rather than to the work area: building-wide fire protection, egress, and in defined circumstances structural evaluation and energy compliance.

The cliff is between Level 2 and Level 3, and it is a cliff rather than a slope. A renovation at 49 percent of building area and a renovation at 51 percent are separated by a requirement set that can include sprinklering a building that has never been sprinklered, upgrading the fire alarm building-wide, and evaluating the lateral system. On a gut repositioning, you are at Level 3, and it must be underwritten as such from the first budget.

3.5 Change of Occupancy

The largest single trigger in existing-building work, and the one adaptive reuse projects cannot avoid, because adaptive reuse *is* a change of occupancy by definition.

A change of occupancy classification requires the building to comply with the requirements applicable to the new classification. Depending on the direction of the change, this can reach:

- **Heights and areas** permitted for the new occupancy in the existing construction type - this can be an absolute bar, because the existing construction type is not changeable in any practical sense
- **Fire protection systems**, including automatic sprinkler thresholds that differ by occupancy
- **Means of egress**, sized for the new occupant load, with new travel distance, corridor rating, and stair requirements
- **Fire resistance ratings** and separation of occupancies in mixed-use buildings
- **Structural**, where the new occupancy carries higher live loads or a higher risk category
- **Accessibility**, where the new occupancy is subject to requirements the old one was not
- **Energy**, where the new use changes the applicable provisions
- **Light and ventilation**, which residential occupancies are subject to in ways commercial occupancies are not

The IEBC organizes much of this around **relative hazard**: where the new occupancy is of equal or lesser hazard than the old in a given category, the existing condition may be permitted to remain; where it is greater, compliance is required. This is why some conversions are far cheaper than others, and why a general statement like "conversions are expensive" is unhelpful. **The specific direction of the occupancy change determines the cost**, and it must be analyzed as a change from a specific classification to a specific classification, not as a generic renovation.

A change of occupancy is also possible for only part of a building. Converting a ground floor while leaving upper floors in their existing classification creates a mixed-occupancy building with separation requirements, and it is frequently the cheaper path.

3.6 Accessibility

Accessibility obligations in existing buildings come from two directions that must both be satisfied: the building code as adopted, and the Americans with Disabilities Act, which is federal civil rights law enforced through litigation rather than through plan review. Compliance with one does not establish compliance with the other.

The core requirements in alteration:

Altered elements must comply, to the maximum extent technically feasible. If you renovate a restroom, the restroom must comply.

Alterations affecting an area containing a primary function trigger an obligation to provide an **accessible route to the altered area**, including the restrooms, drinking fountains, and telephones serving it. This is the requirement that surprises budgets, because the cost is not in the space you are renovating - it is in the path to it. Accessible entrance, accessible route through the building, compliant restrooms, sometimes an elevator.

The 20 percent rule. The path-of-travel obligation is limited: the cost of providing the accessible route need not exceed **20 percent of the cost of the alteration to the primary function area**. This cap is genuinely useful and it is routinely misunderstood in two directions. It is not a cap on accessibility cost generally - altered elements must still comply, outside the cap. And it scales with the renovation: a larger renovation budget produces a larger accessibility obligation, which means the cap does not protect a gut repositioning in the way people assume.

Change of occupancy can bring requirements approaching those for new construction.

Buildings constructed before 1990 are the ones to scrutinize. An entire generation of stock was built with restrooms, entrances, parking, and routes that were compliant when built and are not compliant now, and the moment you touch a primary function area, the path to it comes into scope.

3.7 Structural

Structural provisions apply across the compliance methods and turn on how much you change the demand on existing elements.

The framework uses thresholds. Existing gravity-load-carrying elements may generally remain where the alteration does not increase their demand beyond a small percentage - commonly stated as 5 percent. Existing lateral-force-resisting elements may generally remain where the increase in demand is below a larger threshold - commonly 10 percent. Exceed the threshold and the element must be evaluated and, where deficient, strengthened to current requirements.

Triggers that commonly cross these thresholds in repositioning work:

- **Change of occupancy to a higher live load.** The direction matters enormously. Converting an office to residential generally *reduces* required live load. Converting an office to storage or light industrial substantially *increases* it, and a floor designed for office loading will not carry storage loading without strengthening, which on a multi-story building can be prohibitive.
- **Rooftop equipment**, which adds gravity load and often requires framing reinforcement
- **New openings in shear walls or braced frames** - including loading dock doors, storefront, and new windows - which remove lateral capacity and can trigger evaluation of the entire lateral system
- **Vertical additions**
- **Change to a higher risk category**, which increases seismic and wind design requirements
- **Substantial structural alteration**, which in defined circumstances triggers evaluation of the lateral system as a whole

Seismic evaluation and retrofit are performed to ASCE 41, and in high-seismic jurisdictions the retrofit can exceed the rest of the renovation combined. Some jurisdictions impose mandatory retrofit ordinances on defined building types - unreinforced masonry, non-ductile concrete,

soft-story wood frame - on a fixed schedule, entirely independent of whether any work is proposed.

3.8 Fire Protection and Egress Retroactivity

Worth separating out because the source of the requirement is often not the IEBC.

Sprinkler thresholds vary by occupancy, and a change of occupancy can bring a building across one. A useful example: storage occupancies commonly require sprinklers above a fire area threshold that office occupancies do not share, which means an office-to-storage conversion can require sprinklering a building that never needed it.

Fire area subdivision as an alternative. Where a threshold is stated per fire area rather than per building, subdividing the building with rated fire barriers can keep each area below the threshold. On a project that needs demising walls anyway - multi-tenant flex, for instance - the demising construction can serve both purposes, which is one of the more elegant cost avoidances available in existing-building work.

Retroactive requirements from the fire code and local ordinance. Existing-building fire code provisions and local ordinances impose requirements on existing buildings independent of any alteration: high-rise sprinkler retrofits, fire alarm upgrades, emergency responder radio coverage, elevator recall. These have compliance deadlines and they transfer with ownership.

Egress. Occupant load under the new use drives exit capacity, number of exits, travel distance, common path of travel, dead-end corridor limits, and corridor rating. Residential occupancies have tighter constraints than office in several of these, which is one of the quieter reasons office-to-residential conversions require new stairs.

3.9 The Handshake to Part 2

Everything in this section is the same body of law covered in **Part 2 Module 7**, encountered from the other side.

In Part 2, the code analysis is a design activity. The architect performs it, documents it, and resolves it through design. It happens after the owner is committed and it is treated as a technical problem with a technical answer.

In Part 1, the identical analysis is an **underwriting input**. It happens before the building is bought, it produces a number that goes into the budget, and its output is a price, a renegotiation, or a decision not to proceed.

The framework does not change. What changes is that in Part 1, the answer can still be "then we do not buy it."

The practical implication for a developer: **a code consultant or architect must be engaged during due diligence on any repositioning deal, at concept level, to perform a**

preliminary code analysis. It costs a small fraction of the diligence budget. Skipping it is the single most expensive economy available in this module, and it is made constantly, because the design team is not conventionally engaged until after closing.

Section 4 - Physical Capacity for Reuse

Code determines what you must do. Geometry determines what you can do, and geometry is the one category in this module that does not respond to money.

4.1 Floor Plate Depth and the Window Line

The controlling constraint in residential conversion, and the reason most office-to-residential projects fail before cost is ever discussed.

Residential units require access to exterior windows - for light and ventilation requirements, for habitability, and above all for rentability. The maximum practical depth from window line to unit entry runs roughly 28 to 32 feet before units become dark, awkward, and hard to lease. A double-loaded corridor building therefore works at a total depth of roughly 60 to 70 feet.

Office buildings of the 1960s through the 1990s were designed on the opposite logic. Deep floor plates with central cores maximized rentable area per floor and per curtain wall dollar, and depths of 120 to 180 feet are entirely normal. A central core building 150 feet deep has roughly 60 feet from window line to core on each side - twice what a residential unit can use.

The area beyond reach of a window has three possible fates:

- **Carve it out.** A light well, courtyard, or atrium cut through the floor plates. This works, and it costs: structural modification at every floor, new exterior wall and waterproofing on the new faces, drainage, and the permanent loss of the floor area removed. It is the standard solution and it is why successful conversions concentrate in buildings that are relatively narrow to begin with.
- **Use it for something else.** Amenity, storage, back-of-house, parking. Useful in moderation, and there is a limit to how much unleasable core a residential building can carry.
- **Accept it.** Deep, poorly lit units that lease at a discount or do not lease.

The screening heuristic is crude and effective: **measure the depth from the window line to the core. If it substantially exceeds about 30 feet on both sides, the building is a conversion candidate only with major surgery, and the surgery must be priced before anything else in the analysis.**

Core position matters as much as depth. A building with the core pushed to one side or to an exterior face can work as single-loaded or modified double-loaded residential. A central core in a deep plate is the hardest case.

4.2 Floor-to-Floor Height

Residential conversion adds systems that office buildings do not have, and they all need vertical space in the ceiling cavity: unit-level HVAC, sprinkler mains and branches serving a compartmented plan rather than an open one, and - the difficult one - **sanitary drainage with slope, distributed throughout the floor plate rather than concentrated at a wet core.**

Office buildings concentrate plumbing at the core. Residential buildings need waste piping from every kitchen and bathroom, sloped to a stack. On a floor plate where the stacks do not exist, this means either a large number of new penetrations through the structure or furred chases and raised floors.

Floor-to-floor heights of 12 feet and up give room to work. Heights of 10 to 11 feet, common in some vintages, make it very difficult to fit the systems while retaining acceptable ceiling heights.

A specific hazard worth flagging: **post-tensioned concrete slabs.** Core drilling through a post-tensioned slab without accurate tendon location can sever a tendon, which is a serious structural and safety event. Conversions of post-tensioned buildings require scanning before every penetration, which is a real cost and a real schedule item on a project that needs hundreds of them.

4.3 Structural Capacity by Direction of Conversion

The live load question, which cuts in opposite directions depending on the conversion:

Conversion	Live load direction	Typical outcome
Office to residential	Decrease	Generally not the constraint
Office to storage or light industrial	Substantial increase	Frequently prohibitive on elevated floors
Office to assembly	Increase, plus risk category	Requires evaluation
Retail to office	Roughly neutral	Usually workable
Industrial to office	Decrease	Generally workable
Any to any, with added rooftop equipment	Localized increase	Framing reinforcement

The office-to-industrial case has a clean logical resolution that is worth knowing: **slab-on-grade will carry nearly any loading; elevated floors will not.** A conversion that puts the

heavy-load use on grade and leaves elevated floors in a lighter classification avoids the problem entirely, at the cost of a mixed-occupancy building with separation requirements. On a two- or three-story building this is often the answer.

4.4 Envelope

Curtain wall from the 1960s through the 1980s is typically single glazed or early insulating glass with thermally unbroken aluminum framing. It performs poorly, it will not meet current energy requirements for altered components, and in a residential use it produces condensation and comfort complaints that generate tenant turnover. Replacement is a major cost and frequently a code trigger through the structural implications of a new wall system.

Punched windows in a masonry or precast wall are much better news. They can often be replaced individually, at unit cost, without touching the wall.

Precast and tilt-up panels are generally durable, with the failure mode at the sealant joints rather than the panels. Sealant of a certain era brings the PCB question from Section 2.3.

Operability and ventilation. Residential occupancies require light and ventilation, satisfied either through openable area or mechanically. Office buildings have fixed glazing and central air. Mechanical ventilation usually resolves the code requirement; it does not resolve the market expectation that an apartment has a window that opens.

4.5 Core, Egress, and Vertical Transportation

Stairs. Office stair locations are placed for an open plan with long travel distances through unobstructed space. A compartmented residential plan has shorter travel distance limits and dead-end corridor limits, and the existing stairs are frequently in the wrong places. New stairs mean new floor openings, new structure, and lost area.

Elevators. Office elevator counts are sized for peak-hour office traffic and are usually adequate or generous for residential. Cab size, controls, and accessibility compliance are the issues. Hydraulic elevators of certain vintages have below-grade cylinders in unlined casings, which are both an environmental exposure and a replacement trigger.

The core generally. In a conversion, the office core is a large block of space in the middle of the building that no longer does what it did. Whether it becomes amenity, storage, or waste is one of the first design questions and one of the largest efficiency variables.

4.6 The Efficiency Test

The arithmetic that summarizes this section. Take gross building area, subtract what geometry and code make unusable - core, unlightable interior, area lost to new stairs and shafts and light wells - and compute the net rentable area achievable.

Purpose-built residential runs roughly 80 to 85 percent efficient. Conversions frequently land in the 65 to 75 percent range, and a bad one lands lower. That difference flows directly into cost per rentable square foot, which is the number that determines whether the deal works.

4.7 The Test That Should Be Run First

Before any of the detailed analysis above, run this:

What does it cost, all-in, per unit or per square foot, to convert this building - and what does it cost to build the same thing new on comparable land?

If conversion costs more than new construction, the deal requires a specific justification: a location that cannot be replicated, an entitlement that cannot be obtained, historic tax credits, or a schedule advantage worth paying for. Those justifications are real and they exist. But they must be named explicitly, because "below replacement cost" is a statement about the price of the building and not about the cost of the project, and the two are routinely confused.

A conversion that pencils only because the acquisition was cheap, while the conversion cost exceeds ground-up, is a deal in which the acquisition discount has been consumed and then some.

Section 5 - Historic Status

5.1 The Distinction That Matters

Local landmark or historic district designation is regulation. A preservation commission reviews and approves exterior alterations, additions, and demolition. It can deny. Review is discretionary, adds months, and constrains materials and details - windows, in particular, are the most contested item in preservation review and one of the largest cost variables in an older building.

National Register listing is, for a privately funded project with no federal involvement, largely honorific. It does not prevent alteration or demolition. It carries review obligations only where there is a federal nexus.

Confusing these is common and consequential in both directions: assuming National Register listing prevents you from doing something it does not, or assuming local designation is honorific when it is regulatory.

Also check for **conservation easements**, which are private, permanent, recorded restrictions on alteration - a Schedule B item from Module 3 that behaves like designation and never appears on a historic register.

5.2 The Tax Credit Trade

The federal rehabilitation tax credit provides a credit equal to a percentage of qualified rehabilitation expenditures for certified historic structures, subject to a substantial rehabilitation test and claimed over a period of years. Many states offer additional credits that can stack. The combined benefit is significant enough to change the outcome of a deal.

The price is compliance. Work must be certified as meeting the Secretary of the Interior's Standards for Rehabilitation, through a multi-part application reviewed by the state historic preservation office and the National Park Service. In practice this means retaining and repairing historic fabric rather than replacing it, windows that match rather than windows that perform, and a review process with authority over design decisions.

The trade is real and it is genuinely favorable on some buildings: **the credits pay for compliance you would not otherwise accept.** The failure mode is underwriting the credits without underwriting the compliance cost and schedule, or without engaging a consultant experienced in the certification process. There is also a timing risk - the credits are realized over years and are typically monetized through a tax credit investor, which is a financing structure with its own complexity and its own fees.

5.3 The IEBC Historic Provision

The existing-building code contains provisions for historic buildings permitting some relief from requirements where strict compliance would threaten historic fabric, subject to the building official's judgment and to the building being formally designated or determined eligible. This is genuinely useful and it is discretionary - it depends on the building official, and it should be confirmed in a pre-application conversation rather than assumed in an underwriting.

Section 6 - Income-Producing Assets: Leases and Tenants

When the building has tenants, you are buying a set of contracts as much as a building, and the contracts constrain the renovation.

6.1 The Rent Roll Audit

The rent roll is the seller's summary. The leases are the truth. Auditing means reconciling one to the other, and discrepancies are normal rather than exceptional.

Check every line against the lease document: tenant name and actual entity, square footage as measured versus as stated, commencement and expiration, current rent, escalation schedule, free rent and abatement periods, security deposit held, and the reimbursement structure.

Then reconcile against the operating statements. Rent actually collected versus rent contractually due reveals delinquency the rent roll does not show.

6.2 Lease Abstraction

Each lease is reduced to a summary of the terms that matter. The provisions that most affect a repositioning:

- **Term, options, and notice requirements.** Renewal options, particularly at below-market rates or at fixed rents, constrain your ability to reposition and reprice. An option exercisable years out is an encumbrance.
- **Termination rights.** Tenant termination options, and - critically - **landlord termination or relocation rights**, which are what permit a repositioning to proceed.
- **Expansion, contraction, rights of first refusal and first offer** on adjacent space, which constrain how you can re-demise the building.
- **Exclusive use clauses and co-tenancy**, in retail especially, which restrict who else you can lease to and can give tenants rent relief or termination rights when occupancy drops.
- **Permitted use**, which determines whether your intended repositioning conflicts with a sitting tenant's rights.
- **Operating expense and reimbursement structure:** gross, modified gross, or triple net; base years; caps on controllable expenses; whether capital costs can be passed through and over what amortization period. This determines who pays for the improvements you are about to make.
- **Landlord obligations:** services, repair and maintenance, quiet enjoyment. A renovation that disrupts services or access can breach the lease and generate abatement claims or termination rights.
- **Assignment and subletting**, and **change of control** provisions.
- **Unfunded tenant improvement allowances and outstanding leasing commissions**, which you inherit as liabilities at closing and which are frequently omitted from the seller's summary.

6.3 Estoppel Certificates

A signed statement from each tenant confirming the lease terms, the rent, the deposit, and that no defaults or unfulfilled landlord obligations exist. Delivered at or before closing and typically a condition of the purchase agreement and of the loan.

Estoppels exist because the seller's file may be incomplete. Side agreements, verbal arrangements, abatements granted informally, and promises made to keep a tenant during a soft market may not appear in the lease documents. The estoppel is where a tenant states its understanding, and a discrepancy between the estoppel and the file is a finding, not a formality.

Practical note: **estoppels require tenant cooperation and tenants are not fast.** Circulating, chasing, and reconciling them takes weeks, and a tenant with a grievance may use the estoppel as leverage. This is one of the dependencies driving a longer diligence period.

6.4 SNDAs

A Subordination, Non-Disturbance and Attornment agreement governs the relationship between a tenant and the lender. The tenant subordinates its lease to the mortgage; the lender agrees not to disturb the tenant's possession on foreclosure; the tenant agrees to recognize the lender or its successor as landlord. Required by lenders for significant leases, and - like estoppels - dependent on tenant signature on the tenant's schedule.

6.5 Renovating Around Occupancy

The operational reality that repositioning underwritings routinely omit.

- **Phasing** the work to keep occupied areas functional, which is less efficient and more expensive than a vacant building
- **Maintaining services, access, egress, and parking** for sitting tenants throughout, which is a lease obligation and a code requirement simultaneously
- **Noise, dust, vibration, and hours restrictions**, which can push work to nights and weekends at premium cost
- **Abatement adjacent to occupied space**, which requires containment, monitoring, and often tenant relocation
- **Buyouts and relocation**. Frequently the cheapest path. A tenant with two years remaining may accept a payment to leave early, and that payment is often less than the cost of building around them. This is a budget line that should be estimated during diligence.
- **Holdover**. A tenant who does not leave at expiration, typically at a penalty rent, and who may be difficult to remove quickly. A renovation schedule dependent on a specific vacancy date carries this risk.

Section 7 - Infrastructure and Non-Building Assets

Briefly, to signal the category.

Parking structures have a distinct deterioration profile driven by chloride-induced corrosion of reinforcement from de-icing salts, with repair costs concentrated in slab and post-tensioning repair, joint replacement, and waterproofing. They warrant a specialized condition assessment rather than a general PCA, and deferred maintenance on a parking structure compounds unusually fast.

Industrial process facilities carry embedded process equipment, foundations, pits, and utilities that may have no value to a subsequent user and a material cost to remove. Environmental history is typically the dominant issue, and decommissioning is a scope in itself.

Utility corridors, rail spurs, and easement-based assets raise questions of operating rights, maintenance obligations, and abandonment procedures that run on their own regulatory tracks.

The common principle: the more specialized the asset, the more likely its highest and best use is something other than its current use, and the more likely the existing improvements are a liability rather than an asset in that conversion.

Section 8 - Sequencing and Cost

Study	Typical cost	Elapsed time	Dependency
Property Condition Assessment	\$5k–\$20k	3–5 weeks	Scope against renovation concept
Asbestos / hazmat survey	\$5k–\$40k	3–6 weeks	Needs access and disturbance areas defined
PCB sealant sampling	\$3k–\$15k	3–5 weeks	Needed before envelope pricing
Preliminary code analysis	\$8k–\$40k	3–6 weeks	Requires a renovation concept
Structural evaluation	\$10k–\$60k	4–8 weeks	Requires drawings or field measurement
Test fit / conversion feasibility study	\$15k–\$60k	3–6 weeks	The concept everything else depends on
MEP capacity assessment	\$8k–\$30k	3–5 weeks	Concept-dependent
Elevator consultant	\$3k–\$10k	2–4 weeks	Independent
Lease abstraction	\$150–\$500 per lease	2–4 weeks	Requires complete lease files
Estoppel certificates	Legal time	3–8 weeks	Tenant-dependent; not controllable
Vapor intrusion / soil gas	\$15k–\$60k	4–8 weeks	After Phase I

Ranges vary widely with building size and complexity.

8.1 Why the Period Must Be Longer

The chain is the problem. On a repositioning deal the order is:

Concept → code analysis → hazmat scope → structural scope → cost estimate → decision.

Each link requires the one before it. A concept takes three to six weeks. A code analysis behind it takes another three to six. A hazmat survey scoped from the concept takes three to six more, and an estimate behind all of it takes another two to four. That is twelve to twenty-two weeks in sequence, before tenant estoppels - which run on the tenants' clock, not yours - are counted.

Sixty days cannot hold this. **Ninety to one hundred twenty days is the working minimum on a repositioning, and the request has to be made at Gate 1** - when the purchase agreement is negotiated, before the seller has any reason to accommodate you. This is why Module 7 treats the length and structure of the due diligence period as a substantive negotiation rather than a formality, and why extension provisions are negotiated in advance rather than requested at day 55.

Section 9 - Converting Findings Into the Deal

The same four outcomes as 4A, with one addition specific to existing buildings.

A budget line. Immediate repairs, abatement, code-triggered work, structural strengthening, accessibility path of travel, tenant buyouts.

A yield or efficiency reduction. Area lost to light wells, new stairs, shafts, unlightable core. This is worse than an equivalent cost, because it reduces revenue permanently rather than increasing cost once.

A schedule item. Abatement sequencing, long-lead equipment, phasing around tenants, estoppel and consent timelines, historic review.

A risk to price, insure, or refuse. Unknown concealed conditions, a lateral system that has not been evaluated, contamination with permanent controls, a tenant who may not leave.

A change of thesis. The outcome unique to this module. The findings do not merely price the intended use; they can establish that a different use is the highest and best one. This is not a failure of diligence. It is diligence working. The cost of discovering it before Gate 2 is the diligence budget. The cost of discovering it after is the deal.

Artifacts

Artifact	Prepared by	Purpose
Property Condition Report (ASTM E2018)	PCA consultant	Condition, remaining useful life, immediate repairs, reserves
Asbestos and hazmat survey	Licensed inspector	Presence, quantity, and location of regulated materials
Preliminary code analysis	Architect or code consultant	Compliance method, alteration level, change-of-occupancy consequences
Structural evaluation	Structural engineer	Capacity for the intended use; lateral system implications
Conversion feasibility / test fit	Architect	Unit yield, efficiency, geometric viability
MEP capacity assessment	MEP engineer	Whether existing systems and services support the new use
Elevator survey	Elevator consultant	Condition, modernization, compliance
Lease abstracts	Counsel or abstraction vendor	The obligations you are buying
Estoppel certificates	Tenants	Confirmation of lease terms; discovery of side agreements
SNDAs	Tenants and lender	Lender-tenant relationship
ALTA survey, Phase I, geotech	See Module 4A	Still required on improved sites

Failure Modes

Reasoning from replacement cost. The most common error in existing-building acquisition. Below replacement cost is a fact about the building's price, not about the project's cost, and the market's discount is usually informed.

Underwriting a conversion without a preliminary code analysis. The change-of-occupancy consequences are frequently larger than the renovation itself and are fully knowable for a small fraction of the diligence budget. This is the defining failure of adaptive reuse.

Assuming the deal fails on cost when it fails on geometry. Floor plate depth does not respond to money. Run the depth-to-core measurement before building a budget.

Reading the PCA as a reserve schedule on a repositioning deal. Different question, different document, and the code and accessibility analysis it screens but does not perform is usually the largest line in the budget.

Treating the 20 percent accessibility cap as a ceiling on accessibility cost. It caps only the path-of-travel obligation, it scales with the renovation budget, and altered elements must comply regardless.

Assuming 1978 ended asbestos. Non-friable products continued well into the 1980s. Any pre-1990 building requires a survey, and in most jurisdictions the survey is legally required before disturbance.

Value-engineering abatement. It is a fixed floor. The only lever is reducing disturbed area.

Missing nonconforming use discontinuance. A long-vacant building may have lost its nonconforming use rights under the zoning ordinance, which is a Module 3 question that must be asked about a Module 4B asset.

Accepting the rent roll without the leases. Unfunded TI allowances, outstanding commissions, below-market renewal options, and landlord relocation rights are all in the documents and none of them are in the summary.

Scheduling a renovation around a tenant's expiration date. Holdover is real and eviction is slow. Price a buyout.

Running a repositioning through a 60-day due diligence period. The dependency chain does not fit. This is a Gate 1 failure that manifests at Gate 2.

What Happened Next

Six Corners

The sponsor entered diligence underwriting 66 apartments. Ninety-one days later the deal closed as a small-bay flex and commercial service project. Every fact that produced the change was available at Gate 0.

The PCA. A 1984 building maintained at 40 percent occupancy by an out-of-state fund preparing to exit. The original built-up roof had been overlaid once in 1998 and was at the end of a second life, with ponding and visible saturation over the vacant northeast quadrant. Both rooftop units dated to a 1998 replacement and were past useful life. The single hydraulic elevator was original, with a below-grade cylinder in an unlined casing, requiring replacement rather than modernization. The parking field - 4.1 acres of asphalt - showed alligator cracking across roughly half its area and was at the end of its life. Restrooms, entrance, parking, and routes were built to 1984 standards and did not meet current accessibility requirements.

Immediate and short-term repairs, exclusive of anything related to a renovation: **\$2.4 million.**

That number reframes the acquisition. Module 2 established that the sponsor was paying approximately \$2.0 million, or \$25 per square foot, for the improvements. The improvements required \$2.4 million to remain a functioning office building. The existing structure's contribution to value, before any consideration of reuse, was negative.

Hazardous materials. The survey found asbestos in vinyl floor tile and mastic throughout both floors, in roofing felts, and in joint compound - entirely consistent with 1984 construction and entirely contrary to the sponsor's assumption that a 1980s building was clean. Sealant sampling at the precast panel joints came back positive for PCBs at levels requiring regulated handling on removal, which converted any envelope work from a cost question to a compliance question. Moisture damage and microbial growth were confirmed under the saturated roof section.

Abatement, assuming a full interior renovation: **\$1.15 million**, with a sequence that must precede all other trades and cannot be compressed.

The conversion study. The architect's feasibility study is where the thesis ended.

The building is approximately 260 feet by 150 feet, two stories, 39,000 square feet per floor, with a central core. The distance from window line to core is approximately 60 feet on each of the long elevations - roughly double what a residential unit can use.

Making residential work required cutting a light court approximately 20 feet wide and 200 feet long through both floors. That produced:

- 8,000 square feet of floor area removed
- Approximately 9,600 square feet of new exterior wall on the court faces, with waterproofing, drainage, and structural modification at every floor edge
- Structural framing at the new openings and a lateral system evaluation, because removing that much diaphragm affects load paths

- Net rentable area of approximately 56,000 square feet against 78,000 gross, an efficiency of **72 percent**

At an 850 square foot average unit, that is **66 units**.

Code consequences of the conversion, from the preliminary analysis:

- Work area well over 50 percent of the building: **Level 3 alteration**, with building-wide requirements
- **Change of occupancy from B to R-2**, with sprinklering, a fire alarm system, occupancy separation, and full accessibility applying to the new residential use
- New egress stairs required by residential travel distance and dead-end corridor limits, with the existing core stairs in the wrong locations
- The full curtain wall of punched aluminum windows, single glazed and thermally unbroken, failing energy requirements for altered components and unacceptable for residential comfort - full replacement
- Electrical service from 800 amps to a service requiring a new transformer, at the 52-week lead time established in Module 4A
- Plumbing stacks required throughout a floor plate that had them only at the core
- **Vapor mitigation** - sub-slab depressurization with monitoring - because the soil gas result from 4A exceeded residential screening levels

Structural live load, notably, was not a problem. Office loading exceeds residential loading. This is the one thing that would have worked.

Total conversion cost, including acquisition, abatement, immediate repairs, the light court, code-triggered work, and soft costs: approximately **\$21.3 million**. Against 66 units, that is **\$323,000 per unit**.

Purpose-built garden apartments in the same submarket were delivering at approximately **\$240,000 per unit**.

The conversion cost 35 percent more than building new, produced units with worse light and worse layouts, carried a permanent vapor mitigation obligation, and - per Module 3 - **was prohibited by the reciprocal easement agreement**, requiring consent from two neighbors who had no obligation to grant it.

The sponsor had spent four months underwriting a project that was more expensive than its alternative and that they did not have the right to build.

The pivot. Counsel's reading of the 1986 REA resolved the legal question in a useful direction. The "office and office-supporting uses" language appeared in the recitals; the operative covenant was a prohibited-use list naming residential and manufacturing. Warehousing, distribution, and general commercial service uses were not prohibited. The restriction that blocked apartments did not block flex.

Module 2's buried finding came back: industrial vacancy in the submarket at 3.4 percent, small-bay flex under 20,000 square feet effectively unavailable, rents up 31 percent over three years, and the site 2.3 miles from an interstate interchange with adequate depth for a truck court.

The structural constraint shaped the design. Light industrial and storage occupancies require floor loading far in excess of what a 1984 office second floor will carry. Slab-on-grade will carry it; the elevated floor will not. The resolution:

- **Ground floor**, 39,000 square feet, converted to small-bay flex and commercial service. Slab-on-grade handles the loading. Grade-level and dock doors cut into the precast panels, with a truck court in the parking field.
- **Second floor** remains in its existing occupancy classification as office and flex support, avoiding a change of occupancy on that floor entirely and limiting it to Level 2 alterations.
- Mixed occupancy with rated separation between floors.

Two code findings made this work. First, the ground floor storage occupancy crossed a sprinkler threshold that the office occupancy had not - but the threshold is stated per fire area, and a small-bay flex building needs rated demising walls between tenant bays regardless. Subdividing the ground floor into four fire areas with two-hour barriers satisfied the demising requirement and the fire area threshold with the same construction. Second, cutting dock openings into precast panels that participate in the lateral system required a structural evaluation and added collectors and a new braced line - a real cost, at \$310,000, and an order of magnitude below the light court.

Vapor intrusion required no mitigation at commercial and industrial screening levels. Electrical service was adequate within the existing 800 amps. The transformer was not on the critical path.

The three sitting tenants produced their own findings. Lease abstraction showed one professional services tenant with 26 months remaining and a renewal option at a fixed rent well below market - a real encumbrance. One tenant held over month to month. The third had 14 months remaining with no landlord relocation right, meaning the renovation had to work around them or buy them out. Estoppels surfaced a discrepancy: the holdover tenant produced a 2023 letter from the seller's property manager granting an ongoing rent abatement that did not appear in the lease file or the rent roll. Buyouts for two of the three tenants were negotiated at **\$285,000** total and were cheaper than phasing around them.

Where it landed. All-in cost of approximately \$11.4 million including acquisition, abatement, roof, paving, mechanical, structural work, dock and demising construction, stormwater retrofit, and soft costs. Stabilized net operating income supporting a value meaningfully above that on the strength of the ground floor, with the second floor underwritten conservatively at a low rent and an extended lease-up - an honest treatment of the fact that the second floor remains office space in a submarket with 24 percent office vacancy.

Advanced to Gate 2 and went hard, on a different project than the one the sponsor had been underwriting for four months, at a price renegotiated against the findings. That negotiation is Module 7.

Kipling Row

Demolition of the 12,000 square foot church, so the existing-building track applies narrowly and one finding mattered.

A pre-demolition asbestos survey is required by regulation regardless of the building's fate. It found asbestos in floor tile, mastic, pipe insulation in the basement mechanical room, and transite panels at the boiler flue - normal for a 1948 structure with 1960s alterations. Abatement before demolition: **\$74,000**, with a required notification period and a sequence that had to precede demolition.

Highest and best use as improved was not a serious question. The church has no reuse value under the development thesis and no residual value as a structure. It is a line item: abatement, demolition, disposal, and foundation removal at approximately **\$218,000** total.

The finding worth generalizing: **the pre-demolition survey is mandatory and is frequently omitted from a land acquisition budget**, because a buyer purchasing land does not think of themselves as buying a building. An existing structure on a site you intend to clear is a demolition and abatement cost, and it needs a survey to be priced rather than guessed.

Combined with the \$740,000 from Module 4A, the deal now carries approximately \$958,000 of findings against its original underwriting, and a council vote that has still not happened.

Optional: Try This

Find a vacant or obviously underused building you can actually see - a closed big-box store, a half-empty suburban office building, an old school, a shuttered church. Pick a plausible reuse for it. Then answer four questions.

What is the change of occupancy? From what classification to what classification. That pairing, not the word "renovation," determines most of the cost.

What alteration level does the work reach? If the work area exceeds half the building, you are at Level 3 and the requirements apply building-wide.

Does the geometry work? For any residential reuse, pace off or measure from the window line to the core. If it substantially exceeds 30 feet on both sides, the building needs surgery before it needs a budget.

What does the accessible path of travel cost? The building is probably pre-1990. Where is the accessible entrance, is there a compliant route to the area you are altering, and are the restrooms serving it compliant?

Then run the test from Section 4.7: what would it cost to build the same thing new, on comparable land? If your reuse costs more, name the specific reason it is still worth doing. Location, entitlement, historic credits, and schedule are all legitimate answers. "Below replacement cost" is not one.

Next - Module 5: Concept Feasibility and Cost. Where the real estate deal becomes an architecture problem: test fits, efficiency, conceptual estimating, and the site development costs that break more underwritings than anything else.

1.5 - Concept Feasibility and Cost: Does It Physically Pencil?

This is the bridge module. Everything before it was a real estate problem: market, law, dirt, structure. Everything after it is a finance problem. This module is the crossing, and it is the first point in the deal where an architect, a civil engineer, and an estimator are in the room at the same time.

It is also the third test of highest and best use - financially feasible - split across two modules. Module 5 produces the cost side. Module 6 produces the capital and the return. Neither is meaningful alone.

The work here answers a question the preceding modules cannot: **given what we may build, and what the site will support, what does the thing actually cost, and how much of it is there?** Module 3 established the legal envelope. Module 4 established the physical constraints and their price. Neither tells you how many units fit, how efficient the building is, or what the total development cost will be. Those are design questions with financial answers, and getting them approximately right at concept level is more valuable than getting them precisely right later, because later is after Gate 2.

One framing point worth stating at the outset. The estimates in this module are wrong. They are supposed to be. A concept-level estimate carries an accuracy range wide enough to change the outcome of a deal, and the discipline is not to make it narrower than it is - it is to know how wide it is, to communicate that honestly, and to build the decision around a range rather than a number. Developers who report concept estimates to the dollar are not being precise. They are being misleading, usually to themselves.

Section 1 - Test Fits and Massing Studies

1.1 What They Are

A **test fit** answers whether a program fits a site or a floor plate, and at what efficiency. A **massing study** answers what volume the zoning envelope allows and what building form can occupy it. On a development deal the two are usually done together, at a level of resolution far below schematic design: blocks, bays, corridors, stair and elevator locations, parking layout, and enough dimensional discipline to be trusted.

The output is not a design. It is three numbers - unit count or leasable area, efficiency, and parking count - plus the constraints that produced them.

1.2 What Goes Into One

A test fit that means anything incorporates the findings of Modules 3 and 4 rather than the zoning summary alone:

- Buildable area after setbacks, easements, buffers, and unbuildable land
- The binding dimensional constraint identified in Module 3, which is usually not density
- Grading and retaining implications from the topographic survey
- Foundation type from the geotechnical report, which affects both cost and building footprint
- Land consumed by detention and water quality
- Access location as constrained by spacing standards, which drives the entire circulation layout
- Utility connection points and their locations
- Fire access requirements: apparatus road width, turning radii, and aerial access, which on a multi-building site often determines the layout more than anything in the zoning ordinance

That last item is worth flagging because it is invisible to most people until it reorganizes a site plan. Fire department access roads, turning geometry, and hydrant spacing are code requirements with real dimensional consequences, and they are among the most common reasons a test fit loses units between concept and schematic design.

1.3 The Iteration

A test fit is never done once. The normal sequence runs: maximum theoretical yield from the ordinance, then a layout attempt, then a reduced yield reflecting what actually fits, then a cost check, then adjustments to unit mix or building type, then another layout.

The number that survives three rounds is the number to underwrite. The number from the first round is always too high, and it is the one most likely to end up in a pro forma because it was produced before anyone was paying attention.

Section 2 - Measuring Area

Area terminology is inconsistent across product types and is the source of a remarkable number of underwriting errors. The terms matter because you build gross square feet and you collect rent on something smaller.

Gross building area (GBA). Everything within the exterior walls, all floors. What you build and, roughly, what you pay for.

Net rentable area (NRA) or gross leasable area (GLA). What the tenant pays on. The definition varies by product:

- **Multifamily:** the sum of unit areas. Corridors, stairs, lobbies, amenity, and mechanical space are not rentable. This is why multifamily efficiency is a meaningful number.
- **Office:** measured under a published standard, most commonly BOMA. Tenants pay on **rentable** area, which is their **usable** area plus a proportionate share of building common area. The ratio between them is the **load factor** or common area factor.
- **Retail:** typically the leasable footprint, with common area recovered separately through CAM charges.
- **Industrial:** close to gross, because there is very little that is not leasable.

Load factor. Rentable divided by usable, minus one, expressed as a percentage. A 15 percent load factor means a tenant occupying 10,000 usable square feet pays on 11,500. Load factors run roughly 10 to 18 percent in most office buildings and higher in buildings with large lobbies or inefficient floor plates. It is a real revenue lever and it is also a market-tested one: a building with an unusually high load factor will be discounted by tenants who do the math.

Efficiency. Net rentable divided by gross. Typical ranges:

Product	Efficiency
Industrial / warehouse	95–98%
Garden apartments	82–87%
Wrap or podium multifamily	80–85%
Mid-rise multifamily	78–83%
High-rise multifamily	72–80%
Office (rentable/gross)	82–90%
Adaptive reuse residential	65–78%

Efficiency is the single most leveraged number in a feasibility study. A five-point efficiency swing on a 200,000 square foot building is 10,000 square feet of rentable area - revenue that recurs every year, against construction cost that does not change. Two points of efficiency are frequently worth more than any value engineering exercise the project will ever undertake.

Where efficiency goes. Corridors, stairs, elevators and lobbies, mechanical and electrical rooms, trash rooms, amenity space, structure, and shafts. Double-loaded corridors are more efficient than single-loaded. Larger floor plates are more efficient than small ones. Simple rectangular forms are more efficient than articulated ones, which is the permanent tension between efficiency and the design outcome a design review board wants.

Section 3 - Unit Mix and Program

For residential, unit mix is where market and geometry negotiate.

Smaller units produce higher rent per square foot and lower efficiency. More units per floor means more doors, more corridor, more plumbing walls, and more of everything that is not rentable. A studio-heavy building rents at a premium per square foot and gives back part of that premium in efficiency and construction cost per square foot.

Larger units do the reverse. Better efficiency, lower rent per square foot, and a smaller qualified demand pool at the resulting monthly rent.

The mix is therefore constrained from three directions simultaneously: what the market absorbs (Module 2), what the building form supports, and what the zoning permits - since density is frequently expressed in units, making unit count a regulated quantity and unit size a free one.

For for-sale residential there is an additional revenue layer that outsiders consistently miss. The **base price** is what appears in the comps and in the advertising. Actual closing prices include **lot premiums** for location within the development and **option and upgrade revenue** from the design center. Together these routinely add 8 to 15 percent to the average closing price, and they carry a different cost structure than base construction. A for-sale pro forma built on base prices alone understates revenue materially; one built on optimistic option revenue overstates it. Both errors happen.

Section 4 - Parking and the Cost Cliff

Parking deserves its own section because it is the constraint most likely to convert a workable project into an unworkable one, and because the cost structure is discontinuous rather than linear.

4.1 The Requirement

The governing ratio is the higher of what the zoning ordinance requires and what the market demands, and those are not the same number. In transit-served markets, code minimums frequently exceed what tenants use. In suburban markets, tenant expectations often exceed code. Lenders have their own views. Where the code minimum is the binding number and it exceeds market demand, the project is building parking nobody needs at full cost, which is a real and common value destruction.

Reductions available in most ordinances - shared parking analyses, transit proximity reductions, parking demand studies - are generally discretionary, which puts them back in Module 3's political process.

4.2 The Cost Cliff

Type	Land consumed per stall	Cost per stall
Surface	300–400 SF including drives	\$5,000–\$9,000
Structured, above grade	330–400 SF of structure	\$25,000–\$40,000
Podium / wrapped	350–420 SF	\$30,000–\$50,000
Below grade, one level	350–450 SF	\$45,000–\$75,000
Below grade, multiple levels	350–450 SF	\$60,000–\$100,000+

Ranges vary substantially by market, soil conditions, and water table - the below-grade figures in particular are driven by exactly the geotechnical findings from Module 4A.

The cliff is between surface and structured, and it is roughly a factor of four. The structural consequence is that **projects have a density threshold above which parking must go structured, and crossing it changes the project's cost per unit discontinuously.** As a rough guide, surface parking works up to somewhere around 25 to 35 units per acre depending on ratio and site geometry. Above that, the land runs out.

This produces a counterintuitive result worth internalizing: **increasing density does not always increase value.** Going from 30 to 45 units per acre adds 15 units of revenue and can add \$25,000 to \$35,000 per stall across the entire parking count, including the stalls for the units you already had. The incremental units must carry the cost of restructuring parking for all of them. There is a dead zone - typically in the 35 to 55 units per acre range - where a project is too dense for surface and not dense enough to amortize structure. Projects underwritten into that zone are a recognizable category of failure.

4.3 Secondary Effects

Structured parking brings more than its own cost: a larger building footprint, ventilation and life safety systems, longer construction duration, a different structural system, and - where the podium is concrete under wood-frame residential - a specific construction type and its own code constraints. Below-grade parking brings excavation, shoring, dewatering, and waterproofing, all of which are geotechnical findings rather than parking decisions.

Section 5 - Conceptual Estimating

5.1 Levels of Accuracy

Cost estimates exist at defined levels of resolution, and the relationship between the level and its accuracy range is the most important thing to understand about them. The framework below tracks the classification systems used in the industry (AACE International's Class 5 through Class 1, mirrored in ASTM practice); jurisdictions and firms use varying names for the same ladder.

Level	Basis	Design maturity	Typical accuracy
Order of magnitude (ROM)	Cost per unit, per key, per square foot, drawn from comparable projects	Concept, a program on a page	–30% to +50%
Square foot / parametric	Refined per-SF pricing by building type, adjusted for market, height, quality	Massing and test fit	±20–30%
Assemblies / systems	Priced by building system using UniFormat	Schematic design	±10–20%
Unit price / detailed	Quantity takeoff priced by trade using MasterFormat	Design development to construction documents	±5–10%
Bid / GMP	Actual subcontractor pricing	Construction documents, or defined scope with allowances	The number, subject to its own qualifications

Two things follow.

The accuracy range is asymmetric at concept level. A ROM estimate is more likely to be low than high, because concept estimates capture what is drawn and projects accumulate scope as they are designed. A range of minus 30 to plus 50 percent reflects real experience.

A deal that only works at the bottom of the range does not work. If a project pencils at the estimate and fails at the estimate plus 15 percent, it has not passed the feasibility test at concept level. This is the discipline that concept estimating exists to enforce, and it is the one most often waived.

5.2 UniFormat: Why Estimates Are Organized by System

An estimate at concept and schematic level cannot be organized by trade, because you do not yet know what products will be specified or who will install them. You do know what systems the building needs. **UniFormat** (ASTM E1557) is the classification that organizes construction information by **functional element** rather than by work result:

Level 1 element	Contains
A - Substructure	Foundations, basement construction
B - Shell	Superstructure, exterior enclosure, roofing
C - Interiors	Interior construction, stairs, finishes
D - Services	Conveying, plumbing, HVAC, fire protection, electrical, communications
E - Equipment and Furnishings	Built-in equipment, furnishings
F - Special Construction and Demolition	Special structures, selective demolition
G - Sitework	Site preparation, improvements, civil utilities

The value is that you can price a building by system long before you can price it by product. "Exterior enclosure at \$95 per square foot of wall area" is a statement you can make from a massing model. "Aluminum-framed storefront, Section 08 41 13" is not.

The contrast with MasterFormat matters and is covered in depth in Part 2 Module 4.

MasterFormat organizes by **work result** - the divisions and sections that structure specifications and bidding. It exists because construction is purchased by trade: a drywall subcontractor bids the drywall sections, and a specification has to be organized the way the work is bought.

The two systems answer different questions at different times:

	UniFormat	MasterFormat
Organizes by	Functional element	Work result
Used for	Concept and schematic estimating, elemental cost planning, life-cycle cost	Specifications, bidding, procurement, cost coding during construction
Phase	Pre-design through design development	Construction documents through closeout
Question	What does this building's envelope cost?	Who bids Section 07 42 13 and what does it say?

Estimates migrate from one to the other as design progresses. A developer who understands that migration will not be confused when the concept estimate and the bid tabulation appear to be describing different projects - they are describing the same project through different organizing structures, and reconciling them is a normal and necessary exercise.

5.3 Sources of Concept Pricing

- **Your own recent projects**, adjusted for time and market. By far the best source, and the reason experienced developers in one product type are hard to compete with.
- **Published cost data** such as RSMeans, adjusted by location factor. Useful and generic; it does not know your market's labor conditions.
- **A general contractor or estimator**, discussed in Section 9.
- **Broker and peer intelligence** on what comparable projects delivered for. Directionally useful, frequently incomplete, and rarely reconciled for scope.

The persistent hazard with all per-square-foot benchmarks is that they are only comparable within product type, market, construction type, and scope definition. A "\$185 per square foot" number is meaningless without knowing whether it includes site work, parking, soft costs, and FF&E - and the answer differs by who is quoting it.

Section 6 - Site Development Cost

The most common source of underwriting error in development, and it earns its own section because of a structural reason: **site cost scales with the site, and building cost scales with the building, and per-square-foot benchmarks are expressed per square foot of building.**

A 40,000 square foot building on a flat, clean, fully served three-acre pad and the same building on a sloped, rocky, unserved twelve-acre parcel have the same building cost and radically different project costs. Any benchmark that blends the two is wrong for both.

6.1 The Components

- **Demolition and abatement** of existing improvements, including foundation removal and utility abandonment
- **Clearing, grubbing, and stripping**; topsoil stockpile and respread
- **Erosion and sediment control**, and the construction stormwater permit and SWPPP from Module 4A
- **Mass earthwork**: cut, fill, haul, and the balance question
- **Over-excavation and engineered fill**, which is a geotechnical finding, not a site work decision
- **Retaining structures**, scaling with the square footage of exposed face
- **Rock excavation** where present
- **On-site wet utilities**: water, sanitary, storm
- **Dry utilities**: electrical, gas, telecom, including transformer pads and the utility's charges
- **Off-site improvements**: main extensions, upsizing, road widening, turn lanes, signals, frontage improvements
- **Storm detention and water quality facilities**
- **Paving, curb and gutter, sidewalks**
- **Site lighting, landscape, irrigation, fencing, walls, signage**
- **Fire access roads and hydrants**

6.2 Magnitude

Site development typically runs **10 to 25 percent of hard cost** on a normal site. On a difficult one - steep, rocky, poorly served, requiring significant off-sites - it can exceed 35 percent. On an urban infill site with everything at the property line, it can be under 8 percent.

The corollary is that **the same building is a different project on two different sites**, and the underwriting difference is not a rounding error.

6.3 Why It Gets Missed

Four structural reasons:

It is designed last and committed first. Civil design typically lags architectural design, but site work is the first thing built. The budget line most likely to be an allowance is the one that gets spent first.

It is not in the benchmark. Per-square-foot building costs from comparable projects frequently exclude site work, and the exclusion is not always stated.

It is where every Module 4 finding lands. Over-excavation, retaining, detention, off-sites, tap fees - the physical due diligence findings almost all resolve into site cost. A developer who did not update the site budget after diligence has an underwriting that predates the facts.

It is quantity-driven and the quantities are not known at concept. Earthwork volumes require a grading plan. A grading plan requires a site layout. At concept, the number is an estimate built on an assumption, and it should be carried with a wider contingency than the building.

Section 7 - The Development Budget

7.1 Hard Costs

Everything in the construction contract:

- Site development (Section 6)
- Building construction by system or trade
- General conditions - the contractor's site management, temporary facilities, utilities, cleanup. Time-based, which is why schedule is cost.
- Contractor's fee and overhead
- Bonds and builder's insurance
- Owner-furnished equipment, where applicable
- Construction contingency

7.2 Soft Costs

Everything else required to deliver the project. Typically 20 to 35 percent of hard cost, higher on complicated or slow projects.

- **Design and engineering.** Architecture, structural, MEP, civil, landscape, and specialty consultants. Commonly 4 to 8 percent of hard cost for a straightforward building, higher for complex or renovation work.
- **Due diligence and consultants** from Modules 4A and 4B
- **Entitlement, land use counsel, and community outreach** from Module 3
- **Permits, plan review, and impact fees.** Impact fees can be the largest single soft cost line on residential.
- **Utility connection and tap fees,** and water rights dedication where applicable
- **Legal and closing costs,** including entity formation, purchase, and loan documentation
- **Title and survey**
- **Insurance and property taxes during construction**
- **Financing costs:** origination, lender counsel, appraisal, the inspecting consultant, and the **interest reserve**, which is usually the largest financing line
- **Marketing and leasing:** model units, signage, collateral, leasing commissions, and tenant improvement allowances

- **FF&E and operating equipment**
- **Operating deficit or lease-up reserve** - the carrying cost from certificate of occupancy to stabilization, which is real money and is routinely omitted
- **Developer fee**, where the sponsor charges one to the deal

7.3 Contingency

Three distinct contingencies exist and they are not interchangeable.

Design contingency covers scope that is not yet drawn. It is largest at concept and shrinks as design resolves: on the order of 10 to 15 percent at concept, 7 to 10 percent at schematic, 5 percent at design development, 2 to 3 percent at construction documents. It is an owner-held allowance for the fact that the drawings are incomplete.

Construction contingency covers changed and unforeseen conditions during construction. Typically 3 to 5 percent of hard cost on new construction and **8 to 12 percent or more on renovation**, where concealed conditions are the defining risk.

Contractor's contingency, held inside a guaranteed maximum price, covers the contractor's own risk within the agreed scope. Its treatment - who owns unused contingency at the end - is a contract negotiation, addressed in Part 2 Modules 5 and 6.

The failure mode is universal enough to name plainly: **contingency treated as profit**. A developer under pressure reduces contingency to make a deal pencil, which does not change the project's cost - it only changes who absorbs the overrun and when they find out. A deal that works only at a 2 percent concept contingency does not work.

7.4 Escalation

Cost estimates are priced as of a date. Construction happens later. The gap is escalation, and it is applied at a compound annual rate from the estimate basis date to **the midpoint of construction** - not to the start.

This is a small technical point with a large consequence. A project with a two-year entitlement, a one-year design and permitting period, and an eighteen-month build has its construction midpoint roughly four years past the estimate date. At 4 percent annual escalation, that is roughly 17 percent, not the 12 percent an escalation-to-start calculation would produce. On a \$10 million hard cost, the difference is half a million dollars of pure arithmetic.

Escalation is the most direct mechanism by which the entitlement timeline from Module 3 becomes a cost, and it is the reason a rezoning that takes eighteen months rather than ten is not merely a delay.

7.5 Schedule as a Cost Line

Four channels through which time becomes money:

Interest carry. The construction loan accrues interest on the outstanding balance throughout. A longer build is more interest, and the interest reserve is sized on an assumed duration.

General conditions. The contractor's time-based site costs. On a mid-size project these run a meaningful monthly figure, and every month of extension is that figure again.

Escalation, as above.

Delayed revenue. Rent or sales proceeds that start later, which in a rising market is partly offset by higher pricing and in a flat one is pure loss.

There is also an option cost during entitlement: extension payments under the purchase agreement, property taxes, and the opportunity cost of capital tied up in a deal that has not yet resolved.

Section 8 - Bringing In a Contractor or Estimator

8.1 What You Can Get Without Paying

Contractors provide concept-level input during feasibility, unpaid, in exchange for positioning on the eventual contract. Reasonably available:

- A square-foot budget range for the building type in the local market
- A constructability opinion on the massing and structural approach
- Identification of long-lead items - the transformer problem from Module 4A is exactly the kind of thing a good preconstruction team flags early
- A schedule opinion: duration, sequencing, weather and seasonal constraints
- A view on labor market conditions and subcontractor availability
- On renovation, a view on phasing and on what is likely to be concealed

8.2 What You Cannot

- A quantity takeoff, which requires drawings that do not exist
- Real subcontractor pricing, because subcontractors will not price an unentitled project with no drawings and no schedule
- A binding number of any kind
- The time required to actually study the problem

8.3 The Incentive Problem

Free preconstruction pricing is not neutral. The contractor providing it wants the project to proceed, because if it does not proceed there is no contract. This biases concept numbers low, not through dishonesty but through the ordinary dynamics of an unpaid estimate produced quickly by someone who benefits from the deal advancing.

Three corrections:

Ask what is excluded. The usual answer is site work, off-sites, owner-furnished items, escalation, and contingency. That is most of the gap between a concept number and a real one.

Get more than one. Two independent concept numbers that agree are worth considerably more than one.

Pay for preconstruction on anything complicated. A preconstruction services agreement buys real estimating time, a real schedule, and a team with an obligation to you rather than an interest in the deal closing. On a renovation or any project with significant site complexity, this is among the best-spent money in the entire development budget. It also begins the relationship that determines the delivery method, which is the next section.

Section 9 - Delivery Method Selection Begins Here

The delivery method is conventionally treated as a construction topic. It is not. It is an **owner decision made before design begins**, because it determines who holds the design contracts, and it must be made during feasibility because the capital structure constrains it.

The methods are covered in full in Part 2 Modules 1 and 5. What matters at this stage is the decision logic.

Method	Design contracts held by	Price certainty	Speed	Fits
Design-Bid-Build	Owner	At bid, after complete documents	Slowest	Public work; simple projects; owners wanting maximum design control and competitive pricing
CM at Risk	Owner	GMP during design	Faster; overlap of design and construction	Complex projects; owners wanting preconstruction input while retaining the designer

Design-Build	Contractor	Early, on defined criteria	Fastest	Well-defined programs; owners willing to trade design control for certainty
Progressive Design-Build	Contractor	Negotiated as design develops	Fast	Complex or uncertain scope, with a collaborative owner
CM as Agent	Owner	Varies	Varies	Owners wanting expert representation without construction risk
IPD	Shared	Collaborative, shared risk	Varies	Sophisticated owners and repeat teams

9.1 The Constraints That Actually Decide It

The lender. Construction lenders generally require a guaranteed maximum price or a lump sum before closing, together with an approved contractor. That requirement alone pushes many deals toward CM at Risk or Design-Build, because both produce a price before documents are complete, while Design-Bid-Build cannot produce one until they are.

Schedule. Overlapping design and construction shortens the period of interest carry and reduces escalation exposure. Where the entitlement timeline has already consumed a year or more, methods that recover schedule are worth a premium.

Scope certainty. A well-defined, repeatable building is a Design-Build candidate. A renovation with concealed conditions is not, at least not at a fixed price - the contractor will price the uncertainty, and you will pay for risk that may not materialize.

Owner capability. Methods that shift responsibility to the contractor require less owner-side project management. Methods that retain control require more of it, and an owner without that capacity should either buy it through an owner's representative or select a method that does not demand it.

Public procurement, where applicable, constrains the choice by statute rather than by preference. Out of scope here; see Part 2 Module 5.

9.2 Why It Belongs in Module 5

Because the preconstruction relationship formed during feasibility often becomes the construction contract, and because the delivery method determines what the design fee is, when it is incurred, and how much design risk the owner retains. All three are budget lines in this module. Deciding the delivery method after the design contract is signed means the decision has already been made by default.

Artifacts

Artifact	Prepared by	Purpose
Massing study	Architect	Buildable envelope, form options
Test fit / yield study	Architect, civil engineer	Unit count or leasable area, efficiency, parking
Conceptual site plan and grading	Civil engineer	Earthwork quantities, detention sizing, layout feasibility
ROM cost estimate	Developer, contractor, or estimator	First-pass hard cost
Elemental (UniFormat) estimate	Estimator or contractor	System-level cost, once massing exists
Preliminary development budget	Developer	Hard, soft, contingency, escalation, total project cost
Conceptual schedule	Developer with contractor	Entitlement, design, permitting, construction, lease-up
Preconstruction services agreement	Developer and contractor	Paid, obligated estimating and scheduling
Delivery method recommendation	Developer or owner's rep	The decision, with its rationale

Failure Modes

A beautiful test fit that ignores the geotechnical report. The layout that maximizes yield on a topographic survey may require retaining walls, over-excavation, or a foundation type that costs more than the additional units are worth. The test fit and the geotech must be reconciled by someone who has read both.

Underwriting round-one yield. The first number from the ordinance is always too high. Fire access, grading, detention, parking geometry, and easements take units out. Underwrite the number that survived iteration.

Contingency treated as profit. Reducing contingency to make a deal pencil changes nothing about the project and everything about who absorbs the overrun.

Escalating to the start of construction. Escalate to the midpoint. On a long-entitlement project the difference is a large number produced entirely by arithmetic.

Per-square-foot benchmarks across product types, markets, or scope definitions. The number is meaningless without knowing what it includes. Ask every time.

Site work as a percentage of building cost. It does not scale with the building. It scales with the site, and it is where every Module 4 finding lands.

Treating a contractor's free concept number as the budget. It is unpaid, quick, produced by a party who benefits from the deal proceeding, and usually excludes site work, escalation, and contingency.

Density into the parking dead zone. Too dense for surface, not dense enough to amortize structure. Run the parking cost at the proposed density before committing to the density.

Omitting the operating deficit. The carry from certificate of occupancy to stabilization is real, it is months of debt service and operating expense against partial revenue, and it is left out of first-pass budgets constantly.

Reporting a concept estimate as a single number. It has a range. Communicate the range, and test whether the deal survives the top of it.

What Happened Next

Kipling Row

The test fit confirmed the unit count and the budget broke the deal.

The test fit. Three iterations. The ordinance and the proposed PUD standards supported 34 units; the first layout produced 37, the second 33 after fire apparatus access and turning radii were resolved, and the third recovered 34 by shifting two buildings and accepting a tighter guest parking count. Six buildings of five and six units, two-story attached, surface and garage parking.

Unit mix landed at 8 two-bedroom, 20 three-bedroom, and 6 three-bedroom end units, averaging 1,380 square feet. Gross building area 46,920 square feet. Detention, as established in Module 4A, consumed 0.34 acres.

Revenue. Module 2's comps supported a blended base price of \$437,000 across the proposed mix. The pro forma adds two revenue layers that the comp analysis does not capture: lot premiums averaging \$18,000 for end units and units backing to open space, and design center options averaging \$40,000. Average closing price of **\$495,000**, which is a 13 percent uplift over base and within the normal range for this product.

Gross sales of \$16,830,000, less commissions and closing costs at 5.5 percent, produces net revenue of **\$15,904,000**.

The budget.

Land and closing	\$2,010,000
Demolition and abatement	218,000
Earthwork, over-excavation, retaining	495,000
Wet utilities, on- and off-site	410,000
Storm detention and water quality	265,000
Paving, curb, walks	290,000
Landscape, irrigation, fencing, lighting	185,000
Access easement paving	38,000
<i>Site development subtotal</i>	<i>1,901,000</i>

Foundation premium - piers and structural floors	391,000
Vertical construction, 46,920 SF @ \$158	7,413,000
<i>Hard subtotal</i>	<i>9,705,000</i>
Construction contingency @ 5%	485,000
Hard total	\$10,190,000
Design and engineering	445,000
Entitlement and legal	325,000
Due diligence consultants	148,000
Impact and tap fees @ \$14,200/unit	483,000
Water rights cash-in-lieu @ \$3,200/unit	109,000
Building permit and plan review	96,000
Insurance and taxes during construction	155,000
Financing and interest carry	810,000
Marketing, model, sales administration	215,000
Soft total	\$2,786,000
Total project cost	\$14,986,000

Cost per unit: \$440,765. Net revenue per unit: \$467,775.

Projected profit: \$918,000. Margin on cost: 6.1 percent.

The original underwriting, before the Module 4 findings, showed \$1,876,000 and 13.4 percent - thin for merchant for-sale residential but within range. The \$958,000 of diligence findings consumed more than half the profit.

Two observations the developer made, both of which drive Module 7.

Site development is 19.6 percent of hard cost, and nearly all of the increase from the original underwriting sits there. This is the module's thesis in one line: the findings from Module 4 did not land in the building budget. They landed in the site budget, which is the line that is estimated earliest, verified latest, and spent first.

At a 6.1 percent margin, the deal does not survive its own contingency. A 5 percent construction contingency on \$9.7 million of hard cost is \$485,000. If half of it is consumed - an ordinary outcome - the margin falls under 5 percent. The project would be a two-year, fully guaranteed, entitlement-risked exercise producing less than the cost of the capital.

And this is the profit **assuming the rezoning passes.** Weighted at counsel's 60 to 65 percent probability, against a downside in which \$290,000 of pre-development spend is lost and the land is not purchased, the expected value at the contract price is not a number a rational developer proceeds on.

The test fit was not the problem. The price was.

Six Corners

The flex scheme produced a smaller building problem and a much larger estimating lesson.

The test fit. Ground floor subdivided into four bays averaging 9,750 square feet, each with a grade-level door and two dock positions, demised with two-hour fire barriers. Second floor retained as office and flex support in its existing classification, subdivided into six suites. Net rentable of approximately 74,500 square feet against 78,000 gross - **95.5 percent efficiency**, against 72 percent for the residential scheme studied in Module 4B.

That efficiency delta is worth isolating. The same building, converted to two different uses, produces 74,500 or 56,000 rentable square feet. The 18,500 square foot difference is entirely geometry: light court removal, corridors, and the core that a residential plan cannot use and a flex plan can.

Parking and the REA. The truck court and the retrofit detention basin together consumed 68 stalls from the 312 existing, leaving 244 against an REA obligation to maintain 290 available for shared use. Restriping to current dimensional standards and adding a row along the west edge recovered 44, bringing the count to 288 - two short. Rather than redesign, the two-stall shortfall was folded into the REA amendment already being negotiated for the use restriction. One negotiation, two problems.

The estimate, and what it did. Module 4B carried a concept-level order-of-magnitude figure of approximately \$11.4 million all-in. The assemblies-level estimate produced in this module, organized by UniFormat and priced against the actual scope, came in at **\$13.08 million.**

A 15 percent increase. This is not an error and it is not scope creep. It is the difference between an estimate carrying a minus 30 to plus 50 percent range and one carrying plus or minus 15 percent, and the movement is within the band the ROM estimate declared. The relevant discipline is that the sponsor tested the deal against the top of the ROM range before relying on it, which is why a 15 percent increase was a renegotiation trigger rather than a crisis.

Acquisition and closing	\$4,485,000
Abatement - limited to disturbed areas	310,000
Selective demolition	265,000
Roof replacement	820,000
Envelope: dock and grade openings, sealant	610,000
Structural: collectors, braced line, opening framing	310,000
Ground floor: fire barriers, slab repair, dock equipment	720,000
Second floor: accessible path of travel, restrooms, common areas	480,000
MEP: HVAC, distribution, fire alarm	690,000
Elevator replacement	210,000
Site: truck court, paving, restriping, detention retrofit	1,190,000
<i>Hard subtotal</i>	<i>5,605,000</i>
Construction contingency @ 10%	561,000
Hard total	\$6,166,000
Design, structural, code, MEP consultants	385,000
Due diligence consultants	205,000
Legal, including REA amendment	165,000
Permits and fees	96,000
Tenant buyouts	285,000
Insurance and taxes during construction	145,000
Financing and interest carry	455,000
Leasing commissions and TI allowances	690,000
Soft total	\$2,426,000
Total project cost	\$13,077,000

Two lines deserve comment.

Abatement fell from \$1,150,000 in the residential scheme to \$310,000 in the flex scheme. The material present in the building did not change. The disturbed area did. This is Module 4B's point about abatement made concrete: the only lever on abatement cost is how much of it you touch, and a scheme that leaves floors, ceilings, and partitions largely in place pays a fraction of what a gut pays.

Contingency is 10 percent, double the Kipling Row figure. Existing buildings conceal conditions. A renovation carrying a new-construction contingency is underwritten wrong.

Where it lands. Total cost of \$13,077,000 on 78,000 gross square feet is **\$168 per square foot**. New small-bay flex construction in the submarket runs approximately \$150 per square foot, plus land at roughly \$400,000 per acre - which across 6.1 acres and 78,000 square feet of building adds about \$31 per square foot, for a new-construction total near **\$181 per square foot**.

The conversion beats new construction by roughly 7 percent. That is the honest answer, and it is worth sitting with: it wins, and it wins narrowly. The Section 4.7 test passes, but not by the margin the phrase "a third of replacement cost" implied at Gate 0.

Stabilized net operating income of approximately \$949,000 against \$13,077,000 of cost is a **7.3 percent yield on cost**. Against an exit capitalization rate of 6.75 percent, that supports a value near \$14.06 million and a development margin of roughly 7.5 percent - real, but thinner than the sponsor had been carrying, and thin enough that a \$400,000 reduction in the purchase price is the difference between an adequate deal and a marginal one.

Both deals arrive at Module 7 needing the same thing, for entirely different reasons.

Optional: Try This

Take any building you can see and estimate its cost three ways, in ascending order of effort.

Order of magnitude. Look up a per-square-foot range for that building type in your market, multiply by an estimated area, and write down the range rather than the midpoint.

By system. Break it into UniFormat Level 1 elements - substructure, shell, interiors, services, sitework - and assign each a share of the total. Most building types have recognizable elemental distributions, and learning them is the fastest route to an intuition for where cost actually sits. The result will surprise you: services routinely exceed shell.

The site test. Now imagine the same building on a site with fifteen feet of grade change, bedrock at six feet, no sewer within 800 feet, and a required turn lane. Estimate what changes.

Nothing about the building changed. That gap is the subject of Section 6 and the reason this module exists.

Next - Module 6: The Capital Stack. Whose money, on what terms, and how a lender who never attends a design meeting ends up determining what gets built.

1.6 - The Capital Stack: Whose Money and What Does It Demand?

Every dollar in a development deal arrives with conditions attached, and the conditions are not uniform. Some capital demands to be repaid first. Some demands a minimum return before anyone else earns anything. Some demands approval rights over decisions that appear to be design decisions. Some demands that the building meet a performance standard, house a particular income band, or preserve a particular window.

The capital stack is the ordering of those demands. Understanding it is what lets you predict owner behavior, and predicting owner behavior is most of what a person entering development from the design or construction side actually needs.

This module is vocabulary and intuition.

One concept anchors the module. **Land is a residual.** It is not priced by the market in the way a building is priced; it is worth whatever remains after everything else in the deal has been paid for. Section 2 develops this, and it is the single most useful idea in Part 1 for anyone trying to understand why developers behave the way they do about cost.

Section 1 - Sources and Uses

The foundational document. Two columns that must balance.

Uses is the total development cost from Module 5 - land, hard cost, soft cost, contingency, financing costs, and reserves. Every dollar the project will spend.

Sources is where those dollars come from - senior debt, subordinate debt, equity, and any incentive capital.

Uses	Sources
Land acquisition	Senior construction loan
Hard costs	Mezzanine debt or preferred equity
Soft costs	Common equity - LP
Financing costs and interest reserve	Common equity - GP

Operating deficit / lease-up
reserve

Incentive capital (TIF, PACE,
credits)

Total uses

Total sources

Two things make this document more useful than it looks.

It exposes the equity requirement. Debt is sized by formula (Section 5). Whatever the debt does not cover, equity must. The equity number is therefore a derived figure, and when costs rise, equity absorbs the increase - which is why cost overruns are an equity problem before they are anyone else's problem.

It is the document the lender underwrites against, and it gets locked. At loan closing the budget in the sources and uses becomes contractual. Reallocating between line items requires consent. This is the mechanism explored in Section 9.

Section 2 - Land Basis and Residual Land Value

2.1 The Concept

Ask what a piece of land is worth and the intuitive answer is "whatever comparable land sells for." That is how land is *priced*, but it is not how land is *valued*.

A development site's value to a developer is entirely derived from what can be built on it and what that will be worth. The calculation runs backward:

**Value of the completed project – All development costs other than land –
The developer's required profit or return = Residual land value**

That is what the land is worth *to this project*. Not to any project, but to this specific one, with this program, these costs, and this required return.

2.2 Why It Governs Everything

Three consequences follow, and all three explain behavior that is otherwise baffling from the design or construction side.

Cost increases come directly out of land value. In the residual equation, the project value and the required return are set by the market. Development costs are the only variable in the middle. Every dollar of cost increase reduces residual land value by a dollar. When a developer resists a \$200,000 change, they are not being difficult about \$200,000 of construction - they are watching \$200,000 disappear from a land value they have already committed to pay.

Land price is a negotiation; residual land value is a calculation. The two agree only by coincidence. A seller's asking price reflects what they paid, what they believe, what a broker told them, and what someone offered them in a different market. The residual value reflects arithmetic. **The gap between them is where deals succeed or fail**, and closing that gap is what Module 7 (Site Control and Transaction) is about.

Entitlement creates value by changing the denominator. This is Module 3's economic content stated precisely. A rezoning does not improve the land. It changes what can be built on it, which changes the project value, which changes the residual. This is why land trades on a per-entitled-unit basis rather than a per-acre basis, and why the value created by an approval can exceed the value of the dirt itself.

2.3 The Required Return Is Not Optional

The residual calculation subtracts a required profit before arriving at land value, and that subtraction is what makes it meaningful. A developer who pays residual land value computed on a zero required return has bought a project that consumes two or three years, carries personal guarantees, and returns nothing.

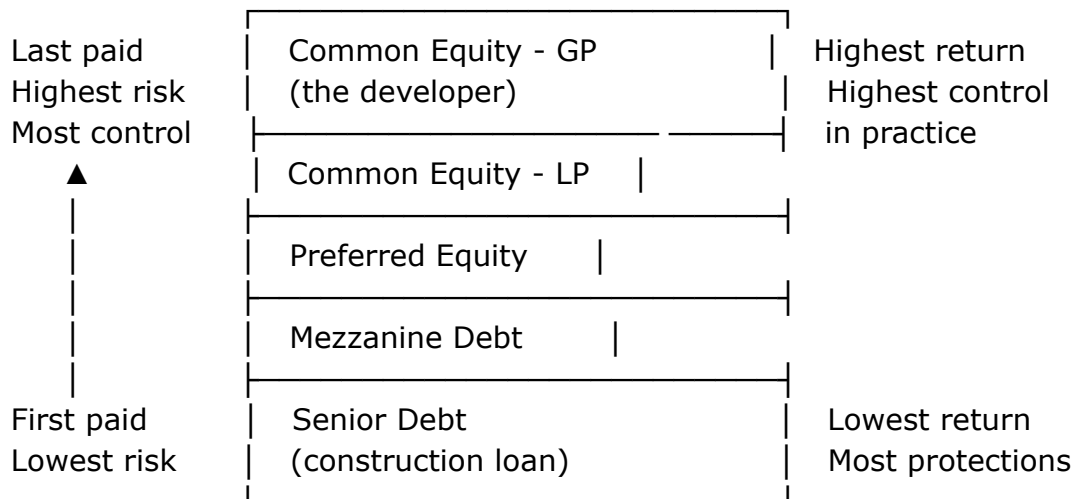
Typical required returns, stated as margin on total cost:

Product	Typical target
For-sale residential, merchant	15–20% on cost
Income property, merchant build	Yield on cost 150–250 bps above exit cap
Value-add repositioning	Varies with hold period and leverage

When a deal fails to clear the target, the correct conclusions are to reduce cost, increase revenue, reduce the land price, or not proceed - in that order of preference and with the last one always available.

Section 3 - The Stack

Capital is layered. Position in the layer determines three things simultaneously: **when you get paid, what you get paid, and what you control.**



The ordering is strict in a downside: senior debt is repaid in full before mezzanine sees anything, mezzanine before preferred, preferred before common, and the GP's promote is the last dollar in the deal. In an upside the order reverses in economic terms - the senior lender gets its interest and nothing more, while the GP captures a disproportionate share of what is left.

Senior debt. The construction loan. Largest single source, lowest cost, first lien on the property. Section 5.

Mezzanine debt. Subordinate debt, typically secured not by a mortgage on the property but by a pledge of the equity interests in the borrowing entity - which means that on default the mezzanine lender takes ownership of the entity rather than foreclosing on the real estate, a much faster remedy. Priced well above senior.

Preferred equity. Equity with a priority return and a defined position ahead of common. Documented in the operating agreement rather than in a loan agreement, and typically carrying rights on non-payment that can include removal of the sponsor and forced sale.

Mezzanine and preferred exist for the same reason: the senior lender will not lend as much as the developer needs, and this layer fills the gap at a cost between debt and equity. Both increase leverage and both increase the probability that a modest underperformance wipes out common equity entirely.

Common equity. The residual owner. Takes the loss first and the upside last. Split between the limited partners who provide most of the cash and the general partner who runs the deal.

Section 4 - The Equity Structure

4.1 GP and LP

The **general partner** - the sponsor, the developer - typically contributes a minority of the equity, commonly 5 to 15 percent, and performs all of the work: sourcing, entitling, designing, building, leasing, and eventually selling. The **limited partners** contribute the remainder and are passive.

The GP also signs the guarantees (Section 5.5), which is a meaningful and frequently overlooked asymmetry. The LP's downside is its invested capital. The GP's downside is its invested capital plus whatever it personally guaranteed.

4.2 The Waterfall, In Concept

Profits are distributed through a sequence of tiers. The shape is this:

1. **Return of capital.** Every dollar contributed comes back, LP and GP alike, before anything is split.
2. **Preferred return.** A minimum annual return - commonly in the high single digits to low teens - paid on contributed capital, usually accruing and compounding if not paid currently.
3. **Catch-up**, in some structures, allowing the GP to draw even with the LP on profits already distributed.
4. **Promote tiers.** Profits above the preferred return are split disproportionately in the GP's favor, with the split becoming more favorable at successive return hurdles. A common shape is something like 80/20 above the preferred, 70/30 above a higher hurdle, 60/40 above a higher one still.

The **promote** is the GP's carried interest - the reason a developer contributing 10 percent of the equity might receive 25 or 30 percent of the profits. It is compensation for sourcing and executing the deal, and it is earned only above the preferred return.

4.3 Why This Explains Owner Behavior

Two implications reach directly into design and construction.

The promote is nonlinear. Below the preferred return, the GP earns nothing beyond its pro rata share. Just above it, the GP's economics improve sharply. A deal hovering near a hurdle makes the developer intensely sensitive to anything affecting cost or schedule, because the marginal dollar is worth several times more to them than it appears to be worth to the project.

LP major decision rights are real. Operating agreements reserve certain decisions to the LP: budget increases above a threshold, material changes to the business plan, additional capital calls, sale, refinance. This is why a developer sometimes cannot approve a change order in a meeting, and why the answer to a straightforward question takes a week. They are not the decision-maker they appear to be.

Section 5 - Construction Loan Mechanics

The single most consequential relationship in the stack for anyone on the design or construction side, because nearly every constraint the project operates under traces to this document.

5.1 Sizing

A construction loan is sized at the **lesser** of several tests:

- **Loan to cost (LTC).** A percentage of total development cost, commonly 55 to 70 percent depending on product, market, and sponsor.
- **Loan to value (LTV).** A percentage of the appraised **as-complete** or **as-stabilized** value.
- **Debt yield.** Projected stabilized net operating income divided by the loan amount, with a minimum threshold.
- **Debt service coverage** on the projected permanent loan.

Whichever test binds first determines the loan. The rest of the capital is equity - which is why an appraisal that comes in low does not merely reduce the loan, it increases the equity requirement dollar for dollar, and equity is the most expensive money in the stack.

5.2 The Draw Process

Construction loans fund in arrears against work completed. Monthly:

1. The contractor submits an application for payment against a **schedule of values** - the contract sum broken into line items, showing percentage complete
2. The architect certifies the application
3. The owner submits a draw request to the lender with supporting documentation
4. The **lender's inspecting consultant** visits the site and verifies that the work claimed has been performed
5. The title company issues a date-down endorsement confirming no intervening liens
6. **Lien waivers** are collected from the contractor and subcontractors for the prior payment
7. The lender funds, less **retainage**

Retainage - commonly 5 to 10 percent, sometimes reduced at a completion milestone - is withheld from each payment and released at or after substantial completion. Its purpose is leverage over completion, and its effect is to push the contractor's and subcontractors' cash flow into deficit for the duration of the project.

The consequence for the field: **payment is a documentation process, not a work process.** A subcontractor who performed the work but whose paperwork is incomplete does not get paid that month. This is the origin of a large portion of the friction between field and office in construction.

5.3 The Interest Reserve

Construction loans accrue interest on the drawn balance, and the project has no revenue during construction. The interest is therefore funded by the loan itself, through a line in the budget called the interest reserve, sized against a projected draw curve and a projected duration.

The reserve is a line item, and it can run out. A project that takes eighteen months longer than underwritten accrues interest for eighteen additional months against a reserve sized for the original schedule. When the reserve is exhausted, the borrower funds interest out of pocket, usually at the exact moment it has the least ability to do so.

This is the most direct mechanism by which schedule delay becomes an existential rather than an economic problem, and it is why lenders and developers treat the construction schedule with an intensity that can seem disproportionate from the field.

5.4 The Inspecting Consultant

The lender engages an independent construction consultant - sometimes called a document and cost review consultant or a fund control consultant - who:

- Reviews the plans, specifications, budget, contract, and schedule **before closing** and opines on adequacy
- Visits monthly and verifies work in place against each draw request
- Reviews and approves change orders above a threshold
- Reports on schedule and on whether the remaining budget is sufficient to complete

This person is not on the project team and has authority that most project teams do not fully register. A negative report can stop a draw. It is worth knowing they exist before they appear on site.

5.5 Guaranties and Recourse

Construction lending is rarely non-recourse in the way stabilized lending can be. Typical structure:

Completion guaranty. The guarantor agrees to complete the project, funding any shortfall personally. Nearly universal.

Payment or repayment guaranty. Personal liability for some or all of the loan balance. Often structured with **burn-off** provisions: full recourse until certificate of occupancy, stepping down at defined leasing or debt service coverage milestones, and sometimes falling away entirely at stabilization.

Carry guaranty. Personal liability for operating shortfalls and debt service during lease-up.

Recourse carve-outs. Even an otherwise non-recourse loan carries personal liability for defined bad acts - fraud, misapplication of funds, unpermitted transfers, environmental liability,

voluntary bankruptcy. These are standard, they are negotiated hard, and they are the source of a category of litigation.

The practical point: **a small developer's personal balance sheet is usually on the line.** A 6 percent margin on a deal with a full repayment guaranty is a different proposition than a 6 percent margin on a non-recourse one, and the difference explains a great deal about risk tolerance.

5.6 Covenants and Conditions

The loan agreement also contains:

- **Conditions precedent to closing** - every item the lender requires before funding: appraisal, survey, title policy, environmental reports, permits, executed construction contract, approved plans, will-serve letters, insurance, and any specific item the lender identifies. On complex deals this list can include curing a title exception or obtaining a third-party consent.
 - **Budget controls** - line item reallocation restrictions, lender control over contingency, and a requirement that the loan remain "in balance," meaning remaining funds must always be sufficient to complete
 - **Change order approval thresholds**
 - **A completion deadline**, with default consequences
 - **Reporting requirements** - monthly draws, updated schedules, leasing reports
-

Section 6 - Exit: Permanent Financing and Sale

A construction loan is short-term and must be retired. There are two exits.

Sale. The building is sold at stabilization. The buyer's price is set by capitalizing the stabilized net operating income.

Refinance. A permanent loan replaces the construction loan, the developer holds the asset, and the difference between the permanent loan proceeds and the construction loan payoff is returned to equity.

6.1 The Metrics

Capitalization rate. Net operating income divided by value. A 6 percent cap rate on \$1,000,000 of NOI implies a \$16.7 million value. Cap rates are set by the market and move with interest rates, product type, market, and asset quality.

Yield on cost (also called development yield or return on cost). Stabilized NOI divided by total development cost. This is the developer's version of a cap rate, measured against what it cost rather than what it is worth.

Development spread. Yield on cost minus exit cap rate, in basis points. **This is the core metric of merchant development for income property.** Building to a 7.5 percent yield on cost and exiting at a 6.0 percent cap rate creates value; building to 6.2 and exiting at 6.0 does not, because the spread does not compensate for two years of execution risk. Targets generally run 100 to 200 basis points, and a thinner spread requires a specific justification.

Debt service coverage ratio (DSCR). NOI divided by annual debt service. Permanent lenders typically require 1.20x to 1.35x. It constrains loan size directly.

Debt yield. NOI divided by loan amount. A leverage test that is independent of interest rates and amortization, which is why lenders adopted it - it does not flatter a loan simply because rates are low. Minimums commonly run 8 to 10 percent.

6.2 The Exit Cap Assumption

The most consequential assumption in any development pro forma, and the least defensible.

The exit cap rate is a guess about market conditions two to four years in the future. It is applied to the largest number in the model, and small changes in it produce large changes in outcome. Conservative practice underwrites an exit cap somewhat above the current going-in rate - commonly 25 to 75 basis points - on the theory that the building will be older and the market may be worse.

A deal that works only at an exit cap tighter than today's should be understood as a bet on cap rate compression rather than as a development project. That bet sometimes pays. It is not the same activity as building something.

Section 7 - Valuation

Three approaches, all three of which appear in a development appraisal.

Sales comparison approach. Value derived from comparable transactions, adjusted using the sequence from Module 2. Strongest for residential and for asset classes that trade frequently. Weakest for special-purpose property.

Income approach. Value derived from the income stream.

- **Direct capitalization** divides stabilized NOI by a market cap rate. Simple, and it assumes a stable income stream.
- **Discounted cash flow** projects cash flow over a holding period, applies a terminal value at the end, and discounts everything to present value at a required rate. Appropriate where the income stream is not stable - lease-up, rollover, or a repositioning.

Cost approach. Land value plus replacement cost of improvements less depreciation. Most relevant for new construction, special-purpose property, and insurance. It is the approach behind "trading below replacement cost," and Module 4B established why that phrase is a statement about price rather than about demand.

7.1 Why It Matters Here

Development appraisals produce three values: **as-is** (the land or existing building today), **as-complete** (the finished but unleased project), and **as-stabilized** (fully leased at market). Loan sizing runs against as-complete or as-stabilized.

Two practical notes. The appraiser is **engaged by the lender**, not the borrower, and that independence is required. The borrower pays and does not direct. And a low appraisal reduces the loan, which increases the equity requirement, which can kill an otherwise workable deal - making the appraisal a genuine risk item between term sheet and closing rather than a formality.

Section 8 - Public and Incentive Capital

Incentives are capital with a specification attached. Every one of them reaches into design, construction, or operations, and the compliance obligation is frequently the reason the incentive is available at all.

8.1 The Major Programs

Tax increment financing (TIF). The incremental property tax generated by the improved property above a frozen base is captured and used to fund eligible costs, usually public infrastructure. Delivered pay-as-you-go or through bonds. Requires a public approval process, typically a finding that the project would not occur "but for" the assistance, and often prevailing wage requirements. Eligible costs are defined narrowly - generally public infrastructure, not the building.

C-PACE. Financing for energy efficiency, water conservation, renewable energy, and in many states resiliency measures, repaid through a special assessment on the property tax bill. Long amortization, fixed rate, transfers with the property on sale. The critical mechanic: **it requires the senior lender's written consent**, because the assessment has priority characteristics that affect the lender's position. Eligible measures are defined and typically require an energy engineering report and sometimes measurement and verification. On a renovation replacing roof, envelope, and mechanical systems, a large share of the scope is often eligible - which makes C-PACE one of the more useful tools in existing-building work.

Property tax abatement. A reduction or freeze in assessed value or tax for a term. Frequently conditioned on job creation, affordability, or investment thresholds, with clawback provisions if the conditions are not met.

Opportunity Zones. Federal capital gains treatment for investments made through qualified funds into designated zones, with benefits tied to holding period and, for existing buildings, to a substantial improvement test requiring investment above a threshold relative to basis within a defined window. The program's terms have been modified legislatively since its creation and continue to be; verify current rules rather than relying on any general description, including this one.

Low-Income Housing Tax Credits (LIHTC). The dominant subsidy for affordable rental housing. Two forms: a competitively allocated credit distributed by state agencies under a qualified allocation plan, and a non-competitive credit paired with tax-exempt bond financing. Strings are extensive: income and rent restrictions, a long compliance period plus extended use obligations, state-specific design and amenity standards embedded in the allocation plan, and in many cases prevailing wage requirements. The credits are syndicated to an investor, which introduces an additional partner with approval rights and a low tolerance for anything that threatens credit delivery.

Historic rehabilitation tax credits. Federal credits for certified rehabilitation of certified historic structures, with many states offering stacking credits. Requires compliance with the Secretary of the Interior's Standards through a multi-part certification process, and a substantial rehabilitation test. Module 4B's framing holds: the credits pay for compliance you would not otherwise accept, and the failure mode is underwriting the credit without underwriting the compliance.

New Markets Tax Credits, utility efficiency incentives, state and local grants, brownfield redevelopment funds, and infrastructure reimbursement districts round out the category.

8.2 The General Rules

Every incentive attaches a condition that reaches the building. Wage requirements, design standards, energy performance, affordability, historic compliance, or reporting obligations. These become specification content and construction obligations, which means they must be transmitted to the design and construction team. They are the same category of thing as Module 3's conditions of approval, and they fail the same way when nobody carries them forward.

Timing rarely matches the need. Most incentives are realized over years while the cost is incurred now, which means they are monetized through an investor or a lender, at a discount and with fees.

Approval is not guaranteed and rarely fast. An incentive underwritten as certain before it is awarded is an entitlement risk in financial clothing.

They introduce partners with approval rights. A tax credit investor, a public agency, and a monitoring authority each have a say and each have a different tolerance for deviation.

Section 9 - How the Lender Shapes the Building

This is the handoff to Part 2, and it is the answer to a question that puzzles people working on the design and construction side: why does the owner behave as though certain things are simply not open for discussion?

Because they are not. They were closed at loan closing.

The budget is locked. The sources and uses submitted at closing becomes contractual. Moving money between line items requires lender consent, and the contingency line is typically under lender control. A design change that is cost-neutral overall but shifts money between budget categories is not automatically permitted.

The loan must remain in balance. Undrawn loan proceeds plus remaining equity must always be sufficient to complete the project. A cost increase that puts the loan out of balance triggers a requirement that the borrower deposit additional equity - immediately, before further draws fund. This is why a mid-project cost increase produces a reaction out of proportion to the dollars.

A guaranteed maximum price or lump sum is usually required before closing. This single requirement drives the delivery method decision from Module 5. A lender that will not close without a price cannot be served by a method that produces a price only after complete construction documents, which pushes deals toward CM at Risk and Design-Build and means the delivery method was effectively selected by the capital structure.

The contractor and architect are approved parties. Replacing either mid-project requires consent.

Change orders above a threshold require lender approval, reviewed by the inspecting consultant. A change the owner and architect agree on is not approved until a third party who has never attended a project meeting signs off.

Retainage is set by the loan, not by negotiation between owner and contractor.

The completion deadline is contractual, with default consequences. This is where schedule pressure originates.

Conditions precedent can reach into the project. A lender may require a title exception cured, a third-party consent obtained, or a specific permit issued before funding. Anything unresolved from Modules 3 and 4 lands here.

Part 2 Module 5 covers procurement and contracting; Part 2 Module 6 covers contract administration - payment applications, change orders, and closeout. **Read both with this section in mind.** The AIA payment application process, the retainage provisions, the change order approval chain, and the substantial completion requirements in those modules are not conventions the industry adopted for its own reasons. They are, substantially, the construction-side expression of the lender's requirements described here.

Artifacts

Artifact	Prepared by	Purpose
Sources and uses	Developer	Capital structure; the equity requirement
Development pro forma	Developer	Cost, revenue, return
Residual land value analysis	Developer	What the land is worth to this project
Term sheet / letter of interest	Lender	Proposed loan terms, non-binding
Loan application and commitment	Lender	Binding terms and conditions precedent
Appraisal	Appraiser, engaged by lender	As-is, as-complete, as-stabilized value
Operating agreement	Counsel	LP/GP economics, waterfall, major decision rights
Subscription documents / PPM	Counsel	Equity raise
Guaranty agreements	Lender's counsel	Personal obligations of the sponsor
Incentive application and agreement	Developer, agency	Award terms and compliance obligations

Failure Modes

Treating land price as an input rather than an output. The residual calculation tells you what the land is worth to the project. Paying more than that is paying for the privilege of working for free.

Underwriting an exit cap tighter than today's. That is a bet on the market, not a development project. If the deal requires it, say so explicitly and size the bet.

Sizing the interest reserve on the optimistic schedule. The reserve runs out at exactly the moment the borrower can least afford to fund interest personally.

Ignoring the appraisal as a risk. A low as-complete value reduces the loan and increases the equity requirement dollar for dollar, between term sheet and closing.

Underwriting incentive capital before it is awarded. An unawarded incentive is a contingency, not a source.

Missing the compliance cost of an incentive. Prevailing wage, design standards, affordability restrictions, and historic compliance are all real costs, and they land in the construction budget rather than in the financing line.

Forgetting that the GP's guarantees are personal. A thin margin on a guaranteed deal is a materially worse proposition than the same margin without recourse.

Failing to model the leverage downside. Mezzanine and preferred layers increase returns in the upside and increase the probability that a modest underperformance eliminates common equity entirely.

Not reading the major decision rights. A developer who cannot approve a change order without an LP call has a schedule problem they did not anticipate.

What Happened Next

Kipling Row

The capital structure was ordinary. The residual land value calculation was not.

Structure. A regional bank construction loan at 60 percent of cost, with unit release provisions - each closing releases a unit from the lien at a defined release price, and proceeds pay down the loan. Equity from four local investors plus the developer. The developer's principals provided a **full repayment guaranty** with burn-off only at loan payoff, which is standard for a three-person shop borrowing from a regional bank and is the single most important fact about this deal's risk.

Sources and uses against the Module 5 budget of \$14,986,000: roughly \$9.0 million of loan and \$6.0 million of equity, of which the developer's own capital was approximately \$600,000.

The residual. Working backward from Module 5:

Net sales revenue of \$15,904,000, less all development costs other than land of \$12,976,000, less a required 15 percent margin on total cost, produces a residual land value of approximately **\$854,000**.

The contract price is \$1,950,000.

The pre-diligence version of the same calculation, before the \$958,000 of Module 4 findings, produced a residual of approximately \$1,811,000 - which is why the developer signed at \$1,950,000 in the first place. They were stretching by about 8 percent on the expectation of finding savings in design, which is a normal and usually survivable stretch. The diligence findings then moved the residual by a million dollars.

Three ways to read the gap:

- At a **15 percent** required margin, the land is worth \$854,000
- At a **10 percent** required margin, it is worth approximately \$1,482,000
- At the contract price of \$1,950,000, the margin is **6.1 percent**

None of those are wrong. They are the same arithmetic at different required returns, and choosing the required return is the developer's decision rather than the market's. What the calculation establishes is the shape of the negotiation: somewhere between \$850,000 and \$1,500,000 of price relief is required for this deal to clear a defensible hurdle, and the seller - a merged congregation that is motivated but not distressed, holding a property with no carrying cost pressure - has no obligation to provide it.

The probability-weighted view. One line of arithmetic the developer had not run.

At the contract price, success produces \$918,000 of profit. Failure at the council vote produces a loss of the roughly \$290,000 of pre-development spend. At counsel's 60 to 65 percent probability:

$$(0.625 \times \$918,000) - (0.375 \times \$290,000) \approx \$465,000$$

That is the expected value of a transaction requiring approximately \$6 million of equity, roughly three years, and a full personal guaranty from the principals. It is a return of under 3 percent per year on the equity, before adjusting for the fact that the guaranty exposes the principals to more than their investment.

The deal does not fail because the rezoning is risky. It fails because the price does not compensate for the risk. That is a different problem with a different solution, and the solution is in Module 7.

Six Corners

A different capital structure, a different exit logic, and a lender condition that reaches back into Module 3.

Structure. A debt fund bridge loan rather than a bank construction loan, priced well above bank pricing, with a two-year initial term and two one-year extension options, an initial funding at closing and a **future funding facility** for capital expenditures, tenant improvements, and leasing commissions drawn as the work and leasing progress. Sized at approximately 65 percent of cost - roughly \$8.5 million against \$13,077,000 - with equity of approximately \$4.6 million from the sponsor's small fund.

Debt fund lending exists for exactly this situation: a transitional asset with no stabilized income, a change of use, and a business plan a bank will not underwrite. The price of that flexibility is rate.

Guaranties. Completion guaranty and carry guaranty from the fund's principals, with recourse carve-outs. No full repayment guaranty, which is one of the ways this deal is structurally safer for its sponsor than Kipling Row is for its developer, despite being the more complicated project.

Conditions precedent, and the one that mattered. The lender's condition list included the usual items - appraisal, title, survey, environmental reports, permits, executed construction contract, insurance. It also included two specific to this deal:

- **The recorded REA amendment**, curing the use restriction from Module 3. The lender would not close on collateral whose intended use was prohibited by a private covenant, regardless of what the zoning permitted. The negotiation with two unrelated neighbors was therefore no longer a business item on the sponsor's list - it was a closing condition, which changed the sponsor's leverage in that negotiation and not in a good direction.
- **A surface use agreement or subordination** addressing the severed mineral estate.

This is Section 9 in operation. A private covenant discovered in a title commitment in Module 3 became, by Module 6, the thing standing between the sponsor and a funded loan.

The exit, and why the metric changed. Module 5 computed a yield on cost of 7.3 percent against an exit cap of 6.75 percent - a **55 basis point development spread**, against a 100 to 200 point target. On a merchant development basis, that is not a deal.

The sponsor is not a merchant developer. The business plan is a five-year hold with a refinance at stabilization, which makes the relevant metrics different:

- **Stabilization and refinance.** Stabilized NOI of approximately \$949,000 supports a value near \$14.06 million at a 6.75 percent cap. A permanent loan at 65 percent LTV is approximately \$9.1 million, against which the NOI produces a debt service coverage ratio near 1.33x and a debt yield above 10 percent. Both clear conventional thresholds comfortably. The refinance retires the bridge loan and returns a modest amount of equity.
- **The hold.** In a submarket with 3.4 percent industrial vacancy and 31 percent rent growth over three years, the return comes from rent growth on a fixed cost basis rather than from the initial spread.

That reframe is legitimate and it is also where the underwriting discipline has to be applied, because it moves the return from something determined by construction cost to something

determined by a rent growth assumption. Stress the assumption and the deal is sensitive in exactly one direction.

The cap rate stress. At \$13,077,000 of cost and \$949,000 of NOI:

Exit cap	Implied value	Versus cost
6.50%	\$14,600,000	+\$1,523,000
6.75%	\$14,059,000	+\$982,000
7.00%	\$13,557,000	+\$480,000
7.25%	\$13,090,000	+\$13,000
7.50%	\$12,653,000	-\$424,000

Seventy-five basis points of cap rate expansion erases the entire development margin. The deal has essentially no cushion against a movement in the capital markets that has nothing to do with the building.

C-PACE. The sponsor identified roughly \$1.8 million of the scope as plausibly eligible: roof replacement, envelope work at the sealant and openings, and mechanical replacement. C-PACE at a long amortization and a fixed rate would reduce the equity requirement and improve leveraged returns meaningfully.

It requires the senior lender's written consent. The debt fund's initial position was no. That negotiation is live going into Module 7, and it is the cleanest available illustration of the general rule: **incentive capital is real money with a real condition, and the condition is often another party's permission.**

Where it lands. The deal works, on a five-year hold, with a thin initial spread, no cushion against cap rate movement, and a closing conditioned on two private-party consents the sponsor does not control.

Both deals arrive at Module 7 needing price relief. One needs it because the arithmetic says the land is worth half what is under contract. The other needs it because a 55 basis point spread is not enough room to absorb anything at all.

Optional: Try This

Take any project you know - one you worked on, or one you can find public numbers for - and run the residual land value backward.

Estimate what it is worth completed. Subtract everything it cost to build and deliver, excluding land. Subtract a 15 percent margin. What remains is what the land was worth to that project.

Then find what the land actually sold for.

The gap, in either direction, is the most informative number in development. If the land sold for less than the residual, someone got a good deal or the entitlement created value after purchase. If it sold for more, someone either accepted a lower return, found revenue you did not model, or made a mistake - and all three of those happen constantly.

Next - Module 7: Site Control and the Transaction. Letters of intent, purchase agreements, earnest money, title, options, and the closing. Where both of these deals get renegotiated, restructured, or abandoned.

1.7 - Site Control and the Transaction

Everything in Modules 2 through 6 was analysis. This module is the only one in Part 1 where the developer signs something.

That distinction matters more than it sounds. A finding is not a result. The claystone at Kipling Row and the reciprocal easement agreement at Six Corners do not change anything by existing - they change something only when they are converted into a price, a term, a condition, or a decision to walk. The purchase and sale agreement is the instrument that performs that conversion, and the negotiation of it is where a developer either recovers the value of their diligence or does not.

Both remaining decision gates live in this module.

Gate 1 - Do we tie it up? A letter of intent is negotiated, a purchase agreement is signed, a refundable deposit is posted, and a due diligence period begins. The developer has bought an option to investigate. This is cheap and it is the entire architecture of the process.

Gate 2 - Do we go hard? The due diligence period expires. The deposit becomes non-refundable. From this point the developer is committed to buy or committed to lose the deposit.

The space between those two gates is the only window in the entire development process in which information is free. Before Gate 1 you do not have the right to investigate. After Gate 2 the investigation is over and the findings are yours to absorb. How wide that window is, what it costs to extend it, and what you are permitted to do with what you learn inside it are all negotiated at Gate 1 - before anyone knows what will be found.

That is the central irony of the transaction, and the reason this module belongs near the end of Part 1 rather than the beginning. You negotiate the terms of your investigation before you have done any of it.

Section 1 - The Letter of Intent

1.1 What It Is

A letter of intent (also term sheet, memorandum of understanding, or letter of understanding) is a short document - two to six pages - that sets out the principal business terms of a proposed transaction before the lawyers draft the real agreement.

The purpose is efficiency. Negotiating a full purchase and sale agreement costs both sides real legal fees, and doing that before the parties agree on price, timing, and structure is wasteful. The LOI establishes that there is a deal worth papering.

Typical contents:

- Purchase price and how it is determined
- Earnest money amount and deposit schedule
- Length of the due diligence period and what happens at its end
- Extension rights and what they cost
- Closing date, or the events that set it
- What the seller will deliver (existing reports, leases, surveys, plans)
- Access rights during diligence
- Any contingencies - financing, entitlement, third-party consent
- Assignment rights
- Broker and commission acknowledgment
- Exclusivity period
- Confidentiality
- An express statement of what is and is not binding

1.2 What Binds and What Does Not

The defining characteristic of an LOI is that most of it is **non-binding**. The price is not binding. The closing date is not binding. Either party can walk away without liability.

Three provisions are normally made expressly binding, and they are the reason the document is worth signing at all:

Exclusivity (the no-shop). The seller agrees not to market the property or negotiate with other buyers for a defined period, commonly 30 to 60 days. This is what the buyer is actually buying with an LOI. Without it, the buyer spends money on legal drafting while the broker continues showing the property.

Confidentiality. Particularly meaningful on income-producing assets, where the buyer will receive rent rolls, leases, and financial statements.

Governing law and expense allocation. Each party bears its own costs.

The binding/non-binding line must be stated explicitly. Courts in some jurisdictions have found LOIs to be enforceable agreements where the language was ambiguous and the parties acted as though bound. The standard protection is a clear clause stating that no obligation arises until a definitive written agreement is executed by both parties, with the enumerated exceptions.

1.3 Why It Still Matters

A non-binding document sets the anchor. Terms conceded in an LOI are extremely difficult to recover in the purchase agreement, because reopening them reads as bad faith even though nothing was binding. The practical rule: **negotiate the LOI as though it binds you, because socially it does.**

The terms most often given away casually at LOI stage and regretted later:

- A due diligence period set by convention (30 or 60 days) rather than by the actual investigation plan
 - No extension rights, or extensions priced only in additional deposit
 - Silence on whether the deposit is refundable and when
 - Silence on seller deliverables, which then arrive late and consume the diligence period
 - Silence on assignment, which matters enormously for entity structuring
 - Silence on what happens if a required third-party consent cannot be obtained
-

Section 2 - The Purchase and Sale Agreement

The PSA is the binding contract. Its anatomy is consistent across deals, and understanding it is how a developer reads their own risk.

2.1 The Economic Terms

Purchase price, and how it is expressed. A fixed number is simplest. Alternatives that appear regularly:

- **Per acre or per square foot**, adjusted by the survey. Ties price to verified area rather than tax records.
- **Per entitled unit or per buildable foot**, adjusted by the actual approval obtained. This allocates entitlement outcome between the parties and is common where the seller believes in the upside.
- **Fixed price with contingent additional payments** triggered by approvals, sales velocity, or profit thresholds.

Earnest money deposit. Held by the title company or escrow agent, not by the seller. Applied to the purchase price at closing. Usually posted in installments: an initial deposit at signing, an additional deposit at the end of diligence, and further deposits at each extension.

Allocation and prorations. Property taxes, rents, security deposits, and operating expenses are prorated at closing. Transfer taxes, title premium, survey, and escrow fees are allocated by agreement or by local custom, which varies substantially by state.

2.2 The Due Diligence Period

The provision that does the most work in the entire agreement.

Length. Negotiated, and it should be negotiated against the investigation plan from Modules 4A and 4B rather than against convention. Thirty days is standard for simple improved property. Sixty is standard for land. Ninety to one hundred twenty is the working minimum for anything requiring a Phase II, a seasonal environmental survey, an existing-building code analysis, or tenant estoppels. A period that cannot physically accommodate the required investigations is not a due diligence period; it is a formality.

Access. The buyer's right to enter and inspect, typically conditioned on notice, insurance, and an obligation to restore. Invasive testing - borings, monitoring wells, test pits - usually requires separate written consent, and the seller may refuse or condition it. **A buyer that cannot drill cannot complete a geotechnical investigation or a Phase II,** so access for invasive work belongs in the LOI, not in a later request.

Seller deliverables. Existing surveys, title policies, environmental and geotechnical reports, plans, permits, leases, service contracts, operating statements, and notices of violation. The clock should be tied to delivery: a diligence period that begins on signing, when the seller delivers three weeks later, is three weeks shorter than it appears. The fix is a deliverables condition - the period begins on the later of execution or delivery of a defined list.

The termination right. The buyer may terminate for any reason or no reason, in the seller's phrasing "in buyer's sole and absolute discretion," before the period expires, and recover the deposit. This is the whole point.

What happens at expiration. Either the buyer terminates, or the deposit becomes non-refundable and the buyer is committed. Note the default direction: in most forms, **silence means the deposit goes hard automatically.** A buyer that misses the deadline has gone hard by inaction. Calendar it, and calendar it twice.

2.3 Going Hard - Gate 2

"Going hard" means the deposit is no longer refundable. From that moment the buyer's choices are to close or to forfeit.

Variations that matter:

- **Full hard.** The entire deposit is at risk. Standard.
- **Partial hard.** A portion goes hard at the end of diligence and the balance at a later milestone.
- **Hard with a carve-out.** The deposit remains refundable for specified failures - a title defect the seller will not cure, a casualty, a failure of a stated condition, a seller default.
- **Applicable to the price.** The deposit is credited at closing. Nearly universal, and worth confirming.

A buyer's protection after Gate 2 is only what the closing conditions give them. This is why closing conditions are negotiated with the care that the deposit deserves.

2.4 Extensions

Extension rights are the cheapest insurance in the agreement and the most commonly omitted.

Typical structure: one or more extension options of 30 to 60 days each, exercisable by notice, in exchange for an additional deposit that is non-refundable but applicable to the price. The developer is buying time with money that still counts toward the purchase.

Two disciplines. **Negotiate extensions at LOI stage**, when you have not yet asked the seller for anything. Requesting an extension at day 55 of a 60-day period is a negotiation you conduct from weakness, and the seller knows exactly why you need it. And **make extension a right, not a request** - an option you may exercise unilaterally, not something requiring the seller's agreement.

2.5 Contingencies

A contingency permits the buyer to terminate, or excuses closing, if a specified condition is not satisfied.

Entitlement contingency. The buyer is not obligated to close unless a defined approval is obtained, final, and beyond appeal. This is the single most valuable provision available to a developer on an unentitled site, and sellers resist it strenuously because it converts a sale into an option. Key terms: what approval qualifies, what density or conditions are acceptable (a rezoning granted with unacceptable conditions should not count as satisfaction), an outside date, who controls the application, and whether the buyer must pursue appeals.

Financing contingency. Common in residential and small commercial transactions, rare in institutional ones. Most sophisticated sellers refuse it, taking the position that financing is the buyer's problem.

Third-party consent contingency. Where the deal requires a consent nobody controls - an REA amendment, an easement release, a ground lessor's approval, an estoppel from a major tenant.

Title contingency. Discussed in Section 6.

Casualty and condemnation. What happens if the property is damaged or taken between signing and closing.

2.6 Seller Representations and Warranties

Statements of fact by the seller about the property, on which the buyer may rely. Typical subjects: authority to sell, no undisclosed litigation, no notices of violation, no undisclosed leases or service contracts, the accuracy of the rent roll, the completeness of delivered documents, no known environmental conditions, no pending condemnation.

Three terms determine whether representations mean anything:

Knowledge qualifiers. "To seller's actual knowledge, without investigation" is a substantially weaker statement than an unqualified one, and narrower still when knowledge is defined as that of a single named individual.

Survival. Representations that do not survive closing are unenforceable the moment the deed records. Survival periods of six to twelve months are typical, sometimes with a portion of the price escrowed.

Caps and baskets. A floor below which no claim may be brought and a ceiling on total liability. A representation with a twelve-month survival and a cap of one percent of the purchase price is a limited remedy.

Against all of this sits "**as-is, where-is.**" Most commercial PSAs include a broad as-is clause disclaiming all representations except those expressly stated, and a release of claims. The practical effect is that **the buyer's protection is diligence, not the seller's promises.** This is not a drafting failure. It is the allocation of risk that the due diligence period exists to make fair.

2.7 Assignment

The right to assign the contract to another entity before closing. It matters for three reasons: the buyer will form a single-purpose entity to take title (Section 7), the equity partner may require a particular ownership structure, and in some deals the contract itself is the asset being sold.

Sellers limit assignment to prevent the buyer from flipping the contract. The common compromise permits assignment to an affiliate controlled by the buyer, without consent, with the original buyer remaining liable.

2.8 Default and Remedies

Buyer default. The seller's remedy is almost always limited to retention of the deposit as liquidated damages. This is the buyer's most important protection: the deposit is the ceiling on exposure.

Seller default. The buyer's remedies are typically return of the deposit, sometimes reimbursement of costs up to a cap, and - critically - **specific performance**, a court order compelling the seller to convey. Specific performance is available for real estate because each parcel is unique and damages are inadequate. A buyer that gives up specific performance has agreed that the seller may sell to someone else and simply return the deposit, which on a deal where the buyer has spent six figures on diligence and entitlement is not a remedy at all.

Section 3 - The Retrade

3.1 What It Is

A **retrade** is a request to reduce the price or change the terms after the contract is signed, based on what diligence revealed. It happens in a meaningful share of commercial transactions and it is, properly done, the system working: the due diligence period exists precisely so that the price can reflect what is actually true about the property.

3.2 What Separates a Legitimate Retrade From a Bad-Faith One

Sellers and brokers distinguish sharply between the two, and a developer's reputation in a market is a real asset that is spent here.

A legitimate retrade is based on **a specific, documented finding that was not knowable at signing**, quantified by a third party, presented with the report attached, and sized to the finding. "The geotechnical report requires drilled piers and structural floors at \$11,500 per unit; here is the report and here is the contractor's pricing" is a legitimate position.

A bad-faith retrade is based on nothing new. The buyer's underwriting changed, the market moved, the buyer overbid and is now using the diligence period as leverage. "We need another \$300,000" with no finding attached is a different conversation, and a seller who has been through it once will require a larger non-refundable deposit from that buyer forever.

The practical distinction is whether you can hand the seller the document.

3.3 The Alternatives to Price

Price is one variable. A seller who cannot or will not reduce price may accept something else, and often the alternative is worth more to the buyer than the dollars:

- **A seller credit at closing** for a specific documented item. Easier for a seller to accept than a price reduction, because the headline price is preserved - which matters when the seller must report to a board, a partnership, or a congregation.
- **An escrow holdback** for an unresolved item, released when resolved.
- **A seller indemnity** for a defined risk, backed by escrowed funds.
- **A longer or extended diligence period** to resolve an open question.
- **Seller cooperation obligations** - pursuing a consent, curing a title exception, terminating a service contract.
- **A restructure** - converting a purchase into an option, extending the closing, adding seller financing.

That last one is the most powerful and the least used. When the gap is caused by risk rather than by cost, the right answer is usually not a lower price but a structure in which the buyer stops paying for a risk it cannot control. Section 4 is that toolkit.

3.4 Knowing the Seller's Position

Everything above is leverage, and leverage depends on facts about the seller that a buyer should establish before the conversation:

- **What is their alternative?** If the next-best buyer pays materially less, the seller's floor is lower than their stated one.
 - **Do they have carrying pressure?** Debt service, taxes, an expiring listing, a fiscal year-end, a fund wind-down, a distribution to make.
 - **Is the price supported by any use that is currently legal?** If the asking price only works for a buyer taking entitlement risk, the pool of buyers who can pay it is very small.
 - **Who must approve?** An individual decides quickly. A board, an investment committee, or a congregation has a process, a calendar, and a need to explain the decision to someone else - which is why headline price and reported upside can matter more to them than net proceeds.
-

Section 4 - Structures Beyond a Simple Purchase

A straight purchase transfers fee title at closing for cash. Many development deals do something else, because the fundamental problem of land acquisition is that the buyer wants to pay for entitled land and the seller owns unentitled land.

4.1 Option Agreement

The buyer pays option consideration for the **right, not the obligation**, to buy at a fixed price within a stated period. Option payments are typically non-refundable and often credited to the purchase price. The buyer may terminate at any time, losing only the payments made.

The economics are the point. Under a PSA with a hard deposit, the buyer is committed to buy or forfeit. Under an option, the buyer is never committed. For a site requiring a discretionary approval, this is the difference between carrying entitlement risk on the full purchase price and carrying it on the option payments.

Sellers price the difference, and should - they are giving up the right to sell to anyone else while retaining all the carrying cost. Option consideration is usually structured as periodic payments so that a buyer who stops believing stops paying.

Options are also the standard structure where the buyer needs a long entitlement period, where the lender will not finance unentitled land, and where the buyer wants to control a site while deciding whether to proceed.

4.2 Rolling Takedown

A single contract under which the buyer purchases the land in phases on a schedule, typically with a deposit securing the whole and a per-phase price that escalates over time. Standard in

for-sale residential land development, where a builder does not want to own four years of lot inventory.

Benefits the buyer's capital efficiency and the seller's total proceeds. Requires careful attention to what happens if the buyer defaults mid-schedule, and to who builds the infrastructure serving later phases.

4.3 Land Banking

A third party buys the land and holds it, granting the developer or builder an option or takedown right. The land banker earns a return on the carry; the developer keeps the land off its balance sheet and out of its capital stack. Common in production homebuilding and increasingly in other sectors as a way to manage the cost of holding unentitled or pre-development land.

4.4 Ground Lease

The owner retains fee title and leases the land long-term - commonly 50 to 99 years - to the developer, who builds and owns the improvements for the term. At expiration, improvements typically revert to the landowner.

Used when the landowner will not or cannot sell: institutions, families with basis or legacy concerns, public agencies, religious organizations. The developer avoids the land purchase entirely, which materially reduces capital requirements, and pays ground rent instead.

The complications are real. Leasehold financing requires the lender to be comfortable with the ground lease, which means **lender protection provisions** are essential: notice and cure rights on default, the right to a new lease if the ground lease terminates, and consent to leasehold mortgages. A ground lease that has not been drafted with financing in mind may be unfinanceable. Term length matters too - lenders generally want the lease to extend well beyond the loan term, and residual value declines as the term shortens.

Subordinated ground leases, where the landowner subordinates its fee to the leasehold mortgage, are far more financeable and far less common, because the landowner is putting its land at risk for the developer's loan.

4.5 Seller Financing

The seller takes back a note for part of the purchase price. Useful when conventional financing is unavailable or expensive, when the seller wants to spread taxable gain over time through an installment sale, or when the parties cannot agree on price and can agree on terms.

For a developer, seller financing on land is particularly valuable because construction lenders will often not lend against unentitled land at all. The critical negotiation is **subordination**: will the seller subordinate its note to the construction loan? An unsubordinated seller note usually must be paid off at construction loan closing, which limits its usefulness. Partial release provisions matter equally in a phased deal.

4.6 Joint Venture With the Landowner

The landowner contributes the land to a joint venture in exchange for an equity interest rather than cash. The developer contributes the development expertise and typically the cash equity.

Attractive in principle: it reduces the cash required at closing and aligns the parties. Difficult in practice, because the landowner is now a partner with approval rights, a different risk tolerance, and frequently no development experience. The land contribution must be valued, the waterfall must be negotiated, and the landowner must be willing to accept a return that comes years later and is not guaranteed.

Most landowners who can sell for cash, do. The structure appears most often where the land is unusually valuable, the owner is sophisticated, or no buyer will pay the owner's price.

4.7 1031 Exchange

A like-kind exchange under Section 1031 of the Internal Revenue Code permits an owner of investment or business real property to defer capital gains tax by exchanging into other like-kind real property, using a qualified intermediary and observing strict deadlines - commonly 45 days to identify replacement property and 180 days to close.

For a buyer, the relevance is usually on the other side of the table. **A seller in a 1031 exchange has a deadline**, which affects what they will trade for timing certainty. A buyer able to close fast, or to accommodate the seller's identification calendar, has leverage that has nothing to do with price. Conversely, a buyer who is themselves completing an exchange is on a clock and should not advertise it.

Note that a seller who is a tax-exempt entity, a fund in liquidation, or a dealer holding property as inventory is generally not a 1031 candidate. Knowing which seller you have tells you what they actually want.

4.8 Choosing Among Them

If the problem is	Consider
Entitlement risk on unentitled land	Option, or PSA with entitlement contingency
Too much capital tied up in land inventory	Rolling takedown, land banking
Seller will not sell at any price	Ground lease, JV
Price gap that is really a risk gap	Option, contingent payments, holdback

No financing available for the land

Seller financing, ground lease

Seller needs timing certainty

Fast close, larger deposit, fewer conditions

Seller needs a headline price

Credits and holdbacks instead of price reduction

Section 5 - Entity Formation

The entity that signs the PSA is frequently not the entity that closes, and neither is necessarily the entity that will operate the property.

5.1 The Single-Purpose Entity

Standard practice is to hold each project in its own **single-purpose entity** (SPE), almost always a limited liability company. The reasons:

Liability isolation. Claims against the project reach only the project's assets, not the sponsor's other projects.

Lender requirement. Construction lenders generally require a **bankruptcy-remote** SPE whose organizational documents prohibit it from owning other assets, incurring other debt, or filing for bankruptcy without the consent of an independent manager or member. The lender is protecting its collateral from being dragged into an unrelated insolvency.

Clean capital structure. The operating agreement of the SPE is where the LP/GP economics from Module 6 live - the waterfall, the major decision rights, the capital call mechanics.

Transferability. Selling the entity rather than the property can avoid transfer tax in some jurisdictions and can permit a sale without triggering loan assumption provisions. It also transfers the entity's liabilities, which is why buyers of entities diligence differently than buyers of property.

5.2 Why the Signing Entity Differs

Deals are frequently signed by an existing entity or by the sponsor personally, before the project entity exists, because:

- The equity partner has not been finalized, and the partner's tax and structural requirements determine how the entity should be formed
- Forming an entity costs money and time that are wasted if diligence kills the deal
- The lender's SPE requirements are not known until a term sheet exists

The contract is then assigned to the SPE before closing - which is why the assignment provision in Section 2.7 is not a boilerplate item.

5.3 The Consequence for Part 2

The "Owner" that signs the agreement with the architect and the contract with the general contractor is typically an LLC formed weeks or months earlier, holding one asset, with no operating history and no assets beyond the project. It was designed that way.

The consequences appear throughout Part 2:

- **Lien rights** matter because there may be no other assets to pursue
- **Payment and performance bonds** exist partly because of it
- **Guaranties** from the sponsor's principals are what give the lender and sometimes the contractor a solvent party
- **Joint check agreements** and **lien waiver discipline** are how subcontractors manage the risk

A contractor who understands that the Owner is a project-specific entity funded by a construction loan will read the payment provisions of their contract differently, and better.

Section 6 - Title

6.1 The Commitment

The title commitment is the title company's offer to insure, and it has a defined structure.

Schedule A - the facts being insured. The proposed insured, the policy amount, the type of estate (fee simple, leasehold), the current record owner, and the legal description. Read the legal description against the survey. If they disagree, something is wrong that must be resolved before closing.

Schedule B, Part I - requirements. What must happen before the policy issues: the deed must be recorded, the existing mortgage released, corporate authority documented, taxes paid, a specific lien satisfied. This is the closing checklist's spine.

Schedule B, Part II - exceptions. What the policy will **not** insure against. Everything encumbering the property: easements, covenants, restrictions, mineral reservations, leases, and matters a survey would show.

Schedule B Part II is the same list examined in Module 3 Section 5. The obligation there was to read the underlying recorded documents. The obligation here is to decide what to do about each one before the diligence period ends.

6.2 Standard Exceptions and How They Are Removed

Every commitment includes standard or pre-printed exceptions: rights of parties in possession, unrecorded easements, boundary and encroachment matters that a survey would disclose, mechanic's liens, and taxes not yet due.

Most are removable by providing what the title company needs:

- Delivering an acceptable **survey** removes or modifies the survey exception
- An **owner's affidavit** addressing parties in possession and recent work removes those exceptions
- A **lien waiver** package handles mechanic's lien exposure where work has been performed

Removing standard exceptions is what distinguishes a standard policy from an **extended coverage** policy, and lenders require extended coverage.

6.3 Endorsements

Endorsements add specific affirmative coverage. The ones that recur in development:

- **Zoning**, insuring that the property is zoned as stated and, in its broader form, that a specified use and structure comply with the ordinance. Requires a zoning report and is limited in what it will affirm.
- **Access**, insuring that the property has actual vehicular and pedestrian access to a named public street.
- **Contiguity**, insuring that multiple parcels adjoin without gaps.
- **Survey**, insuring against matters a survey would disclose.
- **Restrictions, encroachments and minerals**, insuring against violations of covenants and against damage from mineral extraction - meaningful where a mineral estate is severed.
- **Easement and encroachment** endorsements insuring against specific interference.
- **Construction loan / pending disbursement**, keeping coverage current as the loan funds and priority as against mechanic's liens.
- **Tax parcel**, confirming the property is assessed as a separate parcel with no other land included.

Endorsements cost money and they are the cheapest risk transfer available in the transaction. They do not remove an encumbrance - they shift the financial consequence of a defined problem to the insurer.

6.4 Survey–Title Reconciliation

Covered in Module 4A and repeated here because it is the step most often skipped.

Take Schedule B Part II and check each item against the ALTA survey. Each exception should be plotted, or explicitly noted as not plottable, or identified as not affecting the land. The three things you are looking for:

- An easement crossing the buildable area
- An exception described as **blanket in nature**, meaning it exists somewhere on the property without a defined location
- An encroachment, in either direction

This reconciliation is the buyer's job. The title company insures; it does not design around the encumbrances.

6.5 Title Insurance

An **owner's policy** insures the buyer's title up to the policy amount, for as long as the insured holds title. A **lender's policy** insures the lender's lien priority. Both are one-time premiums paid at closing. Who pays is a matter of local custom and negotiation.

Two limits worth understanding. Title insurance is **indemnity for defects in title**, not a warranty that the property is suitable for anything. And it does not cover matters listed as exceptions - which is precisely why the exceptions are the document that matters.

Section 7 - Closing

7.1 Conditions Precedent

Closing occurs when every condition is satisfied or waived. Typical buyer conditions:

- Seller's representations remain true as of closing
- Title company is prepared to issue the policy in the agreed form
- Required estoppels and consents delivered
- Required approvals obtained and final
- No material casualty or condemnation
- Seller has performed its covenants

And the one that appears in every financed deal, whether stated or not: **the lender's conditions precedent must also be satisfied**, because a buyer whose loan does not fund cannot close. The lender's list, from Module 6, includes the appraisal, title, survey, environmental reports, executed construction contract, permits, insurance, guaranties, and any deal-specific condition the lender imposed.

The practical consequence is that **any unresolved item from Modules 3 and 4 lands on the closing checklist**. A title exception not cured, a consent not obtained, a permit not issued: each becomes a condition, and each becomes a reason closing slips.

7.2 The Closing Checklist

The working document of the final weeks. A typical land acquisition closing:

Transaction documents

- Deed, in recordable form
- Bill of sale and general assignment of plans, permits, reports, and warranties
- Assignment and assumption of leases, contracts, and intangibles
- Closing statement / settlement statement
- Assignment of the PSA to the SPE

Title and survey

- Final title commitment with all requirements satisfied
- Owner's affidavit and gap indemnity
- Recorded releases of prior liens
- Final survey, certified to buyer, lender, and title company
- Endorsements ordered and priced

Entity and authority

- Formation documents and good standing for the SPE
- Operating agreement, executed
- Resolutions authorizing the transaction and the loan
- Seller's authority documents

Financing

- Loan agreement, note, mortgage or deed of trust
- Guaranties
- Collateral assignments of the design and construction contracts, with consents from the architect and contractor
- Insurance certificates naming the lender
- Equity funded

Property-specific

- Tenant estoppels and SNDAs
- Third-party consents and recorded amendments
- Utility will-serve letters
- Reliance letters on diligence reports
- Approvals and permits, with any appeal periods expired

Prorations and funds

- Tax, rent, and expense prorations

- Deposit credited
- Wire instructions verified by phone to a known number

That last item is not trivia. Wire fraud in real estate closings is common and losses are frequently unrecoverable. Verification by voice to an independently known number is standard practice.

7.3 Mechanics

Most commercial closings are **escrow closings**: documents and funds are delivered to the escrow agent, who confirms all conditions are met, records the deed, disburses funds, and issues the policy. Nobody sits at a table. In attorney-closing states the process runs through counsel instead.

The **gap** is the interval between the title search date and the recording of the deed, during which an intervening lien could attach. The seller's gap indemnity and the title company's willingness to insure over it handle the risk.

Funding order matters: equity typically funds first, then loan proceeds, then the seller is paid and the deed records simultaneously with the mortgage.

7.4 What Closing Actually Is

At closing, the option to walk away ends. Gate 4 in Module 1's framework is the loan closing and notice to proceed; for a land acquisition without immediate construction, the purchase closing is the point at which the developer owns a problem rather than an opportunity.

Everything after this is Part 2.

Artifacts

Artifact	Prepared by	Purpose
Letter of intent	Buyer or broker	Business terms; exclusivity
Purchase and sale agreement	Counsel	The binding contract
Option agreement	Counsel	Right without obligation
Escrow agreement / instructions	Title company	Deposit handling

Title commitment	Title company	What will and will not be insured
Exception documents	Title company	The recorded instruments themselves
ALTA survey	Surveyor	Reconciled against Schedule B
Estoppel certificates and SNDAs	Tenants	Confirmation of lease terms
Third-party consents	Consenting parties	REA amendments, easement releases
Formation documents and operating agreement	Counsel	The SPE and its economics
Assignment of contract	Counsel	Contract to the SPE
Closing checklist	Counsel	Every condition, with an owner and a date
Closing statement	Title company	Funds and prorations
Deed	Counsel	Conveyance

Failure Modes

Accepting a conventional diligence period. Thirty or sixty days is a convention, not an analysis. Build the period from the investigation plan in Modules 4A and 4B, and remember that Phase II follows Phase I and estoppels run on the tenants' clock.

Failing to tie the diligence clock to seller deliverables. A period that starts at signing and a seller who delivers in week three has cost you three weeks.

No extension rights. Negotiate them at LOI stage. Asking at day 55 is negotiating from visible weakness.

Missing the expiration. In most forms, silence goes hard automatically. Inaction is a decision.

Not securing invasive access. Without the right to drill, there is no geotechnical report and no Phase II.

Giving up specific performance. A seller who can walk by returning your deposit can sell to someone else after you entitled the property.

Reading Schedule B summaries instead of the recorded documents. Repeated from Module 3 because it is repeated in practice.

Treating title insurance as a substitute for reading exceptions. The policy does not cover what it excepts.

Assuming the contract can be assigned. If the SPE cannot take the contract, the structure the lender and equity require may not be available.

Retrading without a document. A price request without a report attached is a reputational transaction, and markets are small.

Negotiating only price. When the gap is caused by risk rather than cost, structure is the answer, and it is usually worth more than the dollars.

Forgetting the appeal period. An approval is not final until the period for challenging it has expired. Lenders know this even when developers forget it.

What Happened Next

Kipling Row

The developer went back to the seller with a binder: the geotechnical report, the abatement scope, the utility fee schedule, the revised civil plan, and a one-page residual land value analysis. Every number in the retrade had a document behind it.

The ask. A reduction from \$1,950,000 to \$1,350,000, on the basis of \$958,000 of diligence findings that had moved the residual land value from approximately \$1,811,000 to approximately \$854,000 at a 15 percent required margin.

The seller's position. The congregation's board was motivated - the merged congregation needed cash for its own building program - but it was not distressed. No debt, no carrying pressure, and a two-year-old decision to sell that nobody was in a hurry to revisit. The board's chair made the point that boards make: the property had been listed at \$1,950,000, the buyer had signed at \$1,950,000, and the board had told its congregation to expect \$1,950,000.

The fact that changed the conversation. The broker, at the developer's suggestion and with the board's consent, tested the market for a by-right sale. Two production builders priced the site for the 11 detached lots the R-1 zoning actually permits. The bids came in at **\$910,000 and \$975,000.**

This is the arithmetic from Module 2 arriving in the room. Detached single-family is legally permissible, physically possible, and not financially feasible at the asking price. **The only buyer who could pay anything near \$1,950,000 was one taking entitlement risk**, and there is exactly one of those in the room.

The restructure. The board would not take \$1,350,000 on a straight sale - partly economics, partly not wanting to report a 31 percent reduction. So the deal changed shape instead of only changing price.

The PSA was terminated and replaced with an **option agreement**:

- **Option price: \$1,465,000**, fixed
- **Option payments of \$45,000 per 90-day period**, up to four periods, each non-refundable and each fully credited against the purchase price
- **Closing within 45 days** of a final, non-appealable PUD approval
- **Developer controls the application** and may terminate at any time, forfeiting only payments made
- **Contingent additional payment of \$60,000** if the approval permits more than 30 units
- Seller cooperates with the application as landowner and does not oppose

What each side got. The board could report a transaction valued at up to \$1,525,000 - a 22 percent premium to the best by-right bid - with a participation in the upside, and no obligation to do anything but wait. The \$185,000 of non-refundable deposit already posted under the PSA was credited to the first option periods.

The developer got the thing that actually mattered, and it was not the \$485,000.

The risk transformation. Under the original PSA, the developer would have been obligated to close on \$1,950,000 of unentitled land, funded by roughly \$6 million of equity that had to be committed and personally guaranteed, with the rezoning still unresolved. Under the option, the exposure before the council vote is the option payments plus soft costs already spent - approximately **\$245,000** - and the \$6 million of equity is not called until an approval exists.

The revised economics:

	Original	Restructured
Land and closing	\$2,010,000	\$1,585,000
Total project cost	\$14,986,000	\$14,561,000
Net sales revenue	\$15,904,000	\$15,904,000
Projected profit	\$918,000	\$1,343,000
Margin on cost	6.1%	9.2%

Equity at risk before the
vote

~\$6,000,000
committed

~\$245,000

Nine point two percent is thin. It is not what the developer underwrote at Gate 0 and it does not survive a large construction overrun. But the deal is no longer a bet in which \$6 million of guaranteed equity rides on a council vote. It is a \$245,000 bet on a council vote, with \$6 million deployed only if the bet wins.

That is the module's entire point. The price moved by 25 percent. The risk moved by 96 percent. When a developer cannot close a price gap, the question to ask is not how much lower the seller will go - it is what risk the buyer is being asked to pay for, and whether the structure can stop asking.

Gate 1 recommitted. Gate 2 deferred to the approval itself, which is now the only remaining gate before closing.

Six Corners

A shorter negotiation, because the sponsor had two independent numbers pointing at the same answer.

The position. The lender's appraiser valued the property as-is at \$4,050,000. The sponsor's own residual, working backward from a 7.5 percent target yield on cost, supported approximately \$3,970,000. The contract price was \$4,400,000.

Two methods, computed by parties with opposing interests, within two percent of each other. The sponsor presented both, along with the PCA, the hazmat survey, the conversion feasibility study, and the code analysis.

The seller's position. An out-of-state fund in wind-down, with a fiscal year-end nine weeks out and a stated intention to be out of the asset. Not a 1031 candidate - a liquidating fund has no replacement property to exchange into - so what they wanted was not timing precision but **certainty of closing**. They had already had one buyer terminate during diligence.

That is the leverage. The sponsor was not the highest bidder; they were the one most likely to close.

The settlement.

- **Price reduced to \$4,065,000**, essentially the appraisal
- **Seller credit of \$140,000** at closing against the documented roof replacement, which preserved a headline price above \$4 million for the fund's reporting
- **Diligence extended 45 days** to complete the REA negotiation, in exchange for an **increased non-refundable deposit of \$400,000**
- **Seller obligated to use commercially reasonable efforts** to obtain the REA consents, as a twenty-year party to the agreement with standing relationships

- **Closing conditioned** on the recorded REA amendment and a mineral estate subordination, with an outside date and a return of deposit if not satisfied

That last provision is the one that mattered. The REA amendment had become a lender condition precedent in Module 6 - no amendment, no loan, no closing. Making it a buyer's closing condition with a deposit carve-out meant the sponsor was not going hard on a consent it did not control.

The consents. The medical group next door wanted certainty about parking and took a recorded covenant guaranteeing 290 shared stalls in perpetuity - which the restriping plan from Module 5 already satisfied at 288, with two stalls added in the amendment itself. The family partnership wanted the deteriorating shared entrance drive fixed; the sponsor agreed to fund repaving at **\$95,000**. Both executed. The amendment recorded eleven days before closing.

The mineral estate was resolved with a surface use agreement and a non-disturbance from the mineral owner, who had no active leasing in the area and charged a nominal fee.

C-PACE did not happen. The debt fund's position on the roughly \$1.8 million of eligible scope evolved from a flat refusal to a conditional consent: it would permit C-PACE only if the bridge loan were reduced dollar for dollar. That reduces cheap senior debt to make room for C-PACE - the benefit is captured by the lender's consent leverage rather than by the sponsor. The sponsor dropped it and preserved the option to pursue C-PACE at refinance, when the bridge lender is being replaced and no longer has a consent right. **An incentive is only worth what the party whose permission you need will let you keep.**

The revised economics:

	Module 5/6	At closing
Acquisition and closing	\$4,485,000	\$4,010,000
Shared drive repaving	-	\$95,000
All other costs	\$8,592,000	\$8,592,000
Total project cost	\$13,077,000	\$12,697,000
Stabilized NOI	\$949,000	\$949,000
Yield on cost	7.26%	7.47%
Value at 6.75% exit cap	\$14,059,000	\$14,059,000
Development margin	\$982,000 / 7.5%	\$1,362,000 / 10.7%

The \$380,000 of price relief moved the yield on cost by 21 basis points and the development margin by more than three percentage points. That is the leverage inherent in a residual: land is the smallest large number in the deal and it absorbs everything.

Seventy-two basis points of spread is still thin - the sponsor's own target was 75 - and the cap rate sensitivity from Module 6 has not improved much. What changed is that the deal now clears its own hurdle rather than missing it, and it does so on a five-year hold in a submarket with 3.4 percent vacancy.

Gate 2 cleared. The deposit went hard, and the deal closed 45 days later into a newly formed single-purpose LLC, with the fund as managing member, a completion guaranty from the fund's two principals, and a collateral assignment of the architect's agreement that the architect signed at closing without asking what it was.

Optional: Try This

Find a purchase and sale agreement. They are not hard to locate - commercial PSAs appear as exhibits in SEC filings for public REITs and in municipal records where a public entity is a party, and state bar associations publish annotated forms.

Read for four things, in this order:

The due diligence period. How long, when does it start, and what does the buyer have to do to terminate?

What happens at expiration. Does the deposit go hard automatically? Are there carve-outs?

The seller's representations. Are they knowledge-qualified? Do they survive closing, and for how long? Is there a cap?

The remedies. What does the seller get if the buyer defaults, and what does the buyer get if the seller does? Is specific performance available?

Those four answers tell you the entire risk allocation of the transaction. Everything else is mechanics.

Then, for the parcel you have been following since Module 1: pull its deed history from the county recorder. Look at what it last sold for and when. If you have been doing the exercises, you now have a rough sense of what it is worth to a developer today. The gap between those two numbers, adjusted for time, is either the entitlement value someone created, the carrying cost someone absorbed, or a mistake someone made - and the deed will not tell you which.

Next - Module 8: The Handoff to Design. What the owner actually transmits to the design team, what gets lost, what was already promised to a city on their behalf, and why the last page of Part 1 is the first page of Part 2.

1.8 - The Handoff to Design

At the Six Corners closing, the architect signed a collateral assignment of their own agreement without asking what it was.

It was a routine document in a stack of routine documents. What it says is that if the borrower defaults, the lender may step into the owner's position under the architect's contract and require the architect to continue working - for the lender, on the lender's terms, with the lender's priorities. The architect had spent four months on a feasibility study, a conversion study that killed the client's original thesis, and a preliminary code analysis. They had never seen the loan agreement, did not know what the completion guaranty required, and could not have said what the conditions of approval obligated the project to build.

None of that is unusual. It is the normal condition of the design team at the moment a project is handed to them.

This module is about that moment. Part 1 has spent seven modules establishing that a project arrives at the design team pre-loaded with constraints - a site, a program, a budget, a schedule, a delivery method, and a set of commitments already made to a city. The question this module answers is what actually gets transmitted, what does not, and what the consequences are of the difference.

Gate 4 lives here. Land is purchased, the construction loan closes, equity funds, guaranties are signed, and the design team is engaged under a real contract with a real scope. Everything before Gate 4 was optional. Everything after it is Part 2.

Section 1 - What the Owner Hands Over

1.1 The Program

In development, the program is the revenue model: unit count, unit mix, rentable area, parking count, bay sizes, clear height. It was established in the test fit in Module 5, validated against the market study in Module 2, constrained by zoning in Module 3, and locked into the pro forma in Module 6. By the time the design team receives it, it has been through an approval process and is carried in a lender's underwriting.

In CDT terms, the equivalent document is the **Owner's Project Requirements (OPR)** - the statement of what the owner needs the building to do, including spaces, adjacencies, performance criteria, and quality expectations.

These are not the same document, and treating them as though they are is a common handoff failure in the industry.

A development program says *34 attached units averaging 1,380 square feet, 2.2 parking spaces per unit, targeting a blended closing price of \$495,000*. It is a revenue specification.

An OPR says what those units must achieve: acoustic separation between party walls, thermal performance, accessibility scope, durability of finishes, expectations for maintainability, the standard to which the building envelope must perform. It is a performance specification.

Most developers hand over the first and assume the second. Most design teams receive the first and infer the second. The inference is usually approximately right and occasionally expensively wrong, and it is the origin of a recognizable class of dispute in which the owner says "that isn't what I wanted" about something that was never written down.

The practical fix is a written OPR, produced at the start of design, that the developer reviews and signs. On sophisticated and institutional projects this is standard. On merchant development it frequently does not happen, and the design team should produce one anyway and get it acknowledged, because the alternative is discovering the owner's unstated expectations during construction.

1.2 The Site Constraints Package

This is the design team's entire inheritance from Modules 3, 4A, and 4B, and it is where transmission most often fails - not because anyone withholds anything, but because the consultants who produced it are gone.

The geotechnical engineer, the surveyor, and the environmental consultant did their work during due diligence, delivered reports to the developer, and were released. The architect is engaged months later. Nobody is left who was present when the findings were discussed.

A complete constraints package contains:

Document	From	What it constrains
ALTA survey with Table A items	Module 4A	Boundaries, easements, encroachments, access
Topographic survey	Module 4A	Grading, drainage design
Geotechnical report, with boring logs	Module 4A	Foundations, slabs, pavement, earthwork
Phase I ESA, and Phase II if performed	Module 4A	Contamination, controls, vapor
Wetland delineation and jurisdictional determination	Module 4A	Buildable area, permit obligations

Utility will-serve letters and capacity correspondence	Module 4A	Service sizing, off-site scope, schedule
Fire flow test results	Module 4A	Sprinkler design, on-site storage
Traffic impact study and required mitigation	Module 4A	Access geometry, off-site improvements
Property Condition Report	Module 4B	Scope of repair and replacement
Hazardous materials survey	Module 4B	Abatement scope and sequencing
Preliminary code analysis	Module 4B	Compliance method, alteration level, occupancy
Structural evaluation	Module 4B	Capacity, lateral system, permitted modifications
Title commitment and all Schedule B exception documents	Modules 3, 7	Private restrictions on what may be built
Recorded REAs, CC&Rs, easement agreements, amendments	Modules 3, 7	Use, design review, shared facilities
Zoning report or opinion	Module 3	Dimensional and use limits
Entitlement approvals with all conditions	Module 3	Binding project obligations
Reliance letters	Modules 4A, 4B	The design team's right to rely on the reports

Three items on that list are the ones most often missing.

The boring logs, as opposed to the geotechnical report's summary recommendations. The recommendations are the consultant's conservative conclusion; the logs are the data, and they show variability across the site that a single recommendation flattens.

The Schedule B exception documents themselves, as opposed to the title commitment's list of recording references. A developer who read them in diligence often transmits only the commitment.

Reliance letters. Reports are addressed to the party that commissioned them. A design team relying on a geotechnical report addressed to someone else - a prior owner, the seller, a different entity in the sponsor's structure - has no recourse against the consultant. Reliance is inexpensive, routine, and must be requested.

1.3 The Budget

The design team receives a number. What it usually does not receive is the structure behind the number, and the structure is what determines how much freedom the number actually has.

From Modules 5 and 6, the budget the developer hands over is:

- Locked into a loan agreement as an exhibit
- Divided into line items that cannot be freely reallocated without lender consent
- Carrying a contingency that is under lender control and flows down the equity waterfall if unspent
- Escalated to a construction midpoint that assumes a specific schedule
- Supporting a yield on cost that was already thin before anything went wrong

A design team told "the budget is \$10.19 million" hears a target. A design team told "the budget is \$10.19 million, it is an exhibit to the loan agreement, reallocation between line items above \$50,000 requires the lender's consent, and the sponsor's promote does not activate until the deal clears a 9 percent margin" understands why the owner responds the way they do.

Transmitting the budget's structure, not just its total, is one of the highest-value things a developer can do at handoff, and it costs nothing.

1.4 The Schedule

Same principle. The design team receives dates. What matters is which dates are hard and why.

- **The loan's outside completion date** is a default trigger, not a preference
- **The interest reserve** was sized on an assumed duration; overrunning it is a cash call on the sponsor
- **Incentive deadlines** - an Opportunity Zone substantial improvement window, a tax credit placed-in-service date, a grant expenditure deadline - are statutory and do not move
- **Approval expirations.** A conditional use permit or preliminary plat that is not acted on within a stated period lapses
- **Seasonal constraints** on earthwork, paving, and planting
- **Lease commencement dates** in signed leases, with delivery penalties
- **Tenant buyout and holdover dates** on occupied buildings

A schedule handed over as a bar chart communicates duration. A schedule handed over with the consequence of each milestone communicates risk.

1.5 Entitlement Commitments

The most under-transmitted category, and the one with the most direct construction consequences.

From Module 3: when an approval is granted with conditions, those conditions become binding obligations on the project. A rezoning, a PUD, a conditional use permit, or a site plan approval can carry dozens of them - a masonry percentage on the facade, a fence height and material along a specific property line, a landscape buffer with specified caliper trees, restricted delivery hours, a required pedestrian connection, a public art contribution, an affordable unit set-aside, a funded share of a traffic signal.

Additionally binding, and coming from outside the approval itself:

- **Development agreement** obligations, including phasing, infrastructure, and dedication
- **Private design review** under CC&Rs or an REA, which may have its own approval authority over your elevations
- **Utility agreements** and their cost-sharing terms
- **Recorded covenants** created during the transaction - the parking covenant at Six Corners is an example
- **Incentive program requirements:** prevailing wage, domestic content, energy targets, historic standards, income restrictions

Here is why this matters more than it appears. **The building official does not enforce conditions of approval, because they are not code.** Planning enforces them, and planning typically enforces them at certificate of occupancy. A project can be complete, code-compliant, inspected, and unable to obtain a CO because a condition nobody transmitted was never built.

The fix is mechanical and rarely done: **a conditions-of-approval matrix**, listing every condition, the responsible party, where it is addressed in the drawings and specifications, and how compliance will be verified at closeout. It belongs in the handoff package and it belongs in Division 01.

1.6 What Gets Lost

A short list of the recurring omissions:

- Reports transmitted without reliance letters
- Geotechnical recommendations without boring logs
- Title commitment without exception documents
- Conditions of approval as a PDF of the approval letter, never converted into a scope matrix
- Utility correspondence showing a capacity or lead-time problem, filed and not surfaced

- The reasoning behind the test fit, so the design team optimizes away a constraint that was load-bearing
 - The rejected alternatives and why they were rejected, so the design team proposes them again
 - Lease obligations on occupied buildings
 - Anything discussed in a meeting and not written down
-

Section 2 - The Entitlement Chicken-and-Egg

2.1 The Problem

A discretionary approval requires drawings. A city cannot approve a rezoning or a PUD on a description; it needs a site plan, elevations, a landscape concept, often a grading and utility concept, and sometimes renderings for a public hearing. Depending on the jurisdiction and the application, that package can approach schematic design.

Those drawings must exist **before the approval**, which means before the land is owned, which means before the construction loan exists, which means they are paid for with equity - at a point when the probability of the project proceeding is, at Kipling Row, roughly 60 percent.

This is the structural reason that pre-development design is unglamorous and underpaid. The money funding it is the riskiest money in the deal.

2.2 How It Gets Paid For

Four arrangements, in descending order of how well they work:

Paid, at a reduced scope. The developer engages the architect for a defined entitlement scope at a normal rate, with the full commission contingent on the project proceeding. Cleanest, and standard for developers with a repeat relationship.

Paid at risk with a success premium. A reduced rate during entitlement, with a higher fee or a bonus if the project proceeds. Prices the risk explicitly.

Unpaid or nominal, against a promise. Common, and the source of most of the friction in this phase. The architect produces entitlement drawings for little or nothing, in exchange for positioning on the commission. The developer receives drawings whose thinness reflects what was paid for them.

Contingent on the deal. The architect is paid at closing, out of loan proceeds. The architect is now an unsecured creditor of a single-purpose entity with no assets.

For a design professional, the relevant disciplines are to define the entitlement scope narrowly and in writing, to establish what the full commission is before doing the free work, and to

understand that **entitlement drawings are not schematic design** - they are a public exhibit, produced under different constraints, and should not be priced or scoped as though they are the first phase of the real project.

2.3 The Trap

The most consequential and least discussed consequence: **entitlement drawings become the design.**

They were produced quickly, cheaply, before the geotechnical report was complete, before any real code analysis, before an estimator priced anything. Then they were presented at a public hearing, published in a staff report, shown to neighbors, and approved - sometimes with the elevations attached to the approval as an exhibit, which makes them binding.

Now the real design begins, and the building cannot easily change. Not because the design is good, but because changing it means going back to the body that approved it.

Three specific failure patterns:

A unit count that the site cannot actually deliver. The entitlement site plan showed 34 units. The geotechnical report, the fire access requirements, and the detention sizing arrive afterward and the honest number is 31. The pro forma, the loan, and the approval are all built on 34.

Elevations attached to the approval. Material changes, fenestration changes, and massing changes may require an amendment. A PUD with approved elevations has converted an aesthetic decision into a regulatory one.

Conditions written against a drawing that no longer exists. A condition referencing "the access drive as shown on the approved site plan" is a problem when the access drive moves.

The mitigation is to build the entitlement package with the knowledge that it will constrain the real design: get the geotechnical and civil input in before the submittal where possible, request approval of a **dimensional envelope** rather than a specific building where the jurisdiction permits it, and negotiate administrative-amendment authority for minor modifications as a condition of approval.

Section 3 - Who Manages the Project

The "Owner" in the contract documents is a single-purpose LLC. Someone has to be that entity's voice, and which arrangement is in place determines a great deal about how the project will run.

Developer as project manager. The developer's principal runs the project directly. Universal in small shops. Fast decisions, real authority, and deep knowledge of why every constraint exists

- because they created them. The limitations are bandwidth and technical depth: a principal running acquisitions, capital, and entitlements on three deals is not reviewing submittals, and may not know what they are looking at when they do.

Internal project manager. The developer employs construction management staff. Common at merchant builders above a certain size and at institutional owners. Consistent process and real availability, with authority that stops at a threshold defined by the LP agreement and the loan.

Owner's representative. A hired project management function, engaged by the owner and independent of the design and construction team. Standard for owner-occupiers, institutions, and any owner without in-house capability. The owner's rep translates between the owner's business language and the project team's technical language, and does the work the owner cannot: reviewing pay applications, managing change orders, tracking the schedule, and holding the consultants to their scopes.

The distinction that matters to a design and construction team is **authority**. A good owner's rep has defined decision authority up to a threshold. An owner's rep who must route every decision to a principal who routes some of them to an investment committee has added a step rather than removed one. Establish at the kickoff who can decide what, at what dollar value, and how long approvals take.

Note the vocabulary collision. The developer's PM, the architect's PM, the contractor's PM, and the owner's rep may all be in the same meeting, all called "the project manager," holding four different jobs with four different loyalties.

Section 4 - The Delivery Method, Finalized

Module 5 introduced this as an owner decision that begins during feasibility. It is finalized here, before the design contract is signed, because the method determines who holds the design contracts. These methods will be discussed in detail in Part 2 - The Design and Construction Process.

4.1 The Methods

Design-Bid-Build (DBB). Owner contracts separately with architect and contractor. Design completes, then bids, then builds. Maximum design control, competitive pricing on a complete scope, no price until documents are complete, no contractor input during design, and the owner holds the risk that the documents are coordinated.

Construction Manager at Risk (CMAR / CM/GC). Owner contracts separately with architect and CM. The CM joins during design, provides preconstruction services, and converts to a guaranteed maximum price. Design and construction overlap. The owner keeps the designer

relationship and gains cost input during design. The GMP is set before documents are complete, which means it carries qualifications and allowances that must be read carefully.

Design-Build (DB). Owner contracts with a single entity that holds both design and construction. Fastest, earliest price certainty, single point of responsibility. The architect works for the contractor, which changes the design team's loyalties and the owner's control. Best where the program is well-defined and can be specified through performance criteria.

Progressive Design-Build. Design-build in two phases: the owner selects the team on qualifications, the team develops the design collaboratively, and the price is negotiated as the design matures, with an off-ramp if the parties cannot agree. Retains design-build's single-point responsibility while handling scope uncertainty better.

CM as Agent (CMa). The CM advises the owner for a fee and holds no construction risk. Used alongside another method.

Integrated Project Delivery (IPD). Multi-party agreement with shared risk and reward among owner, designer, and builder. Requires sophistication and repeat relationships on all sides. Rare, and effective where it fits.

4.2 How the Capital Structure Decides It

The honest account of how this decision usually gets made, which is not the account in most textbooks.

The lender requires a price before closing. This is the dominant constraint on privately financed work. A construction lender generally will not close without a GMP or lump sum and an approved contractor. DBB cannot produce a price until the documents are complete - so a deal that needs to close on a schedule cannot use DBB without carrying the entire design period on equity. **This single requirement pushes most private development to CMAR or Design-Build.**

The interest reserve rewards overlap. Methods that compress the schedule reduce carry and escalation. Where entitlement has already consumed a year, schedule recovery has quantifiable value.

Scope certainty determines whether a fixed price is a good idea. A repeatable building is a Design-Build candidate. A renovation with concealed conditions priced at a lump sum means the contractor prices the uncertainty and the owner pays for risk that may not materialize. Renovation work generally wants CMAR or progressive design-build with a contingency the owner controls.

Equity approval rights. LP agreements frequently require approval of the construction contract and sometimes the method.

Incentive and public funding bring procurement requirements. Prevailing wage, domestic content, and competitive procurement obligations attach to the money, not to the project.

Owner capability. DBB demands the most owner-side management. An owner without that capacity either buys it through an owner's rep or chooses a method that needs less of it.

Public owners are constrained by statute rather than preference - qualifications-based selection for design services, competitive sealed bidding above thresholds, and explicit authorization requirements for alternative delivery. Out of scope here; Part 2 Module 5 covers the mechanics.

4.3 The Decision Summary

If	Then
Lender requires GMP before closing	CMAR or DB
Schedule compression has real value	CMAR, DB, or progressive DB
Scope is uncertain or concealed conditions likely	CMAR or progressive DB
Program is well-defined and repeatable	DB
Design control is paramount	DBB or CMAR
Owner has no in-house project management	DB, or CMAR with an owner's rep
Public funds or statutory procurement	As authorized by statute

Section 5 - Long-Lead Items

5.1 Why This Belongs in Part 1

Conventional practice identifies long-lead items during buyout, after the contractor is engaged and the documents are substantially complete. For a meaningful set of equipment, that is too late.

Electrical service equipment is the clearest case. Utility-side transformers, switchgear, and distribution equipment have carried lead times running from many months to well beyond a year, with the utility's own design and construction queue on top. A project that identifies its transformer requirement at construction documents may find the equipment arriving after the building is otherwise finished.

The identification belongs at acquisition, in the utility research from Module 4A, because that is when the load can first be estimated and when the utility's process can first be started. **The order does not need complete drawings; it needs a load letter and a deposit.**

5.2 The Recurring Long Leads

- Utility transformers, switchgear, and distribution equipment
- Generators and paralleling gear
- Large chillers, air handlers, and custom mechanical equipment
- Elevators, particularly where the hoistway is existing
- Structural steel, where mill order timing governs
- Specialty glazing and custom curtain wall
- Dock equipment, overhead doors, and large rolling equipment
- Utility company work itself: service extensions, road cuts, and the utility's construction queue
- Anything requiring a long permit - a Corps permit, a DOT access permit, a floodplain map amendment

5.3 Procurement Strategies

Early release packages. A partial permit and early contract for foundations, site work, or structure while the rest of the design continues. Requires the AHJ to issue a partial permit and the lender to fund against an incomplete design.

Owner-direct purchase. The owner buys the equipment and assigns the purchase order to the contractor. Locks price and place in queue before a contractor is engaged. The owner takes on storage, insurance, warranty, and delivery risk, and the contractor will not warrant equipment it did not buy.

Preconstruction procurement. A contractor engaged early under a preconstruction agreement places orders before the GMP is final, with the cost carried into the GMP.

Design for what is available. Where a piece of equipment cannot arrive in time, design around it. Temporary power, a different system, a phased energization. This requires knowing the lead time during design rather than during submittals.

The handoff obligation is simple: **the long-lead list, with the load letter status and the utility's stated timeline, goes into the package.** It is one page and it prevents a category of schedule failure that cannot be recovered by working faster.

Section 6 - The Risk Register

6.1 Unresolved at Close

The clean version of development says that all diligence findings resolve before Gate 2. The real version is that some do not, and the deal proceeds with them open because the alternative was not proceeding.

Items that routinely close with a project unresolved:

- A permit still in process - a Corps permit, a DOT access permit, a floodplain amendment
- A condition of approval whose interpretation is ambiguous
- A utility capacity commitment that is conditional or expires
- A tenant who has not been relocated, or a holdover
- A third-party consent obtained in principle but not recorded
- An environmental control with an ongoing operating obligation
- Concealed conditions in an existing building, carried as contingency rather than scope
- An off-site improvement whose cost allocation is still being negotiated
- An incentive award that is applied for and not yet granted

Each one is a live risk the design and construction team is going to encounter, usually without being told it was known.

6.2 The Register

A risk register is a table: the risk, its owner, its probability and impact, the mitigation, the trigger date, and its current status. It is maintained through acquisition, and at handoff it is transmitted rather than archived.

This is the single most useful document a developer can give a design team, and it is almost never given, because it reads like an admission. It is the opposite - it is the difference between a design team that is solving a known problem and a design team that is discovering one.

The entries that matter most to the design team:

- **What is assumed and not verified.** "Detention sized on a 10 percent site area assumption pending final hydrology."
- **What is conditional.** "Will-serve letter expires in 14 months and is conditioned on funding the main upsize."
- **What is contested.** "The condition requiring masonry on 'street-facing facades' has not been interpreted; staff's informal view is that it includes the north elevation."
- **What is deferred.** "Vapor conditions below commercial screening levels; no mitigation required for the current use. A future change to residential occupancy would require mitigation."

That last kind of entry is the one that pays off years later, when someone proposes a use change and nobody remembers why it is a problem.

Section 7 - Gate 4

The last gate in Part 1, and the point at which optionality ends.

What happens, typically within days of each other:

- The purchase closes and the deed records
- The construction loan closes; the note, mortgage, and loan agreement are executed
- Equity is funded
- Guaranties are signed by individuals: completion, carve-out, carry, environmental
- Collateral assignments of the design and construction contracts are executed, with consents from the architect and contractor
- The construction contract is executed with a GMP or lump sum
- Insurance binds
- Notice to proceed is issued

After Gate 4, the option to walk away is gone. The land is owned, the loan is drawn against, the guaranties are personal, and the completion deadline is contractual. Every decision from this point is about execution rather than selection.

This is where Part 2 begins.

Section 8 - The Reconciled Glossary

Development and construction share a vocabulary and disagree about what a significant number of words mean. This is the reconciliation promised in Module 1.

8.1 The Collisions

Term	In development (Part 1)	In construction (Part 2)
Program	The revenue configuration: unit mix, rentable area, parking count	The Owner's Project Requirements: spaces, adjacencies, performance criteria
Feasibility study	Financial analysis of whether the deal produces an acceptable return	Pre-design investigation of physical, code, or site viability

Due diligence	A contractually defined period with a hard expiration, after which money is at risk	Generic care and investigation; no fixed date
Owner	A single-purpose entity answerable to LPs, a lender, and guarantors	A single party with decision authority in the contract documents
Project manager	The developer's employee running the deal	The GC's office-side manager, or the architect's PM, or the owner's rep
Closing	The purchase and loan transaction; occurs before construction	Closeout; occurs after construction
Contingency	An underwriting reserve, lender-controlled, flowing down the equity waterfall	A line within the construction estimate; design, construction, and contractors are distinct
Development	The entire enterprise of creating the asset	Sometimes the site work only; "design development" is a design phase
Approval	An entitlement granted by a legislative or quasi-judicial body	A permit issuance or a submittal review action
Inspection	The lender's inspecting consultant verifying work in place for a draw	The AHJ's field inspection, or a special inspection under IBC Chapter 17
Complete	Satisfying the completion guaranty; lien-free and per the approved plans	Substantial Completion, then Final Completion, as defined in the contract
Draw	The monthly funding request to the lender	The Application for Payment that feeds it

Retainage	A loan requirement protecting the lender's completion cushion	A contract provision withheld from the contractor
Change order	A budget event requiring lender and sometimes LP approval	A contract modification executed on a standard form
Schedule	Entitlement, design, construction, lease-up, and exit	The construction schedule
Hard costs / soft costs	The two halves of the development budget	Not standard usage; estimates use UniFormat or MasterFormat
Site plan	The entitlement exhibit presented at a public hearing	The civil and architectural site drawings in the construction set
Stabilization	Sustained occupancy triggering permanent financing	No equivalent; the project ended at closeout
Value engineering	Cost reduction to restore a return	A structured function-analysis process, though used loosely for the same thing
Consultant	A diligence vendor: geotech, environmental, market, PCA	A design sub-consultant under the architect's agreement
Basis	Land basis: what the land cost	Basis of Design: the documented technical approach

8.2 Terms That Cross Cleanly

Worth noting, because they are the vocabulary the two halves genuinely share: **certificate of occupancy, permit, lien, lien waiver, schedule of values, easement, setback, occupancy classification, alteration level, change of occupancy, special inspection, substantial completion, allowance, UniFormat, MasterFormat.**

8.3 The One to Watch

Program, for the reason given in Module 1 and worth repeating at the seam.

When a developer says the program is fixed, they mean the revenue model is fixed and cannot move without re-underwriting the deal and, frequently, without the lender's and the LPs' consent.

When an architect hears the program is fixed, they hear that the space list is settled and design may proceed.

Those are compatible statements until someone proposes a change that alters rentable area, unit count, or parking - at which point one party is discussing a design refinement and the other is discussing a breach of their loan covenants. Establish at the kickoff which quantities are locked, which are flexible, and what the approval path is for changing one.

Artifacts

Artifact	Prepared by	Purpose
Owner's Project Requirements	Owner or owner's rep, with the design team	Performance expectations, as distinct from the revenue program
Site constraints package	Developer	Transmission of all diligence findings with reliance
Conditions-of-approval matrix	Developer with design team	Every binding commitment, mapped to where it is addressed
Development budget, with structure	Developer	The number and its constraints
Master project schedule	Developer with contractor	Milestones and the consequence of each
Long-lead procurement list	Developer with contractor	Items whose lead time exceeds the design phase
Risk register	Developer	What is unresolved, who owns it, and when it triggers

Delivery method decision memo	Developer or owner's rep	The method and its rationale
Design agreement	Owner and architect	Scope, phases, fee, and CA obligations
Preconstruction or construction agreement	Owner and contractor	Preconstruction scope, then GMP or lump sum
Collateral assignments and consents	Lender's counsel	The lender's step-in rights
Kickoff decision-authority memo	Owner's rep or developer	Who decides what, at what threshold, in how long

Failure Modes

Handing over a revenue program as though it were an OPR. The design team infers performance requirements that were never stated, and the inference is discovered during construction.

Transmitting reports without reliance letters. The design team has no recourse against the consultant whose report they are designing from.

Transmitting recommendations without data. Boring logs, not just the foundation recommendation.

Conditions of approval left as a PDF. They are binding obligations, planning enforces them at CO, and the building official does not know they exist.

Letting entitlement drawings become the design. Produced fast and cheap, then approved, then binding. Build them knowing they will constrain the real work.

Underwriting a yield the entitlement drawings cannot deliver. The site plan showed 34 because nobody had the geotech yet.

Identifying long leads at buyout. For utility equipment, buyout is too late by a year.

Archiving the risk register instead of transmitting it. The design and construction team will encounter every open item; the only question is whether they are told first.

Leaving decision authority undefined. Three people called "project manager" and no statement of who can approve what.

Choosing a delivery method after signing the design agreement. The decision has already been made by default.

Assuming the design team has read the loan agreement. They have not seen it. They have signed a consent to its collateral assignment.

What Happened Next

Kipling Row

The handoff happened backwards, which is normal for a deal with entitlement risk and is the reason Section 2 exists.

The architect had been engaged eight months earlier - before diligence was complete, before the price was renegotiated, and before there was any certainty the project would exist - to produce the PUD submittal. Site plan, building elevations for six buildings, a landscape concept, a grading and utility concept from the civil engineer, and two renderings for the hearing. The scope was defined and paid at a reduced rate, with the full commission contingent on the project proceeding. Total entitlement design spend: approximately **\$118,000**, funded entirely by equity, on a project with a 60 to 65 percent approval probability.

The council approved the PUD 5-2, after one continuance, with the protest petition never filed because two of the six adjacent owners were persuaded by the buffer and fence commitments offered in the staff report. The approval carried **twenty-two conditions**.

The conditions the design team inherited, in the categories that matter:

- Masonry on all elevations facing public rights-of-way, with a minimum percentage defined by the approved elevations
- A six-foot masonry wall along the east property line, with a specified landscape buffer and minimum caliper trees
- The access point located substantially as shown on the approved plan - the location that moved 180 feet east during Module 4A
- A pedestrian connection to the arterial sidewalk
- Construction hours restricted, and a haul route specified
- Detention designed to a standard above the base requirement, offered voluntarily at the hearing
- Approved elevations attached as an exhibit, making material and fenestration changes an amendment matter

That last one has consequences the developer did not fully register at the hearing. The elevations were produced in three weeks, nine months before anyone priced them. The masonry percentage shown on those drawings is now a regulatory requirement rather than a design decision, and value engineering the exterior is no longer available as a response to a cost overrun. On a deal with a 9.2 percent margin, removing that lever matters.

The handoff package was assembled with more discipline than the developer's prior projects, specifically because the diligence findings had cost them \$958,000 and the lesson had landed. It contained the full survey and topo, the geotechnical report **with boring logs**, the Phase I and the limited Phase II with the tank closure documentation, the state no-further-action letter, the recorded access easement negotiated in Module 4A, the title commitment with all exception documents including the 1961 deed restriction, the utility correspondence and the water rights cash-in-lieu calculation, the PUD approval, and a **conditions matrix** listing all twenty-two conditions against a responsible party and a drawing reference.

Reliance letters were requested on the geotechnical and environmental reports, and the geotechnical consultant charged \$1,200 for it.

Delivery method. The bank's term sheet required a fixed price before closing and would not fund against an incomplete design. Design-Bid-Build was therefore unavailable without carrying the full design period on equity, which on a deal with this margin was not viable. The developer selected **CM at Risk** with a local builder who had provided the concept pricing in Module 5, under a preconstruction agreement executed at approval, converting to a GMP at roughly 70 percent construction documents.

The GMP will carry qualifications and allowances. Reading them is Part 2 Module 5.

Long leads. The transformer and service equipment for 34 units, and the utility's own construction queue. The load letter was submitted the week after the council vote rather than at construction documents, on the preconstruction contractor's recommendation. That single sequencing decision is worth more to this project than anything else in the handoff package.

The risk register carried five open items into design: the detention sizing pending final hydrology, the masonry percentage interpretation, an off-site sidewalk connection whose cost allocation was still under discussion with public works, the haul route condition's effect on the earthwork price, and a note that the structural floor system assumption from the geotechnical report had been priced at concept level only.

Gate 4 is scheduled for 45 days after the approval becomes final, when the option is exercised, the land closes, and the construction loan funds simultaneously. The developer's two principals will sign a full repayment guaranty that day.

Six Corners

The opposite pattern: the deal closed first, and the design team was engaged into a project whose constraints were already fully formed.

Gate 4 occurred in a single morning. The purchase closed at the renegotiated price with the roof credit applied. The bridge loan closed with its future funding facility. The fund's two principals signed a completion guaranty, a carry guaranty, carve-outs, and an unlimited environmental indemnity. The architect signed the collateral assignment.

The handoff package was good on the technical material and initially silent on the commitments, which is the characteristic pattern.

Transmitted without prompting: the PCA, the hazardous materials survey and the PCB sealant results, the structural evaluation, the preliminary code analysis, the Phase I and the soil gas data, the ALTA survey, and the MEP capacity assessment. All with reliance letters, because the debt fund had required them.

Not transmitted until the architect asked, three weeks into design:

- **The recorded REA amendment**, including the parking covenant guaranteeing 290 shared stalls in perpetuity - a binding constraint on the site plan that the developer thought of as a closed transaction item rather than a design input. The restriping plan satisfying it existed only as a Module 5 sketch.
- **The shared drive repaving obligation**, \$95,000 of scope owed to a neighbor under the consent agreement, which was in the budget and not in any drawing.
- **The mineral estate surface use agreement**, restricting where certain surface work could occur.
- **The remaining tenant's lease**, with 26 months of term, a below-market renewal option, no landlord relocation right, and landlord obligations for services and access that the renovation schedule had to honor.

None of this was concealed. All of it lived in the transaction file, in the heads of the sponsor's principal and their counsel, and none of it had been converted into a design constraint.

The code path was the part that transmitted well, because the preliminary code analysis had been the decisive document in the deal. The design team inherited a defined strategy: ground floor change of occupancy to S-1 and F-1 with four fire areas separated by two-hour barriers doing double duty as tenant demising; second floor remaining in its existing B classification, avoiding a change of occupancy entirely and limiting that floor to Level 2 alterations; mixed-occupancy separation between floors; and a structural scope of collectors and a new braced line to replace the lateral capacity removed by the dock openings.

The design team's job was to execute a strategy, not invent one. That is what a preliminary code analysis buys, and it is the argument for spending 2 percent of the diligence budget on one.

Delivery method. The scope was renovation with concealed conditions in a 1984 building, which is the profile that prices badly under a lump sum. The sponsor selected **CM at Risk**, with an owner-controlled contingency at 10 percent rather than a contractor-priced allowance for unknowns, and a preconstruction agreement engaging the contractor before the code strategy was finalized so the fire area layout could be priced against the demising walls it was doing double duty as.

Long leads. The 52-week transformer from Module 4A was not required for the flex scheme - the existing 800-amp service was adequate, which is one of the reasons the pivot worked. The elevator replacement, the dock equipment and overhead doors, and the roofing material went on the long-lead list at acquisition.

The risk register carried the item that matters most for the next decade of this building's life:

*Soil gas beneath the slab exceeds residential vapor intrusion screening levels and is below commercial and industrial screening levels. No mitigation is required for the approved S-1, F-1, and B occupancies. **Any future change of occupancy to residential will require sub-slab depressurization, monitoring, and disclosure.** Source is an off-site dry cleaner release; the state file remains open.*

That paragraph is the single most valuable sentence in the handoff. In eight years, when someone proposes apartments on this site because office and flex have both moved on, that entry is the difference between a diligence finding and a discovery.

Gate 4 cleared. Notice to proceed issued 60 days later.

Part 1 Ends Here

Eight modules and two deals ago, the premise was that every project arrives at the design team pre-loaded with constraints that nobody explains.

They have now been explained.

The site was selected because a trade area had demand the existing supply did not serve. The use was determined by a four-part test: legally permissible, physically possible, financially feasible, maximally productive - applied in that order, eliminating uses as it went. The unit count is what survived three rounds of test fit against setbacks, fire access, detention, and parking geometry. The budget is an exhibit to a loan agreement. The schedule has a date in it that is a loan default trigger. The delivery method was chosen by a lender who required a price before closing. The massing was approved in a public hearing by a body that attached twenty-two conditions, and the elevations are an exhibit to that approval. And the "Owner" is an LLC formed six weeks ago, holding one asset, whose principals have personally guaranteed its completion.

The five gates:

G	Question	Resolved in
0	Is this worth looking at?	Modules 1, 2
1	Do we tie it up?	Module 7
2	Do we go hard?	Modules 3, 4A, 4B, 5
3	Can we build what we underwrote?	Module 3
4	Do we close and start?	Modules 6, 8

Part 2 begins at Gate 4, with the design team engaged, the program fixed, the budget locked, and the first drawing not yet made.

Optional: Try This

If you work on the design or construction side, take a project you are on now and ask four questions you have probably never asked:

What are the conditions of approval, and where are they addressed in the documents? Ask the owner for the approval letter. Count the conditions. Then find them in the drawings.

What does the loan agreement require? Not the whole document - three things: the change order approval threshold, the rules for reallocating budget line items, and the outside completion date.

What does the geotechnical report's boring log show, as opposed to its recommendation? Look at the variability across the site.

What was on the risk register at acquisition? If the answer is that there wasn't one, you have learned something about how the project was underwritten, and about which surprises are still ahead.

If you work on the development side, run the reverse: assemble the handoff package from Section 1 for a project you have done, and count what you would have had to go find.

Part 2 - The Design and Construction Process begins with the project team and the delivery methods, from the other side of the table.

Part 2 — The Handoff: From Pro Forma to Project

Why This Module Exists

Part 1 ended with the handoff. The developer tied up the site, survived diligence, secured the entitlements, chose a delivery method, and handed the design team a program, a budget, a schedule, and a stack of reports. From the developer's side of the table, the deal is done.

From the other side of the table, nothing has happened yet.

Every number in the pro forma is still a promise. The hard cost line is a target that no design has been tested against. The schedule is an assumption about how long things take, not a plan anyone has built. The unit count is what a test fit suggested, not what a coordinated floor plan has proven. The job of everyone in Part 2 is to turn those promises into a building without breaking them.

Part 1 Module 8 covered the handoff from the owner's seat: what to hand over, whether to hire an owner's representative, and which delivery method to choose. This module sits in the other seats. It covers what the architect, the engineers, and the contractor do with the handoff in the first ninety days, who actually does the work inside each firm, how each delivery method runs day to day once it's been chosen, and how the team writes down what "done" and "on budget" mean before anyone argues about them.

The Reference Project

Part 2 follows a project: a 200-unit garden community in a Sun Belt secondary market. If you carried your own parcel from Part 1 Module 2, use it instead. The method is the same.

Product	200 units, three-story garden walk-up, wood frame, surface parking
Land	\$3,000,000
Hard costs	24,500,000(122,500 per unit)
Soft costs	\$5,000,000

Owner contingency	\$1,250,000
Financing costs	\$1,250,000
Total development cost	\$35,000,000
Stabilized NOI	\$2,280,000
Yield on cost / market cap rate	6.51% / 5.25%, 126 bps of spread
Construction loan	\$21,000,000 at 60% loan-to-cost
Construction / lease-up / sale	24 months / 12 months / month 38
Delivery method	Construction Manager at Risk, GMP

The three numbers the team has to carry

A pro forma has dozens of lines. The design and construction team needs to internalize three of them, because nearly every decision in Part 2 either protects them or spends them.

1. How much cost the deal can absorb: \$2.73 million. The owner carries \$1,250,000 of contingency. Behind that sits another \$1,480,000 of headroom before the development spread falls to the 100 bps floor below which the investors would not have approved the deal. That's the whole cushion, and it has to cover design growth, bid results, change orders, and delay. On a \$24.5 million hard cost, it's about 11%. That sounds like room until you remember that a Schematic Design estimate can plausibly miss high by 15%.

2. What a month costs: roughly \$118,000 or more. At closeout, a one-month slip in the Certificate of Occupancy costs about \$117,660 in interest, operating costs that are already running, and deferred leasing. Two months cost about \$300,000. Earlier in the job, the math is different but never small: the construction loan's interest reserve has a cushion of \$1,133, so any delay of about three months forces the equity to put in more cash.

3. What design control is worth: \$5.88 per unit per month. Design-build won on expected cost by \$268,590. It lost only if its cost-driven design choices reduced rents or raised operating costs by more than \$5.88 per unit per month. That number converts "design quality" into money: $\$5.88 \times 200 \text{ units} \times 12 \text{ months}$ is \$14,112 of NOI a year, which at a 5.25% cap rate is about \$268,800 of value at sale. It's the reason the owner chose CMAR. It's also the standard every design decision in this project should be held to: does this choice cost more than it protects in rent and operating cost?

Concepts

1. Receiving the Handoff: Targets Become Requirements

What arrives

Part 1 Module 8 lists what the owner hands over. From the receiving side, each item comes with a question the team has to answer before designing anything.

What the owner hands over	The team's question
Program (unit count, mix, sizes, amenities, parking)	Which of these are fixed, and which are negotiable if the budget requires it?
Budget (\$24.5M hard cost)	Is this a target the design must be engineered to meet, or an estimate of a design that exists? (It's always a target.)
Schedule (month 0 start, 24-month build, CO at month 25)	Which dates are driven by financing, leasing season, or investor commitments, and which are soft?
Site constraints package (survey, geotech, environmental, utilities, traffic, stormwater)	Which findings become design scope, and who owns each?
Entitlement commitments (conditions of approval, development agreement)	Which conditions are already binding on the design?
Business plan (merchant build, sell at stabilization)	Where do exit value and long-term operating cost conflict, and which wins?
Risk register (carried from diligence)	Which risks are resolved, which are open, and who owns each one now?

Diligence findings become design scope

The site constraints package is where most handoff failures start, because diligence findings are written for underwriters, not designers. The team's first job is to translate each finding into scope, assign it to a discipline, and check that the budget carries it.

Diligence finding	Becomes design scope for	What the team has to do	Budget allowance
Expansive clay at 4–8 ft	Geotechnical and structural engineers	Design foundations and slabs for over-excavation and replacement; define the earthwork spec	\$920,000
Underground detention vault in lieu of surface pond	Civil engineer	Design the vault; coordinate with site layout and utilities	\$610,000
1,400 LF off-site sanitary extension	Civil engineer, utility	Design the extension; confirm who approves and who inspects	\$385,000
Unrecorded 20-ft drainage easement	Architect, civil	Shift building footprints; re-check parking count and setbacks	\$85,000
Right-turn deceleration lane	Civil, traffic engineer	Design to the road authority's standards; separate permit	\$165,000
14-month transformer lead time	Electrical engineer, utility	Start the utility service application early (see Exercise, Scenario 2)	\$135,000 (the delay cost, if not solved)

Two things about this table matter. First, **every row has a discipline that now owns it**. If a finding doesn't have an owner, nobody designs it, and it reappears as a change order. Second, **the allowances are the underwriter's numbers, not the engineer's**. The civil engineer's first real vault design may cost more or less than \$610,000. The team has to test each allowance against the design as it develops, which is exactly what Module 2.3 (Cost Through Design) is for.

Targets, estimates, and budgets

Three words get used interchangeably in handoff meetings, and they shouldn't be.

- **A target** is what the owner needs the building to cost. \$24.5 million is a target.
- **An estimate** is what a specific design is expected to cost, with a range around it. There is no estimate yet, because there is no design.
- **A budget** is the target plus contingency, allocated to line items and controlled by named people.

The most common handoff failure is treating the target as an estimate. The pro forma says \$24.5 million, so everyone behaves as if the building costs \$24.5 million, and nobody is assigned to prove it. The first real estimate arrives at Design Development, comes in 8% high, and consumes most of the \$2.73 million cushion before construction starts. Nobody made a bad decision. The target just never became a requirement that the design was tested against.

2. The Project Team, From the Inside

Part 1 Module 1 introduced the development ecosystem: developer, equity, lender, brokers, consultants. This section opens up the design and construction firms, because the reader who has only seen them from the outside usually doesn't know who actually does the work, or what each person is paid to care about.

The owner side

Role	What they do	What they're paid to care about
Development manager	Runs the project for the developer. Often manages three or four projects at once.	Budget, schedule, investor reporting. Their promote depends on the return.
Owner's representative	Manages design and construction on the owner's behalf. Reviews estimates, the GMP, pay applications, and changes.	Being seen to protect the owner. Paid a fee, typically 1–3% of construction cost.
Future property manager	Will operate the building. Often not engaged until construction is almost done.	Leasing velocity, operating cost, tenant complaints.

The property manager is the most under-used person on most teams. They know which trash room layout causes complaints, which package room size is too small by year two, and which

appliance brands generate service calls. Engaging them during design costs almost nothing. Changing the same decisions after construction is expensive.

The design team

The architect's firm isn't one person. On a project this size, the team typically looks like this:

Role	What they do
Principal in charge	Owns the client relationship and the contract. Present at major milestones.
Project manager	Runs the fee, the schedule, and the consultants. Your day-to-day contact.
Project architect	Leads the technical design and the document set. Often the person who knows the building best.
Designers and drafters	Produce the drawings and model.
Specification writer	Writes or edits the project manual. In-house or a consultant.
CA architect	Handles submittals, RFIs, and site visits during construction. Sometimes a different person than the project architect.

The architect of record holds contracts with the consulting engineers. On a garden multifamily project, that typically means:

- **Civil engineer:** grading, drainage, stormwater (the vault), site utilities (the sewer extension), paving, and often the site plan approval
- **Structural engineer:** foundations (the clay), wood framing, and the lateral system
- **MEP engineer:** mechanical, electrical, and plumbing, usually one firm on multifamily
- **Landscape architect:** planting, amenity areas, and anything the entitlement conditions require
- **Specialty consultants as needed:** geotechnical (retained by the owner, continuing into design), energy modeling, acoustics, accessibility review, interior design

How the design team is paid: architectural fees on garden multifamily commonly run in the range of 3–6% of construction cost with the full consultant team included, lower when the developer repeats a prototype. The fee is allocated across the design phases, and the architect's profit depends on not redesigning. That gives the design team a strong interest in the owner making decisions early and sticking to them. It also means the design team rarely volunteers extra estimating or extra options, because both cost them hours they aren't paid for.

The tension to watch: each consultant optimizes their own system. The structural engineer adds capacity for safety. The MEP engineer sizes equipment conservatively. The architect wants the clubhouse to photograph well. None of them sees the total cost, and none of them is paid to. Design drifts upward unless someone is estimating the whole building at each phase.

The construction team

Under CMAR, the construction manager's team has two halves, and they're often different people.

Role	Phase	What they do
Preconstruction manager	Design	Leads estimating, constructability review, and early subcontractor outreach.
Estimator(s)	Design	Produce the estimates at each design phase. Do the quantity takeoff.
Project executive	Both	Owns the relationship and the contract. Signs the Guaranteed Maximum Price (GMP).

Project manager	Late design and construction	Runs the contract: buyout, cost, changes, pay applications, schedule.
Superintendent	Construction	Runs the site: trades, sequencing, safety, quality, daily coordination.
Project engineer(s)	Construction	Process submittals and RFIs, track logs, maintain documents.
Assistant superintendent (s)	Construction	Manage specific buildings, trades, or phases on site.

The handoff between the preconstruction team and the operations team is one of the least discussed and most expensive transitions in the project. The estimators know every assumption in the GMP. The superintendent who has to build it often hasn't seen those assumptions. A good CM brings the superintendent into preconstruction by DD. A cheap one doesn't.

How the construction manager is paid: a fee, often 3–5% of the cost of the work on multifamily, plus general conditions (the CM's site staff, trailers, temporary utilities, equipment), which often add another 6–10%. Under a GMP, the CM's real margin depends on how well it buys out subcontracts and how it prices changes.

Trade subcontractors

On a garden community, 80–90% of the hard cost flows through subcontractors: sitework, concrete, framing, roofing, windows, siding, plumbing, HVAC, electrical, fire sprinkler, fire alarm, drywall, painting, cabinets, flooring, and appliances. Their availability, their pricing, and their willingness to bid decide whether the \$24.5 million target holds.

Some trades are brought in during design under **design-assist** agreements (Section 3.7). On wood-frame multifamily, the usual candidates are framing, plumbing, HVAC, electrical, and fire sprinkler, because their layouts drive cost and the engineers' drawings rarely reflect the most efficient way to build them.

Third parties the team answers to

Party	Hired by	Why the team cares
Authority Having Jurisdiction (building department, fire marshal, public works)	Public	Controls permit issuance and the Certificate of Occupancy, two of the most schedule-critical dates in the project. Has no stake in the budget.
Road authority and utilities	Public or regulated	Control the deceleration lane, the sewer extension, and the transformer. Separate permits, separate timelines.
Special inspector and testing agency	Owner	Perform the code-required inspections of structural and fire-protection work. Their reports gate the CO.
Lender's construction monitor	Lender (paid by the owner)	Reviews the budget, the GMP, and every pay application. Can hold a draw.
Commissioning agent	Owner	Verifies that building systems work as designed.

3. How Each Delivery Method Actually Runs

Part 1 Module 8 chose the delivery method. This section is about what each method looks like from inside the team once it's chosen: who is under contract with whom, when each party shows up, who holds which risk, and when the owner finally knows the price.

Two decisions, not one

"Delivery method" gets used loosely for two separate decisions:

1. **Delivery method:** how design and construction are organized, meaning who contracts with whom and when the builder gets involved.
2. **Compensation method:** how the builder is paid.
 - o **Lump sum (stipulated sum):** a fixed price for a defined scope. The contractor keeps savings and absorbs overruns within that scope.

- **Cost of the work plus a fee, with a Guaranteed Maximum Price (GMP):** the owner pays actual cost plus a fee, up to a ceiling. Unused contingency and buyout savings are usually shared.
- **Cost plus a fee, no GMP:** no ceiling. Rare outside emergencies and genuinely unknowable scopes.
- **Unit prices:** payment per unit of work. Common for earthwork and utilities, where quantities are uncertain. The clay over-excavation is a natural unit-price item.

The two decisions combine, and the combination is the risk allocation.

The rule underneath all of them: who warrants the drawings

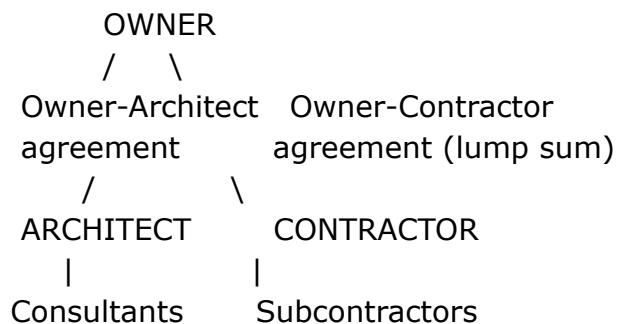
In the United States, when an owner gives a contractor plans and specifications to build from, the owner impliedly warrants that they're adequate. This comes from a 1918 Supreme Court case, *United States v. Spearin*. The contractor built what it was given. The owner provided what it built from.

The owner can try to recover from the architect, but only if the error fell below the architect's professional standard of care, and only up to what the architect's insurance and deductible practically allow.

Every delivery method is, underneath, a decision about who warrants the drawings. In Design-Bid-Build and CMAR, the owner does. In Design-Build, the design-builder does, because it produces the drawings it builds from. That single difference explains most of the behavioral differences between the methods.

Design-Bid-Build (DBB)

Contract map:



How it runs: The architect designs to 100% Construction Documents without any builder involved. The owner puts the documents out to bid, receives lump sums, and awards the contract. The contractor builds what the documents show. The architect administers the contract as the owner's agent. The architect and contractor have no contract with each other.

When the owner knows the price: at bid, after nearly all the design fee is spent.

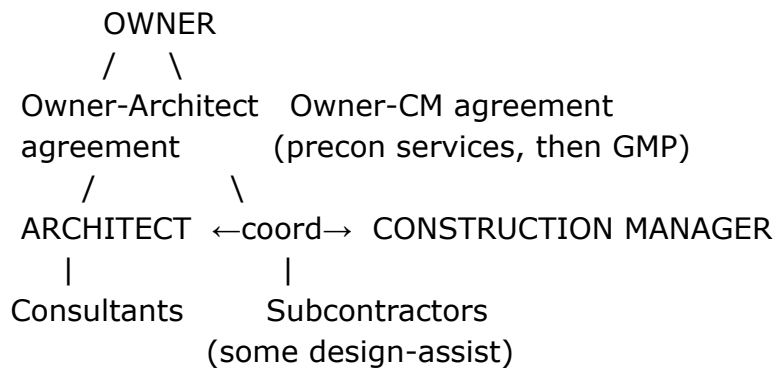
Who holds what: the owner warrants the drawings. The contractor holds cost risk for the scope shown. Anything not shown is a change.

What it looks like from the team's seat: The design team works without cost feedback unless the owner hires a separate estimator. Contractors see the documents for the first time during a three- to five-week bid window and price exactly what's drawn, sometimes with an eye on where the documents are thin, because thin documents produce profitable change orders. The relationship starts adversarial by structure, not by personality.

On the reference project: DBB would put the first reliable price at about month –1. If bids come in 8% high, the owner has about \$2 million of overrun, a \$2.73 million cushion, and no time to redesign without missing the financing close.

Construction Manager at Risk (CMAR)

Contract map:



How it runs: The owner hires the architect and, separately, a CM early in design. During design, the CM provides preconstruction services: estimates at each phase, constructability review, a baseline schedule, subcontractor market intelligence, and early procurement of long-lead items. At an agreed point, the CM commits to a GMP and becomes the builder. Early packages, such as sitework and foundations, can be released before the full GMP.

When the owner knows the price: progressively. An estimate at SD, a better one at DD, a better one at 50% CD, and the GMP near the end.

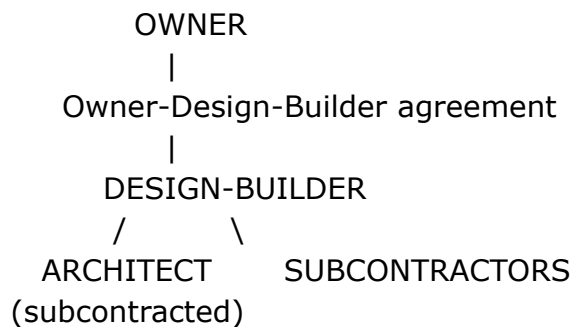
Who holds what: the owner still warrants the drawings. Before the GMP, the owner holds cost risk but can see it coming. After the GMP, the CM holds overruns within the defined scope.

What it looks like from the team's seat: The architect and CM have no contract with each other but work together for months. That relationship is the method's biggest strength and its most common failure point. When it works, the CM's estimates shape the design while changes are cheap. When it doesn't, the architect treats the CM's estimates as criticism, the CM treats the architect's documents as incomplete, and the owner referees.

The catch: a GMP set on incomplete documents is a price plus a list of assumptions. The GMP will contain allowances (placeholders adjusted to actual cost, which leaves that risk with the owner), clarifications (the CM's statement of what it assumed the documents mean), exclusions, and a construction contingency. Module 2.5 covers how to read them. Every assumption the owner doesn't catch becomes a future change.

Design-Build (DB)

Contract map:



How it runs: The owner defines its requirements in a criteria document, sometimes prepared by a separate "bridging" architect. It contracts with one entity that designs and builds, usually a contractor that subcontracts the architect. Price is often set at the proposal stage.

When the owner knows the price: early, often before design is developed.

Who holds what: the design-builder warrants its own drawings. The owner's influence runs through the criteria document. Anything the criteria don't specify, the design-builder decides.

What it looks like from the team's seat: The architect's client is the contractor. There is no independent design professional watching construction for the owner unless the owner pays for one separately. The design is optimized for the design-builder's cost and schedule.

When it fits: repeatable building types with clear specifications (industrial, self-storage, some hospitality, prototype multifamily from a developer with its own standards), and owners who value single-point responsibility over design control.

Progressive Design-Build (PDB)

How it runs: The owner selects a design-builder mainly on qualifications, then develops the design with it collaboratively. At an agreed milestone, often 60–90% design, the design-builder proposes a price. If the owner doesn't accept, it can take the design to another builder. This is called the "off-ramp."

Who holds what: collaborative like CMAR during design, single-point responsibility like DB after pricing.

The catch: the off-ramp is valuable mostly as leverage. Actually using it means switching teams mid-project, which costs months.

When it fits: complex projects where the owner wants single-point responsibility but can't define the building upfront. It's increasingly common in public infrastructure and institutional work.

Integrated Project Delivery (IPD)

How it runs: The owner, architect, contractor, and key trades sign one multi-party agreement (for example, AIA C191 or ConsensusDocs 300). They share a profit pool tied to a target cost, waive most claims against each other, and make decisions jointly.

Who holds what: shared. Come in under target and everyone's profit grows. Go over, and everyone's profit shrinks, usually down to a floor that covers direct costs.

When it fits: large, complex projects with repeat relationships, such as a hospital system that builds with the same team for years. It's rarely practical on a single merchant multifamily project with conventional financing, but its ideas (early trade involvement, target value design, shared incentives) are being borrowed into CMAR and PDB.

Design-assist: what most projects actually do

Real projects rarely fit one box. The most common hybrid is **design-assist**: key subcontractors are brought in during design, usually under a CMAR, to help develop their systems. The engineer of record still designs and stamps the documents. The trade contributes installation knowledge, current pricing, and product preferences.

On the reference project, plumbing, HVAC, and electrical together represent roughly a quarter to a third of hard cost. Stacking bathrooms and kitchens vertically, locating the water heaters, and routing the electrical service are small layout decisions with large cost consequences. They're easy to make well in Schematic Design and expensive to change after DD. That is why design-assist trades should be selected by the end of SD, not at buyout.

Design-assist agreements need three things spelled out: what the trade will deliver during design, how it's paid for that work, and whether it's guaranteed the construction subcontract (usually, subject to a fair price, with the CM's right to bid it out if not).

Comparing the methods, from the team's seat

Who's involved, and who holds the risk

Method	Builder joins	Who warrants the drawings	Architect works for
DBB	At bid	Owner	Owner
CMAR	Early design	Owner	Owner
DB	At proposal	Design-builder	Design-builder
PDB	Early design	Design-builder	Design-builder
IPD	Day one	Shared	All parties

What the owner gets, and what it costs the owner to manage

Method	Price known	Independent design oversight in construction	Owner management effort
DBB	At bid	Yes	Medium
CMAR	Estimates each phase; GMP at 60–100% CD	Yes	High
DB	At proposal	Only if owner pays for it	Low to medium
PDB	At 60–90% design	Only if owner pays for it	High
IPD	Target set early	Shared	Very high

Private owners, like the reference developer, can pick any method and negotiate any terms, subject mainly to what the lender will finance. Public owners often can't.

- **Procurement law:** many states require competitive low-bid DBB for public construction unless specific legislation authorizes CMAR, DB, or PDB, and those authorizations come with required selection procedures.
- **Design services:** the federal Brooks Act and most state equivalents require qualifications-based selection for architects and engineers. Fees are negotiated after selection, not bid.
- **Bonding:** public projects require payment and performance bonds by statute, because subcontractors can't place liens on public property.
- **Prevailing wage:** federal projects under the Davis-Bacon Act, and many state and local projects, require prevailing wages, which can raise labor cost significantly depending on the region and trade.

4. The Contracts That Assemble the Team

Module 2.5 covers contract families in depth. At the handoff, what matters is that the contracts match the delivery method and the owner's needs. Four agreements assemble the team on the reference project.

Owner-architect agreement (the AIA B101 is the common form). Defines the architect's basic services by phase, the fee, and what counts as additional services. The fights later in the project almost always come from the additional services list. Common items that are *not* basic services unless the owner adds them: more than one round of design options, estimating, redesign to meet a budget the owner changes, extra site visits, energy modeling, detailed interior design, and extended construction administration if the schedule slips.

A detail that matters under CMAR: the architect's agreement should require the design team to respond to the CM's estimates and constructability comments at each phase. Without that language, the architect can reasonably treat the CM's input as optional.

Architect-consultant agreements (AIA C401 is the common form). Pass the architect's obligations down to each engineer for their discipline. The owner has no contract with the engineers. If the MEP engineer makes an error, the owner's claim runs against the architect, and the architect's runs against the engineer.

Owner-CM agreement (AIA A133 is the common form under CMAR). Usually two parts: a preconstruction services phase with defined deliverables and a fixed fee (commonly a modest amount relative to construction cost), and a construction phase that begins when the owner accepts the GMP through an amendment. The preconstruction deliverables should be specific: estimates at SD, DD, 50% CD, and GMP; constructability reviews at each phase; a baseline schedule by DD; a subcontractor market report; a long-lead procurement plan.

Design-assist subcontracts. Written by the CM, approved by the owner, and scoped as described in Section 3.

5. Writing Down What "Done" Means: OPR, Basis of Design, and the Charter

Three documents come out of the first weeks after the handoff. Together, they convert the pro forma's assumptions into requirements the team can be held to.

Owner's Project Requirements (OPR)

The OPR is a written statement of what the owner needs the building to do. The term comes from commissioning (Module 2.8 returns to it), but its value is broader: it's the translation layer between the pro forma and the design.

For the reference project, a useful OPR states:

- **Program:** 200 units in the specified mix; minimum and target unit sizes; clubhouse, pool, and fitness amenities; parking count; target net-to-gross efficiency

- **Cost:** \$24.5 million hard cost target, with a requirement that the design be estimated at each phase and reconciled to the target before the phase is approved
- **Schedule:** construction start at month 0; CO by month 25, timed to the leasing season
- **Positioning:** finish level equal to or better than the two newest comparable properties, at a lower finish cost per unit
- **Operations:** package handling, trash and recycling logistics, access control, unit entry hardware, and preferred equipment brands, all from the property manager
- **Performance:** energy target, acoustic separation between units, exterior durability expectations
- **Business plan:** merchant build with a planned sale at stabilization, so exit-value drivers win where they conflict with long-term operating savings.

Every OPR item should trace to a pro forma line or the business plan. If it can't, question whether it belongs. If a pro forma assumption doesn't appear in the OPR, add it.

Basis of Design (BOD)

The BOD is the design team's written response to the OPR: how the design intends to meet each requirement. "Units will be served by split-system heat pumps with air handlers in unit closets, because..." "Water heating will be in-unit electric tank heaters, because..." The BOD makes the design team's assumptions visible early, when the owner and CM can still push back at almost no cost. It's updated at each design phase.

The project charter

The OPR says what the building must do. The charter says how the project will be run. It's short, often five to ten pages, and on a CMAR project it's usually drafted by the owner's rep and agreed by the whole team at kickoff.

A charter contains:

- **Objectives:** the OPR reduced to a handful of measurable goals, including the three numbers above
- **Budget structure:** line items, and who controls each contingency (owner contingency outside the GMP, the CM's construction contingency inside it, and any design contingency the estimator carries)
- **Delivery method brief:** how the chosen method will actually run on this project (the second half of this module's deliverable)
- **Team roster and responsibility matrix:** who is responsible, who approves, who is consulted, and who is informed for each major activity (often called a RACI)
- **Decision authority matrix:** who can approve what, at what dollar threshold
- **Communication plan:** meeting cadence, reporting formats, where documents live, and who speaks to the AHJ, the utilities, the lender, and the investors
- **Milestone schedule:** design phase approvals, GMP, financing close, construction start, substantial completion, CO
- **Risk register:** carried forward from Part 1, with each open risk assigned to a person on the team

A decision authority matrix for the reference project might look like this:

Decision	Approves	Threshold
Design clarification, no cost or schedule impact	Owner's rep	Any
Change within contingency	Owner's rep	Up to \$25,000 each
Change within contingency	Development manager	25,000–100,000
Change within contingency	Developer principal	Above \$100,000
Anything affecting unit count, unit mix, rents, or the CO date	Developer principal	Any amount
Phase approvals (SD, DD, CD) and GMP acceptance	Developer principal, with lender notice	—

The charter isn't bureaucracy. It's how the owner avoids becoming the bottleneck on its own project, and how the team knows what "on budget" means before it's arguing about it.

6. The First Ninety Days

The period right after the handoff sets the rhythm for the rest of the project. On the reference project:

When	What happens
Weeks 1–2	Owner's rep engaged. Handoff package compiled into one location and distributed to every consultant. Charter drafting starts.
Weeks 3–4	Architect's full-services agreement signed (the architect was engaged during diligence for test fits). CM proposals received and interviews held.
Weeks 5–6	CM selected under a preconstruction agreement. Kickoff meeting: the pro forma assumptions are walked through line by line with the full design team, the CM, and the owner's rep. Every consultant hears the three numbers.
Weeks 5–8	Property manager consulted on operations. Utility service applications started. Pre-application meeting with the building department, fire marshal, and public works to confirm the permit path, review timelines, and separate permits (road, sewer, stormwater).
Weeks 7–10	Design-assist trades (plumbing, HVAC, electrical, fire sprinkler, framing) selected through the CM. BOD drafted. Schematic Design underway.
Weeks 10–12	Lender's construction monitor identified and its GMP and contract requirements confirmed. First CM estimate scheduled for the end of SD.

An illustrative design clock, working backward

Milestone	Month
Handoff and team assembly	–10 to –8
Schematic Design, first estimate	–8 to –6
Design Development, second estimate	–6 to –4

Construction Documents, 50% CD estimate	−4 to −2
Permit review (overlapping CDs)	−3 to −1
GMP and loan documents	−1
Financing close, construction start	0

By day ninety, the project should have a charter everyone has agreed to, a full team under contract, a design underway against written requirements, and a date for its first real estimate. Module 2.2 picks up from there.

Artifacts

Artifact	Produced by	When	What it's for
Handoff package (consolidated)	Owner's rep	Weeks 1–2	One source for every report, condition, and assumption
Diligence-to-scope matrix	Owner's rep with the architect	Weeks 2–4	Every finding assigned to a discipline with a budget allowance
Owner's Project Requirements	Owner, drafted by owner's rep	Weeks 2–6	What the building must do, traced to the pro forma
Basis of Design	Architect and engineers	Weeks 7–10, update each phase	How the design intends to meet the OPR
Project charter	Owner's rep, agreed by the team	Weeks 2–6	How the project will be run
Responsibility (RACI) and decision	Owner's rep	Part of the charter	Who does, approves, and decides

authority
matrices

Delivery method brief	Owner's rep with the CM	Part of the charter	How the chosen method runs on this project
Owner-architect, architect-consultant, owner-CM, and design-assist agreements	Owner's counsel, architect, CM	Weeks 3–10	Contract structure matching the delivery method
Risk register (carried forward)	Owner's rep	Ongoing	Open risks with named owners
Milestone schedule	CM, with the architect	Weeks 6–10	Design clock tied to financing close

Phase Approval: Authorize Schematic Design

Part 1 ran on decision gates. Part 2 runs on phase approvals: points where the owner authorizes the team to spend the next tranche of design fee. The first comes at the end of this module.

The question: is the team ready to start designing against the pro forma?

What must be true before the owner says yes:

- ☐ Every consultant has received the full handoff package, and someone has confirmed it
- ☐ Every diligence finding has an owning discipline and a budget allowance
- ☐ The OPR is written and every item traces to a pro forma line or the business plan
- ☐ The charter is agreed, including decision authority and contingency ownership
- ☐ The architect, CM, and key consultants are under contract, and the architect's scope requires response to the CM's estimates
- ☐ The first estimate has a date
- ☐ Long-lead items (the transformer, above all) have a start date and an owner
- ☐ The lender's requirements for the GMP, the contract, and the draw process are known

Whose money is at risk: Schematic Design fees, typically 15–20% of the design fee, are spent at the owner's risk before any estimate exists. That's the price of learning whether the building fits the budget.

Failure Modes

Treating the target as an estimate. The \$24.5 million is a requirement the design has to meet, not a fact about a design that exists. Nobody is assigned to prove it, and the first real estimate arrives too late to fix cheaply.

Assumptions that never became requirements. The pro forma assumed a net-to-gross efficiency. Nobody told the architect. The architect designs generous breezeways and a larger clubhouse. The gross area and cost are the same, but there are fewer rentable square feet.

Diligence that nobody read. The structural engineer starts foundation concepts from "typical conditions for the area" because the geotechnical report never reached their office. The clay shows up in DD as a redesign.

Undefined decision authority. The development manager is running three projects and every decision waits for them. The schedule slips a week at a time, and at about \$118,000 a month, that adds up.

The business plan kept private. The design team assumes a long-term hold and specifies a premium central water heating system. The developer is selling at month 38, and the buyer won't pay for operating savings that haven't been proven. The money is spent on something that doesn't come back at exit.

Design-assist trades brought in at buyout. The plumbing layout is locked in the engineer's drawings before the plumber sees it. The plumber's more efficient layout arrives as a value-engineering proposal after the GMP, when changing the drawings costs more than it saves.

Contracts that don't match the method. The owner chose CMAR but signed a standard architect agreement with no obligation to respond to the CM's estimates. Every cost comment becomes a negotiation about additional services.

Lender requirements discovered late. The lender needs a GMP, a payment and performance bond, and its monitor's review before closing. Nobody asked until month -2.

Where AI Is Changing This

The handoff is an information transfer problem, and that's where a lot of new software is aimed: tools that extract requirements and risks from diligence reports, generate test-fit layouts from a unit mix and zoning envelope, produce conceptual estimates from historical cost data, and flag unusual risk allocation in contracts. Part 3 Module 3.5 (Owner's Program, Budget, and Delivery Method) covers the software categories, and Module 3.4 covers diligence document extraction. The useful question for any of these tools is the same one this module asks of the team: does it

turn an assumption into a requirement someone is accountable for, or does it just make the assumption faster to produce?

Exercise

Decision scenarios

These use the reference project's numbers. None has a single right answer. Reasoning notes follow each one.

Scenario 1: The design-build offer comes back. Three weeks into the CM selection, a regional design-builder with strong garden-multifamily experience approaches the developer. It offers a lump sum of \$24.2 million, \$300,000 below the hard cost target, using its own prototype buildings adapted to the site. It also promises construction start about six weeks earlier, because its prototype is already designed. Finishes would follow its standard package, one level below what the OPR describes.

Reasoning notes: Start with what the offer is worth. \$300,000 of hard cost, plus six weeks of schedule. How much NOI can the finish package cost before the offer loses? At \$450,000, the break-even is $\$450,000 \times 5.25\% \div 2,400$ unit-months, or about **\$9.84 per unit per month**. If the lower finish package costs more than about \$10 a month in achievable rent, the offer destroys value. A disciplined response doesn't accept or reject on the headline. It asks the leasing team for a finish-driven rent estimate against the two newest comps, checks whether the prototype's footprints survive the drainage easement shift, and asks how "lump sum" treats the clay and the detention vault, which are site-specific and not in any prototype.

Scenario 2: The transformer application. The electrical engineer plans to file the utility service application after the permit is issued, around month -1, when the load calculations are final. The utility's process (application review, utility design, easements, and the 14-month transformer lead) runs about 22 months. The CM's schedule needs permanent power by about month 20 for system startup and commissioning. The CM proposes filing at the end of Schematic Design, around month -6, using a conservative preliminary load. The risk is that if the final load comes in lower, the owner may pay for an oversized service or a reapplication, estimated at 30,000-60,000.

Scenario 3: Cutting the owner's rep. The developer's CFO proposes eliminating the owner's rep to save about \$320,000 over the project, roughly 1.3% of hard cost. The development manager would cover the role alongside two other projects.

Scenario 4: The architect's lighter construction administration. To bring its fee down, the architect proposes reducing construction administration: monthly rather than biweekly site visits,

and submittal review limited to "general conformance" with no coordination check across disciplines. The savings are \$110,000.

Deliverable

For the reference project, or for your own project from Part 1, produce two documents.

1. Project charter (5–10 pages)

- Objectives, including your project's version of the three numbers: cost cushion, cost of a month, and value of design control
- Owner's Project Requirements, with each item traced to a pro forma line or the business plan
- Diligence-to-scope matrix: every finding, its owning discipline, and its budget allowance
- Budget structure and contingency ownership
- Team roster with a responsibility (RACI) matrix
- Decision authority matrix with dollar thresholds
- Communication plan
- Milestone schedule
- Risk register, carried from Part 1, with every open risk assigned to a named person

2. Delivery method brief (2–3 pages). This brief documents how that method will run on your project:

- Contract map: who contracts with whom
- Engagement timeline: when each party joins and what it delivers
- Risk allocation table: who holds design, cost, schedule, site, and scope risk, before and after the price is set
- Price-certainty timeline: when each estimate arrives and what class it is
- Design-assist plan: which trades, when, and on what terms
- The conditions under which you'd revisit the decision

Module 2.2 follows the reference project into design. The team is assembled, the charter is signed, and the architect starts drawing. Module 2.2 covers how design moves from concept to construction documents, how decisions lock at each phase, and why the cost of changing your mind rises so quickly.

2.2 — Design as Progressive Commitment

Why This Module Exists

Module 2.1 ended with a team under contract, a charter everyone signed, and a date for the first estimate. Now the architect starts drawing.

Over the next eight months, the design team will make several hundred decisions. Most are small: a door hardware finish, a light fixture, a sealant color. A handful will move the yield on cost more than every negotiation in Part 1 combined: the site layout, the building type, the construction type, the mechanical system, how much of the building the renters actually pay for. This module is about when they get made, how they get locked, and why changing them gets more expensive every month.

Design is not a continuous activity that happens to produce drawings. It's a sequence of commitments. Each phase narrows the options, raises the precision of the documents, and raises the cost of going back. The phases exist because a building can't be designed all at once. The owner approvals at the end of each phase exist because someone has to decide, on the record, that the options just closed are closed for good.

A reader who understands this can walk into any design meeting and answer three questions: what phase are we in, what's already locked, and what would it cost to reopen it?

Where the Project Is

The reference project enters Schematic Design at about month –8. Here's what's already true:

Site	11.8 acres, zoned for up to 259 units
Scheme	Scheme A : 200 units, three-story garden walk-up, surface parking
Construction type (target)	Type V-A, wood frame
Units	200, averaging 887 SF

Net rentable / gross area	177,500 SF / 208,824 SF (85% efficiency)
Parking	320 surface stalls (1.6 per unit)
Hard cost target	\$24,500,000
Delivery	CMAR, with design-assist plumbing, HVAC, electrical, fire sprinkler, and framing
Design clock	SD -8 to -6 · DD -6 to -4 · CD -4 to -2 · Permit -3 to -1 · GMP -1 · Close and start 0

And the three numbers the team carries (Module 2.1):

1. **Cost the deal can absorb:** *2.73million*(1.25M contingency plus \$1.48M headroom).
2. **What a month costs:** about \$118,000 at closeout, and any three-month slip triggers an equity call on the interest reserve.
3. **What design control is worth:** \$5.88 per unit per month in NOI, which is the line between CMAR and design-build.

Every decision in this module gets measured against those three.

Concepts

1. Progressive Commitment

Why changing your mind gets expensive

In Pre-Design, changing the building from three stories to four is a spreadsheet edit. In Schematic Design, the same change means redesigning the building footprints, the site plan, the stair and breezeway layout, and the structural concept: weeks of work across several disciplines. In Design Development, it means re-engineering the structure, resizing the mechanical systems, recalculating egress and accessibility, and revisiting the envelope: months of rework. In Construction Documents, it means revising a coordinated set across every discipline and possibly resubmitting for permit. During construction, it means change orders, demolition of completed work, and a slipped CO date.

The industry often describes this as a cost-of-change curve. The rule-of-thumb ratios vary by source, and nobody should treat them as measured. The shape, though, is consistent: each phase multiplies the cost of the same change.

When the change happens	Rough relative cost	What has to be redone
Pre-Design	1x	A test fit and a spreadsheet
Schematic Design	~5x	Concept drawings across 3–4 disciplines
Design Development	~15x	Engineered systems, coordinated layouts, estimates
Construction Documents	~50x	A coordinated document set; possibly the permit
Construction	~150x or more	Built work, plus schedule, plus disputes

Why the curve is steep: the cascade

The cost rises so fast because a construction document set is interdisciplinary. A change doesn't touch one drawing. It propagates.

Take a small one on the reference project: during CD, the owner asks to move the stair in the prototype building from the end of the breezeway to the center, so it's closer to the parking. That single change touches:

- **Architectural:** unit layouts adjacent to the stair, breezeway geometry, exterior elevations, stair details
- **Structural:** the floor framing around the new opening, the stair's support, the lateral system if the stair wall was a shear wall
- **Plumbing:** any stack that ran where the stair now goes
- **Electrical:** exterior and breezeway lighting, the meter or panel location if it was near the stair
- **Fire protection:** sprinkler layout at the breezeway and stair
- **Civil:** sidewalks, accessible routes from parking to the new stair location, grading at the building entrance
- **Code analysis:** exit travel distances and common path of travel, which may now exceed limits at the far units
- **Every copy of that building on the site.** If the prototype is used nine times, the change is made nine times.

Every one of those disciplines coordinated its documents around the original stair location. Moving the stair unravels that coordination. Each consultant bills for rework, the CM re-prices, and the permit reviewer has to look at it again.

This is the real reason for the phase structure. It's not paperwork. It's a way to make the expensive decisions while they're still cheap.

2. The Phases

The phase names below come from the owner-architect agreement (AIA B101 and its equivalents). Not every project uses AIA contracts, but nearly every U.S. project uses this vocabulary. Completion percentages are conventions, not rules.

Each phase below covers five things: what gets decided, what gets produced, what the estimate can tell you, what the owner's approval locks, and how to recognize a set at this phase if someone hands you one without a title block.

Pre-Design: already done

On a developer-led project, most of Pre-Design happened during Part 1: the market study and program (Module 2), the zoning analysis (Module 3), diligence (Modules 4A and 4B), the test fit and preliminary budget (Module 5), and the handoff package (Module 8).

The design team enters SD with a program, a target, a site constraints package, and a scheme. What it doesn't have is a design anyone has tested. Pre-Design produced requirements and constraints, not solutions.

Schematic Design (SD)

What gets decided: the big moves. The site plan (where the buildings, parking, amenities, and stormwater vault go), the building prototype (unit layouts, stair and breezeway configuration, building footprint), the construction type and fire protection strategy, the structural system concept, the mechanical and electrical system types, the envelope concept, and the clubhouse program.

What gets produced:

- Site plan with building footprints, parking, drive aisles, amenities, and the vault location
- Floor plans of each building type and the clubhouse, showing units, stairs, and breezeways, but not wall construction
- Exterior elevations showing massing and material zones
- A few building sections showing floor-to-floor heights and roof form
- Unit plans for each unit type
- Structural, mechanical, electrical, plumbing, and civil narratives (written descriptions, not engineered drawings)
- A preliminary code analysis confirming the fundamental strategy is viable
- The first Basis of Design (Module 2.1)
- **The first real estimate**, from the CM

What the estimate can tell you: an SD estimate is an AACE Class 3 estimate, with a typical range of –8% to +15%. On a \$24.5 million target, that's roughly \$22.5 million to \$28.2 million. Notice the problem: **the upside of that range is \$3.7 million, and the deal can absorb \$2.73 million.** At SD, the estimate can't yet prove the building fits the budget. What it can do is show whether the design is trending toward the target or away from it, and which systems are driving the gap.

What the owner's approval locks: the site plan, the building footprints and prototype, the unit mix and count, the construction type, and the major system types. After SD approval, changing any of these is a redesign, not a refinement.

How to recognize an SD set:

- Walls drawn as single or double lines with no difference between wall types
- Rooms and units labeled, but few dimensions beyond overall sizes
- No wall sections, no construction details, no door or window schedules
- Engineering disciplines represented by narratives rather than drawings
- No specifications, or a one-page outline
- It looks more like a diagram of a building than instructions for building one

Design Development (DD)

What gets decided: everything specific. Wall assemblies (every layer, from framing to siding), window products and their performance values, roof assembly, floor-ceiling assemblies between units, the specific HVAC equipment and where it sits, the water heater type and location, electrical service size and meter arrangement, plumbing stack locations, the structural member sizes and foundation design, the stormwater vault design, finishes and the appliance package, and the energy code compliance path.

What gets produced:

- Dimensioned floor plans with wall types identified
- Enlarged unit plans, kitchen and bath layouts, and clubhouse plans
- Typical wall sections and key details
- Preliminary door, window, and finish schedules
- Structural foundation and framing plans with preliminary sizes
- MEP plans with equipment located and major distribution routed
- Civil grading, utility, and stormwater plans, including the vault
- **Outline specifications**, naming the products and performance requirements for each major system
- A code analysis that names specific rated assemblies, calculates occupant loads and egress, and identifies the energy compliance path
- An updated Basis of Design
- **The second estimate**, from the CM

What the estimate can tell you: a DD estimate is Class 2, with a typical range of –4% to +10%. On \$24.5 million, the upside is about \$2.45 million, now just inside the \$2.73 million cushion. This is the first point at which the team can say, with reasonable confidence, whether the building fits. That's why the DD estimate is the most important cost milestone in the project, and why Module 2.3 is built around reconciling it.

What the owner's approval locks: materials, assemblies, dimensions, equipment types, and the energy path. After DD approval, the construction documents are being produced on top of these decisions. Changing one ripples through a document set that's being coordinated around it.

How to recognize a DD set:

- Wall types distinguished by line weight, hatch, or tags
- Dimensions throughout
- Door schedules with sizes and hardware groups assigned, even if incomplete
- Typical wall sections and a handful of details
- Structural drawings with member sizes
- MEP equipment shown and major ducts, pipes, and conduits routed
- An outline specification
- Typical conditions detailed, special conditions (transitions, penetrations, expansion joints) not yet

Construction Documents (CD)

What gets decided: almost nothing new, which is the point. CD is about documenting DD decisions completely enough that a contractor can price and build them without guessing, and a building department can approve them.

What gets produced:

- A complete, coordinated drawing set across every discipline
- Hundreds of details: every flashing condition, every transition, every penetration through a rated assembly, every stair and railing
- Complete door, window, finish, and equipment schedules
- Final structural drawings, with calculations submitted separately
- Complete MEP drawings with sizes on every run
- The **project manual**: procurement and contracting requirements (Division 00), general requirements (Division 01), and full technical specifications for every product and system
- The final code analysis, energy code compliance forms, and the statement of special inspections, all submitted with the permit application
- A list of **deferred submittals**: elements the specialty subcontractor designs later, such as the final sprinkler layout and the roof truss engineering, which are reviewed by the design team and the building department during construction
- **The 50% CD estimate and, near the end, the GMP** (Module 2.5)

What the estimate can tell you: a CD estimate is Class 1, with a typical range of –3% to +4%. At this point, the price uncertainty is mostly about the subcontractor market, not the design.

What the owner's approval locks: the documents themselves. Approval at CD means "issue for permit and pricing." After this, changes are addenda (during pricing), revisions (during permit review), or change orders (after the GMP).

How to recognize a CD set:

- A cover sheet with a drawing index and code summary
- A project manual exists
- Details number in the dozens to hundreds
- Every element has a specification reference
- Schedules are complete
- MEP sizes appear on every run
- Title blocks say "For Permit," "For Pricing," "Issued for Construction," or similar

Bidding, permitting, and construction administration

The phases continue after CD, but they're covered elsewhere: the GMP and procurement in Module 2.5, the permit path in Module 2.6, and construction administration in Module 2.7. The design team doesn't stop working at CD. It answers permit comments, reviews the CM's pricing questions, reviews submittals and RFIs, and observes construction. About 20–25% of the architect's fee is spent after CD.

The fee behind each phase

Architect fees are allocated across the phases, and the allocation reflects effort. A common split:

Phase	Share of fee	On an illustrative \$1.0M design fee
Schematic Design	15%	\$150,000
Design Development	20%	\$200,000
Construction Documents	40%	\$400,000
Bidding / GMP support	5%	\$50,000
Construction Administration	20%	\$200,000

Two things follow from this. First, **by the time the owner has a reliable price (DD), it has spent about 35% of the design fee.** By the time it has a Class 1 price (late CD), it has spent about 75%. Second, **redesign is expensive for the architect too.** The fee was sized for doing each phase once. That's why the design team pushes hard for decisions and resists reopening them, and why an owner that changes its mind at DD should expect an additional-services request.

3. Every System Follows the Same Arc

Every building system moves through the same progression: **concept at SD, selection at DD, documentation at CD.** Knowing this tells you what information exists at any phase, and what doesn't.

System	SD	DD	CD
Site and civil	Building and parking layout; vault located; utility connection points identified	Grading plan; vault sized; sewer extension and deceleration lane designed; utilities routed	Full civil set; erosion control; off-site plans for the road authority and utility
Foundations and structure	Wood frame with slab-on-grade ; lateral system concept; over-excavation strategy for the clay	Foundation design from the geotech recommendations; framing layout; shear walls located; preliminary sizes	Complete foundation and framing plans; connection details; truss requirements for deferred submittal
Mechanical and plumbing	System type selected (e.g., split heat pumps, in-unit electric water heaters); equipment zones identified	Equipment selected and located; ducts and stacks routed; plumbing stacks aligned between floors	Full sizing; equipment schedules; controls; energy compliance forms

Electrical	Service concept; meter arrangement; preliminary load for the utility application	Service sized; panels and meter banks located; lighting and power layouts	Full circuiting; panel schedules; single-line diagram; utility coordination drawings
Envelope	Siding and roofing materials; window pattern; insulation strategy	Wall and roof assemblies defined layer by layer; window product and performance values; air barrier strategy	Every condition detailed: sills, heads, jambs, transitions, penetrations, roof edges
Fire protection and life safety	Sprinkler standard selected; construction type confirmed; stair and exit strategy	Rated assemblies selected by tested design number; sprinkler criteria defined	Rated assemblies detailed; penetration firestopping specified; sprinkler layout listed as a deferred submittal

The arc explains why late decisions are so damaging. A decision that should have been made in SD but wasn't, such as the mechanical system type, doesn't just delay mechanical. It delays everything that depends on it. The architect can't finalize the closet sizes in the units without knowing whether there's an air handler in them. The structural engineer can't frame the roof without knowing whether there's equipment on it. The electrical engineer can't size the service, which means the utility application waits, which means the transformer waits.

4. The Decisions That Set the Building

The design tradeoff engine: **constraint** → **design choice** → **cost per SF** → **unit yield** → **NOI** → **return**. Every decision below runs that circuit. This section is about where each one happens in the design process, so it gets made on time and on the record.

Scheme and density: before SD is approved

On most projects, the test fits from Pre-Design are too rough to settle the question. SD is where the architect tests them properly, and where the CM prices them. The owner's SD approval should record which scheme was chosen, what the alternatives yielded, and what market

evidence would reopen the question (for Scheme C, a rent premium of about 17% over garden product).

Code strategy: a design decision, not a compliance check

The building code sets constraints, but within them, the design team makes choices, and those choices cost money. On garden multifamily, the most consequential ones are made together during SD:

- **Construction type.** Type V-A (wood frame with a one-hour fire-resistance rating on the structure) versus Type V-B (wood frame with no required rating on the structure).
- **Sprinkler standard.** NFPA 13R (the residential standard, lower cost, limited to buildings of four stories and 60 feet under the 2021 IBC) versus NFPA 13 (the full standard, higher cost, but it earns larger allowable areas).
- **Building size.** How many units per building, and therefore how large each floor is.
- **Building spacing.** How far apart the buildings sit, which affects both the site plan and, through the frontage increase, how large each building is allowed to be.

These four are linked. Under the unamended 2021 IBC, an apartment building (Group R-2) protected with an NFPA 13R system is limited, before any frontage increase, to about 12,000 SF per floor in Type V-A and about 7,000 SF per floor in Type V-B, and to three stories in Type V-B. The reference project's buildings run roughly 7,500–8,500 SF per floor.

So the choice isn't "V-A or V-B." It's a package:

- **V-A with 13R** allows the planned building size, at the cost of rated construction on the structure.
- **V-B with 13R** saves that cost, but only works if the buildings get smaller (more buildings, more stairs, more exterior wall, more roof) or if the site plan spaces them far enough apart to earn a frontage increase.
- **Either type with NFPA 13** unlocks more area per floor, at a higher sprinkler cost and, in cold climates, the added cost of protecting the system from freezing.

Which package wins depends on the site plan, the local amendments, and current prices. The point for this module is **when** it gets decided: in SD, together with the site plan and the building prototype, because it determines both. Settling construction type at DD means redrawing the site plan.

The code analysis evolves with the design. It starts as a feasibility check in Pre-Design, becomes a strategy in SD, becomes specific in DD, and becomes a permit document in CD:

Phase	What the code analysis establishes
Pre-Design	Which occupancy classifications, construction types, and sprinkler options are viable for the program

SD	The chosen strategy: construction type, sprinkler standard, building sizes, exit and stair concept, separations between uses
DD	Specific rated assemblies by tested design number; calculated occupant loads, exit widths, and travel distances; accessible unit count and routes; plumbing fixture counts; the energy path
CD	Final analysis, energy compliance forms, and statement of special inspections, submitted with the permit application

(Part 1 Module 3 covers where zoning ends and the building code begins. Module 2.6 covers the permit path.)

Systems selection and who pays the meter: SD into DD

The mechanical, plumbing, and electrical systems are selected at SD and specified at DD. On garden multifamily, the major choices are the HVAC system type, the water heating approach, the electrical service and metering arrangement, and the utility billing structure.

That last one is a business decision disguised as a design detail. **Who pays each utility changes what an efficiency investment is worth.** Individually metered electric, common-area electric on the owner's account, and water and sewer either billed back or included in rent all change the math. The metering decision has to be made in SD, because it determines the electrical service design and the plumbing layout, and it has to be made with the business plan in view.

These choices go in the Basis of Design with a reason attached, so the CM can price them and the owner can challenge them before DD.

The energy compliance path: during DD

The energy code offers several ways to comply, and the one chosen changes how the building is specified and how substitutions are handled during construction. The choice is made during DD, once the envelope and systems are specific enough to evaluate.

- **Prescriptive path:** every element meets its own minimum value (wall insulation, window U-factor and solar heat gain, equipment efficiency, lighting power). It's simple to design and easy to document, but it's brittle. Every substitution during construction has to meet the prescriptive value for that element on its own.
- **Trade-off path (envelope only):** allows weaker performance in one envelope element to be offset by stronger performance in another, documented with a calculation (often using the DOE's free COMcheck software for commercial buildings).
- **Performance path:** the whole building is modeled, and the proposed design has to perform at least as well as a code-compliant reference building. It's the most flexible. It lets the team trade envelope against mechanical efficiency, or window performance

against insulation. But it makes the energy model the compliance document, and every specification has to stay consistent with the model's assumptions through construction.

A performance model might show that a better heat pump lets the team use a less expensive window package, or that the \$420,000 envelope upgrade isn't required but is worth doing anyway. The model costs a fee and creates a discipline obligation during construction. Scenario 3 below prices that trade.

(Low-rise residential buildings of three stories or fewer are covered by the residential provisions of the energy code in many jurisdictions. The compliance paths are similar in concept, with different forms and tools. Confirm which apply with the building department during the pre-application meeting.)

How specific to make the specifications: DD into CD

At DD, the specifications are an outline. At CD, they're complete. Along the way, the team decides how tightly to lock each product:

- **Performance specification:** states what the product must do and lets the contractor choose. Maximizes competition and substitution flexibility, but only works if every requirement that matters is written down.
- **Basis-of-design specification:** names a specific product as the standard and allows equivalents that meet the same criteria. The most common approach.
- **Prescriptive or proprietary specification:** names one product and allows no substitutes. Used when the owner has a standard (appliance brands the property manager can service, a door hardware system tied to access control) or when only one product works.

The choice matters most for the items that drive rent and operating cost. The appliance package is one of the few finishes renters price, so locking it is worth more than locking the corridor paint. Module 2.4 covers specifications in depth.

5. Coordination Between Disciplines

A garden community is designed by at least six independent firms working in parallel: architect, civil, structural, MEP, landscape, and the fire sprinkler designer. Each produces its own drawings. Coordination is the work of making sure those independent designs actually fit together, physically and logically.

When coordination happens

- **SD:** high-level compatibility. Is the mechanical closet big enough for the air handler? Can the plumbing stacks align between floors in the chosen unit layouts? Does the vault fit where the civil engineer placed it without conflicting with the building footprints?
- **DD:** active coordination. Systems are routed and conflicts identified. On BIM projects, the combined model is run through clash detection software. On 2D projects, disciplines overlay their drawings.

- **CD:** full coordination. All known conflicts are resolved. Complex areas get coordination drawings that show every discipline on one sheet: the mechanical closets, the attic or truss space, the breezeway ceilings, and the utility rooms.
- **Construction:** residual coordination through RFIs. The goal is to keep this small.

Three kinds of coordination failure

Spatial conflicts: two things in the same place. A duct and a beam at the same elevation. A plumbing stack and a shear wall. Clash detection catches most of these on BIM projects, and field crews catch the rest, expensively.

Clearance and performance conflicts: nothing physically overlaps, but one discipline's work degrades another's. A duct routed under a beam lowers the ceiling below what the finish schedule assumed. An electrical panel fits on the wall, but its required working clearance conflicts with a door swing. A pipe passes through a rated wall where the gap is too small for the specified firestop system. These don't show up in automated clash detection because the objects don't touch. They need people who understand each discipline's requirements.

Sequencing conflicts: everything fits in the finished building, but it can't be built in that order. The sprinkler main has to be installed before the trusses close off access. A rated ceiling has to be inspected before it's covered, but the plumbing above isn't finished. The drawings show the finished condition, not the construction sequence. These surface through the CM's constructability review during design, or through RFIs during construction. Catching them in design is one of the main reasons the owner chose CMAR.

Who owns coordination

The architect is responsible for coordinating its consultants' documents so the design set is internally consistent. That doesn't mean the architect guarantees every conflict has been found, and it doesn't mean the owner is protected when one hasn't. Under *Spearin* (Module 2.1), the owner warrants the drawings to the contractor. A coordination error that reaches the field is the owner's cost first, and the architect's only to the extent it falls below the standard of care.

On a BIM project, the **BIM execution plan** assigns coordination responsibility explicitly: who models what, who runs clash detection, how clashes are grouped and prioritized, who resolves each type, and how often. Under CMAR with design-assist, the trades participate in that process, bringing installation knowledge to conflicts that the engineers' models alone wouldn't reveal.

BIM and level of development

On BIM projects, "how complete is the model?" has a standard answer: the **Level of Development (LOD)** framework, published by BIMForum. LOD describes how reliable the information in a model element is, not how detailed it looks.

LOD	Meaning	Typical phase	What you can rely on
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100	Conceptual placeholder	Pre-Design, early SD	That something exists roughly here
200	Approximate size, shape, location	SD	Rough quantities and spatial fit
300	Specific size, shape, location	DD and CD	Dimensions and quantities for documents and estimates
350	Includes connections and interfaces with other systems	Late CD	Detailed coordination and clash resolution
400	Fabrication-level detail	Construction (shop drawings)	Fabrication and installation

Two practical points. First, **an estimate or code check run on an LOD 200 model is only as reliable as LOD 200**. A model can look finished and still be approximate. Second, on most U.S. projects, **the drawings and specifications are the contract documents, not the model**. If they conflict, the drawings control, unless the contract says otherwise.

6. Change Control During Design

The owner's approvals are the gates

Each phase ends with an owner approval. It's the owner's formal statement that the decisions made in that phase are final.

- **SD approval** means "yes to this site plan, this building, this construction type, and these system types. Develop it."
- **DD approval** means "yes to these materials, assemblies, dimensions, and equipment. Document it."
- **CD approval** means "yes, issue these documents for permit and pricing."

An approval is only as useful as the owner's understanding of what it's approving. That's why each approval should come with the estimate for that phase, the updated Basis of Design, an updated code analysis, and the decision log that this module's deliverable builds.

Scope creep

Scope creep is the gradual expansion or modification of requirements after they've been set. It comes from everywhere: the owner adds a dog park, the leasing team asks for a larger package

room, the architect refines the clubhouse, the fire marshal interprets a provision differently than expected.

The damage is rarely the direct cost of the change. It's the cascade. Converting a storage room in the clubhouse to a coworking lounge sounds minor. It changes the occupant load (and possibly egress), the ventilation requirement, the electrical and data outlets, the finish level, and possibly the plumbing fixture count for the building. Each consultant spends hours, the CM re-prices, and the coordinated set gets re-coordinated.

The protection is the phase approval. When a change is genuinely necessary after an approval (markets shift, lenders impose conditions, codes get amended), it goes through a controlled process:

1. The change is written down, with the reason
2. Every affected discipline estimates its redesign effort and schedule impact
3. The CM prices the construction cost
4. The owner decides, using the decision authority matrix from the charter
5. The decision is logged, and the Basis of Design and estimate are updated

What must not happen is an informal change: a verbal request in a meeting, a sketch emailed to one consultant, a "we'll sort it out in CDs." Informal changes are how coordination errors are born.

Design contingency

Estimators at SD and DD usually carry a **design contingency**: an allowance for things the design hasn't shown yet but inevitably will, such as the details that haven't been drawn, the blocking that isn't on the framing plans, and the transitions nobody has resolved. It's typically largest at SD and shrinks toward zero at CD as the documents fill in. It is not owner contingency and it is not the CM's construction contingency. Module 2.3 covers the three contingencies and why confusing them is one of the most expensive mistakes in a development budget.

Artifacts

Artifact	Produced by	Phase	What it's for
SD drawings and narratives	Design team	SD	The concept, documented well enough to estimate and approve
DD drawings and outline specifications	Design team	DD	Specific systems and assemblies, documented well enough to price reliably
Construction documents and project manual	Design team	CD	Complete instructions for permit, pricing, and construction
Phase estimates (Class 3, 2, 1)	CM	End of SD, DD, 50% CD	What the design costs at each phase, with its range
Basis of Design, updated each phase	Design team	SD-CD	How the design meets the OPR, with reasons
Code analysis, evolving	Architect	Pre-Design-CD	From feasibility check to permit document
Energy compliance documentation	MEP engineer or energy modeler	DD-CD	The chosen path, and forms or model for permit
BIM execution plan and clash reports	Design team, CM	DD-CD	Who coordinates what, and what's been found
Constructability reviews	CM and design-a	Each phase	Sequencing and installation issues caught on paper

assist
trades

Design change log	Owner's rep	All phase s	Every change after an approval, with cost, time, and decision
Phase-gate decision log	Owner's rep	All phase s	The record of what was decided, why, and what reopening it would cost

Phase Approvals

Part 1 ran on decision gates. Part 2 runs on phase approvals, and this module has three.

Approve Schematic Design (about month –6)

The question: is this the building we're going to build?

- [] Scheme chosen, with the alternatives' results recorded
- [] Site plan resolves the diligence findings: vault located, easement cleared, deceleration lane and sewer extension accommodated
- [] Construction type, sprinkler standard, and building sizes chosen together, with the site plan
- [] Mechanical, water heating, electrical service, and metering approach chosen, with the utility billing structure confirmed against the business plan
- [] Unit count, mix, and net-to-gross efficiency confirmed against the pro forma
- [] Class 3 estimate received, and any gap to the target has a reconciliation plan
- [] Preliminary electrical load issued to the utility (the transformer clock, Module 2.1)
- [] Decision log updated

Whose money is at risk: DD fees, about 20% of the design fee, spent on a design whose cost is still uncertain by as much as the whole cushion.

Approve Design Development (about month –4)

The question: does the building fit the budget?

- [] Class 2 estimate reconciled to the target, with any value engineering ranked by net value per dollar, not dollars saved
- [] Wall, roof, and floor-ceiling assemblies defined; rated assemblies identified by tested design
- [] Equipment selected and located; plumbing stacks and ducts routed

- ☐ Energy compliance path chosen, with the reason recorded
- ☐ Outline specifications issued; the owner's standards (appliances, hardware, access control) locked
- ☐ Design-assist trades' input incorporated
- ☐ Constructability review and first clash detection completed
- ☐ Decision log updated

Whose money is at risk: CD fees, the largest share of the design fee, plus the pursuit costs of the financing and GMP process that follows.

Approve Construction Documents (about month –2)

The question: are these documents ready for the permit and the GMP?

- ☐ Documents coordinated across all disciplines; clash detection clean or remaining items tracked
- ☐ Code analysis, energy forms, and statement of special inspections complete
- ☐ Deferred submittals listed
- ☐ 50% CD estimate reconciled; no open design questions the CM would have to price as assumptions
- ☐ Permit submitted, or ready to submit
- ☐ Decision log updated

Whose money is at risk: the remaining design fee, the permit fees, and every change after this point, which is priced as an addendum, a permit revision, or a change order.

Failure Modes

Approving a phase without an estimate. The owner approves SD because the drawings look right. The first number arrives at DD, and it's 8% high with 35% of the fee spent.

Choosing the scheme at DD. By then, one scheme has most of the drawings and all of the momentum. The comparison becomes a defense.

Deciding construction type separately from the site plan. The team chooses V-B at DD to save money, and discovers the buildings are too large. The site plan gets redrawn in CD.

Leaving the metering decision to the MEP engineer. Utility billing is a business decision. Made by default, it quietly determines whether every efficiency investment in the building is worth something or nothing.

Choosing the energy path by habit. The team uses the prescriptive path because it always does, and gives up a trade that would have paid for itself. Or it uses the performance path

without anyone committing to keep the specifications consistent with the model during construction.

Informal changes. A verbal request in an OAC meeting, a sketch sent to one consultant. The coordinated set stops being coordinated, and nobody knows.

Treating an LOD 200 model as a finished design. It looks complete. It isn't. Estimates and code checks run on it inherit its uncertainty.

Coordination left to the field. The owner warrants the drawings. Every conflict discovered in framing is priced at construction rates and paid for by the owner.

Where AI Is Changing This

Design is where software is moving fastest. Generative massing and test-fit tools can produce and compare dozens of site layouts in the time it used to take to draw one. BIM authoring tools are automating sheet production and routing. Coordination platforms group and triage clashes, and newer tools check drawings for cross-sheet inconsistencies. Code-checking tools can flag objective requirements early in design. Part 3 covers these in Modules 3.6 (conceptual design), 3.7 (BIM authoring and engineering analysis), 3.8 (coordination and drawing QA), and 3.10 (code compliance and plan review).

The useful test for any of them is this module's frame: **does the tool move a decision earlier on the cost-of-change curve, or does it just produce documents faster?** Producing more options in SD is valuable only if the owner is set up to choose among them before DD. Faster drawing production in CD saves fee, but it doesn't change what the building costs. Catching a coordination conflict in the model is worth what it would have cost in the field, which on the reference project was \$696,189.

Exercise

Decision scenarios

These use the reference project's numbers. Reasoning notes follow each one.

Scenario 1: Twelve more units at DD. Midway through DD, the civil engineer reports that moving detention into the underground vault freed enough area on the site for one more 12-unit building. Zoning allows up to 259 units, so the count is legal. The architect estimates \$75,000 of additional design fees and about one month added to the design schedule. The entitlement conditions reference the approved site plan at 200 units.

Reasoning notes: Run the circuit. Non-land cost on this project is about 160,000 *per unit* (32 million ÷ 200), so twelve more units cost about \$1.92 million. At \$11,400 of NOI per unit, they add \$136,800 a year, worth about \$2.61 million at a 5.25% cap. That's a 7.1% yield on the incremental cost, above the project's 6.51%, and roughly \$690,000 of value before fees and delay. Now subtract: \$75,000 of fee, and a month of schedule that pushes the CO toward or away from the leasing season. The number still works. What might not is the entitlement. If the approved site plan fixes the count at 200, adding units means an amendment, which is a discretionary approval with its own timeline and its own chance of failure. Part 1 Module 3 insists entitlement risk is binary, not a schedule problem. The disciplined answer: pursue it only if the amendment can run in parallel without holding the permit, and design the extra building as a separable add-alternate so the base project doesn't wait for it.

Scenario 2: The SD estimate is high. The CM's Class 3 estimate at the end of SD comes in at \$25.6 million, \$1.1 million (4.5%) over the target. The architect points out that the range on a Class 3 estimate is -8% to +15%, so "the number could still come down." The developer wants to approve SD and deal with cost at DD.

Reasoning notes: The architect is technically right and practically wrong. A Class 3 estimate is uncertain, but its uncertainty is skewed high: estimates miss high far more often than low, because the undesigned parts of a building are usually things that get added. At \$25.6 million, the Class 2 estimate at DD could reasonably land anywhere from about \$24.6 million to \$28.2 million. The upper part of that range is beyond the \$2.73 million cushion. Approving SD now means spending the DD fee on a design that's trending away from the budget. The disciplined answer is to approve SD with conditions: identify which systems drive the \$1.1 million, rank the options by net value per dollar, and make the reconciliation a required deliverable at the start of DD rather than a hope at the end of it.

Scenario 3: Pay for the energy model? The MEP engineer can comply using the prescriptive path with the baseline envelope. Alternatively, for an additional \$35,000 in modeling fees, the performance path would let the team pair higher-efficiency heat pumps with a less expensive window package, saving an estimated \$150,000 in net construction cost. The CM warns that under the performance path, any substitution during construction on windows, insulation, or HVAC equipment will have to be checked against the model.

Reasoning notes: On construction cost alone, \$150,000 saved for \$35,000 spent is an easy call. Two more things belong in the decision. First, the NOI effect: if the heat pump efficiency gain lowers owner-paid utilities (common areas, the clubhouse), it's worth dividing by the cap rate, not just counting it. If the tenant pays for unit electricity, the tenant gets most of the benefit. Second, the construction-phase obligation: the model becomes a compliance document. Someone has to own it through submittals and substitutions, or the as-built building won't match what was permitted. Record that owner in the charter. If nobody owns it, the \$150,000 can turn into a failed inspection at closeout, where a month costs \$117,660.

Scenario 4: V-B instead of V-A? At the end of SD, the framing design-assist subcontractor suggests building in Type V-B instead of V-A, which it estimates would save about \$400,000 by eliminating the rated construction on the structure. The architect responds that the buildings, at about 8,000 SF per floor, exceed the unincreased V-B area limit with an NFPA 13R system.

Reasoning notes: This is a package decision, not a line item. Three paths exist. The team could break the buildings into smaller ones, which adds stairs, exterior wall, roofs, and probably site area; those costs may exceed the savings. It could space the buildings to earn a frontage increase, which is a site plan question the civil engineer and architect can answer quickly, and it might cost units. Or it could keep V-A. The right move is to price all three in the next two weeks, while SD is still open, and record the answer. Deciding this at DD means redrawing the site plan in CD, and at that point the \$400,000 is gone in redesign and delay.

Scenario 5: EV charging at 70% CD. At 70% CD, the developer's leasing team asks for EV chargers at 20% of the parking spaces (64 spaces). The electrical engineer reports that the added load would push the service beyond the preliminary load submitted to the utility at the end of SD.

Reasoning notes: The direct cost is chargers, conduit, and panels. The hidden cost is the utility clock. If the added load forces a larger transformer or a reapplication, the utility process may restart, and Module 2.1 showed that a late utility application is worth about two months of delay, or roughly \$135,000 to \$300,000 depending on when it hits. Before deciding, check two things. Does the adopted code or a local ordinance already require a percentage of EV-capable or EV-ready spaces? If so, some of this should already be in the design, and it's a design correction, not an owner change. And can the building be made EV-ready now (conduit and panel capacity, no chargers) with the load managed by software, so the service application doesn't change? The cheapest version of this request is the one that doesn't touch the transformer.

Deliverable: Phase-Gate Decision Log

For the reference project, or your own project, build the log that documents every consequential design decision from SD through CD.

Each entry records:

Field	What goes in it
Phase	SD, DD, or CD
Decision	What was decided, in one sentence
Alternatives	What else was considered
The number	The cost, NOI, value, basis points, or schedule figure that decided it
Approved by	Name and role, per the charter's decision authority matrix
What it locks	Which downstream decisions now depend on it

Cost to reopen	What changing it would cost at the next phase
What would reopen it	The specific evidence that should trigger another look

A starter log for the reference project:

1. Scheme A: 200 units, three-story garden, surface parking · SD

- **Alternatives:** Schemes B and C
- **The number:** 126 bps, vs. (100) bps for B and 9 bps for C
- **Locks:** site plan, building type, parking, construction type
- **Cost to reopen at DD:** full redesign; months of schedule
- **What would reopen it:** comps showing a rent premium of 17% or more for four-story elevator product

2. Type V-A with NFPA 13R, buildings about 8,000 SF per floor · SD

- **Alternatives:** V-B with smaller buildings or a frontage increase; NFPA 13
- **The number:** the three-path pricing from Scenario 4
- **Locks:** building sizes, site plan, rated assemblies
- **Cost to reopen at DD:** site plan redraw
- **What would reopen it:** a frontage increase proven feasible without losing units

3. Individually metered unit electric; owner pays common areas · SD

- **Alternatives:** master-metered with billback
- **The number:** determines whether efficiency investments create value
- **Locks:** electrical service design, meter banks, every efficiency decision after it
- **Cost to reopen at DD:** electrical redesign; possible utility reapplication
- **What would reopen it:** the business plan changing to a long-term hold

4. Preliminary electrical load issued to the utility at month –6 · SD

- **Alternatives:** file at permit, month –1
- **The number:** about \$135,000 of delay avoided (Module 2.1)
- **Locks:** service size
- **Cost to reopen at DD:** possible utility reapplication
- **What would reopen it:** load growth beyond the submitted estimate

5. Energy performance path · DD

- **Alternatives:** prescriptive path
- **The number:** \$150,000 net construction savings for a \$35,000 modeling fee
- **Locks:** window and HVAC specifications; a named owner for the model during construction

- **Cost to reopen at CD:** re-specification of every energy-related product
- **What would reopen it:** substitutions that break the model

6. Keep the \$420,000 envelope upgrade (owner-paid utilities) · DD

- **Alternatives:** delete it in value engineering
- **The number:** +\$303,810 of value
- **Locks:** wall assemblies, window package
- **Cost to reopen at CD:** wall redesign and re-modeling
- **What would reopen it:** utility billing moving to the tenants

7. Value engineering ranked by net value per dollar · DD

- **Alternatives:** the naive list, ranked by dollars saved
- **The number:** a \$628,809 swing between the two lists
- **Locks:** finish schedule, clubhouse program
- **Cost to reopen at CD:** finish and amenity redesign
- **What would reopen it:** leasing evidence that the second amenity building supports rent

8. EV-ready at 20% of spaces; chargers deferred · CD

- **Alternatives:** install chargers now
- **The number:** protects the utility application and its schedule
- **Locks:** panel capacity, conduit
- **Cost to reopen during construction:** service upsizing
- **What would reopen it:** a code requirement or leasing demand

Complete the log for your project with at least ten entries, including at least one decision in each of these categories: scheme and density, code strategy, systems and metering, energy path, value engineering, and a late owner change.

Module 2.3 follows the reference project's estimates. The SD number just came in \$1.1 million high. Module 2.3 covers how estimates mature through design, what quantity takeoff actually is, the three kinds of contingency, and how a team closes a budget gap without destroying the value it was trying to protect.

2.3 — Cost Through Design: Estimating and the Budget Gap

Why This Module Exists

Module 2.2 ended with the Schematic Design estimate coming in \$1.1 million over the target. The owner approved SD anyway, on the condition that the gap would be closed early in Design Development. It wasn't. Four months later, the DD estimate arrives at **\$27.44 million, 12% over the \$24.5 million target**, and the gap is now larger than everything the deal can absorb.

This is the most common crisis in development, and it's the most direct point of contact between Part 1 and Part 2. The pro forma in Part 1 was built on a hard cost number. Part 2 is where that number meets a real design and a real estimate. When they disagree, someone has to close the gap without destroying the value the building was supposed to create.

This module covers how that number is produced and tested, from the first estimate to the one the GMP is built on:

- What an estimate is made of, and who produces it
 - How estimates mature as the design develops, and how they change their organization along the way
 - What quantity takeoff is, and how quantities become dollars
 - Target value design: designing to a budget instead of estimating after the fact
 - Value engineering as it's meant to work, rather than as a list of cuts
 - The DD gap on the reference project: where it came from, what it does to the deal, and how the team closes it
-

Where the Project Is

Phase	End of Design Development, about month –4
Scheme	200 units, three-story garden walk-up, Type V-A, surface parking
Buildings	Seven 24-unit buildings (Type A) and two 16-unit buildings (Type B), plus a clubhouse and pool
Hard cost target	\$24,500,000
SD estimate (Module 2.2)	\$25,600,000, which was \$1.1 million over (4.5%)
DD estimate	\$27,440,000, \$2.94 million over (12.0%)
What the deal can absorb (Module 2.1)	2.73million (1.25M owner contingency plus \$1.48M headroom to the 100 bps spread floor)

The gap is \$210,000 larger than the entire cushion, and construction hasn't started.

Concepts

1. What an Estimate Is Made Of

An estimate isn't one number. It's a stack of numbers, and most arguments about estimates are really arguments about one layer of the stack.

The layers

Direct cost of the work. What it costs to build the building: materials, labor, equipment, and subcontractor pricing, organized by building system or trade. This is where takeoff and unit pricing live, and it's usually 75–85% of the total.

General conditions and general requirements. The contractor's cost of running the job: superintendent and project manager salaries, site trailers, temporary power and water, cleanup, dumpsters, safety, surveying, small tools, and equipment used across trades. Often 6–10% of direct cost on multifamily. The terms get used loosely. Some contractors split them into *general conditions* (staff and time-driven costs) and *general requirements* (site-driven costs). What

matters is knowing which costs sit in this layer and which sit in the trades, so nothing is counted twice or missed.

Insurance and bonds. The contractor's liability insurance, any subcontractor default insurance, and the payment and performance bond. Typically 1.5–3% combined.

Fee. The contractor's overhead and profit, usually 3–5% on multifamily under a CMAR contract.

Design contingency. An allowance for what the design hasn't shown yet. Large at SD, smaller at DD, and supposed to reach zero by the GMP.

Escalation. The cost of prices rising between the estimate date and the midpoint of construction. It's a forecast, not a cushion.

Here's the reference project's DD estimate, summarized:

Layer	Amount
Direct cost of work	\$22,090,000
General conditions (7.5%)	1,656,750
Insurance and bond (2.2%)	522,428
Fee (4.0%)	970,767
Design contingency (3.0%)	757,198
Escalation to construction midpoint (5.55%)	1,442,841
DD estimate	\$27,439,986 (reported as \$27,440,000)

Notice that each layer is a percentage applied to the layers above it. **Every \$1.00 of direct cost becomes about \$1.24 in the estimate.** That multiplier matters when closing a gap. Removing \$100,000 of direct cost saves about \$124,000. Adding \$100,000 of scope costs about \$124,000.

The fine print: qualifications, assumptions, exclusions, and allowances

Every serious estimate comes with a written list of what it does and doesn't include. This list is often more important than the number.

- **Qualifications and assumptions:** what the estimator assumed where the design was silent. "Assumes 2x6 exterior studs at 16 inches on center." "Assumes over-excavation

depth of 4 feet per the geotechnical report." "Assumes the utility provides the transformer at no cost to the project."

- **Exclusions:** what isn't in the number at all. "Excludes utility company fees." "Excludes permit fees." "Excludes rock excavation." "Excludes furniture, fixtures, and equipment in the clubhouse."
- **Allowances:** placeholders for scope that isn't designed yet. "Landscaping allowance: \$380,000." "Signage allowance: \$45,000." Allowances are adjusted to the actual cost later, which means the owner holds that risk.

Read this list before reading the number. Two estimates that differ by \$1 million often differ mostly in what they assumed or excluded.

2. Who Estimates, and Why Estimates Disagree

The people

The owner's cost consultant (or independent estimator). Hired by the owner to produce an estimate that doesn't come from the party that will build the project. On a CMAR project, this is the owner's check on the CM's numbers. Some owners rely on the owner's rep for this; larger or more cost-sensitive projects hire a dedicated firm.

The CM's preconstruction team. Produces the estimates the project actually runs on during design. The preconstruction manager leads; estimators do the takeoff and pricing. They price the self-performed work (if any) and gather subcontractor input on the rest.

Subcontractors and design-assist trades. On a CMAR project, the CM asks key trades to price their scope as the design develops. Design-assist trades (Module 2.1) go further, proposing layout and product changes that lower their own costs.

The design team. Architects and engineers generally don't estimate construction cost on projects like this, and most owner-architect agreements treat estimating as an additional service. They do, however, need to respond to the estimates, which is why the architect's contract should require it (Module 2.1).

Why two estimates of the same drawings disagree

When the owner's cost consultant and the CM estimate the same DD set, they rarely agree. Reconciling them is a routine task, and it follows a predictable order. Differences almost always fall into five buckets:

1. **Scope differences.** One estimate includes something the other doesn't. One estimator assumed the sewer extension was in the civil package; the other assumed it was a utility cost. These are the largest and easiest to resolve, because the answer is in the drawings or the contract.
2. **Quantity differences.** Both included the scope, but measured it differently. One took off exterior siding net of windows; the other didn't deduct openings. These are resolved by comparing takeoffs line by line.

3. **Unit price differences.** Both have the same quantity but different costs per unit. One used a subcontractor quote; the other used historical data. These are judgment calls, resolved by evidence: current quotes, recent bids, and published cost data adjusted for location and date.
4. **Markup differences.** Different general conditions, insurance, or fee assumptions. These should be settled by the CM's contract, not negotiated estimate by estimate.
5. **Timing differences.** Different escalation assumptions or different construction midpoints. These are settled by agreeing on one schedule and one set of escalation rates.

The reconciliation isn't a negotiation toward the middle. It's a search for the reasons behind each difference. When it's done, the two estimators should agree on scope and quantities, and any remaining differences should be understood and documented.

3. How Estimates Mature Through Design

The methods change as the design changes

This section describes them by how they're built, because the method has to match the information that exists.

Phase	Method	Organized by	Class
Screening and test fit	Cost per unit or per square foot, from comparable projects	Whole building	5 or 4
Schematic Design	Assemblies (systems) estimate: quantities of major elements times assembly costs	UniFormat	3
Design Development	Assemblies with growing detail, plus trade input on key systems	UniFormat, starting to map to trades	2
50% CD through GMP	Detailed unit-cost estimate: itemized quantities times unit costs, priced by trade	MasterFormat trade packages	1

Parametric estimates (cost per square foot, cost per unit) work before there's a design, because they rely on what similar buildings cost. Their weakness is that the reference project's

site isn't average: the clay, the vault, and the deceleration lane don't appear in a comparable project's cost per square foot.

Assemblies estimates work once there's a concept. Instead of pricing every stud and screw, the estimator measures major elements (square feet of exterior wall, square feet of roof, number of units, number of bathrooms) and multiplies by a cost for the whole assembly. An exterior wall assembly price includes the framing, sheathing, weather barrier, insulation, siding, and interior drywall together.

Detailed estimates work once the documents are specific enough to count individual items. They're organized the way the work will be bought: by trade.

Two organizing systems, and why the estimate switches between them

Part 1 Module 5 introduced **UniFormat**, which organizes a building by *systems and functions*: substructure, shell, interiors, services, equipment, special construction, and sitework. Module 2.4 covers **MasterFormat**, which organizes construction by *work results and trades*: concrete, wood framing, thermal and moisture protection, openings, finishes, plumbing, HVAC, electrical, and so on.

The two systems serve different purposes:

- **UniFormat is how designers think and how early estimates are built.** It answers "what does the exterior wall cost?" regardless of how many trades build it. It lets the owner compare design alternatives by system while the design is still flexible.
- **MasterFormat is how the building is bought and specified.** It answers "what does the siding subcontractor cost?" because the siding subcontractor is the one who signs a contract. Specifications, subcontracts, schedules of values, and pay applications are all organized this way.

So the estimate changes organization partway through design. The exterior wall that was one UniFormat line at SD becomes several MasterFormat lines at CD: wood framing (Division 06), sheathing and weather barrier (Divisions 06 and 07), insulation (Division 07), siding (Division 07), windows (Division 08), and drywall (Division 09). A good estimator maintains a crosswalk between the two, so the owner can still see "what does the exterior wall cost?" after the estimate has been reorganized for buyout.

The reference project's DD estimate, still in UniFormat, looks like this:

UniFormat system	What's in it	DD direct cost
A Substructure	Foundations, slab-on-grade, over-excavation for the clay	\$1,770,000
B Shell	Framing, floors, exterior walls, windows, roofs, breezeways and stairs	6,630,000
C Interiors	Partitions, doors, finishes, cabinets, stair finishes	4,420,000
D Services	Plumbing, HVAC, fire protection, electrical	5,300,000
E Equipment and furnishings	Appliances, window coverings, clubhouse equipment	880,000
F Special construction	Pool and amenity structures	440,000
G Sitework	Earthwork, paving, utilities, the detention vault, sewer extension, deceleration lane, landscape	2,650,000
Total direct cost		\$22,090,000

4. Quantity Takeoff

What takeoff is

Takeoff is measuring the drawings to find out how much of everything the building contains: how many cubic yards of concrete, square feet of drywall, linear feet of pipe, and how many doors, toilets, and light fixtures. It's called takeoff because the estimator "takes off" quantities from the drawings.

Takeoff is the foundation of every assemblies and detailed estimate. An estimate is, at bottom, **quantity x unit cost**, repeated a few thousand times. If the quantities are wrong, no amount of pricing skill fixes the number.

How things are measured

Every item is measured in the unit it's bought and installed in:

Measurement	Unit	Examples on the reference project
Count	Each (EA)	Doors, windows, plumbing fixtures, appliances, light fixtures, HVAC units, parking stall stripes
Length	Linear feet (LF)	Pipe, conduit, curb and gutter, baseboard, handrail, gutters, sewer main
Area	Square feet (SF) or squares (100 SF, for roofing)	Drywall, flooring, siding, roofing, sheathing, paving, slab
Volume	Cubic yards (CY)	Concrete, excavation, over-excavation and fill, gravel base
Weight	Tons or pounds	Reinforcing steel, structural steel, asphalt

A worked example: the exterior wall of a Type A building

Take the exterior wall assembly on one 24-unit building. The architect's DD drawings show a footprint of roughly 250 by 34 feet, three stories, with fiber-cement lap siding.

Step 1: Measure the gross wall area. Perimeter: $2 \times (250 + 34) = 568$ LF Wall height from slab to top plate: about 31 feet Gross wall area: $568 \times 31 = \mathbf{17,608 \text{ SF}}$

Step 2: Deduct openings. Windows: 72 at about 15 SF each = 1,080 SF Patio doors: 24 at about 40 SF each = 960 SF Net opaque wall: $17,608 - 2,040 = \mathbf{15,568 \text{ SF}}$

(Openings are deducted from surfaces that stop at the opening, such as siding and sheathing. Windows and doors are counted separately, as "each," and priced in their own lines.)

Step 3: Apply waste factors and price each layer. Materials don't install at exactly the measured area. Sheathing gets cut around openings. Lap siding overlaps and has offcuts. The estimator adds a waste factor to each material, typically 5–15% depending on the material and how complicated the wall is.

Layer	Quantity with waste	Unit cost (installed)	Extension
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OSB sheathing, 7/16" (10% waste)	17,125 SF	\$2.10/SF	\$35,962
Weather-resistive barrier (10% waste)	17,125 SF	\$0.85/SF	14,556
Fiber-cement lap siding and trim (12% waste)	17,436 SF	\$7.40/SF	129,028
Batt insulation, R-21 (5% waste)	16,346 SF	\$1.35/SF	22,068
Interior gypsum board, taped (12% waste)	17,436 SF	\$2.60/SF	45,334
Wall assembly, excluding framing and openings		\$15.86 per SF of net wall	\$246,948

The unit costs here are illustrative. Real unit costs come from subcontractor quotes and local data (Section 5).

Step 4: Multiply by the prototype, carefully. There are seven Type A buildings. If they're truly identical, the wall assembly costs about $7 \times \$246,948 = \$1,728,636$. This is how garden multifamily is estimated efficiently: take off one prototype in detail, then multiply.

The trap is the word "identical." On the reference project, two of the Type A buildings sit on the sloped portion of the site and have walk-out conditions on one end. Their foundations step down, their exposed wall height is taller on one side, and their siding quantity is higher. A takeoff that multiplies the flat-site prototype by seven will miss that. A good estimator takes off the prototype once, then takes off the *differences* for each building that isn't typical.

Step 5: Roll up. The wall assembly becomes one line in B Shell at DD. At CD, it splits into its trades: the sheathing and framing go to the framing package, the weather barrier and siding go to the siding subcontractor, the insulation and drywall go to the drywall or insulation subcontractor, and so on. The quantities stay the same. Only the organization changes.

What takeoff misses

Takeoff measures what's drawn. It doesn't measure:

- **What isn't drawn yet.** Blocking for cabinets and grab bars, fire caulking at every penetration, transitions between materials, and flashing at conditions that haven't been detailed. This is what design contingency is for.

- **What's drawn inconsistently.** If the floor plan and the building section show different wall heights, the estimator has to pick one or ask. Every assumption belongs on the qualifications list.
- **Means and methods.** Takeoff says how much siding there is, not how the subcontractor will reach the third floor. Scaffolding, lifts, and access are priced in the trade's unit costs or in general conditions.
- **Site-specific conditions.** The slope, the clay, and the tight laydown area near the vault all affect productivity. A takeoff doesn't know about them unless the estimator does.

Who does takeoff, and with what

Estimators do takeoff, usually in software that lets them measure directly on PDF drawings or extract quantities from a BIM model. Subcontractors do their own takeoffs when they bid, which is why a subcontractor's number sometimes catches something the CM's estimate missed. Takeoff is also one of the tasks changing fastest with software. Part 3 Module 3.11 covers the tools.

5. Pricing the Quantities

A quantity becomes a cost when it's multiplied by a unit cost. Estimators get unit costs from four places:

Subcontractor quotes. The most current and most reliable, because the subcontractor is the one who will do the work. Early in design, subcontractors give "budget" numbers that aren't binding. At buyout, they give bids.

The contractor's historical cost data. What the CM paid per unit on recent similar projects, adjusted for time and location. Good for common items, but only as good as how carefully the CM tracked actual costs.

Published cost data. Databases such as RSMeans (from Gordian) publish unit costs for thousands of items, with adjustment factors for location and date. They're useful for checking numbers and for items nobody has priced recently. They're less reliable than local quotes for the items that drive the budget.

Labor productivity, for self-performed work. If the CM does some work with its own crews (concrete, rough carpentry, or general labor, for example), it prices it as material plus labor hours times wage rates. Labor hours come from productivity rates: how many square feet of formwork a crew installs per day, for instance. Productivity is where site conditions show up. The same crew is slower on a slope, in heat, or in a tight laydown area.

Two adjustments apply to every unit cost:

- **Location.** Construction costs vary widely by market, driven mainly by labor rates and subcontractor availability. Published data includes location factors for this reason.
- **Time.** A unit cost from last year's project needs escalating to today, and today's estimate needs escalating to the construction midpoint..

6. Target Value Design

The traditional sequence, and its problem

On most projects, cost and design follow a familiar loop: the architect designs, the estimator estimates, the estimate comes in over budget, and the team cuts until it fits. The cutting happens late, under pressure, after most of the design fee is spent. It usually removes the most visible dollars rather than the least valuable ones.

The alternative: design to the budget

Target value design (TVD) reverses the sequence. The budget comes first, divided among the building systems, and the design is developed to fit it. The practice comes from lean construction and is most associated with Integrated Project Delivery, but its core ideas work under CMAR too.

The core practices:

1. **Set the allowable cost.** The owner's business case determines the most the building can cost. On the reference project, that's the underwriting budget: \$24.5 million.
2. **Set a target below it.** Many TVD teams set the target a few percent below the allowable cost, to force innovation and create room for surprises. This project had no room to do that, which is part of how it got into trouble.
3. **Divide the target by system.** Each UniFormat system gets a budget. This is the target allocation.
4. **Assign each system to a small team.** The designer, the engineer, the estimator, and the design-assist trade for that system work together and own its budget.
5. **Estimate continuously.** Each team estimates its system as the design develops, weekly or biweekly, not just at phase ends.
6. **Trade across systems, openly.** If the shell team needs \$200,000 more for the envelope upgrade, it has to be found somewhere else, with the owner's approval, before the design proceeds.

The effect is that cost becomes a design constraint, like the zoning setback or the code's area limit, instead of a surprise that arrives with the estimate.

The target allocation for the reference project

Here's the underwriting budget divided by UniFormat system, next to the DD estimate:

System	Target direct cost	DD direct cost	Variance
A Substructure	\$1,280,000	\$1,770,000	+\$490,000
B Shell	5,900,000	6,630,000	+730,000
C Interiors	4,293,000	4,420,000	+127,000
D Services	4,640,000	5,300,000	+660,000
E Equipment and furnishings	880,000	880,000	—
F Special construction	440,000	440,000	—
G Sitework	2,290,000	2,650,000	+360,000
Direct cost	\$19,723,000	\$22,090,000	+\$2,367,000
With markups (× about 1.24)	\$24,500,000	\$27,440,000	+\$2,940,000

With target value design, this table exists from the first week of SD, and each row has a named owner. The variances would have been visible, system by system, months before the DD estimate. Without it, the same table gets built for the first time in a crisis meeting.

The budget and the estimate use the same structure, so they can be compared line by line.

7. Value Engineering, Done the Way It Was Meant

What VE is supposed to be

Value engineering began as a discipline in manufacturing in the 1940s and was later adopted in construction, notably for public projects. Its formal definition is about **function**: analyze what each element of a building has to do, then find the lowest-cost way to do it without losing that function. A classic VE study happens early in design, with a facilitated team, and it asks questions like "what does this wall need to do, and is there a cheaper wall that does all of it?"

Value is often expressed as function divided by cost. You can improve value by keeping the function and lowering the cost, or by improving the function at the same cost.

What VE usually becomes

In practice, "value engineering" usually means a list of deletions and substitutions produced after an estimate comes in high, sorted by dollars saved, and cut from the top until the number fits. That isn't value engineering. It's cost reduction, and it often reduces value, because it deletes the largest items first whether or not renters care about them.

Three kinds of savings, and why the difference matters

When a team is closing a gap, every proposed saving falls into one of three kinds:

- **Real value engineering:** same function, lower cost. Aligning plumbing stacks so fewer risers are needed. Right-sizing HVAC equipment from actual load calculations instead of rules of thumb. These are almost free in value terms, and design-assist trades are the best source of them.
- **Scope reduction:** less function, lower cost. Deleting the second amenity building. Reducing the parking ratio. These cost something in NOI or leasing and have to be priced through the cap rate.
- **Risk transfer:** the cost doesn't go away, it moves. Converting a line item to an allowance, shifting scope to the owner, or excluding something from the CM's number. These lower the estimate without lowering the cost, and they're the most dangerous items on a VE list because they look like savings.

A disciplined VE log labels every item with its kind. The owner should expect most real savings to come from the first kind, accept the second only with the value math, and treat the third as a warning sign.

Timing

VE is most valuable when it's earliest. A plumbing layout change proposed during SD costs the MEP engineer a few hours. The same change proposed after the GMP costs a redesign, a re-coordination, and possibly a permit revision, and much of the savings goes to paying for those. That's the cost-of-change curve from Module 2.2 applied to VE, and it's why design-assist trades should be engaged by the end of SD.

8. The DD Estimate Comes In 12% Over

What it does to the deal

Accept the \$27.44 million as it is, and the development spread falls below the floor before any construction risk has materialized:

	At underwriting	DD estimate accepted as is
Total development cost	\$35,000,000	\$37,940,000
Stabilized NOI	\$2,280,070	\$2,280,070

Yield on cost	6.51%	6.01%
Spread over the 5.25% cap rate	126 bps	76 bps

At 76 bps, the investors would not have approved the deal. The owner's contingency is still untouched at \$1.25 million, but it now has to cover every construction risk with no headroom behind it.

Where the gap came from: the estimate walk

The first question in a gap meeting is usually "who blew the budget?" The honest answer is almost always "several people, for different reasons." The tool that shows this is an **estimate walk** (also called a bridge or reconciliation): a line-by-line explanation of how one estimate became the next.

From the SD estimate to the DD estimate, with markups included:

Item	Change	Kind
SD estimate	\$25,600,000	
Foundations and slab as designed for the expansive clay	+510,000	Design growth
Detention vault as designed, vs. the \$610,000 diligence allowance	+285,000	Design growth
Deceleration lane and sewer extension as designed	+160,000	Design growth
MEP: larger electrical service, HVAC tonnage, sprinkler layout	+600,000	Design growth
Detailing: blocking, transitions, firestopping now drawn	+150,000	Design growth
Design contingency drawn down (from about 6% to 3%)	-690,000	Contingency
Envelope upgrade retained (it was an SD alternate)	+520,000	Owner decision
Acoustic floor-ceiling upgrade requested by leasing	+195,000	Owner decision

Escalation re-forecast for lumber and electrical	+110,000	Market
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DD estimate	\$27,440,000
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Now combine the walk with the gap that already existed at SD:

Source of the \$2.94 million gap	Amount	Share
SD gap that was never closed	\$1,100,000	37%
Design growth beyond the design contingency (\$1,705,000 – \$690,000)	1,015,000	35%
The owner's own decisions	715,000	24%
Market escalation	110,000	4%
Total	\$2,940,000	100%

This table changes the conversation. The design team's net overrun is about a third of the gap. The largest single piece is the SD gap the owner approved over and chose not to close. A quarter came from decisions the owner made during DD, both of which may be worth making (the envelope upgrade creates \$303,810 of value, and the acoustic upgrade is a leasing requirement). None of this makes the overrun acceptable. It does make the fix a shared project rather than a blame exercise, and it shows exactly where process broke down: the SD reconciliation was never done, and the owner's DD decisions were never traded against anything else.

What the range says

The DD estimate is Class 2 , with a typical range of –4% to +10%. On \$27.44 million, that's about \$26.3 million to \$30.2 million. The DD estimate is the first one reliable enough to act on, and it's telling the team the building doesn't fit.

Closing the gap

The team's options fall into a handful of categories. Here's the plan the reference project adopts, ranked roughly in the order the team should pursue them:

Lever	Cost reduction	NOI effect per year	Condition
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VE List, ranked by net value per dollar	\$885,000	–\$29,000	Owner approves deleting the second amenity building and the window change
Design-assist plumbing: stacked wet walls, PEX with manifolds, fewer risers	310,000	—	Architect adjusts unit plans in two unit types
HVAC equipment right-sized from room-by-room load calculations	240,000	—	MEP engineer confirms comfort and code compliance
Alternative slab and foundation design for the clay	380,000	—	Geotechnical engineer approves in writing; structural engineer redesigns
Detention vault resized with a bioretention and infiltration credit	150,000	—	City stormwater review accepts it without delaying the permit
Early purchase of the lumber package and switchgear	180,000 (expected)	—	Owner funds deposits; savings depend on the market
Total	\$2,145,000	–\$29,000	

That leaves **\$795,000** of the gap open. The team has three choices for the remainder:

1. **Cut further.** The remaining items on the VE list destroy value: reducing parking costs more in NOI than it saves, and the standard appliance package destroys \$1.86 of value per dollar saved. The envelope upgrade is the largest remaining target, and cutting it destroys \$303,810.
2. **Absorb it.** Accept a hard cost of \$25.3 million, using headroom.
3. **Defer it.** Carve items that could be added later (the second amenity building, EV chargers) into bid alternates or post-completion projects, so they can be restored if buyout comes in favorable.

The team recommends absorbing the \$795,000 and pricing the second amenity building as an add-alternate at GMP. Here's what that does to the deal:

	At underwrit ing	DD as is	After the plan
Hard cost	\$24,500,000	\$27,440,000	\$25,295,000
Total development cost	\$35,000,000	\$37,940,000	\$35,795,000
Stabilized NOI	\$2,280,070	\$2,280,070	\$2,251,070
Yield on cost	6.51%	6.01%	6.29%
Spread	126 bps	76 bps	104 bps
Owner contingency	\$1,250,000	\$1,250,000	\$1,250,000
Headroom to the 100 bps floor	\$1,481,120	-\$1,458,880	\$222,120
What the deal can absorb	\$2,731,120	-\$208,880	\$1,472,120

(Soft costs are held constant for simplicity. In practice, fees tied to construction cost would move slightly.)

The deal survives. It clears the floor by 4 basis points, keeps its full contingency, and heads into CD with about half the cushion it started with. That's a realistic result. Most gap-closing exercises end with a deal that still works but has less room for error, and the owner has to decide with open eyes whether that's acceptable.

Two notes on the plan:

- **Three of the six levers carry conditions outside the team's control.** The slab redesign needs the geotechnical engineer's approval. The vault change needs the city's approval without delaying the permit. The early purchase depends on the market. The plan should list each condition, name an owner, and set a date by which the team knows whether it held. If the slab redesign fails, the deal is back below the floor.
- **The owner's two DD decisions stayed in.** The envelope upgrade creates value on its own. The acoustic upgrade costs \$195,000 and needs to protect about \$4.27 per unit per month in rent or retention to pay for itself, which the leasing team believes it does. Neither was cut to make the arithmetic easier, which is the point of ranking by value instead of by dollars.

What should have happened

Nothing in this sequence required a crisis. With target value design, the substructure, services, and sitework variances would have been visible within weeks of SD. With the SD gap reconciled at SD, as the owner's approval required, the design-assist savings and the slab alternative would have been designed in from the start, rather than retrofitted at the end of DD with 35% of the fee spent. And the owner's two DD decisions would have been traded against other scope in the moment, rather than discovered in the walk.

Artifacts

Artifact	Produced by	When	What it's for
Target allocation by system	Owner's rep with the CM	Start of SD	The budget each system is designed to
Phase estimates (SD, DD, 50% CD)	CM	End of each phase	What the design costs, organized by system, then by trade
Qualifications, assumptions, exclusions, and allowances	CM	With each estimate	What the number does and doesn't include
Independent estimate	Owner's cost consultant	DD and GMP, at minimum	A check on the CM's number
Estimate reconciliation	Both estimators	After each independent estimate	Differences explained by scope, quantity, price, markup, and timing
Takeoff records	Estimators	Each estimate	The quantities behind the number, auditable
Estimate walk (bridge)	CM, reviewed by the owner's rep	Each new estimate	How and why the number changed since the last one

UniFormat-to-MasterFormat crosswalk	CM	DD into CD	Keeps system-level cost visible after the estimate is reorganized for buyout
VE log	Owner's rep	Ongoing	Every proposed saving, its kind, its value effect, and its status
Budget reconciliation package	Owner's rep	End of DD	The estimate reconciled to the pro forma, with the plan to close any gap

Phase Approval: Approve the DD Budget

Module 2.2 set out the checklist for approving Design Development. The cost portion is the one most often skipped, so it gets its own gate here.

The question: does the building fit the budget, and if not, is there an approved plan that makes it fit?

- ☐ DD estimate received, with qualifications, exclusions, and allowances
- ☐ Independent estimate received and reconciled to the CM's estimate
- ☐ Estimate walk from SD to DD, with each change labeled by kind
- ☐ Variance by system against the target allocation
- ☐ Gap-closing plan, ranked by net value per dollar, with each item labeled as real VE, scope reduction, or risk transfer
- ☐ Every conditional saving has an owner and a date by which it's confirmed
- ☐ Effect on the pro forma recalculated: total cost, NOI, yield on cost, spread, and remaining cushion
- ☐ Items deferred to alternates identified, so they can be priced at GMP
- ☐ Approved by the developer principal, per the decision authority matrix (Module 2.1), with notice to the lender

Whose money is at risk: the Construction Documents fee, which is the largest single share of the design fee, and every dollar of cushion the plan uses up. After this approval, the documents are built on the reconciled budget.

Failure Modes

Approving a phase over an unreconciled gap. The SD gap on this project was \$1.1 million, and it became 37% of the DD gap because nobody closed it when it was cheap to close.

Reading the number before the fine print. Two estimates can differ by \$1 million because one excluded the utility fees. Qualifications and exclusions come first.

Splitting the difference between two estimates. Reconciliation looks for the reasons behind each difference. Averaging hides them, and whichever estimate was wrong stays wrong.

Trusting the prototype multiplier. Seven identical buildings on a sloped site aren't identical. Take off the prototype, then take off the differences.

Losing the system view at CD. Once the estimate is reorganized by trade, nobody can see what the exterior wall costs anymore unless someone maintains the crosswalk.

Designing first and estimating later. Without target allocations by system, cost growth is invisible until the phase estimate arrives, and by then it's expensive to fix.

Calling risk transfer a saving. Moving scope to an allowance or to the owner lowers the estimate without lowering the cost. It usually comes back later, with markup.

Cutting by dollars saved. The biggest line on the VE list is often the most valuable thing in the building. On this project, it was the envelope upgrade.

Counting conditional savings as certain. The slab redesign, the vault change, and the early purchase each depend on someone outside the team. A plan that assumes all three is a plan with no margin.

Blaming the architect by default. The estimate walk on this project shows the design team's net overrun was about a third of the gap. A process that doesn't find the other two-thirds will repeat them on the next project.

Where AI Is Changing This

Estimating is one of the areas changing fastest. Automated takeoff tools can count and measure from drawings in minutes, conceptual estimating tools can price a massing model against historical cost data, and model-based estimating pulls quantities directly from BIM. Part 3 Module 3.11 covers the estimating and takeoff market, and Module 3.5 covers conceptual cost tools.

The useful questions for any of these tools come from this module. **Does it make cost visible earlier in design**, when changes are cheap, or does it just produce the same late estimate

faster? **Does it produce a qualifications list**, so someone can see what it assumed? And **does it know what isn't drawn?** Automated takeoff measures the drawings precisely. It doesn't know about the blocking, the slope, or the clay unless someone tells it. Faster takeoff makes an estimator more productive. It doesn't replace the judgment about what the drawings leave out.

Exercise

Decision scenarios

These use the reference project's numbers. Reasoning notes follow each one.

Scenario 1: Two estimates, \$1.2 million apart. The owner's cost consultant prices the DD set at \$26.24 million. The CM prices it at \$27.44 million. The developer's first instinct is to tell the CM to "sharpen its pencil" and meet somewhere around \$26.8 million.

Reasoning notes: Averaging doesn't tell you which number is right. Reconcile by bucket. Start with scope: does the consultant's number include the sewer extension, the deceleration lane, and the vault at the designed size, or at the diligence allowances? Then quantities: did both deduct openings, apply similar waste factors, and account for the two walk-out buildings? Then unit prices: which ones come from current quotes? Then markups and escalation. On many projects, most of a \$1.2 million difference turns out to be scope and assumptions, and the "cheaper" estimate is cheaper because it left something out. Pressure on the CM to lower its number without a reconciliation tends to produce exactly the wrong result: the scope goes into allowances and exclusions, and the owner pays for it later.

Scenario 2: The fast way to close the gap. In the gap meeting, the architect proposes deleting the envelope upgrade, deleting the second amenity building, and switching to the cheaper window package. Together they remove about \$980,000, a third of the gap, in one move.

Scenario 3: The CM's offer. The CM offers to commit now to a GMP of \$25.0 million, \$2.44 million below its own DD estimate, if the owner accepts a \$1.9 million sitework allowance and moves the vault and the sewer extension into owner-direct contracts.

Reasoning notes: Label it by kind: this is mostly risk transfer. The sitework allowance is owner exposure inside the CM's number, adjusted to actual cost later. Owner-direct contracts for the vault and sewer move that cost, and its coordination risk, entirely to the owner, outside the GMP. The GMP number goes down; the owner's cost probably doesn't, and its risk goes up. The right comparison isn't \$25.0 million against \$27.44 million. It's the owner's expected total cost under each structure, including the allowance's likely overrun and the coordination risk of managing separate contractors on the busiest part of the site.

Scenario 4: The prototype. The CM's DD estimate took off one Type A building in detail and multiplied it by seven. The site plan shows two Type A buildings on the slope at the east end,

each with a stepped foundation and a taller exposed wall on one side. The CM's estimator hasn't seen the grading plan.

Reasoning notes: Ask for a differential takeoff for the two buildings: the added foundation, the added wall area, and any retaining. On a building like this, the difference might be tens of thousands of dollars per building, which is small against a \$2.94 million gap and large against a \$222,120 headroom after the plan. It also tells you something about the process. If the estimator hadn't seen the grading plan, the estimate walk should include a line for each discipline's documents the estimator actually used, so gaps like this are visible.

Deliverable: Budget Reconciliation Package

For the reference project, or your own project from Part 1, produce the package the owner would use to approve the DD budget.

1. Estimate summary. The DD estimate by UniFormat system, with each markup layer shown separately, and the qualifications, exclusions, and allowances.

2. Target allocation and variance. The underwriting budget divided by system, next to the DD estimate, with the variance for each system.

3. Estimate walk. From the previous estimate to this one, with every change labeled as design growth, contingency, owner decision, market, or scope correction. Then the sources of the total gap, with shares.

4. One takeoff, shown. Take off one assembly from real drawings (a public project's drawings from a building department or procurement portal work well). Show the measurements, deductions, waste factors, unit costs, and extension. Then identify one condition on the site or in the drawings that the prototype multiplier would miss.

5. Gap-closing plan. Every lever, labeled as real VE, scope reduction, or risk transfer, with its cost reduction, its NOI effect, its capitalized value effect, its condition, its owner, and its confirmation date. Ranked by net value per dollar.

6. Effect on the deal. Total development cost, NOI, yield on cost, spread, contingency, headroom, and total cushion: at underwriting, with the estimate as received, and after the plan.

7. Recommendation. One page to the owner. Lead with the recommendation. State which conditional savings the plan depends on, what happens to the spread if each one fails, and which items are carried as alternates for the GMP.

Module 2.4 turns to the documents themselves. The budget is reconciled and the team is producing Construction Documents. Module 2.4 covers how drawings and specifications work together as one set, how to navigate a set you've never seen, and what to look for when you do.

2.4 — Reading the Document Set

Why This Module Exists

Module 2.3 got the budget back under the floor. Now the design team spends the next two months producing the Construction Documents: roughly 150 drawing sheets and a project manual close to a thousand pages. That set is what the building department approves, what the CM prices the GMP from, what forty subcontractors bid, and what every field decision for the next two years gets checked against.

Most people who work around construction can read part of a set. An architect reads the architectural sheets fluently and skims the electrical. A superintendent reads everything that affects sequence and skips the specifications until a problem sends him there. A developer reads the site plan and the unit plans and trusts everyone else with the rest. Almost nobody is taught to read the set as one connected document, and that's where the expensive mistakes live: in the places where one sheet says something different from another sheet, or where the drawing says one thing and the specification says another.

This module is about reading the set that way. It covers:

- How drawings and specifications split the work, and what controls when they disagree
- A routine for getting oriented in a set you've never seen, in about thirty minutes
- The cross-reference chain: following one element from a plan to its details, schedules, and specifications
- What to look for, depending on whose interests you're protecting
- How to tell whether you're looking at the current version

It doesn't try to teach every symbol and every MasterFormat section number. Those are reference material, and they're in the appendices at the end. The skill is navigation and judgment, and it's learned by tracing real sets, which is what this module's deliverable asks you to do.

Where the Project Is

Phase	Construction Documents, about month –4 to –2
Budget	Reconciled at \$25.3 million hard cost, 104 bps of spread (Module 2.3)
Next milestones	Permit submission, 50% CD estimate, GMP (Module 2.5)

An illustrative CD set for the reference project:

Discipline	Sheets	What's in it
General (G)	4	Cover, sheet index, code summary, symbols and abbreviations
Civil (C)	16	Site plan, grading, utilities, the detention vault, sewer extension, deceleration lane, erosion control
Landscape (L)	6	Planting, hardscape, pool deck, site furnishings
Structural (S)	18	Foundation plans for each building type, framing plans, shear walls, details, truss requirements
Architectural (A)	58	Site-related architectural plans, building and unit plans, elevations, sections, wall sections, details, schedules, clubhouse
Fire protection (FP)	4	Sprinkler design criteria, riser details, fire department connection
Plumbing (P)	12	Unit plumbing plans, riser diagrams, site water and gas connections, fixture schedules
Mechanical (M)	10	Unit HVAC plans, equipment schedules, clubhouse systems
Electrical (E)	20	Site electrical, meter centers, unit power and lighting, panel schedules, single-line diagram, fire alarm

Plus the project manual: procurement and contracting requirements (Division 00), general requirements (Division 01), and roughly 90 technical specification sections.

Concepts

1. One Set, Two Halves

How drawings and specifications split the work

The old rule of thumb is that **drawings show quantity, location, and relationship, and specifications describe quality, standards, and workmanship**. It's not a perfect rule, but it's the right starting point for reading a set.

On the reference project, the demising wall between two units shows both halves at work:

The question	Answered by
Where is the wall? How long is it?	Floor plan
What kind of wall is it?	Wall type tag on the plan, defined in the wall type legend
How tall does it go, and how does it meet the floor and roof?	Wall section and head and base details
What fire rating and sound rating does it need?	Wall type legend and code summary
What exact products, in what configuration?	Wall type legend (the tested assembly) and the gypsum board specification
How are they installed? What tolerances? What inspections?	Gypsum board, insulation, and firestopping specifications
What happens where a pipe passes through it?	Firestopping details and specification

Neither half is complete on its own. A contractor who builds from the drawings alone knows where the wall goes but not which gypsum board, what fastener spacing, or what the inspector

will check. A contractor who reads only the specifications knows the products but not where they go. The set is designed to be read together, and the contract says so.

The contract says the documents are complementary

Under the AIA general conditions (A201) and most equivalents, the contract documents are **complementary: what's required by one is as binding as if required by all**. A requirement shown only on the drawings is still a requirement. A requirement written only in the specifications is still a requirement. The contractor can't ignore a specification section because the drawings didn't mention it.

This is why the documents have to be coordinated with each other, not just within each discipline. And it's why a conflict between them is a real problem, not a technicality.

When they disagree

They will. On a set this size, there are always conflicts: a plan dimension that disagrees with a schedule, a detail that shows a product the specification doesn't include, a specification section left over from another project. What controls?

The answer starts with the contract. Many projects include an **order of precedence** clause, usually in the supplementary conditions or Division 01, that ranks the documents. A common version:

1. The agreement and its amendments
2. Addenda (the most recent governs)
3. Supplementary conditions
4. General conditions
5. Specifications
6. Drawings

Within the drawings, common conventions (often stated in the general notes) include:

- **Specific governs over general.** A note on a detail beats a general note that says "typical."
- **Larger scale governs over smaller scale.** An enlarged plan at $1/2" = 1'-0"$ beats the floor plan at $1/8" = 1'-0"$.
- **Written dimensions govern over scaled measurements.** Never scale a drawing when a dimension is written.
- **Schedules govern over plan notes** for the attributes the schedule covers.

Some contracts instead say that where documents conflict, **the more stringent or more expensive requirement governs**. That rule sounds protective for the owner, and it is at bid time. It also means the contractor prices the more expensive option wherever it finds a conflict, which the owner pays for.

Order of precedence resolves disputes. It doesn't make the conflict go away. The right response to a conflict found during design is to fix the documents. The right response to one found during bidding is an addendum. The right response to one found during construction is an RFI (Module 2.7). The contractor's general conditions typically require it to report conflicts it discovers rather than silently choose one, and a contractor who knowingly builds the cheaper side of a conflict takes on the risk of being wrong.

2. How a Set Is Organized

A set follows a standard organization so that anyone in the industry can navigate it. The details are in Appendix A. What matters for reading is this:

Drawings are numbered by discipline, then sheet type

Most U.S. sets follow the **United States National CAD Standard (NCS)**, maintained by the National Institute of Building Sciences. A sheet number like **A-501** tells you:

- **A**: the discipline (architectural)
- **5**: the sheet type (details)
- **01**: the sequence within that type

So once you know that plans are 1, elevations 2, sections 3, large-scale views 4, details 5, and schedules 6, you can guess where to look before you open the index. **S-101** is probably a foundation plan. **A-601** is probably the door schedule. **E-601** is probably panel schedules. Firms vary (some drop the dash, some use three digits, some use their own schemes), which is why the sheet index is always the real map.

The sheets run in a standard order

General first, then site-related disciplines (civil, landscape), then structural, architectural, and the building systems (fire protection, plumbing, mechanical, electrical). Within each discipline, sheets move from general to specific: notes and legends, then plans, elevations, sections, enlarged views, details, and schedules.

The project manual follows MasterFormat

The specifications are organized by **MasterFormat**, published by the Construction Specifications Institute (CSI). Every section has a six-digit number. The first two digits are the division: 03 is concrete, 06 is wood, 07 is thermal and moisture protection, 08 is openings, 09 is finishes, 22 is plumbing, 23 is HVAC, 26 is electrical. So **09 21 16** is gypsum board assemblies, and **08 71 00** is door hardware. Appendix B lists the divisions.

Each technical section follows **SectionFormat**, the same three-part structure:

- **Part 1, General:** what the section covers, referenced standards, submittals required, quality assurance, warranties

- **Part 2, Products:** the acceptable manufacturers and products, and their performance requirements
- **Part 3, Execution:** preparation, installation, field quality control, cleaning, and protection

Once you know that structure, you can find anything in a section in seconds. Want to know what the contractor has to submit for the windows? Part 1 of 08 53 13. Want the acceptable siding manufacturers? Part 2 of 07 46 46. Want the inspection requirements for firestopping? Part 3 of 07 84 00.

Division 00 and Division 01 are where the rules live

The first two divisions aren't about building materials. They're about how the project runs, and they're the most under-read part of most sets.

Division 00, Procurement and Contracting Requirements, contains the bidding instructions, the bid form, the agreement form, the general and supplementary conditions, bond and insurance requirements, and often the geotechnical report as information available to bidders.

Division 01, General Requirements, applies to every trade. It typically covers:

- Summary of work, phasing, and owner-furnished items
- **Allowances, alternates, and unit prices** (Module 2.5)
- **Substitution procedures:** how and when a contractor can propose a different product
- Contract modification procedures
- Payment procedures and the schedule of values
- Project management, coordination, and meetings
- **Submittal procedures** (Module 2.7)
- Quality requirements, testing, and special inspections
- Temporary facilities and controls
- **Closeout procedures**, operation and maintenance data, warranties, and record documents (Module 2.8)

An owner's rep who wants to know how the project will actually be administered should read Division 01 before any technical section. A subcontractor bidding the siding should read Division 01 too, because its requirements for submittals, cleaning, temporary protection, and closeout apply to the siding scope even though the siding section never mentions them.

3. The First Thirty Minutes With an Unfamiliar Set

Whether you're an owner's rep receiving a set, an estimator pricing it, a lender's monitor reviewing it, or a new project engineer inheriting it, you need a quick way to get oriented before diving into detail. Here's a routine that works on almost any building set.

1. Check what you're holding (2 minutes). Read the title block on the cover. What's the issue status? "Schematic Design," "Design Development," "50% CD," "For Permit," "For Pricing," "Issued for Construction," and "Record Set" mean very different things. What's the date? Is

there a revision block, and what's the latest revision? If there are addenda, do you have them? A careful reading of the wrong version is worse than no reading.

2. Read the sheet index (3 minutes). It's the map. Count sheets by discipline. Look for gaps (a set with no fire alarm sheets, or no clubhouse sheets) and for sheets listed but missing from the PDF.

3. Read the project data and code summary (5 minutes). Usually on the cover or the first general sheet. It tells you the building type, construction type, occupancy, sprinkler system, number of stories, building areas, and the code editions the design follows. This is the frame for everything else. If the code summary says Type V-A with NFPA 13R, every rated assembly in the set should be consistent with that.

4. Skim the general notes and legends (5 minutes). General notes set the defaults that apply everywhere ("all dimensions are to face of stud unless noted otherwise"). Legends define the symbols and wall types. Reading these first saves you from misinterpreting the plans.

5. Look at the site plan (3 minutes). Where are the buildings, the parking, the utilities, the vault, and the access points? On a multi-building project, the site plan also shows which building is which type.

6. Look at one typical building and one typical unit (5 minutes). On a garden project, most of the building repeats. Understand one Type A building and one of each unit type, and you understand most of the architecture.

7. Find the schedules (2 minutes). Door, window, finish, fixture, equipment, and panel schedules hold a large share of the information in a set. Know where they are.

8. Open the project manual's table of contents and Division 01 (5 minutes). See which sections exist. Scan Division 01 for allowances, alternates, unit prices, and the order of precedence clause.

After thirty minutes, you should be able to say what the building is, what version of the documents you have, where to find the major information, and what rules govern the project. Everything after that is tracing.

4. The Cross-Reference Chain

The symbols that link sheets together

A set is a web of references. The symbols that create the web are standardized (Appendix C has the common ones), and nearly all of them follow the same logic: **a number identifying the drawing, and a sheet number telling you where to find it.**

- **Section cut:** a line across a plan or elevation with a symbol at the end, pointing in the direction of view. The symbol reads something like "3 / A-301": section 3, on sheet A-301.

- **Detail callout:** a circle or bubble around an area, with a reference like "7 / A-502": detail 7, on sheet A-502.
- **Elevation marker:** a symbol on a plan pointing at walls, referencing interior elevations on another sheet.
- **Enlarged plan reference:** a boundary drawn around part of a plan, pointing to a larger-scale plan elsewhere.
- **Tags:** labels on individual elements that point to a schedule or legend. Door tags point to the door schedule. Window tags point to the window schedule. Wall type tags point to the wall type legend. Room tags point to the finish schedule. Equipment tags point to equipment schedules.
- **Keynotes:** numbers or codes on a drawing that point to a keynote legend, often organized by MasterFormat section, which links the drawing directly to the specifications.
- **Grid lines:** the lettered and numbered column lines that give every location in a building an address. "Grid C-4" means the same place on every discipline's drawings.

A worked trace: the unit demising wall

The demising wall between two units on the reference project is a good first trace. It matters for code (it's a fire-rated separation between dwelling units), for leasing (renters complain about noise through it more than anything else in the building), and for cost (there are thousands of linear feet of it).

Step 1: Start on the plan. On the Type A second-floor plan, A-112, the wall between units 204 and 205 carries a wall type tag: **W3**.

Step 2: Go to the wall type legend. The legend on A-001 defines W3: a one-hour fire-rated, non-load-bearing partition, built as a double row of 2x4 studs with an air gap, one layer of 5/8" Type X gypsum board on each outside face, and sound batt insulation in one stud row. It lists a tested assembly (a UL design number, or equivalent listing from another agency) and a sound transmission rating. It also says the wall runs from the slab to the underside of the roof sheathing, not just to the ceiling.

Step 3: Go to the wall section and details. Wall section 4 on A-302 shows W3 in its vertical context: how it meets the slab, the floor framing between levels, and the roof. Detail callouts point to the head condition at the roof (7 / A-503) and the joint where W3 meets the exterior wall (8 / A-503). These are where the fire rating is most often lost in the field, because the wall has to stay continuous through spaces that are hard to reach once the ceiling is up.

Step 4: Check the openings in the wall. W3 doesn't usually have doors. But the wall between each unit and the breezeway does, and it's also rated. Its door tags (for example, **204A**) point to the door schedule on A-601, which gives the door's size, material, fire rating, and **hardware group**. The hardware group points to specification section 08 71 00, which lists the closer, the lockset, and the smoke seal.

Step 5: Check the penetrations. The plumbing plan, P-112, shows a waste line passing through W3 between two back-to-back bathrooms. The firestopping details on A-510 show how

that penetration is sealed. Section 07 84 00 says what firestop products and tested systems are acceptable and what the special inspector checks.

Step 6: Go to the specifications. Section 09 21 16 (gypsum board assemblies) specifies the Type X board, fasteners, joint treatment, and installation tolerances. Section 07 21 00 (thermal insulation) specifies the sound batts. Section 07 92 00 (joint sealants) specifies the acoustic sealant at the perimeter, which matters for the sound rating. Section 01 45 00 or the statement of special inspections says who inspects what.

That's one wall, and the trace touched two drawing disciplines, five sheets, and five specification sections. If any one of them is inconsistent with the others (the legend shows one layer of board and the detail shows two, or the specification lists a firestop system that isn't tested for this wall type), the building will be priced one way and built another.

Other chains worth knowing

What you're tracing	Starts on	Links through	Ends in the specifications
A window	Elevation or plan tag	Window schedule, head/sill/jamb details	Windows (08 53 13 for vinyl), glazing, flashing, sealants
An HVAC unit	Mechanical plan tag	Equipment schedule, electrical panel schedule, structural or platform support	HVAC equipment (23 81 26 or similar), controls, testing and balancing
A unit's electrical service	Site electrical plan	Meter center detail, single-line diagram, unit panel schedule	Electrical (26 05 19, 26 24 16, 26 27 13, and others)
The detention vault	Civil site plan	Vault plan and sections, structural details, inspection port details	Earthwork, concrete or precast, stormwater (31 and 33 divisions)
A room finish	Room tag on plan	Finish schedule, finish legend, interior elevations	Flooring, paint, tile, base (09 divisions)

A breezeway stair	Building plan	Enlarged stair plan, stair sections, railing details, structural stair framing	Metal fabrications or stairs (05 51 00), railings (05 52 00)
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5. What to Look For

It depends on whose money you're protecting

Everyone reads the same set. They look for different things.

The owner's representative checks the set against the decisions already made. Does it match the OPR and the Basis of Design (Module 2.1)? Does it reflect every item in the phase-gate decision log (Module 2.2)? Did the value engineering from Module 2.3 make it into every document it touches? On the reference project, vinyl plank replaced LVT in the units. Is that change in the finish schedule, the finish legend, the flooring specification, and the interior elevations, or only in some of them?

The estimator looks for scope: what's shown, what's specified, what's missing, and what's ambiguous. Every ambiguity becomes either a qualification, an allowance, or a risk the estimator prices in.

The superintendent looks for buildability and sequence. Can the sprinkler main be installed before the trusses close off access? Is there room to stage materials near the vault while it's under construction? Are the details workable at the pace a garden project runs at?

The lender's construction monitor looks at whether the documents are complete enough to support the budget and the GMP, and whether anything is left to "design later" that could become a cost.

The future property manager, if the owner is wise enough to ask, looks at what will be expensive to operate: equipment access, trash and package logistics, finishes that won't survive turnover, and meters and valves no one will be able to find.

The building department looks for code compliance. Module 2.6 covers the permit review.

Common defects, and why each one costs money

These appear in almost every set. None is exotic.

Broken references. A detail callout points to a detail that doesn't exist, or to the wrong one. Easy to fix during design. During construction, it's an RFI and a wait.

Plans and schedules that disagree. The plan shows a 3'-0" door, and the schedule shows 2'-8". The room tag says one finish and the finish schedule says another. Each one is small. On a project with 200 units, a mismatch on a typical unit is multiplied 200 times.

A change that didn't reach every document. The most common defect after value engineering. The decision was made, one sheet was updated, and the rest weren't. This is exactly what the owner's rep's check against the decision log is for.

Specification sections that don't match the project. Specifications are usually built from a master template and edited for each project. Leftovers happen: a curtain wall section in a garden walk-up, or a reference to a product the drawings don't use. Leftovers are mostly harmless, but bidders sometimes price them. The more dangerous case is the opposite: something drawn with no specification. A pool barrier gate drawn without a specification for its self-closing, self-latching hardware is a code-required element that nobody is contractually told how to provide.

Conflicting information between disciplines. The architectural plan shows a wall where the mechanical plan routes a duct. The electrical plan shows a panel on a wall the plumbing plan uses for a stack. Module 2.2 covers coordination. From a reader's perspective, the practice is to overlay the disciplines in busy areas: the mechanical closets, the breezeways, the utility rooms, and the attic space.

Outdated references. A specification that cites an old edition of a standard, or a code summary that cites a code edition the jurisdiction no longer uses. These get caught in permit review, which costs time.

Soft language. "Verify in field," "coordinate with owner," "by others," "as required," "match existing." Each of these phrases pushes a decision or a cost to someone else. Some are necessary. A set full of them has deferred design decisions to construction, where they cost more. A useful exercise is to search the PDF for "by others" and confirm that each one actually has an owner.

Deferred design without a list. Some elements are intentionally designed later by a specialty subcontractor: the roof trusses, the final sprinkler layout, sometimes the fire alarm. That's normal, but the set should list them as deferred submittals so the building department and the CM know they're coming (Modules 2.2 and 2.6).

6. Knowing Which Version You Have

A set changes after it's issued, and most expensive mistakes involving documents aren't about misreading a sheet. They're about reading the wrong version of it.

How changes are marked

- **Revision clouds:** scalloped outlines drawn around the part of a sheet that changed.
- **Revision triangles (deltas):** a small triangle with a number next to each change, keyed to the revision block.

- **The revision block:** a table in the title block listing each revision's number, date, and description.

A sheet that has been reissued three times should show three entries in its revision block. The current change is clouded. Earlier clouds are usually removed so the sheet stays readable, which is why the revision block, not the clouds, is the record.

What changes the set

Instrument	When	What it does
Addendum	During bidding or GMP pricing	Changes the documents before the price is set; bidders acknowledge it
Permit revision	During or after plan review	Responds to the building department's comments
Bulletin or ASI (architect's supplemental instructions)	During construction	Clarifies or makes minor changes; may lead to a change order
Change order	During construction	Changes scope, price, or time, by agreement
Record drawings	At closeout	Show what was actually built

Modules 2.5 through 2.8 cover each of these. For reading purposes, the rule is simple: **before you rely on a sheet, confirm it's the current version, and confirm you have every addendum and bulletin that might affect it.**

Many projects produce a **conformed set** after the GMP or bid: a clean reissue of every sheet with all addenda incorporated. It's the most reliable starting point for construction, and it's worth asking for if the project doesn't produce one by default.

The permit set and the construction set

The set the building department approved and the set the contractor builds from should be the same. In practice, they drift apart as addenda, bulletins, and change orders accumulate, and not every change gets sent back to the building department. The field inspector compares what's built against the approved set. When they differ, the inspection fails. Module 2.6 covers how to keep the two aligned.

Where models fit

On a BIM project, the drawings are views extracted from the model. The model is usually more complete than any single sheet, and it's invaluable for coordination (Module 2.2). But on most U.S. projects, the **drawings and specifications are the contract documents, not the model**. If the model and the drawings disagree, the drawings govern unless the contract says otherwise. Read the model for understanding. Rely on the documents for obligations.

Artifacts

Artifact	Produced by	When	What it's for
Drawing set with sheet index	Design team	Each issue	The drawn half of the contract documents
Project manual (Divisions 00, 01, and technical sections)	Architect, specification writer, engineers	Each issue	The written half
Wall type legend, schedules, and keynote legends	Design team	DD onward	The link between plans and specifications
Revision log	Architect	Each reissue	The record of what changed and when
Document review comments	Owner's rep, CM, consultants	Before each issue	Defects found before they're priced or built
Decision log cross-check	Owner's rep	Before permit and GMP	Confirms each approved decision appears in every document it touches
Conformed set	Architect	After GMP or bid	A clean set with all addenda incorporated

Quality Gate: Before the Set Is Issued

Module 2.2's CD approval asks whether the documents are ready for permit and pricing. This checklist is how the owner's rep and CM answer that.

- [] Sheet index matches the sheets actually in the set; no missing or orphan sheets
 - [] Every detail, section, and elevation reference resolves to the right drawing
 - [] Door, window, finish, equipment, and panel schedules match the plans
 - [] Wall types on plans match the wall type legend, and each rated type lists a tested assembly
 - [] Every item in the phase-gate decision log (Module 2.2) and the gap-closing plan (Module 2.3) appears in every document it affects
 - [] Specification table of contents matches the scope: no significant drawn element without a section, and no leftover sections for products the project doesn't use
 - [] Division 01 contains the order of precedence, allowances, alternates, unit prices, and the substitution procedure the owner intends
 - [] Code summary cites the editions the jurisdiction currently enforces
 - [] Deferred submittals are listed
 - [] Busy areas (mechanical closets, breezeways, utility rooms, attic) have been checked across disciplines
 - [] "By others," "verify in field," and similar phrases each have a named owner or have been removed
-

Failure Modes

Reading one discipline at a time. Most costly conflicts are between disciplines or between drawings and specifications. A reader who never crosses those lines won't find them.

Relying on the wrong version. The most careful review of a superseded sheet is worthless. Check the title block, the revision block, and the addenda first.

Skipping Division 01. It sets the rules for submittals, substitutions, allowances, and closeout, and it applies to every trade whether or not the trade's own section mentions it.

Scaling the drawings. Written dimensions govern. A PDF's measurement tool is only as accurate as the scale it's set to, and a sheet reduced for printing is no longer at the stated scale.

Assuming "typical" means identical. A typical detail applies where conditions are typical. The two Type A buildings on the slope aren't typical at their foundations, even though the plans are.

Letting a value engineering decision reach only one document. The decision is made, one sheet changes, and the specification still calls for the original product. The contractor prices whichever one the precedence clause favors, and it's often not the cheaper one.

Treating soft language as harmless. Every "by others" is a scope item that someone assumed someone else was carrying.

Trusting the model over the documents. On most projects, the drawings and specifications are the contract. The model is a tool for understanding and coordinating them.

Where AI Is Changing This

Reading and checking documents is one of the areas where software is improving fastest. Tools can now check sheet indexes and references automatically, compare schedules to plans, flag inconsistencies between sheets, compare specifications against submittals, and draft specifications from templates and product databases. Part 3 covers these in Modules 3.8 (coordination and drawing QA) and 3.9 (specifications).

The honest assessment is that the mechanical part of reading a set, following references and comparing tables, is becoming automated. The judgment part isn't. A tool can find that the finish schedule and the flooring specification disagree. It can't know that the disagreement comes from a value engineering decision in Module 2.3, which document reflects the owner's intent, or whether the cheaper product will hold up at unit turnover. The reader who understands how the set is supposed to connect is the one who can use these tools well, and the one who can tell when they're wrong.

Exercise

Decision scenarios

These use the reference project. Reasoning notes follow each one.

Scenario 1: The value engineering that didn't finish. During the 50% CD review, the owner's rep notices that the finish schedule and finish legend show vinyl plank in the units, as approved in the gap-closing plan (Module 2.3). The flooring specification, 09 65 19, still lists the original luxury vinyl tile as the basis of design. The contract's order of precedence clause says the more stringent requirement governs where documents conflict.

Reasoning notes: Under that clause, a bidder reading the set will likely price the LVT, because it's the more expensive, more stringent product. Fix it now: revise the specification before the set goes out for pricing. Then use it as a signal. If one VE item didn't reach every document, check the rest of the gap-closing plan the same way. The decision log cross-check in the quality gate exists for exactly this.

Scenario 2: The drawn element with no specification. The pool enclosure is drawn on the landscape and site sheets, including a gate. There's no specification section for the fence or the gate hardware. The pool barrier's self-closing, self-latching gate is required by code for residential pools.

Reasoning notes: A bidder will price some fence and some gate, probably a standard one. The inspector at closeout will check whether the gate self-closes and self-latches, and whether the latch height and the fence openings meet the barrier requirements. If the installed gate doesn't, it fails, and the pool can't open. On a merchant project timed to the leasing season, a pool that can't open in spring affects leasing velocity at the worst possible time. The fix during CD is an hour of specification writing. The fix at closeout is a replacement gate, a reinspection, and a delay against a CO where a month costs about \$118,000.

Scenario 3: One layer or two? While tracing the demising wall, a reviewer finds that the wall type legend shows W3 with one layer of Type X gypsum board on each face, but the head detail at the roof shows two layers on each face near the top of the wall.

Reasoning notes: Several things could be true. The detail might be correct and deliberate, because some tested assemblies require added layers at specific conditions. It might be copied from another project. Or the legend might be wrong. Only the architect can say, and the tested assembly listing is the tie-breaker. The cost matters for how urgently to resolve it. The reference project has on the order of several thousand linear feet of demising wall. A second layer on both faces across the full height would add something on the order of \$150,000 to \$250,000. If the second layer is only needed near the roof, it's a small cost. If the discrepancy is found in the field after the drywall is hung, it's rework, reinspection, and schedule on the critical path through the units. Send it to the architect as a design question now, and log it.

Scenario 4: "By others." A search of the PDF set for the phrase "by others" finds 23 instances. Most are on the electrical and site sheets: "Transformer pad by others," "Conduit to meter center by others," "Site lighting pole bases by others."

Reasoning notes: Each one names scope that someone believes someone else is carrying. For the transformer pad and conduit, "others" might be the utility, the civil sitework subcontractor, or the electrical subcontractor, and each would price it differently or not at all. On the reference project, this is also the transformer schedule risk from Module 2.1: if the pad and conduit aren't ready when the utility arrives, the power date slips. The owner's rep should build a list of all 23, assign each to a named party in the contract or the utility agreement, and have the documents revised to say who. It's an afternoon of work that prevents a buyout gap and possibly a schedule delay.

Deliverable: A Traced Navigation of a Real Public Document Set

Find a complete construction document set for a real building. Public projects are the best source: school districts, cities and counties, state facilities agencies, universities, and housing authorities often post full bid sets, including specifications and addenda, on their procurement

portals. A multifamily, residential, or small commercial building is ideal. Choose one with at least 50 drawing sheets and a project manual.

Produce the following:

- 1. Set inventory (1 page).** Project name and type, issue status and date, sheet count by discipline, number of specification sections, and a list of addenda if any were issued. Note anything missing from the index or the files.
- 2. Thirty-minute orientation (1 page).** Follow the routine in Section 3. Record what the building is, its construction type and occupancy from the code summary, the code editions cited, and where the major schedules and legends live. Note anything in Division 01 about order of precedence, allowances, alternates, and unit prices.
- 3. Three traces.** For each, list every sheet and specification section you touched, in order, and what each one contributed:
 - One wall or assembly, from a plan tag to its legend, sections, details, and specifications
 - One opening, from a door or window tag to its schedule, details, hardware or glazing, and specifications
 - One building system element, from an equipment tag on an MEP plan to its schedule, its power or connection on another discipline's sheets, and its specification
- 4. Findings log (at least five findings).** For each finding: where it is, which documents are involved, what the conflict or gap is, which document likely controls under the set's order of precedence (or conventions if there isn't one), and a rough estimate of what it would cost to resolve now versus in the field.
- 5. One-page memo.** Written as if you were the owner's representative receiving this set before pricing. Lead with your overall judgment of how ready the set is. Name the three findings that matter most and why. Recommend what should be fixed before the set goes out for pricing, and what can be handled through addenda.

Appendix A — Drawing Organization (NCS)

Discipline designators

Letter	Discipline
G	General (cover, index, code summary, symbols)
H	Hazardous materials
V	Survey and mapping

B	Geotechnical
C	Civil
L	Landscape
S	Structural
A	Architectural
I	Interiors
Q	Equipment
F or FP	Fire protection
P	Plumbing
D	Process
M	Mechanical
E	Electrical
W	Distributed energy
T	Telecommunications
R	Resource
X	Other disciplines
Z	Contractor and shop drawings
O	Operations

Sheet type designators

Digit	Sheet type
0	General: notes, symbols, legends
1	Plans (horizontal views)
2	Elevations (vertical views)
3	Sections (sectional views)
4	Large-scale views: enlarged plans, sections, elevations
5	Details
6	Schedules and diagrams
7	User defined
8	User defined
9	3D representations

Title block contents

- Project name, address, and number
- Owner
- Architect or engineer of record, with the professional seal and signature
- Consultants
- Sheet title and sheet number
- Scale (or "as noted")
- Issue status and date
- Revision block

Appendix B — MasterFormat Divisions

Procurement and contracting requirements group

Division	Title
00	Procurement and Contracting Requirements

Specifications group: general requirements

Division	Title
01	General Requirements

Facility construction subgroup

Division	Title
02	Existing Conditions
03	Concrete
04	Masonry
05	Metals
06	Wood, Plastics, and Composites
07	Thermal and Moisture Protection
08	Openings
09	Finishes
10	Specialties
11	Equipment
12	Furnishings
13	Special Construction
14	Conveying Equipment

Facility services subgroup

Division	Title
21	Fire Suppression
22	Plumbing
23	Heating, Ventilating, and Air Conditioning (HVAC)
25	Integrated Automation
26	Electrical
27	Communications
28	Electronic Safety and Security

Site and infrastructure subgroup

Division	Title
31	Earthwork
32	Exterior Improvements
33	Utilities
34	Transportation
35	Waterway and Marine Construction

Process equipment subgroup

Division	Title
40	Process Interconnections
41–48	Process equipment, by type

Divisions not listed are reserved for future use.

Sections that come up often on a multifamily project

Section	Title
01 25 00	Substitution Procedures
01 33 00	Submittal Procedures
01 40 00	Quality Requirements
01 77 00	Closeout Procedures
03 30 00	Cast-in-Place Concrete
06 10 00	Rough Carpentry
06 17 53	Shop-Fabricated Wood Trusses
07 21 00	Thermal Insulation
07 25 00	Weather Barriers
07 46 46	Fiber-Cement Siding
07 84 00	Firestopping
07 92 00	Joint Sealants
08 71 00	Door Hardware
09 21 16	Gypsum Board Assemblies
09 65 19	Resilient Tile Flooring
09 91 00	Painting
21 13 13	Wet-Pipe Sprinkler Systems
22 40 00	Plumbing Fixtures
26 05 19	Low-Voltage Electrical Power Conductors and Cables
31 23 00	Excavation and Fill
33 40 00	Stormwater Utilities

Appendix C — Common Symbols and Abbreviations

Reference symbols

Symbol	What it does
Section cut	A line with an end symbol, pointing in the direction of view; "3 / A-301"
Detail callout	A circle or bubble around an area; "7 / A-502"
Elevation marker	Points at walls from a plan; references interior elevations
Door tag	Points to a row in the door schedule
Window tag	Points to a type in the window schedule
Wall type tag	Points to a wall type in the legend
Room tag	Room name and number; points to the finish schedule
Keynote	Points to a keynote legend, often by MasterFormat number
Grid bubble	Labels a column line; gives locations a shared address
Revision delta	A numbered triangle marking a change, keyed to the revision block

Abbreviations worth knowing

Abbreviation	Meaning
AFF	Above finished floor
CLR	Clear (usable dimension)
CONT	Continuous
EQ	Equal
GWB / GYP BD	Gypsum wallboard / gypsum board
NIC	Not in contract
NTS	Not to scale

O.C.	On center
SIM	Similar
TYP	Typical
UNO	Unless noted otherwise
VIF	Verify in field

Every set includes its own abbreviation list. When a set's list and this one disagree, the set's list governs.

Appendix D — SectionFormat at a Glance

Part 1 — General

- Summary and related sections
- References (standards cited)
- Administrative requirements (preinstallation meetings, sequencing)
- Submittals (product data, shop drawings, samples, test reports)
- Quality assurance (qualifications, mockups)
- Delivery, storage, and handling
- Field or site conditions
- Warranty

Part 2 — Products

- Manufacturers (named, basis of design, or performance-based)
- Materials and products, with performance requirements
- Fabrication and finishes
- Source quality control

Part 3 — Execution

- Examination (conditions checked before work starts)
- Preparation
- Installation or application
- Field quality control (testing and inspection)
- Adjusting and cleaning
- Protection

Module 2.5 takes the reference project's documents to market. The set is issued for pricing, the CM builds the GMP, and forty subcontractors bid their scopes. Module 2.5 covers what the GMP

contains, how to read what it excludes, how the contract instruments work, and how the CM turns a GMP into subcontracts through buyout.

2.5 — Buying the Work: Procurement, Contracting, and Buyout

Why This Module Exists

The documents from Module 2.4 describe a building. They don't obligate anyone to build it. That happens in procurement, where the design becomes a price, the price becomes a contract, and the contract becomes 40+ subcontracts with the people who will actually do the work.

Procurement is usually taught as the first two steps: bidding and contracting. This module adds the third, **buyout**, because it's where a large share of the real money on a project is won or lost. A GMP is a set of budgets. Buyout is the process of turning each budget into a signed subcontract. On the reference project, the difference between the GMP's budgets and the actual subcontracts moves hundreds of thousands of dollars, and it happens in a few weeks that most owners never see.

This module covers:

- How design documents become contract documents, and why the difference matters
 - The two ways a price gets set: competitive bidding and a negotiated GMP
 - The contract families and the provisions that govern day-to-day work
 - The tools for scope that isn't fully known: allowances, alternates, and unit prices
 - Bonds and insurance
 - How to read and review a GMP
 - Buyout: bid packages, bid leveling, subcontractor selection, and the buyout log
-

Where the Project Is

Phase	Late CD into GMP, about months –2 to 0
Delivery	CMAR, cost of the work plus a fee with a Guaranteed Maximum Price
Budget	\$25,295,000 hard cost (Module 2.3)
Cushion	\$1,250,000 owner contingency plus \$222,120 headroom = \$1,472,120

Documents	90% CD set, permit submitted (Module 2.6)
What happens next	The CM submits the GMP; the owner reviews and negotiates it; the loan closes; buyout completes; construction starts at month 0

The headroom is thin. A GMP that comes in \$222,121 over the budget puts the deal below the 100 bps floor before a shovel hits the ground.

Concepts

1. From Design Documents to Contract Documents

Two overlapping sets

Bidding documents (or, on a GMP project, pricing documents) are what the contractor prices from. **Contract documents** are what the contractor is obligated to build. They overlap heavily but aren't the same.

In the bidding documents but not the contract:

- Invitation or advertisement for bids
- Instructions to bidders
- Bid form (the price gets carried into the agreement; the form itself doesn't)
- Bid bond (expires once the contract is signed)
- Prequalification documents

In both:

- Drawings
- Specifications (Divisions 01 onward)
- General conditions and supplementary conditions
- Addenda issued before the price was set

In the contract but not the bidding documents:

- The executed agreement
- The payment and performance bonds
- Insurance certificates
- Later modifications: change orders, construction change directives, and minor changes

Why the distinction matters

The contract documents are the legal baseline for what must be built. Everything in them is the contractor's obligation. Everything outside them isn't, regardless of what anyone assumed, discussed, or intended.

This has consequences at every stage:

- **During pricing**, a contractor prices only what's in the documents. If a requirement was in the architect's head but not on the drawings or in the specifications, the contractor didn't price it. After award, it becomes a change order.
- **Addenda bridge the two sets**. An addendum issued during pricing becomes part of the contract. If a contractor misses one, its price doesn't reflect the current documents, and the gap becomes a dispute after award. That's why bid forms require the contractor to acknowledge each addendum by number.
- **During construction**, shop drawings, product data, samples, and the architect's field reports are **not** contract documents. The architect's approval of a submittal confirms that the product conforms to the contract as it already exists. It doesn't change the contract. If an approved submittal turns out not to meet the specification, the specification still controls.
- **Only formal modifications change the contract**. A change order, a construction change directive, or an order for a minor change (often issued as an architect's supplemental instruction). A verbal agreement, an email, a meeting minute, or an approved submittal can't substitute for any of them.

2. Two Ways the Price Gets Set

Competitive bidding (Design-Bid-Build, lump sum)

Most public work and some private work is bought through competitive bidding on complete documents. The process:

1. **Advertisement or invitation**. Public owners must advertise publicly. Private owners usually invite a short list.
2. **Prequalification**. The owner screens bidders for experience, financial capacity, bonding capacity, safety record, and references. Public owners may prequalify or may evaluate these at award.
3. **Pre-bid conference and site visit**. The owner and architect walk bidders through the project. Anything said here is **not binding** unless it's confirmed in writing by addendum.
4. **Questions and addenda**. Bidders submit questions in writing. The architect answers by addendum, issued to every bidder. Addenda can also carry design changes and responses to permit comments. If a significant addendum lands late, the bid date should move.
5. **Bid day**. Bids arrive by a fixed deadline, often opened publicly on public work. Each bid includes the base price, any alternates, unit prices, addendum acknowledgments, and a bid bond.
6. **Evaluation**. The owner checks two things. A bid is **responsive** if it conforms to the bidding requirements: the right form, all addenda acknowledged, the bid bond attached. A bidder is **responsible** if it's capable of performing: qualified, financially sound, and

bondable. Public owners must generally award to the lowest responsive, responsible bidder.

7. **Award and contract.** The agreement is signed, bonds and insurance certificates are delivered, and the owner issues a notice to proceed.

The results are usually recorded in a **bid tabulation**: every bidder's base bid, alternates, and unit prices, side by side. Public agencies often post their bid tabs, which makes them a useful learning resource (see the deliverable).

A negotiated GMP (CMAR)

The reference project doesn't bid the whole building to general contractors. The CM was selected in Module 2.1 through a competitive proposal on fee, general conditions, qualifications, and team. It has been estimating the project since SD. Now it converts those estimates into a Guaranteed Maximum Price.

1. **The GMP documents.** The owner and CM agree on which document set the GMP is based on. Here, it's the 90% CD set plus any clarifications.
2. **Early trade pricing.** The CM has already bid some major trades (sitework, concrete, framing) to real subcontractors. The rest are priced from the CM's estimates and budget quotes.
3. **The GMP proposal.** The CM submits the price, broken into the cost of the work, contingency, general conditions, insurance, bonds, and fee, with a written list of qualifications, assumptions, exclusions, and allowances.
4. **Owner review.** The owner's rep, often with an independent cost consultant, reviews every part of it (Section 7).
5. **Negotiation and amendment.** The agreed GMP is added to the CM's contract by amendment, and the CM moves from preconstruction services to construction.
6. **Buyout.** The CM bids the remaining trades and signs subcontracts (Section 8).

The difference in philosophy matters. In a lump-sum bid, the contractor's price is fixed, and the owner rarely sees how it was built. Under a GMP, the owner pays the actual cost of the work up to the ceiling, which is why it's entitled to see the math, and why buyout results matter to the owner.

3. The Contract Families

AIA documents

The American Institute of Architects publishes the most widely used standard contracts in U.S. construction. They're organized by series:

Series	Relationship	Documents you'll meet
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A	Owner–contractor, contractor–subcontr actor	A101 (lump sum), A102 (cost plus fee with GMP), A133 (CM at risk), A141 (design-build), A201 (general conditions), A401 (subcontract)
B	Owner–architect	B101 (standard owner-architect agreement)
C	Architect–consultant, and multi-party	C401 (architect-consultant), C191 (multi-party IPD)
G	Administrative forms	G701 (change order), G702/G703 (payment application), G704 (substantial completion), G710 (supplemental instructions), G714 (construction change directive)

On the reference project, the owner signed a B101 with the architect (Module 2.1), the architect signed C401s with its consultants, and the owner and CM are signing an A133, which incorporates the A201 general conditions. The CM will use A401-based subcontracts with its trades.

The general conditions carry the day-to-day rules

The agreement (A101, A133) sets the price, the schedule, and the list of contract documents. The general conditions (A201) set the rules for everything else, and they're referenced far more often during construction. The provisions that come up most:

- **Contract documents are complementary** (Article 1). What one requires, all require. Submittals and field reports aren't contract documents.
- **The contractor must study the documents and report errors** (Section 3.2). The contractor can't silently build a condition it knows is wrong and later say it was just following the drawings.
- **Submittal review doesn't relieve the contractor** (Section 3.12). The architect's approval stamp doesn't shift responsibility for errors the contractor introduced.
- **Compliance with laws** (Section 3.7). The contractor must build per the documents *and* comply with applicable law. If it knowingly builds something unlawful, even if the drawings show it, it bears the cost.
- **The architect administers the contract** (Article 4). It's the owner's representative, interprets the documents, and judges performance. It isn't neutral, but it has authority to resolve ambiguities quickly enough to keep the job moving.
- **Changes** (Article 7). Change orders need all three signatures. Construction change directives let the owner and architect direct a change before the price is agreed. Module 2.7 covers the process.
- **Payment and completion** (Article 9). Payment applications, retainage, substantial completion, and final payment. Modules 2.7 and 2.8.

- **Insurance and bonds** (Article 11).
- **Claims and disputes** (Article 15). Usually mediation first, then arbitration or litigation as the agreement specifies.

ConsensusDocs

ConsensusDocs is an alternative family of standard contracts, developed by a coalition of more than 40 industry associations, including the Associated General Contractors of America. The AIA is not part of it. Its general conditions (ConsensusDocs 200) differ from A201 in a few notable ways: the architect doesn't serve as the initial decision maker on disputes, consequential damages are mutually waived by default, and several provisions are generally viewed as more favorable to contractors and subcontractors.

Common equivalents:

ConsensusDocs	Roughly equivalent to	Use
200	A201	General conditions, lump sum
500	A133	CM at risk
750	A401	Contractor–subcontractor
410 series	B101	Owner–design professional

Owner-drafted contracts and the lender

Large developers, institutions, and public agencies often use their own contracts, or heavily modify AIA or ConsensusDocs forms. Either way, the construction lender will review the contract before it closes the loan. Lenders commonly require a GMP or lump sum, a payment and performance bond or equivalent protection, the right to take over the contract if the owner defaults (a consent and assignment signed by the contractor and architect), and limits on the owner's ability to approve change orders without lender consent.

4. Tools for Scope That Isn't Fully Known

No document set is perfect, and some decisions can't or shouldn't be made before the price is set. Four tools handle this. Here's how they appear in the contract.

Allowances. A dollar amount carried in the price for scope that isn't fully designed. The contractor includes the allowance; the final cost is adjusted up or down when the actual cost is known. On the reference project, the landscaping, the access control system, and the site signage are allowances. **The owner holds the risk of an allowance running over.** A GMP with \$695,000 of allowances is a guaranteed price on the rest and an estimate on the \$695,000.

Check how the contract treats markup on allowance overruns, and whether the allowance covers material only or material and installation.

Alternates. A defined change to the base scope, priced separately at bid or GMP. The owner can accept or reject each one, usually within a set time after award. Alternates can be additive (add the second amenity building) or deductive (delete the pool shade structures). They buy the option to decide later at a competitive price. The same scope added after award arrives as a change order with no competition and a markup. The reference project carries the second amenity building as an add-alternate (Module 2.3).

Unit prices. A price per unit for work whose quantity is uncertain, such as rock excavation or unsuitable soil. The contract pays for the actual quantity. On the reference project, over-excavation for the clay is bought by the cubic yard, because the geotechnical borings can't say exactly how deep the bad soil runs.

Owner-furnished items. Items the owner buys directly and the contractor installs, or that the owner buys and installs through a separate contract. Common examples are appliances, some finishes, and long-lead equipment such as switchgear bought early (Module 2.3). Owner-furnished items can save the contractor's markup and lock in prices, but the owner takes on delivery, damage, and coordination risk.

5. Bonds and Insurance

Bonds

A construction bond is a three-party guarantee. The **surety** (a bonding company) guarantees to the **obligee** (the owner) that the **principal** (the contractor) will perform. A bond isn't insurance for the contractor. If the surety pays a claim, it has the right to recover from the contractor.

Bond	Guarantees	Typical amount
Bid bond	The bidder will sign the contract and provide the other bonds if selected	5–10% of the bid
Performance bond	The contractor will complete the work per the contract	100% of the contract
Payment bond	The contractor will pay its subcontractors and suppliers	100% of the contract

Public projects require performance and payment bonds by statute: the Miller Act for federal work and "Little Miller Acts" in the states, because subcontractors can't place liens on public property. On private projects, bonds are optional, but lenders usually require them or an equivalent.

Bonding also works as a screening tool. A surety won't bond a contractor it doesn't trust to finish, and every contractor has a **bonding capacity** that limits how much bonded work it can carry at once. A subcontractor's bonding capacity tells you something about its financial strength before you hire it.

On the reference project, the performance and payment bond costs about \$170,000, about 0.7% of the GMP.

Subcontractor default insurance (SDI) is a common alternative for the CM's subcontractors. Instead of requiring each subcontractor to provide its own bond, the CM buys an insurance policy that covers the cost of a subcontractor default. It's usually cheaper than bonding every trade, and it gives the CM control over the claim. The owner should know which trades are bonded, which are covered by SDI, and how the SDI deductible is treated in the GMP.

Insurance		
Coverage	Who carries it	What it covers
Commercial general liability (CGL)	Contractor and subcontractors	Third-party bodily injury and property damage from their operations, including completed operations
Workers' compensation	Contractor and subcontractors	Their own employees' workplace injuries
Professional liability (errors and omissions)	Architect and engineers	Claims from design errors below the standard of care
Builder's risk	Usually the owner	Physical damage to the building under construction and materials on site
Umbrella or excess	Contractor, sometimes the owner	Limits above the primary policies

Two contract provisions come up in every negotiation:

- **Additional insured.** The owner (and often the architect and the lender) is named as an additional insured on the contractor's CGL policy, so the contractor's coverage protects them against claims arising from the contractor's work.
- **Waiver of subrogation.** The parties waive their insurers' rights to sue each other for losses covered by insurance, particularly builder's risk. Without it, the owner's builder's risk carrier could pay a fire claim and then sue the contractor who caused the fire.

On the reference project, builder's risk deserves attention. Wood-frame buildings under construction are highly vulnerable to fire, and builder's risk premiums for wood-frame multifamily have risen sharply in recent years. Insurers often require site security, fire watch, standpipe and hydrant availability during framing, and limits on hot work. Those requirements cost money and belong in the general conditions or the owner's budget, so they need to be known before the GMP is set.

Some large projects use a **wrap-up** insurance program, where the owner (OCIP) or the contractor (CCIP) buys one policy covering every party on the site instead of each subcontractor carrying its own. Wrap-ups can reduce cost and gaps in coverage but take effort to administer. They're more common on larger projects than this one.

6. Changes Before and After the Price

The way a change enters the documents depends on whether the price has been set.

When	Instrument	What it does
During bidding or GMP pricing	Addendum	Changes the documents before the price is set; every bidder prices it
After award, no change in cost or time	Order for a minor change (often an architect's supplemental instruction)	Clarifies or makes a minor change the architect can order
After award, price not yet agreed	Construction change directive	Owner and architect direct the change; price and time are settled later
After award, price agreed	Change order	Signed by owner, contractor, and architect; changes scope, price, or time

The same change costs much less as an addendum than as a change order. Before the price is set, the change is priced competitively (or at least transparently) as part of the whole. After the price is set, the contractor is the only one who can do the work, and the change arrives with markup and often with a schedule claim. This is the cost-of-change curve from Module 2.2 applied to procurement.

The practical rule for the owner: **anything you know about before the GMP should be in the GMP**. Permit comments, value engineering decisions, owner changes, and fixes from the

document review in Module 2.4 all belong in an addendum or a GMP clarification, not in a change order after the amendment is signed.

7. Reading a GMP (Guaranteed Maximum Price contract)

What a GMP contains

The CM's GMP proposal on the reference project:

Component	What it is	Amount
Cost of the work	Subcontracts, self-performed work, materials, and equipment; includes \$695,000 of allowances and the over-excavation unit price	\$21,530,000
CM construction contingency (2.5%)	The CM's contingency for buyout gaps, coordination, and its own errors	538,250
Design completion contingency	For scope not shown in the 90% CD set	250,000
General conditions	24 months of site staff, trailers, temporary utilities, and equipment	1,650,000
Insurance (1.5%) and bond (0.7%)	CM's liability insurance and the payment and performance bond	529,818
Fee (4.0%)	CM overhead and profit	979,923
GMP as proposed		\$25,477,991

That's \$182,991 over the \$25,295,000 budget from Module 2.3, which leaves \$39,129 of headroom before the deal hits the floor.

The fine print

The GMP comes with a list that matters as much as the number:

- **Qualifications and clarifications:** how the CM interpreted the documents. "Assumes over-excavation to 4 feet, with additional depth paid at the unit price." "Assumes the owner's builder's risk policy covers the CM's materials on site."
- **Assumptions:** conditions the price depends on. "Assumes construction start by month 0." "Assumes one permit for all buildings."
- **Exclusions:** scope the price doesn't include at all. On this GMP: "Excludes utility company fees, the transformer pad, and primary conduit to the transformer."
- **Allowances:** listed individually with their amounts.
- **Contingency use:** what the CM can spend its contingency on, and whether it needs the owner's approval.

What the owner's review found

The owner's rep and the independent cost consultant review the proposal line by line. On the reference project, they find six issues:

1. A contingency that shouldn't still be there. The \$250,000 design completion contingency is for scope not shown in the 90% CD set. But the CM also has a 2.5% construction contingency, and the owner has \$1.25 million outside the GMP. Either the remaining 10% of the documents contain something specific the CM can name and price, or the \$250,000 is a cushion that, under shared savings, returns half to the CM if unused. **Resolution:** removed. Any genuine design completion items get priced as specific line items.

2. An allowance set too low. The access control allowance is \$210,000. The cost consultant's estimate, based on the specified system and 200 units, is \$290,000. A low allowance makes the GMP look cheaper, but the owner pays the overrun later, possibly with markup. **Resolution:** raised to \$290,000, which adds \$80,000 to the cost of the work but gives an honest number.

3. An exclusion that belongs inside. The transformer pad and primary conduit are excluded. Module 2.4 found them marked "by others" on the drawings. The utility won't build them. Excluding them from the GMP leaves them unassigned, and they sit on the critical path to permanent power (Module 2.1). **Resolution:** moved into the GMP as a sitework line item at \$95,000.

4. General conditions for a month too long. The general conditions assume 24 months, but the CM's own baseline schedule shows substantial completion in 23. **Resolution:** reduced by one month, \$68,750.

5. Self-performed work without competition. The CM plans to self-perform the concrete foundations and slabs. That can be efficient, but it's the one trade the owner can't see competitively bid. **Resolution:** the CM will obtain two outside concrete bids for comparison and price its self-performed work open-book, with its labor rates, equipment rates, and material quotes available for audit.

6. Unbought scope and escalation. About 55% of the cost of the work has been bid to subcontractors. The remaining 45% is still carried at estimate, exposed to escalation at roughly \$40,000 a month of delay until it's bought. **Resolution:** a buyout schedule is attached to the GMP amendment, with the long-lead and volatile packages first, and the owner will receive the buyout log (Section 8) monthly.

The negotiated GMP

Component	As proposed	After review
Cost of the work	\$21,530,000	\$21,705,000
CM construction contingency	538,250	542,625
Design completion contingency	250,000	—
General conditions	1,650,000	1,581,250
Insurance and bond	529,818	526,737
Fee	979,923	974,224
GMP	\$25,477,991	\$25,329,837
Over the Module 2.3 budget	\$182,991	\$34,837
Headroom left before the floor	\$39,129	\$187,283

The negotiated GMP is \$148,154 lower than the proposal, and it's more honest: the access control allowance is realistic, the transformer pad has an owner, and the phantom contingency is gone. The deal still clears the floor, with \$187,283 of headroom and the full \$1.25 million owner contingency behind it.

Notice what the review didn't do. It didn't ask the CM to "sharpen its pencil" on the fee or cut subcontractor prices that were already bid. Most of the value in a GMP review comes from scope, contingency, and allowances, not from squeezing the percentages.

8. Buyout

What buyout is

The GMP is a set of budgets. **Buyout** is the process of turning each budget into a signed subcontract or purchase order. It typically starts before the GMP (the major trades are often bid during CD) and continues into the first months of construction.

For the CM, buyout determines how much margin it actually earns and whether its contingency survives. For the owner, buyout determines whether the savings or overruns inside the GMP flow back to the project, and whether the subcontractors doing the work are capable of finishing it.

The steps

1. Package the work. The CM divides the project into **bid packages**, usually along MasterFormat lines and the way local subcontractors are organized: sitework, concrete, framing, trusses, roofing, windows, siding, plumbing, HVAC, electrical, fire sprinkler, drywall, painting, flooring, cabinets and countertops, appliances, and so on. A garden project like this one might have 30 to 45 packages.

2. Write scope sheets. Each package gets a **scope sheet** that lists exactly what the subcontractor is responsible for, what it isn't, and where its work meets the next trade's. This is where buyout is won or lost. The drawings and specifications describe the building, not who does what. The scope sheet has to answer questions like:

- Who provides the blocking in the walls for cabinets, grab bars, and TV mounts: the framer or the cabinet installer?
- Who installs the windows: the window supplier, the framer, or the siding contractor?
- Who does the firestopping at penetrations: each trade that makes a penetration, or a single firestopping subcontractor?
- Who provides the crane and hoisting?
- Who cleans up, and who protects finished work from the next trade?
- Who provides temporary power and temporary heat, and who pays for them?

Every question that doesn't have a written answer becomes a **scope gap**, which is either paid twice (two trades both carry it) or not at all (neither does). Scope gaps are the most common source of early change orders and CM contingency use.

3. Prequalify the bidders. The CM screens subcontractors before inviting them to bid. It checks bonding capacity or SDI eligibility, financial statements, current backlog, safety record (often the **experience modification rate**, an insurance-based safety score), references on similar projects, and whether the firm's crew size can meet the schedule.

4. Solicit bids. The CM sends each package to three to five qualified bidders, with the documents, the scope sheet, the schedule, and the subcontract terms.

5. Level the bids. Bids never arrive on the same basis. One excludes blocking. Another excludes hoisting. Another is priced on an old addendum. **Bid leveling** adjusts each bid to the same scope so they can be compared fairly.

6. Select and negotiate. The CM chooses a subcontractor, usually the lowest leveled bid from a qualified firm, though best value can justify a higher number. Under many CMAR contracts, the owner has the right to review or approve major subcontractor selections.

7. Sign the subcontract. The subcontract flows down the prime contract's obligations to the subcontractor and adds its own terms: scope, price, schedule, payment terms, retainage, insurance, bonding or SDI, warranty, and change order procedures.

A worked example: leveling the framing bids

Framing is the largest single trade package on the reference project, with a GMP budget of \$3,250,000. Four framing subcontractors bid:

Bidder	Base bid	Adjustments to common scope	Leveled bid
A	\$3,020,000	+\$145,000 (excludes blocking, \$85,000; excludes hoisting, \$60,000)	\$3,165,000
B	3,240,000	— (complete as scoped)	3,240,000
C	3,110,000	+\$72,000 (excludes window installation)	3,182,000
D	3,180,000	+\$85,000 (excludes blocking)	3,265,000

The low base bid (A, \$3,020,000) is \$220,000 under B's. After leveling, the spread between A and B is only \$75,000, and C is just \$17,000 behind A. The CM's estimator priced the missing scope from its own estimate and the other bidders' breakouts.

Leveling is only half the evaluation. The rest is the bidder:

- **Bidder A** is the leveled low bid, but it's currently framing three other projects and its bonding capacity is nearly used up. Framing is on the critical path. A framer that can't staff the job loses months, and on this project a month costs about \$118,000 at the end and more in the interest reserve.
- **Bidder C** is \$17,000 higher after leveling and has the crew capacity. Its exclusion (window installation) has to go somewhere: the window supplier offers installation for about the same amount.

Scenario 1 asks you to make the call.

The buyout log

The buyout log tracks every package: its GMP budget, the subcontract amount, and the difference. It's the most important cost report during the first months of construction, and the owner should see it monthly.

The reference project's buyout log at month 2:

Package	GMP budget	Subcontract	Variance
Sitework and utilities, including the vault	\$2,480,000	\$2,395,000	(\$85,000)
Concrete (self-performed, open-book)	1,690,000	1,712,000	22,000
Framing	3,250,000	3,165,000	(85,000)
Roof trusses	610,000	598,000	(12,000)
Windows	720,000	702,000	(18,000)
Siding	1,380,000	1,310,000	(70,000)
Plumbing	1,640,000	1,585,000	(55,000)
HVAC	1,310,000	1,284,000	(26,000)
Electrical	1,720,000	1,860,000	140,000
Drywall	1,590,000	1,548,000	(42,000)
Bought to date	\$16,390,000	\$16,159,000	(\$231,000)
Not yet bought (finishes, cabinets, flooring, sprinkler, fire alarm, appliances, allowances)	5,315,000	—	—

Ten packages, covering about 75% of the cost of the work, have been bought \$231,000 under budget. That's the good news. Read the log more carefully and two things stand out.

The electrical package came in \$140,000 over. Copper and electrical gear prices moved between the estimate and the bid, even though the switchgear itself was bought early (Module 2.3). The electrical subcontractor's other material didn't have the same protection.

The savings are net. The \$231,000 is nine packages under and one over. If the CM uses the savings from framing and siding to cover the electrical overrun, the owner only sees the net. That's legitimate under most GMP contracts, but the owner should see every line, because a pattern of overruns in one trade often predicts trouble later.

Where buyout money is won and lost

Scope gaps and overlaps. The single largest source. A clear scope sheet is worth more than a hard negotiation.

Timing. Every month a package stays unbought, its price floats with the market. Long-lead and volatile items (electrical gear, HVAC equipment, windows, trusses, lumber) should be bought first.

Subcontractor selection. The low bidder is only cheapest if it finishes. A default mid-project costs far more than the difference between bids: a replacement subcontractor at a premium, a delay, and a claim on the bond or SDI policy.

Subcontract terms. Retainage, payment timing, and change order markups all affect the price a subcontractor bids and how it behaves later. A subcontractor paid 60 days after it bills prices that in.

Buyout savings, and who keeps them. Under a lump-sum contract, buyout savings belong to the contractor; that's the point of a fixed price. Under a GMP, the contract has to say what happens to them. Common arrangements include returning them to the owner as they occur, moving them into the CM's contingency, or holding them until the end and splitting them under the shared savings clause. The dark side of shared savings: a GMP padded by \$300,000 returns the padding as "savings" at the end and keeps half. The owner's independent review of the GMP (Section 7) is the defense.

9. Regulatory Approvals and Procurement Timing

The building permit and the price interact. On the reference project, the permit application was submitted at about month -3 (Module 2.6), overlapping the end of CD and the GMP. There are three possible sequences:

- **Permit before pricing (cleanest).** The GMP is based on approved documents, and no permit comments arrive after the price is set. It adds time, because pricing waits for plan review.
- **Permit during pricing (most common).** Plan review comments arrive while the CM is building the GMP. They go into the GMP as clarifications or into the documents as addenda. This is what the reference project does, and it works when the comments are minor.
- **Permit after the price (riskiest).** Comments that arrive after the GMP is signed become change orders. A comment that changes something fundamental can cost several times more than it would have as an addendum.

The GMP amendment should state which document set and which permit comments it includes. Permit conditions, such as special inspections and deferred submittals, become the CM's obligations through the general conditions and Division 01. Module 2.6 covers the permit path in depth.

Artifacts

Artifact	Produced by	When	What it's for
Bidding or pricing documents	Architect, owner's counsel	Before pricing	The basis of the price
Addenda	Architect	During pricing	Changes before the price is set
Bid tabulation (lump sum)	Owner	Bid day	Every bid, alternate, and unit price, side by side
GMP proposal with qualifications, exclusions, and allowances	CM	End of CD	The price and its conditions
GMP review memo	Owner's rep with the cost consultant	Before acceptance	Issues found, and the recommended resolution
GMP amendment	Owner and CM	GMP acceptance	The agreed price added to the contract
Bonds and insurance certificates	CM, sureties, insurers	Before notice to proceed	Performance, payment, and coverage
Scope sheets	CM	Before each bid package	Exactly what each trade carries
Bid leveling sheets	CM	Each package	Bids adjusted to a common scope

Subcontracts and purchase orders	CM	Buyout	The actual obligations of each trade
Buyout log	CM, shared with the owner	Monthly through buyout	Budget versus subcontract for every package

Phase Approval: Accept the GMP

The question: is this a price the owner can accept, and does the contract say what the owner thinks it says?

- ☐ GMP is based on an identified document set, with all addenda and permit comments listed
- ☐ Independent estimate received and reconciled to the GMP
- ☐ Every contingency inside the GMP has a stated purpose, and design contingency is near zero
- ☐ Allowances checked against independent estimates; markup on overruns defined
- ☐ Exclusions reviewed; every excluded item has an owner and a budget somewhere
- ☐ Qualifications and assumptions reviewed; none contradicts the documents or the OPR
- ☐ Alternates and unit prices listed, with acceptance deadlines
- ☐ General conditions duration matches the baseline schedule
- ☐ Self-performed work priced open-book or tested against outside bids
- ☐ Buyout schedule attached, with long-lead and volatile packages first
- ☐ Treatment of buyout savings and CM contingency written into the amendment
- ☐ Bonds, SDI, and insurance requirements confirmed, including builder's risk conditions for wood frame
- ☐ Effect on the pro forma recalculated: total cost, spread, and remaining cushion
- ☐ Lender's requirements met, including its review of the GMP and its consent documents

Whose money is at risk: from this point, the equity is committed and the loan starts funding. The GMP caps the cost of the work within its scope. Everything outside it (owner changes, unforeseen site conditions, design errors, allowance overruns, and excluded scope) is the owner's, and the owner's cushion is now \$1.25 million of contingency plus \$187,283 of headroom.

Failure Modes

Pricing the wrong documents. A GMP that doesn't list its document set and addenda can't be enforced against the set the contractor is actually building from.

Reading the number before the fine print. Exclusions, allowances, and qualifications move more money than the fee percentage.

Leaving a contingency in the GMP that the owner already carries. Two contingencies covering the same risk means paying for it twice, and under shared savings, paying the CM a share of the unused one.

Accepting low allowances. They make the GMP look cheaper and move the risk to the owner.

Excluding scope without assigning it. "By others" on the drawings plus an exclusion in the GMP equals scope that nobody is contracted to build.

Changing the documents after the price is set. Anything known before the GMP belongs in the GMP.

Choosing subcontractors on base bid. The low base bid on the framing package was \$220,000 under the complete bid. The leveled difference was \$75,000, and the low bidder may not have the crews.

Writing thin scope sheets. Blocking, window installation, firestopping, hoisting, and cleanup don't belong to any one specification section. They belong to whoever the scope sheet says they do.

Buying late. Unbought scope floats with the market. The electrical package on this project shows what that looks like.

Letting buyout savings disappear. Under a GMP, savings belong to the project until the contract says otherwise. If the owner never sees the buyout log, it never sees the savings.

Where AI Is Changing This

Procurement is changing in three places. Bid management platforms already handle invitations, prequalification, and bid collection for most large contractors, and newer tools level bids automatically by extracting inclusions and exclusions from each bidder's proposal. Contract review tools flag risky or unusual clauses, like a missing waiver of subrogation or an uncapped indemnity, before signing. And estimating tools make it faster to check a GMP against independent quantities. Part 3 covers these in Modules 3.11 (estimating and takeoff) and 3.12 (bidding, prequalification, and contracts).

The judgment that remains is the part this module emphasizes. A tool can find that one framing bid excludes blocking. It can't tell you whether the leveled low bidder has the crews to hold the schedule, or whether the access control allowance is realistic for the system the owner actually wants.

Exercise

Decision scenarios

These use the reference project. Reasoning notes follow each one.

Scenario 1: Which framer? The framing bids are leveled as shown in Section 8. Bidder A is \$17,000 below Bidder C after leveling, but A is framing three other projects and has limited bonding capacity left. Bidder C has capacity, and the window supplier will install the windows for about the \$72,000 C excluded. The CM's policy is to award to the leveled low bidder unless there's a documented reason not to.

Reasoning notes: \$17,000 is about 0.5% of the package. Framing sits on the critical path for every building, and a framer that can't staff the job costs months. Using this project's numbers, a single month of delay at closeout costs about \$118,000, and a three-month slip triggers an equity call on the interest reserve. If A's probability of a meaningful staffing delay is even 15–20%, its expected cost exceeds C's premium. The documented reason exists. Before awarding to C, confirm the window supplier's installation price and warranty, and make sure the window installation scope sheet covers flashing integration with the weather barrier, which is where window leaks come from.

Scenario 2: Covering the electrical overrun. At month 2, the CM proposes moving \$140,000 of buyout savings to cover the electrical package's overrun, so its construction contingency stays intact. The amendment says buyout savings can only move into the CM's contingency with the owner's written approval.

Reasoning notes: The CM isn't asking for money. It's asking to use project savings to cover a project overrun, which is a reasonable use. The questions are about precedent and visibility.

First, is the electrical overrun a market movement (a legitimate GMP risk the CM priced) or a scope gap (something the CM's estimate missed)? If it's a market movement inside the GMP's scope, the CM arguably should absorb it from its own contingency, because that's what contingency is for. Second, what does approval do to the shared savings split? Savings moved into contingency and later spent are no longer available to split. A reasonable resolution: approve the transfer, document the reason, and record the net buyout position so the final accounting is clear.

Scenario 3: Dropping the bond. The developer's CFO proposes waiving the payment and performance bond to save about \$170,000. The CM is a large, well-capitalized firm, and the CFO argues the risk of default is tiny.

Scenario 4: The design-assist plumber's price. The plumbing subcontractor worked under a design-assist agreement since SD and developed the stacked wet-wall layout that saved \$310,000 in Module 2.3. Its subcontract price comes in at \$1,585,000, \$55,000 under the GMP budget. The CM believes a competitive bid could produce another \$60,000 to \$80,000 of savings and wants to bid the package.

Reasoning notes: Module 2.1 said design-assist agreements should state whether the trade is guaranteed the subcontract, usually subject to a fair price. This price is already under budget, and the trade's design work produced most of the savings. Bidding it out could save \$60,000 to \$80,000 on paper. It would also give away the design-assist plumber's knowledge of the layout, invite a new bidder to find gaps in a design it didn't develop, and signal to every future design-assist trade that the deal isn't honored. If the agreement promised the work at a fair price, and \$55,000 under budget is fair, the right move is to negotiate, not rebid. A useful check: ask the cost consultant to compare the plumber's unit prices against recent local plumbing bids, and negotiate any clear outliers.

Deliverable

Choose one of the two options.

Option A: GMP review memo (the reference project or your own)

Write the memo the owner's rep would send the developer recommending whether to accept the GMP. Include:

1. **Recommendation first.** Accept, accept with conditions, or reject, in one sentence.
2. **GMP summary.** Components, as proposed and as negotiated.
3. **Findings.** Every issue in the contingencies, allowances, exclusions, qualifications, general conditions, self-performed work, and buyout status, each with its dollar effect and resolution.
4. **Buyout status.** Percentage bought, the buyout log, the unbought packages and their escalation exposure, and the buyout schedule.
5. **Contract terms.** Treatment of buyout savings, contingency use, allowance overruns, alternates, unit prices, bonds, SDI, and insurance.

6. **Effect on the deal.** The GMP against the budget, the resulting spread, and the remaining cushion.
7. **Conditions of acceptance.** What must be true before the amendment is signed.

Option B: Bid evaluation (a real public project)

Many public agencies post bid tabulations along with the bid documents. Find a completed bid for a building project with at least four bidders, ideally one where the tabulation shows alternates and unit prices.

1. **The bids.** Tabulate base bids, alternates, and unit prices.
2. **Responsiveness.** From the posted results and the instructions to bidders, identify any bids that were rejected or that show irregularities, such as missing addenda acknowledgments.
3. **What the spread tells you.** Compare the low bid to the others and to the owner's estimate if one was published. Is the low bid an outlier? What might explain it?
4. **Alternates.** Which alternates would you accept, and how does the combination change the ranking of bidders?
5. **Unit prices.** If the quantities were to exceed the estimate by 25%, would the ranking change?
6. **Recommendation.** A one-page memo recommending award, with the evidence you'd want to see before signing, such as prequalification, bonding confirmation, and references.

Module 2.6 takes the reference project from signed GMP to notice to proceed. The permit is in review, the long-lead items are on order, and the CM is building the master schedule. Module 2.6 covers preconstruction planning: the schedule, site logistics, the submittal schedule, the permit path, and the risk register.

2.6 — Preconstruction Planning: Schedule, Logistics, Permits, and Risk

Why This Module Exists

The GMP is signed. The loan is closing. From the owner's seat, it looks like the hard decisions are over and construction is about to "just happen."

It won't just happen. Between the signed contract and the first truck on site, the team makes a set of decisions that determine how long the building takes: the order the buildings are built in, which materials get ordered when, where the lumber sits, which permits have to be in hand before which work starts, and which risks get managed and which get hoped away. Most of these decisions cost little to make well. Getting them wrong costs months.

This module covers the planning that sets that schedule:

- The master schedule: CPM basics, the critical path, and why float matters
 - Pull planning and takt planning, which turn the master schedule into work crews can actually do
 - Long-lead procurement and the submittal schedule, worked backward from the date each item is needed
 - The permit path: which approvals are needed for which work, and when
 - The site logistics plan
 - The risk register
 - The conditions for issuing the notice to proceed
-

Where the Project Is

Phase	GMP signed, about month –1 to 0
GMP	\$25,329,837 (Module 2.5)
Cushion	\$1,250,000 owner contingency plus \$187,283 headroom
Buyout	About 75% bought; framing, sitework, windows, and trusses under subcontract
Permit	Grading and site permit issued; building permits in final review
Utility	Service application filed at month –6 (Module 2.1); first transformer promised for month 17
Target	Construction start at month 0, all buildings complete by month 23, Certificate of Occupancy for the full community by month 25

The site is 11.8 acres with nine residential buildings (seven 24-unit Type A buildings and two 16-unit Type B buildings), a clubhouse, a pool, surface parking, and the underground detention vault. Month 0 falls in January, so sitework starts in winter.

Concepts

1. The Master Schedule and the Critical Path Method

What the master schedule is

The master schedule, often called the **baseline schedule** once it's approved, is the CM's plan for the whole project, from mobilization to final completion. Division 01 of the specifications usually requires the CM to submit it within a set period after notice to proceed, often 30 days, and to update it monthly. On most commercial projects, it's built with the **Critical Path Method (CPM)**, using scheduling software such as Oracle Primavera P6 or Microsoft Project.

The master schedule isn't a to-do list. It's a network of **activities** connected by **logic** (which activities depend on which), each with a **duration**. The software calculates when each activity can start and finish, and which chain of activities determines the project's end date.

The building blocks

- **Activity:** a unit of work with a defined start and finish, such as "Building 1, frame second floor." Good schedules break work down to activities of a few days to a few weeks.
- **Duration:** the working days an activity takes, based on quantity and crew productivity (the same productivity data from Module 2.3's estimate).
- **Logic (relationships):** how activities depend on each other. The most common is **finish-to-start**: framing can't start until the slab is finished. Others include **start-to-start** (drywall on the second floor can start once drywall on the first floor has started) and **finish-to-finish**.
- **Lag:** a required gap between activities. A concrete slab needs curing time before framing loads it.
- **Constraints:** fixed dates imposed from outside, such as "permanent power available no earlier than month 17." Use them sparingly, because they override the logic.

Forward pass, backward pass, and float

Take one Type A building, simplified to ten activities, plus the window procurement that has to happen before the windows can be installed:

Activity	Duration (working days)	Depends on
W. Window submittal, review, and fabrication	70	Starts when the window subcontract is signed
A. Slab on grade	15	Site pad ready
B. Wall and floor framing, three stories	35	A
C. Roof trusses and sheathing	10	B
D. Roofing	10	C
E. Windows and weather barrier	15	C and W
F. MEP rough-in	25	C

G. Rough-in inspections	5	E and F
H. Insulation and drywall	30	D and G
I. Finishes	40	H
J. Punch and final inspection	10	I

The **forward pass** calculates each activity's earliest start and finish by working through the network from the beginning. The **backward pass** works from the end to find each activity's latest start and finish that won't delay the project. The difference between them is **total float**: how much an activity can slip without delaying the finish.

Activity	Early start	Early finish	Late start	Late finish	Total float
W. Windows procurement	0	70	0	70	0
A. Slab	0	15	0	15	0
B. Framing	15	50	15	50	0
C. Trusses and sheathing	50	60	50	60	0
D. Roofing	60	70	80	90	20
E. Windows and weather barrier	70	85	70	85	0
F. MEP rough-in	60	85	60	85	0
G. Rough-in inspections	85	90	85	90	0
H. Insulation and drywall	90	120	90	120	0
I. Finishes	120	160	120	160	0
J. Punch and final	160	170	160	170	0

The building takes **170 working days**, about eight months.

The **critical path** is the chain of activities with zero float. Any delay to them delays the building. This network has two:

- Slab → framing → trusses → MEP rough-in → inspections → drywall → finishes → punch
- **Window procurement** → windows → inspections → drywall → finishes → punch

That second one is the lesson. **Window procurement is on the critical path, with zero float, before anyone has set foot on site.** If the architect's first review of the window shop drawings rejects them and a resubmittal costs two weeks, the first building finishes two weeks late. A schedule that shows only field work would miss it entirely. Roofing, by contrast, has 20 days of float: it can slip four weeks without affecting the finish.

From one building to nine: resource-driven logic

The single-building network is driven by logic. The whole project is driven by **resources**. There's one framing subcontractor, and it can run a limited number of crews. On the reference project, the framer (Bidder C from Module 2.5) commits to crews that can start a new building about every six weeks.

That cadence, not the per-building logic, sets the project's length:

- Sitework, over-excavation of the clay, the vault, and underground utilities take about three and a half months, through a winter.
- Building 1 starts framing around month 4.
- The ninth building starts framing about 8×1.4 months later, around month 15.
- Each building takes roughly seven months from framing start to completion, so the last building finishes around month 22 to 23.

This is why the framer's capacity mattered so much in Module 2.5's selection. A framer that could only start a building every eight weeks would push the last building out by about four months. The cheapest framer on paper would have been the most expensive on the schedule.

Float belongs to someone

Float is valuable, and contracts often argue over who owns it. If a building's roofing has 20 days of float, can the owner use that float for a late design change without owing the CM time? Can the CM use it to absorb its own delays? Many contracts say float belongs to the project and is available to whoever needs it first. Some say it belongs to the owner. Some say the contractor. The A201 general conditions don't address it directly, so the answer is often in the supplementary conditions or Division 01. It matters most when the schedule is tight, which on this project is always.

Schedule updates and what to watch

Each month, the CM updates the schedule with actual progress. The owner's rep should look at:

- **Critical path changes:** did a different chain of activities become critical? That usually signals a new problem.
- **Float erosion:** are near-critical paths (activities with only a few days of float) losing float month over month?
- **Logic changes:** did the CM change relationships to make the finish date hold? A schedule that stays on time only because the logic keeps changing is a schedule that isn't on time.
- **Forecast completion:** the date the updated schedule predicts, compared with the baseline and the leasing season.

Earned-value measures such as the schedule performance index measure dollars, not the critical path, so they can look healthy while the critical path slips.

2. Pull Planning and Takt Planning

Why a CPM schedule isn't enough

A CPM schedule is built by the CM's scheduler in an office. It's excellent for sequencing the whole project, finding the critical path, and measuring delay. It's less good at the level of detail where work actually happens: which crew is on which floor next Tuesday, and whether they have everything they need to be productive when they get there. The people who know that are the trade foremen.

Pull planning and the Last Planner System

The **Last Planner System**, developed by the Lean Construction Institute, adds layers of planning that bring those people in. It's widely used on commercial projects, including under CMAR.

1. **Master schedule:** the CPM schedule, with its milestones.
2. **Phase pull plan:** for each phase (for example, "Building 1, dry-in"), the foremen of every trade involved meet in front of a wall and plan **backward** from the milestone. Each trade writes on sticky notes what it needs to finish, and what it needs from the trade before it. Planning backward from the milestone ("pulling") exposes handoffs and dependencies that a forward plan hides.
3. **Lookahead plan:** a rolling six-week view, used to identify and remove **constraints** before the work arrives: a missing submittal approval, material not delivered, an inspection not scheduled, a design question unanswered.
4. **Weekly work plan:** each Friday, foremen commit to the specific tasks they'll complete next week, and only tasks with no remaining constraints qualify.
5. **Percent Plan Complete (PPC):** the share of weekly commitments actually completed. It's tracked week to week, along with the reasons for misses.

The value is less in any single plan than in the habit. When PPC drops, the reasons tell the team where the process is failing, often in constraints that the CPM schedule never showed.

Takt planning for repetitive work

A garden community is nine nearly identical buildings. That makes it well suited to **takt planning**, a method borrowed from manufacturing. Instead of scheduling each building separately, the team defines a **takt time** (a fixed rhythm, here about six weeks per building) and sequences trades to flow through the buildings one after another: framing moves from Building 1 to Building 2, roofing and windows follow into Building 1, then MEP rough-in, and so on. Each trade gets a predictable rhythm, crews stay continuously employed, and a problem in one building shows up immediately as a disruption to the flow.

Takt planning also makes the framer's cadence visible as what it is: the rhythm every other trade follows.

3. Long-Lead Procurement and the Submittal Schedule

Working backward from the need date

Materials don't arrive when they're ordered. They arrive after a submittal is prepared, reviewed by the architect (sometimes twice), fabricated, and shipped. For each item, the procurement plan works backward from the date it's needed on site:

Latest submittal date = need date – delivery – fabrication – review cycles – preparation

For the reference project's first building, Building 1's windows are needed at the start of activity E, around month 5.5. With a 10- to 12-week fabrication lead after approval, two architect review cycles of two weeks each, and two weeks for the supplier to prepare the submittal, the window submittal is due around month 1.5 to 2, less than two months after notice to proceed.

The key long-lead items on the reference project:

Item	Lead after approval	Needed on site	Latest approval
First transformer and utility service (utility)	~14 months from application	Month 17	Application filed month –6 (done)
Meter centers and electrical distribution	26–36 weeks	Month 8	Bought early, before month 0 (Module 2.3)
Roof trusses (a deferred submittal)	4–6 weeks	Month 5	Month 3.5, after both architect and building department review

Windows	10–12 weeks	Month 5.5	Month 3
Heat pumps and air handlers	8–14 weeks	Month 6	Month 3
Fire alarm panel and devices	About 8 weeks	Month 8	Month 6
Cabinets and countertops	8–10 weeks after field measurement	Month 10	Field measure at framing of Building 1
Appliances	8–12 weeks	Month 11	Month 8

Lead times vary widely by market and year. The electrical equipment and transformer lead times in particular have run long in recent years. The structure is the point: each item has a need date, and each need date implies a latest approval date that often lands in the first weeks of the job.

The submittal schedule

The **submittal schedule** (also called the submittal register or log) lists every submittal the specifications require: product data, shop drawings, samples, test reports, and qualifications. Each technical section's Part 1 says what's required (Module 2.4). A garden project like this one might have several hundred.

For each submittal, the schedule shows:

- The specification section and description
- The responsible subcontractor
- The date it's due from the subcontractor
- The architect's review period, set by the contract (often 10 working days)
- The date the approved submittal is needed to support procurement
- Whether it's a long-lead item or a deferred submittal

Two things make it valuable. First, it's **sorted by need date, not by specification section**, so the window and truss submittals get reviewed before the paint colors. Second, it's **tied to the master schedule**, so a late submittal shows up as a schedule problem before it becomes a field problem.

Deferred submittals take longer

Module 2.2 introduced deferred submittals: elements designed by the specialty subcontractor, such as the roof truss engineering and the final sprinkler layout. They go to the architect and engineer of record **and** to the building department. That second review can add weeks, and the

building department's timeline isn't in the CM's control. On this project, the truss engineering has to be approved by both before the trusses are fabricated. The truss submittal is one of the first things the CM should push through.

4. The Permit Path

More than one permit

"The permit" is usually several permits, from several agencies, each gating different work. On the reference project:

Permit or approval	Agency	Gates what	When it's needed
Grading and site development	City engineering or public works	Clearing, mass grading, over-excavation	Month 0 (issued)
Stormwater (construction general permit and local plan)	State environmental agency, city	Any land disturbance	Month 0 (issued)
Building permits (per building or master)	Building department	Foundations and vertical construction	Month 3 (foundation)
Fire line and fire sprinkler	Fire marshal	Underground fire water, sprinkler installation	Month 2 (fire line)
Right-of-way permit for the deceleration lane	Road authority	Work in the public road	Month 12 or later
Sewer extension	Utility district	Off-site sewer construction	Month 1
Electrical, plumbing, mechanical trade permits	Building department	Each trade's work	Pulled by each subcontractor

Pool	Health	Pool	Month 18
	department	construction	
	and building	and opening	
	department		

Several of these are outside the building department, have their own review timelines, and are run by agencies with no stake in the project's schedule. The deceleration lane is a good example: it's required for the final CO (a condition of the site plan approval), but the road authority's review and construction window may be months long.

How plan review works

A building permit review usually follows the same pattern:

1. **Application and completeness check.** The building department confirms the submission includes everything it requires: drawings, specifications, code analysis, energy compliance forms, the statement of special inspections, and fees. An incomplete submission doesn't start the review clock.
2. **Review.** Plans examiners for each discipline (building, structural, fire, mechanical, electrical, plumbing, and sometimes energy and accessibility) review the documents against the adopted codes.
3. **Comments.** The department issues a comment letter listing required corrections.
4. **Resubmittal.** The design team revises the documents and responds to each comment in writing.
5. **Approval and issuance.** Once comments are resolved and fees paid, the permit is issued. The approved documents are stamped, and a set is kept on site for the inspectors.

Review times vary widely by jurisdiction and workload, from a few weeks to several months per cycle. Two cycles are common. The pre-application meeting in Module 2.1 is where the team learns the local rhythm and what the reviewers focus on.

Phasing the permit to protect the schedule

Many jurisdictions allow the permit to be issued in phases: a grading permit first, then a foundation-only permit, then the full building permit. On the reference project, the grading and stormwater permits were issued first so sitework could start at month 0 while the building permits were still in review. Framing isn't needed until month 4, so the building permits have about three months of float, as long as nothing else slips. That float is what lets the project start construction before every approval is in hand.

Inspections

Once work starts, the permit is enforced through inspections at defined stages. On wood-frame multifamily, the typical sequence includes foundation (before concrete), underslab plumbing and electrical, framing and shear walls, MEP rough-in, insulation, fire-rated assemblies (often before drywall covers them), and final. The special inspections listed in the statement of special

inspections (Module 2.2) run in parallel. Every inspection is a potential stop: work that has to be covered can't be covered until it passes. The schedule should show inspections as activities, like activity G above, not as assumptions.

Keeping the permit set and the construction set aligned

The building department approved a specific set of documents. During construction, addenda, bulletins, and change orders change the documents the contractor builds from. If a change affects something the department reviewed, such as a rated wall, the fire sprinkler layout, or the egress path, it usually needs to go back to the department as a permit revision. Field inspectors compare what's built against the approved set. When they differ and no revision was approved, the inspection fails, and the fix costs more than the revision would have.

The owner's rep should keep a simple log of every document change and whether it affects the approved set, and should make sure the architect submits the revisions that are required. Module 2.7 covers the change process that generates these revisions.

5. The Site Logistics Plan

What it is

The site logistics plan shows how the site will work while the building goes up: where trucks enter and leave, where materials are stored, where trailers and worker parking go, how fire trucks get access, where temporary utilities run, and how stormwater is controlled. It's usually drawn as a set of site plans, one for each major phase, because the site changes as buildings rise and paving goes in.

On a garden project, logistics is less dramatic than on a tight urban site, but it's still where days are won and lost. Nine buildings, a vault, and 11.8 acres of sitework all compete for the same space.

What it covers

Access and circulation. Construction entrances, truck routes through the site, turnaround areas, and the gates that control them. On the reference project, the deceleration lane won't be built until later, so the construction entrance uses the secondary access point to avoid a lane closure on the main road.

Laydown and staging. Where lumber packages, trusses, windows, and other materials are stored near the buildings that will use them. A framing package for a 24-unit building is several truckloads. It needs a flat, well-drained area within reach of the telehandler.

Equipment. Telehandler (forklift) paths, crane locations for truss setting, and concrete truck and pump access.

Temporary facilities. Site trailers, worker parking, toilets, dumpsters, concrete washout areas, and material storage containers.

Temporary utilities. Temporary power for tools and trailers (often generators or a temporary utility service), temporary water, and, critically, water for fire protection.

Fire access and protection during construction. The International Fire Code, as adopted locally, typically requires fire department access roads and a working water supply for firefighting, such as active hydrants, to be in place before combustible construction materials arrive on site. For a wood-frame project, that means the fire access road's all-weather base and at least some hydrants have to be done before framing starts. Builder's risk insurers often add their own requirements, such as site security, fencing, and fire watch (Module 2.5).

Stormwater and erosion control. The stormwater permit requires a stormwater pollution prevention plan (SWPPP): silt fence, inlet protection, stabilized entrances, and regular inspections, especially after rain. Violations can bring fines and, in some jurisdictions, stop-work orders.

Neighbors and the public. Work hours, noise, dust control, deliveries, and street cleaning.

Security. Fencing, lighting, cameras, and access control, which matter for theft (appliances and copper are frequent targets) and for fire risk.

The phases of the reference project's logistics plan

Phase	Months	What's on site	Logistics priorities
Sitework	0–4	Grading, over-excavation, vault, underground utilities	Stabilized entrance, stormwater controls, haul routes, stockpiles; build the fire access road base and hydrants early
Vertical	4–18	Framing and dry-in moving building to building	Laydown near each building, telehandler paths, truss deliveries, trailer and parking compound, fire access and hydrants active
Finishes and turnover	16–25	Interiors, paving, landscape, pool, first buildings turning over	Separating residents from active construction, final paving sequencing, landscape and pool, removing temporary facilities

One detail on the reference project deserves attention. The best laydown area, flat and central, sits on top of where the detention vault goes. If the vault is built later, it takes the laydown area away in the middle of framing. So the logistics plan and the master schedule both put the vault

early, in the sitework phase, and then use the backfilled area above it as laydown. Logistics and schedule are the same decision viewed two ways.

Turning buildings over in phases

On real garden projects, buildings are typically turned over in groups as they're finished, each with its own certificate of occupancy (or a temporary certificate), while construction continues elsewhere on the site.

The master schedule makes this possible physically: at a six-week framing cadence, the first buildings are finished about a year before the last ones. What gates phased turnover isn't construction. It's everything a building needs to be occupied:

- **Permanent power** to that building. On this project, the first transformer, serving the clubhouse and Buildings 1 through 4, is promised for month 17. That date is why the utility application was filed at month -6 (Module 2.1). Filing at permit would have pushed it to about month 22.
- **Site infrastructure** serving those buildings: water, sewer, the fire access road, and the parking and accessible routes that serve them.
- **Life safety separation** from active construction: fencing, protected walkways, and fire access that doesn't route residents through the work zone.
- **The clubhouse and leasing office**, which are usually first because leasing starts there.
- **Final inspections** for each group.

The reference project plans three turnover groups:

Group	Buildings	Units	Target
1	Clubhouse and Buildings 1-3	72	Month 18
2	Buildings 4-6	72	Month 21
3	Buildings 7-9, pool, final site work	56	Month 24, with the full-community CO at month 25

Group 1 is gated by power, not construction. Its buildings are physically complete around month 14 and wait for the transformer. Scenario 3 asks what it's worth to close that gap.

6. The Risk Register

From Part 1 to the field

The risk register started in Part 1 as a list of deal risks. Module 2.1 carried it into the project charter and assigned owners. By notice to proceed, it should be a working construction document: specific risks with a probability, a cost or schedule impact, a named owner, a mitigation, and a trigger that says when to act.

A useful register is short enough to review monthly, specific enough to act on, and honest about the things the team can't control.

Quantifying risk

The simplest useful measure is **exposure = probability × impact**. A 25% chance of a two-month delay that costs \$300,000 has an exposure of \$75,000. Exposure isn't a forecast. It's a way to rank risks and decide which ones deserve attention and money.

The reference project's top risks at notice to proceed

Risk	Exposure	Owner	Mitigation
First transformer arrives late; Group 1 turnover slips	25% × about \$300,000 (two months) = \$75,000	Owner's rep, electrical engineer	Monthly utility coordination meetings; confirm the pad and conduit are ready early (Module 2.5); temporary power plan for model units and the leasing office
Winter sitework weather delays over-excavation and the vault	40% × about \$120,000 (three weeks) = \$48,000	CM superintendent	Start with areas that drain; stockpile imported fill; rain days in the schedule; lime or cement treatment option priced as a unit price
Clay over-excavation quantity exceeds estimate	30% × \$150,000 = \$45,000	CM, geotechnical engineer	Unit price already in the contract (Module 2.5); geotechnical engineer on site to confirm depths as excavation proceeds

Building permits issued late, delaying framing	$20\% \times \text{about } \$118,000 \text{ (one month)} = \$23,600$	Architect	Weekly contact with plan review; comments turned around within five working days; foundation-only permit option
Framer can't sustain the six-week cadence	$15\% \times \text{about } \$250,000 \text{ (six weeks at the end)} = \$37,500$	CM project manager	Crew commitment in the subcontract; weekly pull planning with the framer; second-crew option priced in advance
Window submittal rejected; Building 1 dry-in slips	$25\% \times \text{about } \$60,000 \text{ (two weeks on Building 1)} = \$15,000$	CM, architect	Pre-submittal meeting with the architect; expedited review agreed in writing; temporary weather protection plan
Truss engineering held up in building department review	$20\% \times \text{about } \$60,000 = \$12,000$	CM, truss supplier	Submit trusses first; confirm the department's deferred submittal timeline at the pre-construction meeting
Fire during framing	Low probability; loss far exceeds the cushion	CM, owner (builder's risk)	Fire access and hydrants before framing; site security and fencing; hot work permits; builder's risk insurer's requirements
Deceleration lane approval or construction delays the final CO	$20\% \times \text{about } \$118,000 = \$23,600$	Owner's rep, civil engineer	Road authority submittal at month 0; construction window confirmed; ask whether a bond can substitute for completion at CO

Stormwater violation or stop-work order	10% × about \$50,000 plus delay	CM superintendent	Weekly and post-rain SWPPP inspections; a named SWPPP manager; photos logged
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Total exposure on these ten is roughly \$280,000 to \$300,000 before the fire risk. That's well within the owner's \$1.25 million contingency, which is the point of the register: it shows what the contingency is for, and which risks the team should spend money on now to avoid spending contingency later.

7. From Contract to Notice to Proceed

The preconstruction meeting

Before work starts, the owner, the architect, the CM, key subcontractors, and often the special inspector and the lender's construction monitor meet to set up the job. The agenda typically covers:

- Contacts, roles, and lines of communication
- The master schedule and the first six-week lookahead
- Submittal procedures and the first submittals due
- RFI procedures and response times (Module 2.7)
- Payment application procedures, the schedule of values, and retainage (Module 2.7)
- Change order procedures (Module 2.7)
- Site logistics, safety, security, and work hours
- Permit status, inspection procedures, and the special inspection program
- Quality control, mockups, and pre-installation meetings
- Closeout expectations (Module 2.8)

Notice to proceed

The **notice to proceed** (NTP) is the owner's written authorization for the CM to start work. It typically starts the contract time: the number of days the CM has to reach substantial completion. It shouldn't be issued until the conditions in the phase approval below are met, because once the clock starts, delays caused by missing owner items (a permit, site access, financing) become owner delays with time and cost consequences.

Artifacts

Artifact	Produced by	When	What it's for
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Baseline master schedule (CPM)	CM	Within 30 days of NTP	The plan against which delay is measured
Monthly schedule updates	CM	Monthly	Progress, critical path, and forecast completion
Phase pull plans and six-week lookaheads	CM with trade foremen	Ongoing	The work crews can actually do, with constraints removed
Long-lead procurement log	CM	Before NTP, updated weekly	Every long-lead item with its need date and latest approval date
Submittal schedule	CM	Before NTP	Every required submittal, sorted by need date
Permit matrix	Owner's rep with the architect and CM	Before NTP	Every permit, its agency, what it gates, and its status
Site logistics plan, by phase	CM	Before NTP	How the site works during each phase
Stormwater pollution prevention plan	Civil engineer, CM	Before land disturbance	Permit compliance
Turnover plan	CM with the owner's rep and property manager	Before NTP, refined during construction	Which buildings turn over when, and what each group needs
Risk register	Owner's rep with the CM	Before NTP, reviewed monthly	Risks, exposure, owners, and mitigations

Phase Approval: Issue the Notice to Proceed

The question: is everything the owner is responsible for in place, so the contract clock starts on a plan the team can actually execute?

- ☐ Construction loan closed, and the lender has approved the GMP and the draw process
- ☐ Payment and performance bond and insurance certificates received; builder's risk in force, with the insurer's site requirements known
- ☐ Grading, stormwater, and any early permits issued; building permit status and float confirmed
- ☐ Stormwater pollution prevention plan in place and controls installed before land disturbance
- ☐ Baseline schedule submitted or due within the contract period, with the critical path, long-lead items, inspections, and turnover groups shown
- ☐ Long-lead procurement log reviewed; every item with a latest approval date in the first 90 days has an owner
- ☐ Submittal schedule sorted by need date; the architect has committed to review periods for the first submittals
- ☐ Site logistics plan for the sitework phase approved, including fire access and water supply before framing
- ☐ Utility coordination confirmed for the first transformer and the pad and conduit that must be ready for it
- ☐ Risk register reviewed, each top risk with an owner and a trigger
- ☐ Preconstruction meeting held

Whose money is at risk: from NTP, the contract clock runs. Delays the owner causes (a missing permit, late site access, a late decision) become owner delays, with time extensions and general conditions costs the owner pays out of contingency.

Failure Modes

A schedule of field work only. Window procurement was on the critical path before anyone set foot on site. A schedule without submittals, fabrication, and inspections isn't a critical path schedule.

Ignoring resource logic. One building's network says eight months. The framer's cadence says the project takes about 23. The resource constraint, not the logic, sets the finish.

Treating the permit as one thing. Eight approvals from five agencies gate different work. The one nobody tracked is the one that holds up the CO.

Issuing NTP before the owner's items are ready. The contract clock starts, and the owner's delays become the owner's cost.

Logistics decided in the field. The vault built in the middle of the laydown area in month 8 costs more than any single line in the logistics plan.

Fire protection arriving after the lumber. For wood frame, the fire access road and hydrants come before framing. Late, they stop the job. Absent during a fire, they turn a loss into a catastrophe.

Planning a single turnover by default. Physically complete buildings waiting months for power, a road permit, or the last building are months of lost leasing.

A risk register that doesn't change. If it looks the same in month 8 as at NTP, nobody is using it.

Accepting a schedule that holds only because the logic keeps changing. The finish date looks stable. The project isn't.

Where AI Is Changing This

Scheduling is one of the more active areas for new software. Tools can now generate and compare many schedule scenarios from the same scope, analyze schedule risk by simulating delays, and compare updates to spot logic changes and float erosion that a human reviewer would miss. Pull planning and lookahead tools digitize the sticky-note wall and track PPC automatically. Reality capture and computer vision tools compare photos and scans to the schedule to measure actual progress. Part 3 covers these in Modules 3.13 (scheduling and planning), 3.14 (project management), and 3.16 (reality capture).

The judgment that remains is knowing what the schedule is for. Software can find the critical path in seconds. It takes someone who understands the project to notice that the utility, the road authority, and the framer's crew count are the real constraints, and that none of them is in the default activity list.

Exercise

Decision scenarios

These use the reference project. Reasoning notes follow each one.

Scenario 1: The building permit comes back with a second round of comments. At month 1.5, the building department issues a second round of comments on the building permits, mostly on the fire-rated assemblies and the energy forms. The architect expects a resubmittal in two weeks and approval about four weeks after that, around month 3. Framing on Building 1 is scheduled to start around month 4.

Reasoning notes: The permit has about a month of float left, assuming the foundation work can proceed under a foundation-only permit or the building permit arrives before the slab pour. Check which it is. If slabs need the building permit, the float is smaller, because the slab for Building 1 starts around month 3.5. Protect it: ask whether the department will issue a foundation-only permit, confirm the architect's two-week turnaround in writing, and move the comments that affect the rated assemblies to the top, because they also affect the truss and window submittals. Log it on the risk register with a trigger: if approval isn't in hand by month 3, the foundation-only permit becomes mandatory.

Scenario 2: The windows are late. The window supplier's first submittal is rejected for missing performance data, and the resubmittal adds two weeks. Then the supplier reports that fabrication will take 14 weeks, not 11. Building 1's windows will now arrive about five weeks late.

Reasoning notes: In the single-building network, windows had zero float, so five weeks late means Building 1 finishes five weeks late. But the project's finish is set by the framer's cadence and the last building, not by Building 1. The question is whether the delay stays in Building 1 or propagates to the flow of trades behind it. Two options: temporary weather protection so MEP rough-in and even drywall can proceed with the window openings closed temporarily, or let Building 1 slip and keep the trades flowing to Building 2. Price both. Temporary protection costs thousands of dollars per building. Five weeks of delay to Building 1 would matter a great deal if Group 1 could otherwise turn over sooner. It can't: Group 1 waits for the transformer at month 17 regardless, and Building 1 is still finished well before that. The delay is real, but the power date absorbs it. Protect the trade flow, not Building 1's finish date, and confirm the second and third buildings' windows are on the new lead time.

Scenario 3: Pull the first turnover forward. Group 1 (the clubhouse and Buildings 1–3, 72 units) is physically complete around month 14 but can't turn over until the transformer at month 17 and final inspections at month 18. The utility offers a temporary service that could energize Group 1 by month 15, at a cost of about \$140,000, including temporary equipment, extra coordination, and later changeover. The property manager points out that month 15 is in early spring, the start of the leasing season.

Reasoning notes: Three months earlier for 72 units is a meaningful head start. Rough scale: if those units lease at the project's absorption rate and each earns something like \$1,500 a month in rent, pulling 72 units' leasing forward by three months could bring forward several hundred thousand dollars of revenue, before concessions and the operating costs that also start earlier.

Scenario 4: The fire marshal's condition. At the preconstruction meeting, the fire marshal confirms that the fire access road must have an all-weather surface and the first two hydrants must be active before any framing lumber is delivered. The site schedule currently finishes the water main loop and hydrants at month 4.5, two weeks after the first lumber delivery.

Reasoning notes: This is a hard constraint. Lumber can't arrive until hydrants are active, so the whole framing flow shifts two weeks unless the water main is re-sequenced. Look at the sitework logic: can the water main loop serving Buildings 1 and 2 be completed first, with the rest later? Can the vault excavation, which competes for the same equipment, start earlier or later? A two-week shift at the start of framing moves the last building's finish by about two weeks. At this project's numbers, that's worth tens of thousands of dollars, more than enough to justify added equipment or a weekend crew on the water main. It's also a reminder to confirm every agency's conditions before the baseline schedule is set, not after.

Deliverable: Master Schedule and Logistics Plan

For the reference project, or your own project, produce the preconstruction planning package the owner would review before issuing the notice to proceed.

1. Summary master schedule (1–2 pages). Major phases and milestones from NTP to the final CO: sitework, the vault, each building's framing start and completion, dry-in, the first transformer, each turnover group, paving, the pool, and final inspections. Show the critical path and identify what drives it: logic, a resource, or an outside party.

2. One building's CPM network. Using the example in Section 1 or your own activities, run the forward and backward pass. Show early and late dates, float, and the critical path, including at least one procurement activity. Then change one duration (a longer window lead time, or a failed inspection) and show what moves.

3. Long-lead procurement log. At least eight items, each with its lead time, need date, and latest approval date.

4. Submittal schedule (top 20). The 20 submittals with the earliest need dates, sorted by need date, with responsible party, due date, review period, and whether each is long-lead or deferred.

5. Permit matrix. Every permit or approval, its agency, what it gates, when it's needed, and its status and risk.

6. Site logistics plan, by phase. A marked-up site plan (hand sketch over a public site plan is fine) for each of the three phases: entrances and haul routes, laydown, equipment paths, trailers and parking, temporary utilities, fire access and hydrants, stormwater controls, and the separation between residents and construction in the turnover phase.

7. Turnover plan. Turnover groups, what gates each one, and the target dates.

8. Risk register (top 10). Each risk with its exposure (probability × impact), owner, mitigation, and trigger.

9. What a month costs. One paragraph stating your project's cost of a month of delay at the start, the middle, and the end of construction. This number should appear at the top of the package, because it's the reason every other page exists.

Module 2.7 starts construction. The notice to proceed is issued, the site is mobilized, and the first buildings go up. Module 2.7 follows the physical build sequence (sitework, foundations, framing, dry-in, rough-in, and finishes) and places contract administration where it actually happens: submittals, RFIs, changes, inspections, and monthly pay applications.

2.7 — Building It: Field Execution and Contract Administration

Why This Module Exists

Contract administration is usually taught as a list of forms: submittals, RFIs, change orders, pay applications, field reports. Each form is explained in isolation, and a reader who has never run a site finishes the chapter knowing what a G702 is without any picture of when it shows up or why.

This module follows the reference project through construction in the order the work actually happens: mobilization, sitework, foundations, framing, dry-in, rough-in, drywall and finishes, and site completion. At each stage, it shows what's happening in the field, who's on site, what can go wrong, and which contract administration tools come into play. Each tool is explained in detail the first time it matters, and then it keeps reappearing, because that's how the job works: the same handful of processes run every week for two years.

The goal is a mental movie of construction. By the end, a reader should be able to walk onto the reference project in any month, look at what's being built, and know which documents are moving, which inspections are next, and where the money is going.

Module 2.8 picks up at turnover and closeout.

Where the Project Is

Notice to proceed	Month 0, in January
GMP	\$25,329,837
Owner contingency	\$1,250,000 (outside the GMP)
Buyout savings to date	\$231,000 net, tracked separately, split 50/50 at final accounting (Module 2.5)
Schedule	Framing starts about every six weeks per building; turnover groups at months 18, 21, and 24; full-community CO at month 25 (Module 2.6)

The People on Site

Before the sequence, a quick cast list. On a typical day during framing, the site might have:

- **The CM's superintendent**, who runs the site: coordinating trades, sequencing work, enforcing safety, walking quality, and making dozens of small decisions a day. On a nine-building project, often supported by one or two assistant superintendents.
- **The CM's project manager**, usually in the site trailer or the office, running the contract: buyout, cost, changes, pay applications, and the schedule.
- **The CM's project engineers**, processing submittals and RFIs, maintaining logs, and chasing documents.
- **Trade foremen and crews**: sitework, concrete, framing, roofing, and so on, as the sequence moves.
- **The building inspector**, on scheduled inspections.
- **The special inspector and testing technicians**, testing soil compaction, concrete, and specific structural and fire-related work.
- **The architect's construction administration staff**, visiting every week or two.
- **The owner's representative**, weekly or more often.
- **The lender's construction monitor**, monthly, before each draw.

The owner's rep isn't there to supervise the work. That's the CM's job, and the contract makes the CM responsible for means, methods, sequencing, and safety. The owner's rep is there to see whether the project is on track, whether the money matches the progress, and whether decisions are being made on time.

Mobilization (Month 0)

In the field

The site goes from empty lot to active job site in a couple of weeks. The CM installs perimeter fencing and gates, the stabilized construction entrance, and the stormwater controls required by the stormwater pollution prevention plan (silt fence, inlet protection, sediment traps). Trailers arrive and get temporary power, water, and data. A surveyor stakes out the building corners, the utility lines, and the grades. The first equipment arrives: dozers, excavators, scrapers, compactors.

Contract administration begins

The first pay application. Within the first month, the CM submits its first payment application. Before that happens, the owner's rep and the lender's monitor need to approve the **schedule of values**: the CM's breakdown of the GMP into line items, each with a dollar value. Every future pay application bills against it by percent complete.

The schedule of values deserves a careful review, because the CM has a reason to **front-load** it: assigning more value to early work (mobilization, sitework, general conditions) and less to late work, so that more of the contract is billed early. A 6% front-load leaves the owner about \$1.47 million overpaid by month 11 on this project. If the contractor failed during that window, the owner would have paid for work that doesn't exist. The fix is simple and unglamorous: compare each line to the estimate and the buyout log, and check that mobilization and general conditions are billed as they're actually incurred.

Meetings and records. The rhythm of the job starts now:

- **Weekly owner-architect-contractor (OAC) meetings**, covering safety, the schedule and lookahead, submittals, RFIs, changes, quality, and upcoming inspections. The CM usually keeps the minutes. They aren't contract documents, but they're the record of what was agreed, and they're where disputes are later reconstructed from.
- **Daily logs** kept by the superintendent: weather, crews and headcounts on site, work performed, deliveries, inspections, visitors, and problems. Daily logs are the single most important evidence in any later delay claim.
- **Field reports** from the architect after each site visit.
- **Weekly pull planning** with the trade foremen (Module 2.6).

Sitework (Months 0–4)

In the field

Sitework is heavy equipment moving a lot of dirt. On the reference project, it's also where the diligence findings from Part 1 become physical:

- **Clearing and mass grading** reshape the site to the civil engineer's grading plan.
- **Over-excavation of the expansive clay** removes the bad soil under each building pad, down to the depth the geotechnical engineer specifies, and replaces it with compacted engineered fill. This is paid by the cubic yard under the unit price in the contract (Module 2.5).
- **The detention vault** is excavated, built, and backfilled early, so the area above it can serve as laydown (Module 2.6).
- **Underground utilities**: the water main loop with hydrants, the fire line, sanitary sewer, storm drainage, and conduit for electrical and communications. The off-site sewer extension runs in parallel under the utility district's permit.

- **The fire access road** gets its all-weather base, and the first hydrants go live, before any framing lumber arrives (Module 2.6).
- **Building pads** are brought to finished grade and compacted.

In January, weather is the enemy. Rain turns clay into mud that can't be compacted, and compacting fill that's too wet fails the tests.

Testing and inspection: the first of many

Compaction testing. Every lift of engineered fill is tested for density and moisture by the testing agency, often using a nuclear density gauge, to confirm it meets the specification (typically a percentage of maximum dry density from a laboratory test). Failing tests mean re-working the fill. The results are the evidence that the pads are ready for foundations.

Unit price quantities. Because over-excavation is paid by the cubic yard, the quantity has to be measured and agreed, not estimated. The usual method is surveyed before-and-after elevations, with the geotechnical engineer on site confirming how deep the bad soil actually goes. The owner's rep should see these measurements every month. A unit price contract that nobody measures is a blank check.

Utility inspections and testing. Water lines are pressure-tested and disinfected, sewer lines are tested and often camera-inspected, and the building department, fire marshal, or utility inspects each system before it's backfilled. Once it's buried, it's expensive to check again.

The first RFI

Sitework is usually where the first RFIs appear. On the reference project, the utility crew finds an existing storm pipe, not shown on the survey, crossing where the new water main is supposed to run.

How an RFI works. A **Request for Information** is the contractor's written question to the architect about a conflict, ambiguity, or gap in the documents. It includes:

- A number, the date, and the question
- References to the specific sheets, details, and specification sections involved
- The contractor's suggested resolution, if it has one
- The date an answer is needed to avoid delay
- Photos, sketches, or markups

The architect (here, routing it to the civil engineer) investigates and answers in writing. The answer might clarify the existing documents, provide a sketch, or say that the issue requires a change to the contract. **An RFI response clarifies the documents. It doesn't change them.** If the answer changes the scope, price, or time, it has to be followed by a formal change (see Foundations, below).

Contracts usually set an expected response time, often 5 to 10 working days. The owner's rep should watch two numbers in the RFI log: how many RFIs are open and how old they are. A slow architect response on a critical-path item is an owner delay, because the owner provides the documents.

The money in sitework

Sitework is a large share of early spending: on this project, the sitework and utilities package is about \$2.4 million. The first two or three pay applications are dominated by it. Each month:

1. The CM submits the pay application (the AIA G702 summary and G703 continuation sheet are common forms), showing each schedule-of-values line's value, work completed to date, materials stored, and retainage.
2. The architect reviews it against its observations and certifies the amount it believes is due.
3. The lender's construction monitor inspects the site and reviews the application for the lender.
4. Lien waivers arrive from the CM and its subcontractors for the previous payment.
5. The owner funds, using equity first and then the construction loan.

Retainage is withheld from each payment: on this project, 10% until the work is 50% complete, then 5%. It's the owner's leverage to get the work finished, and it's released after completion. Subcontractors carry it, and they price it into their bids.

Foundations and Slabs (Months 3–16, Building by Building)

In the field

Foundations are where the building starts. On the reference project, each building gets a slab-on-grade with thickened edges and interior footings where the design requires them. The work repeats nine times, about every six weeks:

1. **Layout.** The surveyor marks building lines, and the crew sets forms.
2. **Underslab rough-in.** Plumbing drains and water lines, and electrical conduit, are installed in the ground inside the forms.
3. **Underslab inspection.** The building inspector checks the plumbing (usually with a water or air test) and the electrical before anything is covered.
4. **Vapor retarder and reinforcement.** A vapor retarder goes down over the compacted base, and reinforcing steel or post-tensioning tendons are placed on chairs.
5. **Pre-pour inspection.** The building inspector and, where required, the special inspector verify reinforcement size, spacing, and cover, and the location of embedded items like hold-down anchors for the shear walls.

6. **The pour.** Concrete trucks arrive on a tight schedule; the testing agency takes samples at the truck to test slump, air content, and temperature, and casts cylinders that will be crushed at 7 and 28 days to confirm strength.
7. **Finishing and curing.** The slab is finished to the specified flatness and cured before framing loads it.

A problem at step 5 is cheap. The same problem found after step 6 means cutting or demolishing concrete.

Submittals: the first real wave

Foundations bring the first wave of **submittals**: concrete mix designs, reinforcing steel shop drawings, vapor retarder product data, anchor bolts and hold-downs.

How submittals work. A submittal is information the contractor provides to show it intends to meet the documents. There are three main types:

- **Shop drawings:** drawings made by a subcontractor or fabricator showing exactly how something will be built or fabricated. Reinforcing steel placement, roof trusses, windows, and HVAC equipment layouts are common examples.
- **Product data:** manufacturers' data sheets, test reports, and certifications showing a product meets the specification.
- **Samples:** physical samples of finishes and materials for the architect's selection or approval.

The process:

1. The subcontractor prepares the submittal.
2. The CM reviews it and stamps it, confirming it has checked the submittal against the documents and coordinated it with other work. A CM that passes submittals through without reviewing them is failing at one of its core jobs.
3. The architect (and the relevant engineer) reviews it for **general conformance with the design intent**, not for dimensions, quantities, or construction methods, which remain the contractor's responsibility.
4. The architect returns it with one of four actions, in some variation:

Action	Meaning	What happens next
Approved / no exceptions taken	Conforms	Proceed
Approved as noted	Conforms with the noted corrections	Proceed, incorporating the notes
Revise and resubmit	Doesn't conform enough to proceed	Revise and resubmit; don't fabricate

Rejected

Doesn't conform

Submit something that
does

Two principles to remember. **Submittals aren't contract documents.** An approved submittal doesn't change the contract; if it conflicts with the specification, the specification still governs (Module 2.5). And **the architect's approval doesn't relieve the contractor** of responsibility for errors it introduced (A201 Section 3.12). But an architect that approves a submittal that plainly doesn't meet the documents takes on some responsibility too. Submittal review is a real checkpoint, not a rubber stamp.

The submittal schedule from Module 2.6 is what keeps the long-lead submittals, the windows and the trusses, moving ahead of the paint colors.

The first change order

On the reference project, excavation for the footings at Building 3 hits rock at month 8, which the geotechnical borings didn't show between their locations. The GMP excluded rock excavation (Module 2.3). Work at Building 3 can't wait for a negotiated price, because the framer is scheduled to arrive in three weeks.

How changes work. There are three instruments, depending on whether the change affects cost or time and whether the price is agreed:

Instrument	When it's used	Who signs
Order for a minor change (often an architect's supplemental instruction)	No change to cost or time	Architect
Construction change directive (CCD)	The work must proceed before the price is agreed	Owner and architect
Change order	Scope, price, and time agreed	Owner, contractor, and architect

Change Order Process:

1. **The change is identified.** It can come from anyone: an unforeseen condition, a design error found through an RFI, an owner request, or an inspector's requirement.
2. **Proposal request.** The architect describes the change, with sketches or revised documents, and asks the CM to price it.

3. **Change order proposal.** The CM prices it: labor, materials, equipment, subcontractor costs, markups, and any time impact, with backup.
4. **Review and negotiation.** The architect and the owner's rep check quantities, rates, and markups against the contract. Markups on changes are usually set in the contract, so the negotiation is about quantities, rates, and time.
5. **Decision.** The owner approves according to the decision authority matrix (Module 2.1) and the change order is signed.
6. **Documents updated.** The change is reflected in the drawings or specifications, and, if it affects something the building department reviewed, submitted as a permit revision (Module 2.6).

For the rock at Building 3, the architect and owner issue a **construction change directive** so work continues immediately, with the cost tracked on a time-and-materials basis or at an agreed unit rate. Once the quantity is known, the directive becomes a change order: \$105,600, charged to owner contingency.

Every change gets a category. Example five: unforeseen conditions, owner scope, design errors, code and inspector requirements, and credits. The category matters because each points to a different fix. Unforeseen conditions are what contingency is for. Owner scope is a decision the owner made and can control. Design errors trace back to coordination (Module 2.2) and, under *Spearin*, are the owner's cost first.

Claims

When the owner and CM can't agree whether something is a change, or what it's worth, it becomes a **claim**. The general conditions set the procedure: written notice within a set period, supporting documentation, a decision by the initial decision maker (often the architect under A201), then mediation, and then arbitration or litigation as the contract specifies. Most claims settle before they get far. The documents that decide them are the ones kept every day: daily logs, meeting minutes, RFIs, photos, and schedule updates.

Framing and Structure (Months 4–17)

In the field

Framing is when the buildings appear. For a three-story garden building, the framer builds:

- **First-floor walls**, either stick-built on site or delivered as prefabricated wall panels
- **Floor systems** for the second and third floors, typically engineered wood joists or open-web floor trusses with plywood or OSB subflooring
- **Walls for the upper floors**
- **Shear walls**, the specific walls sheathed and nailed to resist wind and earthquake loads, anchored to the foundation with hold-downs
- **Stairs and breezeway framing**
- **Roof trusses**, set with a crane or telehandler and braced, then sheathed

A Type A building takes the framer about seven weeks. Crews then move to the next building, and the flow of trades follows behind (Module 2.6).

Inspections: the framing inspection

The framing inspection is one of the most important on a wood-frame project. The inspector checks the structure against the approved drawings and the code: wall and floor framing, headers, shear wall sheathing and **nailing** (the specified nail size and spacing is what makes a shear wall work), hold-downs, fireblocking and draftstopping in concealed spaces, and truss installation and bracing. Many jurisdictions also require special inspection of certain structural elements.

Framing inspections often happen after rough-in (so everything in the walls can be seen at once) or in stages. Either way, **nothing gets covered until it passes**. A failed framing inspection on one building delays drywall on that building, and every trade behind it in the flow.

Deferred submittals arrive

The roof trusses are a **deferred submittal**: the truss manufacturer's engineer designs them to the architect's and structural engineer's criteria, and the design goes to the engineer of record and the building department for approval before fabrication (Modules 2.2 and 2.6). The truss drawings also carry installation and bracing requirements the framer must follow. Temporary bracing during setting is a safety issue as much as a structural one.

A design error surfaces

At month 10, framing the breezeway stair at Building 2 reveals that the architectural and structural drawings disagree on the stair opening dimension. The framer stops and the CM submits an RFI. The architect's answer requires re-framing the opening and resizing a header. Change order: \$70,400, categorized as a design error.

Under *Spearin*, the owner warrants the drawings to the CM, so the owner pays first. Whether any of it is recoverable from the architect depends on whether the error falls below the standard of care. It's also the prototype, so the owner's rep's first question should be: **is the same condition at every other building?** On a nine-building project, catching a repeated error at Building 2 instead of Building 7 is worth a lot.

Quality: first-work inspections

The CM's quality program should include a **first-work inspection** for every major trade: the first building framed, the first shear wall nailed, the first window installed is inspected in detail by the superintendent, the architect, and sometimes the manufacturer's representative before the trade repeats it eight more times. Problems caught at Building 1 are fixed once. Problems caught at the Building 9 inspection were built nine times.

Fire risk peaks

A framed, unsheathed wood building is at its most vulnerable to fire. This is when the builder's risk insurer's requirements (Module 2.5) matter most: site security, fencing, no unattended hot work, working hydrants, and in many jurisdictions, fire watch or temporary standpipes. A fire at this stage can destroy a building in minutes and cost months.

Dry-In (Months 5–18)

In the field

Dry-in means the building is closed to the weather, so interior work can start and materials inside won't be damaged. It involves:

- **Roofing:** underlayment, then shingles or the specified roof system, with flashing at edges, penetrations, and walls
- **Weather-resistive barrier:** the wrap or coating over the exterior sheathing
- **Windows and exterior doors,** installed and flashed into the weather barrier
- **Siding and trim,** usually following a few weeks behind

The quality risk that matters most: water

More long-term problems in wood-frame multifamily come from water intrusion than from almost anything else, and most of it enters around windows, doors, and penetrations where the flashing sequence is wrong. The details in the documents (Module 2.4) show the correct sequence. Getting it built correctly depends on the installer.

The standard defense has three parts:

- **A pre-installation meeting** before the first window is installed, with the window installer, the siding installer, the weather barrier installer, the superintendent, the architect, and ideally the window manufacturer's representative. Everyone walks through the flashing sequence together.
- **A mockup:** a section of wall built with a window, flashing, weather barrier, and siding, approved before production installation starts.
- **Water testing:** field testing of installed windows to a standard such as ASTM E1105, often required on the first installations and on a sample afterward.

These cost a few thousand dollars. A leak discovered after drywall costs orders of magnitude more, and water damage discovered after occupancy costs even more.

Delays and weather

At month 21, water enters Building 8 during a storm before it's fully dried in, damaging framing and insulation. Change order: \$59,400, categorized as unforeseen. Some of this is the CM's risk (protecting the work is the contractor's responsibility under most contracts), and some may be claimed as weather. The contract language and the daily logs decide which. It's also why the windows' place on the critical path (Module 2.6) matters: every week a building sits framed but not dried in is a week of exposure.

MEP Rough-In (Months 6–19)

In the field

With a building dried in, the mechanical, electrical, plumbing, and fire protection trades run their systems through the framing before the walls are closed:

- **Plumbing:** drains, vents, and water supply lines to every fixture location; "top-out" once the piping reaches the roof
- **HVAC:** air handlers set in unit closets, ductwork, refrigerant lines to the outdoor condensers, bath and kitchen exhaust, dryer vents
- **Electrical:** boxes, wiring, panels in each unit, feeders from the meter centers
- **Fire sprinklers:** piping and heads, per the sprinkler subcontractor's approved layout (another deferred submittal)
- **Low voltage:** data, TV, access control, and fire alarm wiring

This is the most crowded phase inside the buildings. Five or six trades work in the same units, running pipes, ducts, and wires through the same joist spaces and walls. It's where the coordination work from Module 2.2 either pays off or doesn't.

Coordination problems appear

Coordination failures that weren't caught during design show up here. At month 19, the HVAC subcontractor can't route a duct as drawn because the floor framing and a plumbing drain occupy the same space. The fix requires rerouting ductwork in multiple units. Change order: \$95,700, a design error.

Design-assist trades (Modules 2.1 and 2.5) and a field coordination process (the trades walking the first unit together before rough-in starts) prevent many of these. A **first-unit walk** at Building 1, with every rough-in trade present, is one of the cheapest quality investments on the job.

Inspections pile up

Rough-in brings the heaviest concentration of inspections:

- **Rough plumbing, mechanical, electrical, and sprinkler inspections** by the building department and fire marshal
- **The framing inspection**, often combined with rough-in
- **Firestopping and fire-rated assembly inspections**: where pipes and wires pass through rated walls and floors, the openings must be sealed with tested firestop systems, and many jurisdictions require special inspection before the assembly is covered
- **Insulation inspection** under the energy code, after insulation is installed and before drywall

Each building needs all of them passed before drywall. On a nine-building project running on a six-week cadence, there are inspections almost every week. The superintendent schedules them in the lookahead, and a missed or failed inspection is a constraint that shows up immediately in pull planning (Module 2.6).

The inspector adds a requirement

At month 14, during an inspection, the fire marshal requires an additional fire department connection and additional signage at two buildings, beyond what the approved drawings showed. Change order: \$46,200, categorized as a code and inspector requirement. Inspectors do sometimes interpret the code differently in the field than the plans examiner did in review. The owner's rep should ask the architect whether the requirement is actually in the adopted code or is a field interpretation that can be discussed. Sometimes it's worth the conversation. Often it isn't, and the faster resolution is to do it.

Drywall and Finishes (Months 8–22)

In the field

Once a building passes its cover inspections, it moves into finishes, the longest and most labor-intensive phase:

1. **Drywall:** hang, tape, mud, sand, and texture. Rated assemblies (the demising walls from Module 2.4) are inspected before the joints are finished.
2. **Paint**
3. **Cabinets and countertops,** built from field measurements taken during framing
4. **Flooring:** the vinyl plank from Module 2.3, carpet, tile
5. **Trim and doors**
6. **Plumbing fixtures, electrical devices, light fixtures, and HVAC grilles:** the "trim-out"
7. **Appliances**
8. **Final cleaning**

On a garden project, this is a unit-by-unit production line. Each building's 24 units move through the same sequence, with trades following each other from unit to unit.

Samples and selections

Finishes bring the **samples** submittals: paint colors, flooring, cabinet door styles and finishes, countertops, tile. These are usually the owner's and the interior designer's decisions, and they have to be made on time. A late color selection holds the paint subcontractor, and the flooring waits behind the paint.

Quality: the unit punch

Most CMs run their own **quality control punch** as each unit finishes: a room-by-room check for damage, missing items, incomplete work, and defects. The architect's and owner's formal punch lists come later (Module 2.8). A disciplined internal punch is what keeps the formal punch list short.

Owner changes add up

At month 22, the leasing team asks for upgraded finishes in the model units. Change order: \$39,600, owner scope . Each owner request is reasonable on its own. Together, they're the largest category in the change log: 36.6% of all change order costs. The owner's rep's job is to test each one with a question: does this add enough NOI, or leasing velocity, to be worth its cost? The model units are a reasonable yes. The EV chargers, requested at month 17, are the harder call (see the deliverable).

Site Completion (Months 18–24)

In the field

As the buildings finish, the site work that was held back comes back:

- **Final grading, curbs, sidewalks, and paving**, sequenced so that finished paving isn't damaged by construction traffic
- **Parking lot striping and signage**, including accessible spaces and routes
- **Landscaping and irrigation**, largely under an allowance (Module 2.5)
- **The pool, pool deck, and amenities**
- **Site lighting and site furnishings**
- **The deceleration lane** in the public road, under the road authority's permit and inspection

Site work also has to support the phased turnover: Group 1's buildings need finished paths, parking, lighting, and accessible routes while construction continues on the rest of the site (Module 2.6).

The last inspections before turnover

At month 23, the building department's inspection of the accessible routes finds curb ramps and striping that don't meet the requirements. Change order: \$31,900, a code requirement. Accessibility at the site level (curb ramps, slopes, cross slopes, accessible parking and routes to building entrances) is one of the most common items found at final inspection, because it's often the last work built and the most sensitive to small grading errors.

Module 2.8 picks up with substantial completion, punch lists, and the certificate of occupancy.

The Owner's View: Reporting Every Month

What the owner's rep tracks

Across the whole sequence, the owner's rep keeps a handful of logs current and summarizes them each month for the developer, the investors, and the lender:

- **Schedule:** actual progress by building, the critical path, forecast completion, and forecast turnover dates by group
- **Changes:** every change order, CCD, and pending change, with its category and cost
- **Contingency:** owner contingency remaining, and whether it's enough for the risk still ahead
- **Buyout:** GMP budgets versus subcontracts, and the savings line (Module 2.5)
- **Pay applications:** billed to date, retainage held, and whether billing matches progress

- **RFIs and submittals:** counts, ages, and anything holding critical work
- **Inspections and quality:** failed inspections, re-inspections, and repeated defects
- **Safety:** incidents and near misses
- **Risks:** the risk register from Module 2.6, updated
- **Decisions needed:** anything the owner has to decide, with a deadline

Two ways to measure contingency

A note that belongs in every owner report: **judging contingency against percent complete is misleading.** The naive test asks whether contingency remaining is proportional to work remaining. If the project is 70% complete and 40% of contingency is left, it looks fine.

The better test asks whether contingency remaining is enough for the **risk** remaining. Risk isn't spread evenly through a project. The riskiest phases for this project are rough-in, finishes, inspections, and closeout, which are late. At month 17, a naive test shows \$329,215 of cushion. Measured against the risk still ahead, contingency is \$5,550 short. Same numbers, opposite conclusion.

Forecast to complete

A monthly report looks backward at what was spent and forward at what's left. The **forecast to complete** is the owner's best estimate of final cost: the GMP, plus approved changes, plus pending changes, plus a realistic allowance for the risk remaining, minus any expected savings. Compared against the budget, it tells the owner whether the project is still inside its cushion.

Artifacts

Artifact	Produced by	When	What it's for
Schedule of values	CM	Before the first pay application	The basis for every payment
Payment applications and certificates	CM, certified by the architect	Monthly	Billing against progress
Lien waivers	CM and subcontractors	With each payment	Protecting the property from liens

Daily logs	Superintendent	Daily	The factual record of the job
OAC meeting minutes	CM	Weekly	The record of decisions and commitments
Architect's field reports	Architect	Each site visit	Observed conformance and issues
Submittal log	CM	Ongoing	Status of every submittal
RFI log	CM	Ongoing	Every question, its age, and its answer
Change order log	Owner's rep, CM	Ongoing	Every change, its category, and its cost
Inspection and testing reports	Building department, special inspector, testing agency	Ongoing	Evidence the work meets the code and specifications
Buyout log	CM	Monthly through buyout	Budget versus subcontract
Schedule updates	CM	Monthly	Progress and forecast completion
Monthly owner report	Owner's rep	Monthly	Everything above, summarized with decisions needed

Owner Decisions During Construction

Construction doesn't have phase approvals the way design does. It has a recurring set of decisions, every month:

Approve the pay application? Does the work billed match the work in place? Are the lien waivers in? Does the lender's monitor agree?

Approve this change? Is it a real change or within the CM's scope? What category? Is the price right? Does contingency have room for it, measured against the risk remaining, not against percent complete?

Pay to protect the schedule? When a delay threatens a turnover group or the leasing season, what does acceleration cost, and what does the delay cost?

Whose money is at risk: during construction, every owner change, unforeseen condition, and design error comes out of the owner's \$1.25 million contingency first. After that, it comes out of headroom. On this project, headroom is \$187,283.

Failure Modes

Approving the first pay application without reviewing the schedule of values. A front-loaded schedule of values leaves the owner paying ahead of the work for months.

Unmeasured unit prices. A unit price nobody measures is a blank check.

Treating RFI responses as changes. An answer that changes scope, price, or time needs a change order. Without one, it becomes a dispute at closeout.

Letting submittals pass through without review. The CM's review and the architect's review are real checkpoints. The water intrusion, coordination, and code problems they miss are paid for in the field.

Covering work before it's inspected. Anything covered before it passes has to be uncovered.

Finding a prototype error at the last building. First-work inspections and first-unit walks catch repeated problems once instead of nine times.

Skippping the window mockup and water test. A few thousand dollars before installation, or much more after drywall or after occupancy.

Approving owner changes one at a time. Each is reasonable. Together, they were the largest category of change on this project.

Judging contingency by percent complete. It showed \$329,215 of cushion at month 17. Against the risk remaining, contingency was short.

Thin daily logs. Every delay claim is decided by the records kept when nothing seemed to be going wrong.

Where AI Is Changing This

Construction administration is where AI agents are arriving fastest. Tools can draft RFIs and responses, check submittals against specifications, generate meeting minutes and daily logs, and extract data from pay applications and lien waivers. Reality capture tools compare site photos and scans to the schedule and the model to measure progress. Part 3 covers these in Modules 3.13 through 3.17.

Part 3 also raises the workforce question this module makes concrete. The project engineer's job (processing submittals, RFIs, and logs) is how most project managers learned construction. If agents absorb that work, the industry needs a new way to build the judgment this module is about: knowing, when a framer stops work at a stair opening in month 10, that the first question is whether the same error is in the other eight buildings.

Exercise

Decision scenarios

These use the reference project. Reasoning notes follow each one.

Scenario 1: Rock at Building 3. At month 8, excavation for the Building 3 footings hits rock. The GMP excludes rock excavation. The framer is scheduled to start Building 3 in three weeks. The CM proposes a time-and-materials construction change directive and estimates \$90,000 to \$120,000.

Reasoning notes: The work has to proceed: holding Building 3 delays the framer, which delays every building behind it in the flow. A CCD is the right instrument, because it authorizes the work before the price is agreed. Protect the owner in how it's written: require daily time-and-materials tickets signed by the owner's rep or the special inspector, use equipment rates from a published schedule or the contract, have the geotechnical engineer confirm the rock is actually rock (not dense material that could be excavated with standard equipment), and survey the quantity. Then check the other buildings: is there any reason to think rock extends under Buildings 4 or 5? A few test pits now are cheap compared with another surprise.

Scenario 2: The failed shear wall nailing. At Building 5's framing inspection, the inspector finds shear wall nailing at wider spacing than the drawings require on several walls. The framer argues it can fix the nailing later and wants to let the MEP trades start rough-in so the flow doesn't stop.

Reasoning notes: The shear walls resist wind and earthquake loads; the nailing is what makes them work. The inspector won't pass the framing, and the fix has to happen before anything

covers the walls. Letting rough-in proceed around unfinished shear walls risks covering them. The fix is the framer's cost, because it's the framer's error. The schedule effect matters more: if the fix takes three days and the MEP trades can work on other parts of Building 5 in the meantime, the flow barely moves. The bigger question is why it happened. If Building 5 has the problem, check Buildings 3 and 4, which passed inspection but may have been inspected less thoroughly. And if the framer's crews are stretched thin (the risk Module 2.5 flagged), this is an early sign.

Scenario 3: The first schedule of values. Before the first pay application, the CM submits its schedule of values. Mobilization is \$420,000, general conditions are billed 30% in the first three months, and sitework is weighted 8% above the buyout amount while finishes are 6% below.

Scenario 4: The EV chargers. At month 17, the leasing team asks to add EV chargers at 20 parking spaces, at a quoted cost of \$178,200. The site was built EV-ready (Module 2.2), so the conduit and panel capacity are in place. The developer asks for a recommendation in this month's owner report.

Deliverable: Monthly Owner Report

Write the month 17 owner report for the reference project, or a monthly report for your own project. Keep it to about four pages. Lead with what the owner needs to know and decide.

1. Summary (half a page). Status in three sentences: schedule, cost, and the decision needed this month.

2. Schedule. Progress by building (framing, dry-in, rough-in, finishes, complete), the critical path, the forecast date for each turnover group, and the forecast full-community CO compared with month 25. Note the first transformer's status, since Group 1 depends on it.

3. Change log. Every change order and CCD to date, with its month, category, and cost, and pending changes with their estimated costs. For month 17 on the reference project:

Month	Change (category)	Owner cost
8	Rock at Building 3 footings (unforeseen)	\$105,600
9	Upgraded amenity AV package (owner scope)	52,800
10	Stair opening conflict, architectural vs. structural (design error)	70,400
11	Utility-required transformer pad relocation (unforeseen)	41,800
13	Deleted unused conduit sleeves (credit)	(22,000)

14	Fire marshal field requirement: added FDC and signage (code)	46,200
16	Omitted roof overflow scuppers (design error)	63,800
	Approved through month 17	\$358,600
17	EV chargers at 20 spaces (owner scope) — pending decision	178,200

Summarize by category and by share.

4. Contingency. Owner contingency remaining: \$891,400 before the chargers, \$713,200 if they're approved.

5. Buyout and pay applications. The buyout log summary, including the \$231,000 net savings and how it's being treated under the contract. Billed to date, retainage held, and whether billing matches progress.

6. Forecast to complete. GMP plus approved changes plus pending changes plus the risk remaining, minus expected savings, compared with the budget and the headroom.

7. Open items. RFIs and submittals that could affect the critical path; failed or pending inspections; quality issues and whether they repeat across buildings; safety.

8. Risks. The top five from the risk register, updated.

9. Decisions needed. Each decision, the recommendation, the reasoning in two or three sentences, and the deadline. For month 17, that's the EV chargers.

Module 2.8 follows the reference project through the finish line: substantial completion, punch lists, commissioning, the certificate of occupancy, warranties, and the handover to the property manager.

2.8 — Closeout, Turnover, and Handback to Operations

Why This Module Exists

Closeout has a reputation as the quiet end of a project: punch lists, binders, and paperwork after the real work is done. Financially, it's the opposite. It's the hinge of the whole project, the moment construction money stops and operating money starts. Every week it slips, the project pays for both.

A one-month slip in the certificate of occupancy costs about **\$117,660**. Two months cost about \$300,000, and three months about \$530,000. A poor turnover package, with missing warranties, missing equipment data, and uncommissioned systems, costs about **\$390,000** in value at sale, against **\$60,000** to do it right. Retainage of roughly **\$1.8 million** is waiting to be released, and releasing it too early or too late both cost money.

This module follows the reference project across the finish line:

- What "done" means, contractually and legally: substantial completion, final completion, and certificates of occupancy
 - Commissioning: proving the building works, not just that it's built
 - The inspections and tests that gate the certificate of occupancy
 - The punch list, at the scale of 200 units
 - Closeout documents, warranties, final payment, and the final accounting of the GMP
 - The handover to the property manager, and what they need on day one
-

Where the Project Is

Phase	Months 16–26
Turnover plan (Module 2.6)	Group 1 (clubhouse and Buildings 1–3, 72 units) at month 18; Group 2 (Buildings 4–6, 72 units) at month 21; Group 3 (Buildings 7–9, 56 units, pool and final site work) at month 24; full-community certificate of occupancy at month 25

First transformer	Energized at month 17
Owner contingency	\$891,400 remaining at month 17 (Module 2.7)
What a month costs at closeout	About \$117,660, rising with each additional month

Concepts

1. What "Done" Means

Three different milestones get called "done," and they're controlled by different people under different rules.

Substantial completion (contractual)

Under the A201 general conditions, **substantial completion** is the stage when the work, or a designated portion of it, is complete enough that the owner can occupy or use it for its intended purpose. The architect determines it after an inspection, and it's documented with a certificate of substantial completion (the AIA G704 is a common form) that attaches the punch list of remaining items.

Substantial completion is one of the most consequential dates in the contract. It typically:

- Stops the contract time, so liquidated damages (if any) stop accruing
- Starts the contractor's one-year correction period (A201 Section 12.2.2) and many warranties
- Shifts responsibility for security, maintenance, utilities, and insurance to the owner, as stated on the certificate
- Triggers release of most retainage, less a holdback for the punch list

A201 also allows **partial occupancy** before the whole project is complete, with the owner and contractor agreeing on the terms. With three turnover groups, the reference project will have three certificates of substantial completion, each covering its buildings.

Final completion (contractual)

Final completion comes when all the work is done, including the punch list, and all closeout documents have been delivered. It triggers final payment, the release of the remaining retainage, and the end of the contractor's active obligations, apart from warranties.

Certificate of occupancy (regulatory)

The **certificate of occupancy** (CO) is the building department's determination that the building, as constructed, complies with the approved documents and the adopted codes, and can be legally occupied. The building department doesn't care about the contract, the budget, or the leasing season. It cares whether the building passed every required inspection and test.

A **temporary certificate of occupancy** (TCO) allows occupancy of all or part of a building before everything is finished, when the incomplete work doesn't create a hazard to occupants. TCOs are time-limited (often 30 to 90 days, depending on the jurisdiction), usually conditioned on specific remaining work, and sometimes require a bond or deposit for incomplete site work. Phased turnover often relies on them.

When the two systems don't line up

Substantial completion is the architect's determination under the contract. The CO is the building department's determination under the law. They should arrive together. When they don't, the project has a problem.

The dangerous case is when the architect certifies substantial completion, but the building department withholds the CO. The contractor argues it has substantially completed the work: the warranty period starts, insurance and utilities shift to the owner, and retainage should be released. The owner can't use the building. It pays carrying costs on units it can't lease.

The misalignment usually comes from one of three causes:

- A required test or inspection hasn't happened yet, such as the fire alarm acceptance test the fire marshal must witness
- What was built differs from the approved permit documents, because a change wasn't submitted as a permit revision (Module 2.6)
- A system doesn't perform at final testing

Experienced owners fix this with one sentence in the supplementary conditions: **"Substantial completion shall not be deemed achieved until a certificate of occupancy, or a temporary certificate of occupancy acceptable to the owner, has been issued."** It ties the contractual milestone to the regulatory one, so the contractor can't claim substantial completion, with its financial consequences, until the building can actually be occupied. The reference project's contract includes it.

2. Commissioning: Proving It Works

What commissioning is

Commissioning is a quality process that verifies the building's systems are designed, installed, tested, and able to be operated and maintained according to the owner's requirements. It isn't the same as startup (turning equipment on for the first time), testing and balancing (adjusting

air and water flows), or the building department's inspections. Commissioning checks that those things were done and that the systems work together.

It started in Module 2.1, with the **Owner's Project Requirements** and the design team's **Basis of Design**. Those documents are what commissioning tests against.

The process

1. **Commissioning plan.** The commissioning agent, often an independent firm hired by the owner, writes the plan during design: which systems, what tests, who participates, and when.
2. **Design review.** The commissioning agent reviews the design for operability and maintainability, ideally during DD and CD.
3. **Submittal review.** For commissioned systems, the agent reviews key submittals.
4. **Prefunctional checklists.** As equipment is installed, the trades complete checklists confirming it's installed correctly, connected, labeled, and ready to start.
5. **Startup.** Equipment is started by the installer, often with the manufacturer's representative.
6. **Functional performance testing.** The agent directs or witnesses tests of each system through its operating modes (heating, cooling, economizer, occupied and unoccupied schedules, alarms, and failure modes) to confirm it works as designed.
7. **Issues log.** Deficiencies are logged, assigned, fixed, and retested.
8. **Training.** The property manager's maintenance staff are trained on each system.
9. **Commissioning report.** A record of every test and issue, and the final status.
10. **Seasonal testing and the post-occupancy review.** Systems that couldn't be fully tested in the season of turnover (heating tested in summer, for example) are retested later, typically before the end of the one-year correction period.

Commissioning on a garden community

Commissioning is often associated with large commercial and institutional buildings, and on a garden project the scope is narrower. It commonly covers:

- **The clubhouse:** HVAC, controls, lighting controls, domestic hot water, and the fitness and leasing spaces
- **The pool:** pumps, filtration, chemical treatment, heating, and safety systems
- **Fire alarm and fire sprinkler systems,** together with the acceptance tests the fire marshal witnesses
- **Domestic water:** backflow prevention, pressure, and hot water delivery
- **A sample of units:** heat pumps, air handlers, thermostats, bath and kitchen exhaust, and water heaters, tested in a statistically meaningful sample of each unit type
- **Energy code testing:** many jurisdictions require duct leakage testing and building envelope air leakage testing for residential units, on all units or a sample, as a condition of final inspection
- **Site lighting and access control**

3. The Inspections and Tests That Gate the CO

What has to be complete

Requirements vary by jurisdiction, but on a project like this, the CO for each turnover group typically requires:

Inspections passed:

- All building, mechanical, electrical, and plumbing finals for the buildings in the group
- Fire sprinkler final, including the underground fire line and the fire department connection
- Energy code finals, including duct and envelope leakage testing where required
- Accessibility at the buildings and along the accessible routes serving them (Module 2.7 found curb ramp problems at month 23)
- Site and civil items required for the buildings: paving or all-weather access, fire lanes and signage, water and sewer connections, and stormwater facilities

Tests completed and accepted:

- Fire alarm acceptance test, witnessed by the fire marshal, in which every initiating device and notification appliance is tested and each programmed function is verified
- Backflow preventer testing
- Any smoke control or emergency systems, if present

Documents submitted:

- The special inspector's final report, with all nonconformances resolved
- Letters from the engineers of record, where the jurisdiction requires them, stating the work conforms to the approved design
- Energy code compliance documentation reflecting what was actually installed

Outside agencies:

- Utility service energized and metered
- Road authority acceptance of any public improvements required for occupancy, such as the deceleration lane. Some jurisdictions let the owner post a bond for incomplete public improvements so they don't hold the CO.
- Health department approval before the pool opens

Building the CO path into the schedule

Every item above is a schedule activity with a duration and a responsible party, and several depend on people the CM doesn't control: the fire marshal's availability for the acceptance test, the utility's metering crew, the road authority's final inspection. The master schedule (Module 2.6) should show them explicitly for each group, working backward from the target turnover date. The fire alarm acceptance test in particular often requires booking the fire marshal weeks

in advance and requires the system to be fully programmed and pretested before the marshal arrives.

4. The Punch List

What it is

The **punch list** is the list of work that's incomplete or doesn't conform, found at the substantial completion inspection. It's attached to the certificate of substantial completion and has to be finished before final completion.

On a single commercial building, a punch list might have a few hundred items. On 200 apartment units, it can have thousands: paint touch-ups, a missing cabinet door bumper, an outlet cover not installed, a scuffed floor, a door that doesn't latch, a grout joint not sealed. Each is small. Together, they decide whether units are ready to rent.

How it works at unit scale

A well-run process on a garden project has four layers:

1. **The CM's internal quality punch** as each unit is finished (Module 2.7). This catches most items before anyone else sees them.
2. **The contractor's punch list**, a room-by-room walk of every unit when the CM believes a building is complete.
3. **The architect's and owner's punch walk**. The architect, the owner's rep, and ideally the property manager walk the units and add items. Punch-list software with photos and locations tied to unit and room makes this manageable at 200 units.
4. **The back-punch**: a verification walk after the items are fixed, confirming each one is complete before it's closed.

The property manager's participation matters more than it seems. The architect checks conformance with the documents. The property manager checks whether a resident will call about it in the first week.

Holding back money for the punch list

At substantial completion, most retainage is released, but enough is held to get the punch list finished. On the reference project, with a \$150,000 punch list, that's \$225,000 held and the rest released with lien waivers.

Holding back too little leaves the owner without leverage for the last 5% of the work, which is often the hardest to get done because crews have moved to other jobs. Holding back far too much invites claims and slows the closeout the owner is trying to finish.

Punch in occupied buildings

With phased turnover, punch work in Group 1 continues while residents are moving in, and construction continues on Groups 2 and 3 next door. That changes how the work gets done: access to occupied units has to be scheduled with notice, trades have to work around residents, and safety barriers between residents and active construction have to be maintained (Module 2.6). It's much easier to finish the punch list before a resident arrives than after.

5. Closeout Documents and the Money

What the contractor delivers

Division 01's closeout section (Module 2.4) lists what the contractor must deliver. On the reference project, the major items are:

- **Operation and maintenance (O&M) manuals:** for every system and piece of equipment, including manufacturer's instructions, maintenance schedules, parts lists, and contacts. Increasingly delivered as searchable digital files organized by building and system, rather than binders.
- **Warranties:** the contractor's general warranty, plus manufacturers' warranties for the roof, windows, siding, appliances, HVAC equipment, water heaters, and more, each with its start date, term, and claim procedure.
- **Record drawings:** the documents updated to show what was actually built, including field changes, relocated utilities, and concealed work. Underground utility locations matter most, because they're invisible and expensive to find later.
- **Equipment list or asset register:** every major piece of equipment with its location, model, serial number, installation date, and warranty expiration.
- **Commissioning report and test reports:** fire alarm, sprinkler, backflow, duct and envelope leakage, testing and balancing.
- **Attic stock:** extra materials specified for future repairs, such as flooring, paint, tile, and shingles.
- **Keys, access credentials, and codes:** a keying schedule and the access control system's programming, transferred to the owner.
- **Training records:** who was trained on what.
- **Final lien waivers** from the CM and every subcontractor and major supplier.
- **Consent of surety** to final payment, so the bond remains in force.
- **Final certificates and approvals:** the CO, inspection records, and any outside agency acceptances.

Warranties: which clock is running?

Warranty start dates are a common source of confusion and lost money, especially with phased turnover:

- The contractor's **one-year correction period** under A201 starts at substantial completion, which on this project means three different start dates, one per turnover group.
- **Manufacturers' warranties** often start at installation or shipment, not at substantial completion. The roof on Building 1 was installed around month 7. If its manufacturer's

warranty started then, the owner has already lost almost a year of it by the time the building is occupied.

- **Some warranties require registration** within a set period, or they revert to a shorter term.

The fix is in the specifications, written during design: require manufacturers' warranties to start at substantial completion of the building where the product is installed, and require the contractor to register them. At closeout, the owner needs a **warranty log**: every warranty, its start and end date, what it covers, and who to call. **Final payment and retainage**

Retainage is one of the largest single payments on the project. For the example budget, this computes to about \$1.79 million on its budget, released after completion. With phased turnover, retainage can be released by group as each reaches substantial completion, less each group's punch holdback.

Final payment is made when:

- The architect certifies final completion
- All closeout documents have been delivered
- Final lien waivers and the surety's consent are in hand
- The lender's construction monitor and title company are satisfied that no liens are

The final accounting of the GMP

Under a GMP, the project's final cost has to be reconciled: approved change orders, final allowance amounts, unused contingency, and buyout savings. On the reference project:

Item	Amount
GMP (Module 2.5)	\$25,329,837
Approved change orders	739,400
Buyout savings, final (net across all packages)	(268,000)
Unused CM construction contingency (\$542,625 less \$395,000 used)	(147,625)
Savings pool, split 50/50 under the contract	\$415,625
Owner's share returned	(207,813)
Final hard cost to the owner	\$25,861,424

(The chargers from Module 2.7's month 17 decision were approved later, after MEP rough-in retired its risk, and are included in the change order total. The CM keeps its half of the savings pool as an incentive, as the contract intended.)

What that means for the deal:

	At underwriting	After the DD crisis (Module 2.3)	Final
Hard cost	\$24,500,000	\$25,295,000	\$25,861,424
Owner contingency unused	—	—	\$718,413
Total development cost	\$35,000,000	\$35,795,000	\$35,111,424
Stabilized NOI	\$2,280,070	\$2,251,070	\$2,251,070
Yield on cost	6.51%	6.29%	6.41%
Spread	126 bps	104 bps	116 bps

(Soft costs are held constant for simplicity, and this assumes the CO arrives on time.)

The project finishes with a spread of 116 basis points: better than the 104 bps it was left with after the DD crisis, and 10 bps short of underwriting. It spent \$531,587 of its \$1.25 million owner contingency. Nothing about that outcome was luck. It came from the gap closed at DD without cutting value, a GMP review that removed a phantom contingency, a buyout that came in under budget, a framer chosen for capacity, and a schedule that held the CO date.

6. Handover to Operations

The property manager takes over

At turnover, the building stops being a construction project and becomes a business. The property manager, whether a third-party management company or the owner's own team, becomes responsible for leasing, residents, maintenance, and operating costs. Module 2.1 argued for engaging them during design. This is where that pays off.

On day one of each turnover group, the property manager needs:

- **Units that are rent-ready**, not just construction-complete. A unit that passed the architect's punch walk can still need cleaning, blinds, smoke detector checks, and a final walk before a resident moves in. The CM and the property manager should agree on a **unit acceptance walk** and a definition of rent-ready before the first group turns over.
- **Utilities transferred**. Common-area utilities move to the owner's accounts. Individually metered unit electric (Module 2.2) moves to residents as they move in, and the property manager needs a process for vacant units. Utility accounts left in the CM's name are a common, avoidable mess.
- **Access and keys**. The access control system programmed with the property manager's administrators, unit keys or credentials cut and logged, and the construction access codes changed.
- **Service contracts in place**, starting the day the group turns over: fire alarm monitoring (required once the building is occupied), trash and recycling, pest control, landscape maintenance, pool service, and internet or cable if the owner provides it.
- **The inspection, testing, and maintenance schedule for fire protection systems**. The fire code requires ongoing inspection and testing of sprinklers, fire alarms, extinguishers, and backflow preventers on set schedules. The property manager needs to know what's due when.
- **The asset register loaded into a maintenance system**, so preventive maintenance (filter changes, water heater flushing, HVAC service) starts on schedule, not after the first failure.
- **Warranty procedures**: who to call for what during the one-year correction period (usually the CM for construction defects) and for manufacturer warranties afterward.
- **Training**, completed and documented.

The model units and the leasing office

Leasing starts in the clubhouse and the model units, which should be finished, furnished, and photographed before Group 1 turns over. The model unit upgrades change order in Module 2.7

was part of this. Leasing velocity in the first weeks depends heavily on how the leasing office and models show, which is why the clubhouse turns over with Group 1.

Operating costs start before revenue does

Fixed operating costs (staff, insurance, taxes as a completed building, utilities for common areas and vacant units, marketing) arrive with the CO, while revenue arrives slowly as units lease, often with a free month as a concession. NOI is negative for the first several months of lease-up. With phased turnover, the cost starts earlier, at month 18 for Group 1, but so does leasing. The property manager's budget has to reflect the actual turnover schedule.

The eleven-month walk

About eleven months after each group's substantial completion, before the contractor's one-year correction period ends, the owner's rep, the property manager, and the CM walk the buildings and review the maintenance and resident request logs for defects covered by the correction period: roof leaks, drywall cracks, doors that stopped latching, HVAC problems, drainage issues. The commissioning agent's seasonal testing happens around the same time. Anything found after the correction period ends becomes the owner's cost, unless a manufacturer's warranty covers it.

7. Where Part 2 Ends

Part 2 started in Module 2.1 with the handoff: the developer had closed on the land and handed the team a program, a budget, a schedule, and a stack of diligence reports. It ends here with the handback: the team hands the owner and the property manager an occupied building, a turnover package, and a set of operating obligations.

That decision circles back to Part 1. The buyer at month 38 will run its own diligence, and one of its first requests will be a property condition assessment (Part 1 Module 4B). The assessor will read the turnover package: the warranty log, the equipment list, the commissioning report, the record drawings, the maintenance history. A complete package is evidence that the building's operating costs are what the rent roll says they are. A missing one is a reason to lower the price.

The turnover package is sold with the building.

Artifacts

Artifact	Produced by	When	What it's for
Commissioning plan, issues log, and report	Commissioning agent	Design through post-occupancy	Proving the systems work as intended
CO path schedule, by turnover group	CM with the owner's rep	Months 12 onward	Every inspection, test, and approval needed for occupancy
Punch lists and back-punch records	CM, architect, owner's rep, property manager	Each turnover group	What remains, and proof it's done
Certificates of substantial completion	Architect	Each turnover group	Starts warranties and shifts responsibilities
Certificates of occupancy or TCOs	Building department	Each turnover group	Legal permission to occupy
O&M manuals and asset register	CM, subcontractors	Before each turnover	What's installed and how to maintain it
Warranty log	CM, reviewed by the owner's rep	Before each turnover	Every warranty, its dates, and who to call
Record drawings	CM and architect	Final completion	What was actually built
Final lien waivers and consent of surety	CM, subcontractors, surety	Before final payment	Clear title and bond protection
GMP final accounting	CM, reviewed by the owner's rep	Final completion	The final cost and savings split

Turnover package	Owner's rep, compiled from all of the above	Before each group turns over	Everything the property manager needs
Occupancy readiness plan	Owner's rep with the CM and property manager	Months 12 onward	What has to be true for each group to open, and who's responsible

Phase Approval: Accept Each Turnover Group

The question: is this group ready for residents, legally, physically, and operationally?

- ☐ Certificate of occupancy or an acceptable TCO issued for every building in the group
- ☐ Fire alarm acceptance test witnessed and passed; fire alarm monitoring active
- ☐ Commissioning complete for the group's systems, or open items listed with dates
- ☐ Punch list walked with the property manager, and a holdback of 150% of its value set
- ☐ Units rent-ready by the agreed definition, confirmed by a unit acceptance walk
- ☐ Site work serving the group complete: parking, lighting, accessible routes, fire access
- ☐ Separation between residents and ongoing construction in place
- ☐ Utilities transferred; access control and keys transferred
- ☐ Service contracts started
- ☐ O&M manuals, asset register, and warranty log delivered for the group, with warranty start dates confirmed
- ☐ Maintenance staff trained
- ☐ Certificate of substantial completion issued, retainage released less the holdback, and lien waivers received

Whose money is at risk: from this point, the owner pays the fixed operating costs of the turnover group whether or not units are leased. Every month the group isn't ready costs rent it can't collect. Every defect the turnover package doesn't document eventually becomes the owner's cost, or the next buyer's discount.

Failure Modes

Letting substantial completion arrive before the CO. Without the supplementary condition tying them together, the contractor can start the warranty clock and claim retainage on a building nobody can occupy.

Scheduling the fire marshal last. The fire alarm acceptance test needs a fully programmed, pretested system and a marshal's appointment. Both take weeks.

Treating commissioning as optional on residential work. It finds problems before the inspector does, and its absence shows up in higher operating costs the buyer capitalizes.

Punch lists without the property manager. The architect checks the documents. The property manager checks what residents will call about.

Holding back all or none of the retainage. Too little leaves no leverage. Too much invites claims. 150% of the punch list value is a common, defensible middle.

Warranties that started at installation. The roof on Building 1 lost a year of warranty before anyone lived under it.

Record drawings that don't show the underground. The most expensive thing to find later is a pipe nobody can locate.

Construction-complete units handed over as rent-ready. The resident's first impression is a unit that still needs cleaning.

Utility accounts, access codes, and service contracts left for "later." Each takes a week to fix and makes the first week of leasing harder.

Skipping the eleven-month walk. Everything found after the correction period ends is the owner's cost.

Treating the turnover package as paperwork. It's evidence of the building's operating costs, and the next buyer will read it.

Where AI Is Changing This

Closeout is where construction data either survives or disappears. Newer tools assemble O&M manuals and warranty logs automatically from approved submittals, verify as-built conditions against scans, and transfer the asset register into facilities management and maintenance systems. The larger shift is toward digital twins and operations platforms that carry design and construction data into the fifty years a building is operated. Part 3 covers these in Modules 3.18 (closeout, commissioning, and turnover) and 3.19 (handover to operations and digital twins).

The value of those tools depends on the discipline this module describes: warranty start dates written into the specifications, equipment data required in the submittals, record drawings actually updated in the field. Software can assemble a turnover package from the project's data. It can't recover data nobody collected.

Exercise

Decision scenarios

These use the reference project. Reasoning notes follow each one.

Scenario 1: The fire alarm test fails. Two weeks before Group 1's target turnover at month 18, the fire marshal's acceptance test fails: several devices weren't programmed to their correct zones, and some notification appliances don't meet the required sound levels in the bedrooms. The fire alarm subcontractor says it can fix everything, but the marshal's next available test date is five weeks out. For \$18,000, the subcontractor can put an extra technician on the job, and the owner's rep thinks a direct request to the fire marshal's office might secure a date in two weeks if the system is ready

Scenario 2: Substantial completion for Group 3. At month 24, the CM requests substantial completion for Group 3. The buildings have their COs. The pool isn't finished: the health department's final inspection is scheduled for month 25. The architect's punch walk found an average of about 35 items per unit.

Reasoning notes: The supplementary condition ties substantial completion to the CO. The buildings have theirs, so substantial completion can be granted for the buildings. The pool can't be occupied until the health department approves it, so it gets its own substantial completion later. That's what partial occupancy and designated portions are for. On the punch list, 35 items per unit on 56 units is about 2,000 items. That's not unusual, but it's too many to accept without a plan. Estimate the punch list's value, hold 150% of it, and require a completion schedule with a back-punch. Make sure the property manager's unit acceptance walk is part of the process, because these units are about to be leased.

Scenario 3: Warranty start dates. Preparing the Group 1 turnover package, the owner's rep finds that the roofing manufacturer's warranty on Buildings 1 through 3 started at the roof's completion, around months 7 to 9. The specifications required warranties to start at substantial completion, but the roofing subcontractor registered them at installation.

Reasoning notes: The specification is a contract requirement. The roofing subcontractor didn't comply, and it's the CM's responsibility to deliver warranties that meet the specifications. The practical fix is usually to have the manufacturer reissue or extend the warranty to start at substantial completion, which many manufacturers will do for a fee or after an inspection. The CM should pay for it. The owner's rep should also check every other warranty in the package for the same problem, because it's rarely only one. If the owner had never specified the start date, the lost year would simply be the owner's loss.

Scenario 4: Cutting the commissioning agent. At month 12, with contingency looking tight, the developer's CFO proposes eliminating the commissioning agent and closeout coordinator to save \$60,000. The CM says it will do its own startup and testing.

Deliverable: Turnover Package and Occupancy Readiness Plan

For the reference project, or your own project, produce the two documents the owner's rep would deliver before the first turnover group opens.

1. Occupancy readiness plan (3–4 pages)

For each turnover group:

- The buildings, units, and amenities in the group, and the target date
- Every requirement for occupancy: inspections, tests, outside agency approvals, site work, utilities, and commissioning, each with a responsible party and a date
- The critical path to the CO for that group, and what gates it
- The rent-ready definition and the unit acceptance walk process
- Life safety and access separation from ongoing construction
- Property management readiness: staffing, service contracts, utility transfers, access control, and training
- What a month of delay costs for that group, and what it would be worth paying to avoid one

2. Turnover package (table of contents and responsibility matrix)

A complete table of contents for the package the property manager receives, with each item's responsible party, format, and due date:

- O&M manuals, organized by building and system
- Asset register with model, serial number, location, installation date, and warranty expiration for every major piece of equipment
- Warranty log, with start dates tied to each group's substantial completion
- Record drawings, including underground utilities
- Commissioning report and all test reports
- Attic stock inventory
- Keying schedule and access control programming
- Training records
- Fire protection inspection, testing, and maintenance schedule
- Service contract list and start dates
- The CO or TCO, inspection records, and outside agency approvals
- Contacts for the CM, subcontractors, and manufacturers during the correction and warranty periods
- The eleven-month walk date for each group

End with a short memo to the developer summarizing the final accounting of the GMP and the project's final spread, and naming the three items in the turnover package a buyer's property condition assessment would ask for first.

Part 2 Conclusion

Part 2 followed one building from the day the developer closed on the land to the day the property manager took the keys.

Module 2.1 received the handoff and turned the pro forma's assumptions into requirements. Module 2.2 showed design as a sequence of commitments, each more expensive to reverse. Module 2.3 met the first real estimate, found the building didn't fit the budget, and closed the gap without destroying the value the budget was meant to protect. Module 2.4 taught the document set as one connected thing. Module 2.5 bought the work and turned the GMP into subcontracts. Module 2.6 planned the job and discovered that the utility, the framer's crews, and the fire marshal set the schedule more than any drawing did. Module 2.7 built it. Module 2.8 finished it and handed it back.

Two threads ran through all of it.

The first is **the document chain**. A requirement, whether it comes from the code, the owner, the lender, or the market, becomes a design decision, then a drawing or specification, then a contractual obligation, then built work, then an inspection, and finally a record in the turnover package. Every requirement follows that chain, and every expensive problem in Part 2 was a break in it: a value engineering decision that reached one document and not another, an exclusion nobody owned, a warranty that started on the wrong date.

The second is **money over time**. Part 2 carried three numbers from Module 2.1 to the end: the cost the deal could absorb, what a month costs, and what design control is worth. Nearly every decision in Part 2 protected one of them or spent it. The reference project started with 126 basis points of spread, fell to 76 when the DD estimate arrived, recovered to 104, and finished at 116. That isn't the story of a lucky project. It's the story of a team that knew, at each decision, what that decision was worth.

Part 1 decided whether to build. Part 2 built it. **Part 3** looks at the software and AI reshaping every step of what Part 2 described, from generative design and automated takeoff to agents that draft RFIs and platforms that carry construction data into operations, and asks, for each one, which break in the document chain it actually fixes.

Part 3 - Technology and AI

Why This Part Exists

Parts 1 and 2 cover how the work gets done: how sites are analyzed and bought, and how buildings are designed and delivered. Part 3 covers the software that runs each of those phases, who sells it, and what AI is actually changing about the work.

Vendor marketing describes AI as transforming everything. Skeptics on the jobsite say nothing has changed. The truth is somewhere in the middle, especially in this industry. The things that slow it down matter as much as the technology itself: professional licensure, liability, contract risk, the authority of the building official, and the physical reality of a jobsite.

How Part 3 Is Organized

Part 3 follows the same lifecycle as Parts 1 and 2, one phase at a time, so you can move between them. Read how feasibility works in Part 1, then read what software runs it and what AI is doing to it here.

Section	Chapters	Lifecycle phase
A. Real Estate & Project Conception	1–5	Market research through the owner's program
B. Design	6–8	Concept through coordinated models
C. Construction Documents & Approvals	9–10	Specifications, code compliance, permitting
D. Procurement	11–12	Estimating, bidding, contracts
E. Construction	13–17	Scheduling, project management, financials, field, workforce
F. Closeout	18	Punch, commissioning, turnover
G. Synthesis	19–22	Data, liability, workforce, technology adoption

Every chapter follows the same structure:

1. Where the phase sits.
2. Who does the work.
3. The software landscape.
4. Where AI is landing.
5. What changes for jobs and process.
6. Friction and risk.
7. A residential callout.
8. Signals to watch.
9. A self-check exercise.

Three scope decisions shape the whole part.

The focus is the United States. International vendors appear only where they hold real U.S. share.

The part ends at closeout. Operations and facility management are where the industry's biggest platforms are heading next, and they get their own treatment later. Chapter 18 points you forward.

Residential gets a callout in every chapter. Residential construction is roughly 42% of U.S. construction spending, so it isn't a footnote. However - it runs on a largely separate software market, and in several phases it barely resembles commercial practice at all. Where that's true, the chapter says so plainly, because knowing what one side skips is how you learn the other.

What This Part Is Not

These chapters are not product reviews or buying guides, and no vendor sponsored, reviewed, or approved any of the content.

Vendor names appear to show how a market is structured: who leads, who challenges, and where the money is moving. A company's presence in a table is not an endorsement, and its absence is not a judgment.

The Shape of the Market

AEC software splits into two kinds of companies.

Platforms try to own a phase of the lifecycle, or several phases, and become the system of record for it. They win through integration, switching costs, and data that accumulates over years of projects.

Point solutions do one job exceptionally well, such as AI takeoff, contract review, or foot-traffic analytics, and integrate with the platforms around them.

Most of the market's history is point solutions proving a category, followed by platforms acquiring them.

On the design and construction side, six companies define the landscape:

Company	Home turf	Where it has been expanding
Autodesk	Design authoring (AutoCAD, Revit), Autodesk Construction Cloud	Field labor data, and operations and maintenance
Procore	Construction project management for general contractors	Owners, financials, AI agents, reality capture
Trimble	Survey and geospatial hardware, SketchUp, Tekla, Viewpoint construction ERP	AI contract analysis, site-to-office connectivity
Nemetschek	Archicad (CAD), Vectorworks, Bluebeam (Document markup)	Heavy civil estimating and construction
Bentley Systems	Infrastructure engineering and digital twins	Geospatial and infrastructure data
Oracle	Primavera P6 scheduling, Aconex, Textura payments	Large owners and capital programs

The real estate side looks different. Its leaders are mostly data businesses rather than workflow tools. CoStar Group dominates commercial real estate information and marketplaces. Altus Group owns ARGUS, the standard for commercial cash-flow modeling and valuation. Yardi and MRI run the property management and accounting systems that most commercial owners operate on. As Chapter 1 shows, that difference matters a great deal for AI. Workflow software gets disrupted differently than proprietary data does.

2025–2026: The Consolidation Year

If you want one fact that captures the current market, it's this: in the first half of 2026, the largest AEC platforms bought AI-native companies to embed AI directly into their core workflows. Procore acquired Datagrid, an agentic AI platform. Autodesk acquired Rhumbix, a field labor data company. Trimble acquired Document Crunch, which uses AI to analyze construction contracts.

The deals got bigger from there:

- Nemetschek agreed to acquire HCSS, a heavy civil estimating and operations software company, from Thoma Bravo for about \$2.4 billion.
- Autodesk agreed to buy MaintainX, a maintenance and operations platform, for about \$3.6 billion in cash, the largest acquisition in its history. The deal closed in August 2026.
- Procore agreed to acquire DroneDeploy, the drone and ground reality capture platform, for about \$845 million. It closed in September 2026.

One useful industry analysis groups these deals into two logics.

Buying capability. The acquirer buys something it can't build fast enough itself, usually AI talent and product.

Buying data. The acquirer is really paying for years of accumulated operational data. Nemetschek gained decades of heavy civil estimating outcomes and bid-to-actual cost history. Autodesk gained ongoing data on how physical assets behave in operation.

Keep both logics in mind as you read every chapter. When a platform acquires a company, ask which one it was buying. Often the answer is data, and data is what trains and grounds useful AI.

Why the Market Stays Fragmented

Given all this consolidation, why don't most firms simply run on one platform? Because the construction industry is structured to resist it.

Every project is a temporary company. An owner, an architect, several engineering consultants, a general contractor, and dozens of subcontractors come together for a few years and then disband. Each firm chooses its own software, answers to its own accounting system, and brings its own habits. No single firm controls the stack.

Every phase handoff loses information. The feasibility model doesn't feed the design model. The design model gets flattened into PDF drawings for bidding. Construction data rarely survives into operations. Each handoff is a point where data gets re-entered, reinterpreted, or lost. That

handoff tax is one of the largest hidden costs in the industry, and much of the software market exists to reduce it.

Margins are thin and risk is high. General contractors often work on low single-digit net margins. A failed software rollout mid-project can cost more than the software would ever save, so firms are cautious and slow to switch.

The result shows up in survey data. ServiceTitan's 2026 report on commercial specialty contractors found that only 20% of contractors run their business on a single platform.

Residential and commercial are separate software economies. Residential software is mostly sold to small businesses on price and ease of use, often anchored to QuickBooks. Commercial software is sold to enterprises on integration, controls, and risk management. Some vendors serve both, but most successful companies pick a side. Multifamily construction usually sits closer to the commercial side.

How to Judge AI Impact

AI discussion in AEC tends toward two unhelpful questions: "Will AI replace architects?" and "Is this tool any good?" Part 3 uses a more practical approach.

Rate tasks, not job titles

Every job is a bundle of tasks. An estimator doesn't just "do estimating." An estimator reads plan sets, performs quantity takeoff, builds assemblies, solicits subcontractor pricing, levels bids, identifies scope gaps, applies markup strategy, and defends the number in front of a project executive. AI affects each of those tasks differently. Some are nearly automated, some are faster, and some are untouched. The job survives, but its shape changes, and so does the number of people needed to do it.

Every chapter rates tasks on this four-level scale:

Level	Meaning	Example
Automated with sign-off	The machine does the task and a person reviews it	Quantity takeoff on clean plan sets
Augmented	A person does the task, much faster or better	Schedule scenario planning
Assisted	AI helps at the edges, such as search or drafting	Drafting RFI language
Resistant	The task depends on authority, liability, relationships, or physical work	Stamping drawings, running a site

Notice that "automated" always comes with sign-off. In AEC, someone remains accountable for nearly every output. That's the defining feature of this industry's relationship with AI.

The five constraints

Across every phase, five constraints slow AI down no matter how capable the technology becomes:

1. **Licensure and responsible charge.** Architects and engineers stamp drawings and carry personal professional responsibility for them. A model can't hold a license.
2. **Standard of care.** Professionals are judged against what a reasonably careful peer would do. That standard shifts slowly, and nobody wants to be the test case.
3. **Authority having jurisdiction.** Building officials, not software vendors, decide what gets permitted and what passes inspection.

4. **Contract risk allocation.** Construction contracts assign who pays when something goes wrong. Until contracts address AI-generated work, firms carry that risk themselves.
5. **Physical work.** Buildings are assembled by people in weather, on uneven ground, with materials that don't match the drawings.

Where AI moves fastest

Put the constraints together and a pattern appears. AI is moving fastest on tasks that meet four conditions:

- **Document-heavy:** reading, extracting, summarizing, drafting.
- **Verifiable:** a person can check the output quickly.
- **Low cost per error:** a mistake gets caught before it's expensive.
- **Contained within one firm's data and control:** no cross-company handoff is needed.

That's why lease abstraction, takeoff, and market reports are moving quickly, while design liability, permit approval, and field supervision are not.

A caution on adoption statistics

You'll see many AI adoption figures, and they rarely measure the same thing. "Uses AI" can mean anything from one estimator trying a chatbot to an agent running in production across a firm. As one data point, ServiceTitan's 2026 survey of more than 1,000 commercial contractors found that 38% reported measurable business impact from AI, up from 17% the year before. Cost estimating and bid management were the most common uses. Read every adoption statistic by asking what was actually measured.

How to Read the Vendor Tables

Each chapter includes a table of market leaders and major competitors. A few ground rules apply.

"Leader" is a judgment. It weighs installed base, revenue where known, and industry mindshare. Reasonable people will disagree at the edges.

Every table is dated. This market changes quickly. In 2026 alone, several category leaders changed owners. In the PDF edition, detailed vendor lists live in an appendix that gets refreshed, while the analysis stays in the chapters.

Lists are illustrative, not exhaustive. There are hundreds of AEC software companies. The tables show market structure, not every option.

What's Next

Chapter 1 starts where every project starts: before there is a project. It covers market intelligence and site sourcing, meaning how developers, investors, and brokers find opportunities and decide whether a market is worth pursuing. It's also the clearest example in the entire lifecycle of a lesson that shows up again later. When AI makes analysis cheap, proprietary data becomes more valuable, not less.

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3.1 - Market Intelligence & Site Sourcing

1. Where This Sits

Every building starts as a question someone asks before there's a site, a design, or a budget. Where should we build or buy? What will this market support? Who owns the land we want, and would they sell?

This is the very beginning of a project, before the owner has decided whether the project should exist at all. Part 1 of this series covers how that analysis is done: market studies, supply and demand, rent and sale comparables, and site screening. This chapter covers the software and data behind that work, and what AI is changing about it.

This phase is the best place to start for a reason that will come up again throughout Part 3. Market intelligence is almost entirely information work. There's no jobsite, no stamp, and no building official involved. That makes it the phase where AI can move fastest. It also shows most clearly what AI can't replace, which is proprietary data and local knowledge.

2. Who Does the Work

Market intelligence is produced and consumed by several groups, often in the same deal.

Investors and developers. Acquisitions analysts and associates at private equity real estate funds, REITs, family offices, and development firms screen markets, pull comparables, and build the case for or against a deal. At smaller firms, the principal does it personally.

Brokerages. Brokers need market knowledge to win listings and advise clients. Large brokerages run research departments that publish quarterly market reports, while individual brokers and their analysts do the day-to-day prospecting and comp work.

Corporate real estate and retail site selection teams. Retailers, restaurant chains, healthcare systems, and logistics companies analyze trade areas, foot traffic, and competitor locations before committing to a site.

Lenders and appraisers. Banks and debt funds evaluate the collateral market. Appraisers need defensible comparables to support a valuation.

Data companies. A substantial industry exists to collect, clean, and sell the underlying information: public records, transactions, leases, listings, and location data. Much of this chapter is really about them.

3. The Software Landscape

Market intelligence isn't one software category. It's a stack of data sources and tools, each answering a different question.

Category	The question it answers	Leader	Major competitors
CRE information platform	What exists, who's in it, what's it worth, what's available?	CoStar	Crexi Intelligence, Reonomy (Altus)
Listing marketplace	What's for sale or lease right now?	LoopNet (CoStar)	Crexi
Property and ownership data	Who owns this parcel, and what's recorded against it?	Fragmented	Reonomy, Regrid, LightBox, ATTOM
Lease and sale comparables	What did comparable deals actually trade for?	CoStar	CompStak
Sector research	How is this specific asset class performing?	Varies by sector	Yardi Matrix (multifamily), brokerage research
Location analytics	How many people visit this place, and where do they come from?	Placer.ai	Other mobile-data analytics providers
Data integration	How do we combine all of these with our own data?	Fragmented	Cherre, in-house data teams

CoStar: the incumbent and its data moat

CoStar Group is the dominant incumbent in commercial real estate information, and it's worth understanding how it got there. For decades it has paid field researchers to call brokers, photograph buildings, and verify tenants and transactions. That's slow and expensive data acquisition that competitors have found hard to replicate. It also owns LoopNet, the largest commercial listing marketplace, which feeds it a continuous stream of listing activity.

The business is durable. In its second quarter of 2026, CoStar reported that subscribers to its core platform grew 19% year over year to 327,000, with renewal rates of 93%. The company has also been adding AI on top of its data. That quarter it released a rent benchmarking feature built with AI that extracted key terms from about 4 million actual lease agreements.

That example captures this chapter's central idea. The AI is valuable because of the 4 million leases behind it. Anyone can license a language model. Very few can supply that dataset.

CoStar has also pushed well beyond commercial real estate. It owns Homes.com, Apartments.com, and Matterport, the 3D capture company. In February 2026 it launched Homes AI, a conversational home search assistant built on Microsoft Azure OpenAI, and said it plans to extend the same capability to CoStar and LoopNet.

The challenger: Crexi

Crexi, founded in 2015, competes with a different model. Its listing marketplace is free to list on and free to search, and it earns revenue through data subscriptions (Crexi Intelligence) and broker tools. It has been aggressive with AI. In April 2026 it launched Crexi Market Analytics, which uses AI deep research to generate sourced market reports by combining Crexi's own transaction data with third-party sources. In May 2026 it announced Crexi Create, which turns uploaded property financials, rent rolls, and leases into an editable offering memorandum draft.

The data moat fight

The rivalry between CoStar and Crexi is also a long-running legal battle, and it's worth following because it's fundamentally about who owns real estate data. CoStar sued Crexi in 2020 for copyright infringement over listing images. In 2025 the district court found that Crexi had copied and cropped thousands of CoStar photos. That same month, the Ninth Circuit revived Crexi's antitrust counterclaims, finding it plausible that CoStar held monopoly power and used broker contracts and technical barriers to exclude competitors. CoStar disputes those claims and has sought further review.

The details of the litigation matter less for your purposes than the lesson. In this category, access to data is the competitive battleground, and AI raises the stakes.

Specialists worth knowing

CompStak built a lease comparables business on a trade. Brokers, appraisers, and researchers contribute the comps they know, and in exchange they get access to the database. Its data captures what deals actually signed for, including free rent and tenant improvement allowances, rather than asking rents. In December 2025 it launched CompStak AI, which includes a Rent Predictor tool.

Reonomy focuses on property and ownership intelligence, especially identifying the real owner behind an LLC and surfacing off-market prospects. Altus Group acquired it in 2021 for about US\$201.5 million.

Placer.ai estimates foot traffic, visitor origins, and dwell time for physical locations using anonymized mobile device data. It's a staple of retail site selection. Its outputs are statistical estimates extrapolated from a device panel, which makes them strong directional evidence, not ground truth.

Regrid, LightBox, and ATTOM aggregate parcel, ownership, and property records from thousands of county and municipal sources. That's unglamorous, essential work, because U.S. property records are scattered across more than 3,000 counties with no common format.

Cherre doesn't sell a dataset of its own. It helps large owners and investors connect the many datasets they license with their internal data so the whole thing can be queried in one place.

4. Where AI Is Landing

Here is how the core tasks of market intelligence rate on the Part 3 impact lens.

Task	Level	What's happening
Drafting a market overview report	Automated with sign-off	AI deep-research tools produce sourced market summaries in minutes. The analyst's job becomes checking and editing.
Pulling and normalizing comparables	Automated with sign-off	AI extracts comps and normalizes concessions to an effective-rent basis. Judging true comparability still needs a person.

Abstracting data from leases and documents at scale	Automated with sign-off	Extracting terms from thousands of leases is now routine, and it's how new data products get built.
Querying market databases	Assisted to augmented	Natural-language questions replace filter menus and exports. The answers are only as good as the underlying data.
Screening many sites against criteria	Augmented	Analysts can screen hundreds of parcels or markets against zoning, demographics, and traffic in the time it used to take to screen a handful.
Identifying the real owner behind an entity	Augmented	Entity resolution across records is faster, but ownership chains still need verification before anyone makes a call.
Predicting which owners are likely to sell	Augmented	Models flag signals such as loan maturities, long hold periods, and distress. They generate leads, not deals.
Trade area and foot traffic analysis	Augmented	Machine learning makes the estimates possible. Interpreting them for a specific site is still analyst work.
Forming an investment thesis	Assisted	AI can argue both sides and surface risks. Deciding what to believe about a market's future is judgment.
Getting access to off-market deals	Resistant	Relationships, reputation, and local presence. An owner decides who gets the first call.
Walking the site and the neighborhood	Resistant	Physical presence still catches what the data misses.

The central shift: analysis gets cheap, data gets expensive

Before generative AI, a market report represented real labor: pulling data from several subscriptions, reconciling it, building charts, and writing the narrative. That labor was part of what clients paid for. Now the assembly is nearly free. What remains valuable is what the AI draws from, meaning proprietary transaction data, verified comps, and fresh listing activity.

That's why the data incumbents aren't panicking about general-purpose AI. They're embedding it. Crexi's pitch for its analytics tool is explicit about this. It argues that its live marketplace captures market signals a general AI query can't reach. CoStar's rent benchmarking feature makes the same point from the other direction.

For someone building a career in AEC tech, the practical takeaway is to ask one question of any AI tool in this space: **where does its data come from, and can a competitor get the same data?** If the answer is public web sources, the tool is easy to copy. If it's a proprietary dataset built over years, the tool has a moat.

5. Jobs and Process Change

What compresses. The junior research and analyst work that used to fill most of a first-year acquisitions analyst's week is shrinking fast: assembling market reports, pulling comps, building prospect lists, and formatting data. Firms will need fewer hours for it, and possibly fewer junior hires.

What shifts. The analyst's role moves from assembling information to validating and interpreting it. The valuable skill becomes knowing when a comp isn't really comparable, when an AI-generated market narrative misses something local, and when a data source is stale.

What grows. Not much - data engineering teams, to some extent. These engineers use AI to write code to make data interpreters and managers less needed.

What stays human. Relationships with owners and brokers, local market knowledge, physical site visits, and the final judgment call on a deal.

An open question worth watching. The analyst job has traditionally been how people learn a market: by building comps and reports until patterns become intuitive. If AI does that assembly, the next generation of acquisitions leaders will need a different way to develop that intuition. This apprenticeship problem shows up again in Chapter 14, where it affects project engineers in construction.

6. Friction and Risk

Data licensing terms. Most commercial data subscriptions restrict how the data can be used, shared, and exported. Pasting licensed data into a general-purpose AI tool, or letting an agent pull from a subscription, may violate those terms. Check the contract before building a workflow around it.

Confident errors. AI-generated market reports read smoothly whether or not the numbers are right. A fabricated comp or an outdated vacancy figure in an investment memo is exactly the kind of mistake that survives a quick review. Every figure that drives a decision needs a traceable source.

Stale and lagging data. Commercial real estate data lags reality. Leases get reported months after signing, and many deals never get reported at all. AI doesn't fix lag, and it can hide it by presenting old data in a fresh-looking report.

Model and panel bias. Location analytics are estimates from device panels, and ownership prediction models learn from past patterns. Both can be systematically wrong in unusual markets, in rural areas, or during sudden shifts.

Privacy. Location intelligence rests on mobile device data. Reputable providers aggregate and anonymize it, but regulation and public expectations around location data continue to evolve.

Market concentration. When one company controls much of a category's data, pricing and access terms become strategic risks for every firm that depends on it. The CoStar and Crexi litigation is the most visible version of that tension.

7. Residential Callout

Residential market intelligence runs on different sources, and it's where AI valuation has been in public use the longest.

Different data providers. Homebuilders and residential investors rely on housing-focused research firms such as Zonda and John Burns Research and Consulting for new-home sales, lot supply, and builder activity. MLS data drives resale analysis. ATTOM and similar providers supply property records, and parcel tools like Regrid support land sourcing.

Land acquisition is the core workflow. For a production homebuilder, "site sourcing" means finding land that can be entitled and turned into finished lots at a price that works for the homes the builder sells. Chapter 2 covers lot yield and entitlements, and Chapter 3 covers the pro forma.

Automated valuation models are the mature AI story. Consumer home value estimates like Zillow's Zestimate have been machine-learning products for years, and they offer a clean lesson about AI accuracy. Zillow publishes a nationwide median error of roughly 2% for homes currently listed for sale and roughly 7% for homes that aren't. The gap exists partly because a listed home has fresh signals, including an asking price the model can anchor to. An off-market home has only older data. The general principle applies across AEC: **AI accuracy depends on how fresh and specific the input signal is.**

The commercial giant is moving into residential. CoStar's residential segment, built on Homes.com, Apartments.com, and its 2025 acquisition of Australia's Domain, grew revenue 33% in the second quarter of 2026 and reported positive adjusted EBITDA for the first time. That's a sign the data-moat playbook from commercial real estate is being applied to housing.

A fair housing caution. Using AI to target residential marketing, screen buyers or tenants, or select neighborhoods can create fair housing exposure if models reproduce historical patterns of discrimination. Residential firms need to review how their tools make those decisions.

8. Signals to Watch

- **The CoStar and Crexi litigation.** Its outcome could reshape how listing data and images can be shared across platforms.
 - **Data vendor policies on AI access.** Watch whether major providers open sanctioned access for customers' AI agents, or restrict it to protect their own AI products.
 - **CoStar extending conversational AI** from its residential portals into the CoStar and LoopNet commercial platforms, as it has said it plans to do.
 - **AI-native research startups.** Watch whether any secure proprietary data, or whether they remain easy-to-copy layers on top of public information.
 - **Public records digitization.** Better county and municipal data lowers the cost of building a competing dataset, which weakens incumbents' moats.
 - **Hiring patterns for junior analysts** at brokerages and investment firms, as an early indicator of how much of this work is actually being absorbed.
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9. Self-Check Exercise

The one-hour market brief. Pick a U.S. metro and one asset class you don't know well, such as industrial in a secondary market or suburban medical office.

1. Using only free sources and a general-purpose AI assistant, produce a one-page market brief in one hour. It should cover inventory, vacancy, rent trends, recent notable transactions, and major owners.

2. Find a published brokerage research report for the same market and quarter. Most large brokerages publish these free.
3. Compare the two. List every place they disagree, and for each one, identify which source is more likely right and why.
4. Rate each step of your process using the four-level impact lens.

Questions to answer afterward:

1. Which parts of your brief would you trust in an investment committee memo, and which would you need to verify with paid data?
2. What information did the brokerage report have that no free source could provide? Where do you think it came from?
3. If you were an investor evaluating an AI-native market intelligence startup, what's the first question you'd ask about its data?
4. Why is the off-market error of an automated valuation model so much larger than the on-market error, and where else in AEC would you expect the same pattern?
5. Which task in this chapter would you least want to hand entirely to AI, and why?

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3.2 - Zoning, Entitlements & Site Feasibility

1. Where This Sits

Chapter 1 asked which markets and sites are worth pursuing. This chapter asks the next question: what can actually be built on a specific piece of land, and does it pencil?

The owner still hasn't committed to a project. Part 1 covers the analysis itself: zoning review, highest and best use, and entitlement strategy. The work comes down to three questions:

1. **By right:** What does current zoning allow without discretionary approval?
2. **With approvals:** What could be built through a rezoning, variance, or special permit, and how long and uncertain would that process be?
3. **Feasibility:** Does the building that fits produce a return worth the price of the land?

Getting these answers wrong is expensive. A misread setback or parking requirement discovered after closing can wipe out a deal's economics.

2. Who Does the Work

Development and acquisitions teams run early feasibility, often dozens of sites for every one they pursue.

Architects produce test fits: quick massing studies showing how many units or square feet fit on a site. This has traditionally been unbilled or low-fee work done to win the commission.

Land use attorneys and entitlement consultants interpret ambiguous code language and guide projects through discretionary approvals, public hearings, and negotiations with planning staff.

Civil engineers identify physical site constraints like grading, drainage, and utilities.

City planners answer zoning questions, review applications, and staff the hearings.

Zoning report providers produce formal zoning reports that lenders and title companies require at closing.

3. The Software Landscape

This category has no clear leader, for a structural reason. The United States has no national zoning system. Zoning is written by thousands of local jurisdictions, mostly as prose, each with its own structure and definitions, and amended constantly. Before software can analyze zoning, someone has to turn that prose into data, and that is slow, expensive work.

The market splits into three layers.

Zoning data companies convert code text into parcel-level rules: allowed uses, height, floor area ratio (FAR), and setbacks. Zoneomics and Gridics work here, alongside parcel data providers like Regrid and LightBox.

Massing and yield tools generate buildable options for a site and tie them to cost and returns. TestFit, Autodesk Forma, Deepblocks, and Archistar are the main names.

Municipal-side tools help cities publish and apply zoning, including prescreening permit applications. Gridics, Archistar, and Symbium all sell to cities, and Chapter 10 covers this side further.

A few companies worth knowing:

TestFit, founded in Dallas in 2016, has become a standard feasibility tool for developers and architects. It generates site plans with live feedback on unit counts, parking, and cost. It added a generative design feature in 2024 and subdivision layout features in 2026. One 2026 reviewer described it less as open-ended generative design and more as a constrained massing solver connected to a live pro forma, which is an accurate description of what makes it useful.

Autodesk Forma, formerly Spacemaker, which Autodesk acquired in 2020, is Autodesk's cloud tool for early-stage site planning and environmental analysis, positioned as the front door to its design ecosystem.

Archistar, founded in Australia in 2010, combines site feasibility for developers with AI permit prescreening for cities. It has expanded into the U.S., including a residential zoning prescreening pilot with the City of Austin and use in Los Angeles fire rebuilding.

Symbium grew out of a Stanford project to translate laws into computer code that produces deterministic answers. Bakersfield, California uses its platform for instant rooftop solar permitting.

Gridics and **Zoneomics** focus on the data layer. They sell parcel-level zoning data, reports, and APIs to developers, brokers, lenders, and cities.

4. Where AI Is Landing

Two ways to teach software zoning

Understanding this category starts with a technical distinction that will reappear in Chapter 10.

Rules as code. People read the zoning ordinance and encode its rules into structured logic. The output is deterministic: the same parcel always gets the same answer, and every answer traces back to a rule. It's reliable where it has been built, but expensive to extend to each new jurisdiction and to maintain as codes change.

Language model interpretation. An AI model reads the code text directly and answers questions about it. This scales across jurisdictions quickly and handles unusual language. It can also misread a cross-reference, miss an exception, or state a wrong number with total confidence.

The strongest products increasingly combine the two, using language models to speed up the extraction of rules and human review to verify them before they're relied on.

Task ratings

Automated with sign-off: zoning lookups. Finding a parcel's zoning district and its basic standards is instant where the data exists. The answer still needs checking against the current official code.

Automated with sign-off: massing and yield options. Test fits that took an architect days now take minutes, and teams explore far more options.

Augmented: linking massing to cost and returns. Live feedback shows how a design change affects the pro forma. The cost assumptions behind it still need an estimator's judgment.

Augmented: screening many parcels. Portfolio-scale screening is now practical for firms that used to evaluate sites one at a time.

Assisted: interpreting ambiguous code language. AI can surface possible readings and the relevant sections. The call belongs to a land use attorney or the zoning administrator.

Assisted: entitlement risk and timeline. Some tools track rezoning and variance history. Predicting how a hearing will go is still local judgment.

Resistant: securing discretionary approvals. Public hearings, neighborhood negotiation, and political relationships stay human.

The trap: the zoning code isn't the whole rulebook

The most important lesson in this chapter is that a tool reading only the local zoning ordinance can give a clean, confident, wrong answer. What can actually be built on a parcel also depends on:

- **State laws that override local zoning.**
- **Overlay districts** such as historic, flood, and airport overlays.
- **Pending code amendments.**
- **Conditions attached to past approvals.**
- **Private restrictions** such as easements and deed covenants.

California shows how big the gap can get. SB 79 took effect July 1, 2026. It requires cities in qualifying counties to allow housing near major transit stops at state-set heights and densities regardless of local zoning, reaching 95 feet and 160 units per acre in the highest tier. Near those stops, a parcel's local code no longer describes what's allowed.

The practical question for any feasibility tool is **what sources it checks beyond the local ordinance, and how quickly it reflects changes.**

5. Jobs and Process Change

What compresses. Test-fit production and routine zoning research are shrinking fastest. Architects spend far fewer hours on speculative massing studies, and analysts no longer spend days reading codes to screen a single site. Standardized zoning report preparation is also becoming largely automated.

What shifts. Because screening is cheap, firms evaluate many more sites, so the bottleneck moves from analysis to decision-making. Architects' early involvement shifts from "how many units fit" toward the design and entitlement questions a tool can't answer.

What grows. Nothing here - as tools get better, especially in the codes and zoning, employment and business opportunities will shrink.

What stays human. Discretionary entitlement work: land use law, community engagement, negotiation with planning staff, and reading the politics of a city council. As by-right analysis gets automated, this work becomes a larger share of what developers pay for.

6. Friction and Risk

Verification and liability. If a tool misreads a setback and a developer overpays for land, who is responsible? Most software terms disclaim that risk. Before acquisition, critical zoning conclusions still need confirmation from official sources, a zoning letter from the jurisdiction, or a land use attorney.

Currency. Codes change through amendments, new state laws, and court decisions. A zoning database is only as good as its last update, and users often can't see when that was.

Coverage gaps. Data quality is strongest in large metros and weakest in small jurisdictions, which is exactly where local knowledge is hardest to find.

Constraints beyond zoning. Utilities capacity, flood risk, soils, environmental conditions, and title issues can make a zoning-compliant project infeasible. Those are covered in Chapter 4.

False precision. A crisp 3D massing with a unit count and a projected return looks authoritative. At this stage, it rests on assumptions that deserve skepticism.

7. Residential Callout

Lot yield, not building massing. For homebuilders, feasibility means how many lots a piece of land can produce after roads, open space, stormwater, and setbacks, and what product fits those lots. Subdivision layout tools are growing for exactly this reason.

State reforms are creating high-volume, simple checks. Laws allowing accessory dwelling units, lot splits, and duplexes or fourplexes on formerly single-family lots have spread across states. They produce large numbers of small, repetitive feasibility questions, which suit automation well. Some cities pair reform with tooling. New York City's ADU program, for example, includes site feasibility and cost-estimating tools and a library of pre-approved ADU plans.

Owners, not just developers. Residential feasibility tools increasingly serve individual homeowners asking whether they can add a unit. That's a very different buyer, with less tolerance for ambiguity and no attorney on retainer.

Entitlement risk is often political. Subdivisions and infill projects frequently face neighborhood opposition regardless of what zoning allows. Software can map that history but can't manage it.

8. Signals to Watch

- **State preemption of local zoning.** Every new state law that overrides local codes widens the gap between what an ordinance says and what's allowed, and raises the value of tools that track both.
- **Machine-readable zoning.** Watch whether cities and code publishers begin maintaining zoning as structured data rather than prose. That would lower costs for everyone and weaken data-layer moats.
- **Convergence of rules-as-code and language models.** The winners will likely combine speed with verifiable answers.

- **Feasibility tools reaching into entitlement tracking**, using rezoning and variance history to estimate approval risk.
 - **Platform absorption.** Zoning data and massing tools are natural acquisition targets for design and real estate data platforms.
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9. Self-Check Exercise

The two-way zoning read. Pick a real parcel in a city you know, ideally one zoned for multifamily or mixed use.

1. **Manually:** Find the parcel's zoning district on the city's official map, then read the relevant sections of the code. Determine maximum height, FAR or density, setbacks, and parking requirements. Record how long it took.
2. **With AI:** Ask a general-purpose AI assistant the same questions for the same parcel. If you have access to a zoning data tool, use that as well.
3. **Compare:** List every difference. For each one, determine which answer is correct and why the other was wrong.
4. **Look beyond the code:** Check whether any overlay district, state housing law, or pending amendment applies to the parcel.

Questions to answer afterward:

1. Where did the AI answer go wrong, and would you have caught it without doing the manual read first?
 2. Would you rely on the AI answer to set a land offer? What would you need before you did?
 3. What is the difference between a rules-as-code answer and a language model answer, and when would you require one over the other?
 4. Which part of the entitlement process for your parcel would no software meaningfully change?
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3.3 - Underwriting, Deal & Capital Management

1. Where This Sits

Chapter 1 found the opportunity and Chapter 2 determined what can be built. This chapter covers the decision: what the property or project is worth, what it will return, whether the firm should pursue it, and how the capital gets raised and managed.

This is the owner's go or no-go point, where the owner commits to a project and funds it. Part 1 covers the analysis itself: pro formas, discounted cash flow, cap rates, and investment committee process. This chapter covers the software that carries that work and what AI is doing to it.

Underwriting runs on documents: offering memorandums (OMs), rent rolls, trailing twelve-month operating statements (T12s), leases, and third-party reports. That makes it one of the most AI-exposed phases in the entire lifecycle, and also one where the limits of AI are especially visible.

2. Who Does the Work

Acquisitions analysts and associates build the underwriting model, turning a pile of seller documents into a set of assumptions and a projected return.

Vice presidents and principals challenge those assumptions and present the deal to an **investment committee**, which approves or rejects it.

Asset managers own the property after purchase and re-underwrite it continuously against the original plan.

Development managers run the budget and draws for ground-up projects.

Investor relations and fund finance teams raise capital from limited partners (LPs), process capital calls and distributions, and produce investor reporting.

Lenders and appraisers underwrite the same asset from their own side of the table, focused on downside risk and defensible value.

3. The Software Landscape

The honest place to start is that **Excel is still the most widely used underwriting tool** in commercial real estate. It survives because an underwriting model is an argument. Every assumption needs to be visible, adjustable, and defensible in front of an investment committee, and Excel lets each firm structure that argument its own way.

ARGUS, owned by Altus Group, is the institutional standard for lease-by-lease commercial cash-flow modeling and valuation. When a sponsor, lender, and appraiser all look at the same office or industrial building, their numbers often reconcile back to an ARGUS model. That shared language is ARGUS's real moat. As of 2026, ARGUS Enterprise is no longer sold as a standalone product. It's offered only as part of Altus's broader ARGUS Intelligence Platform. In April 2026, Altus added ARGUS Assist, an agentic AI layer that runs portfolio and deal analysis from plain-English questions. Its roadmap targets acquisition and development modules for late 2026 and 2027.

Multifamily has its own tools, because apartment underwriting runs on unit-level rent rolls rather than long commercial leases. Radix holds a position in multifamily benchmarking similar to ARGUS's in commercial. Archer offers multifamily underwriting with AI document parsing and comps benchmarking.

Document intelligence tools extract structured data from OMs, rent rolls, and T12s and write it into an existing model. This is the fastest-growing category and includes both standalone tools like V7 Go and Clik.ai and extraction features built into larger platforms.

Dealpath is the leading deal management platform for institutional investors. It tracks the pipeline from sourcing through closing and centralizes deal data. It reports more than 300 institutional clients, and its network brings brokerage listings directly into investors' pipelines. It launched an expanded suite called Dealpath AI in May 2026.

Juniper Square manages the capital side for general partners (GPs): fundraising, investor onboarding, fund administration, and LP reporting. It reports serving more than 2,500 GPs and 600,000 LPs, and its JunieAI features draft investor communications and surface fundraising priorities.

Northspyre focuses on development management: budgets, invoices, draws, and cost forecasting for ground-up and renovation projects. In January 2026 it launched a deal management product that includes an AI tool for generating investment memo decks from underwriting data.

Yardi and MRI are the property management and accounting systems that most commercial owners operate on. They don't underwrite deals, but they're where actual operating data lives. When a seller hands over a T12, it was almost certainly exported from one of them, and asset managers compare performance against the original underwriting using their data.

4. Where AI Is Landing

The underwriting pipeline

It helps to think of underwriting as a pipeline with five stages: collecting documents, extracting data, building the model, analyzing the deal, and writing the investment memo. AI affects each stage differently.

Automated with sign-off: extraction and screening. Pulling rent rolls, T12 line items, and property details out of seller documents is now largely automated. Dealpath reports its extraction tool pulls more than 90 fields from an OM in under a minute at about 95% accuracy. Deal screening is following close behind. Tools now read an OM and its financials and produce a one-page tear sheet showing portfolio fit and key metrics in seconds, work that used to take an analyst hours.

Augmented: model population and analysis. AI can populate an existing model, run sensitivities, find comparables, and answer questions across a portfolio, as ARGUS Assist and similar tools now do. Building a structurally sound model from scratch with a general-purpose AI tool is still unreliable, which is why most firms point AI at their own trusted templates instead.

Augmented to assisted: the investment memo. AI can draft the narrative, charts, and summary sections of an investment committee memo from the model. The thesis itself, meaning why this deal and why now, is still the sponsor's argument to make.

Assisted: setting key assumptions. Rent growth, exit cap rate, and capital expenditure reserves drive most of a deal's return. AI can show what comparable deals assumed, but choosing the assumptions is judgment, and it's where committees push hardest.

Resistant: the decision and the capital. Investment committee approval, LP relationships, and negotiation with sellers and lenders stay human.

The verification tax

The most useful number in this chapter comes from a simple piece of arithmetic. An extraction tool that's 95% accurate across 90 fields will get roughly four or five fields wrong on every deal, and it doesn't tell you which ones. If one of those is a lease expiration or a major expense line, the model is wrong. So every output gets checked.

Dealpath's 2026 survey of institutional real estate investors put numbers on this. Nearly all respondents, 97%, said AI is integrated into their investment process. But only 51% said it actually saves time once verification is counted, and 41% said AI-assisted work takes longer than doing it manually. Dealpath calls this the "verification tax." Respondents were also much more willing to let AI summarize a diligence document than to let it score a deal, and 90% said bad or fragmented data limits what AI delivers. The survey comes from a vendor with a product to sell, so read it that way, but the pattern matches the arithmetic.

The practical lesson: **AI saves time only when checking its output is faster than doing the work.** That depends on tools that show exactly where each number came from, down to the page or cell, so verification is a quick check instead of a redo.

AI on both sides of the table

A new dynamic is emerging. Listing platforms now offer AI tools that draft OM's from seller financials. Buyers use AI to extract data from those same OM's. When AI writes the marketing document and AI reads it, the underlying source documents, including the actual rent roll, the actual leases, and the accounting system export, become even more important to review directly.

5. Jobs and Process Change

What compresses. Data entry and first-pass screening, the work that defined the first year of an acquisitions analyst's job, is shrinking fast. So is routine investor reporting, since quarterly narratives and variance commentary can be drafted from fund data.

What shifts. The analyst's role moves from building models to auditing them: spotting the wrong lease date, the seller's optimistic expense normalization, or the comp that isn't really comparable. Firms screen far more deals with the same team, so selectivity becomes the skill.

What grows. Firms need clean, connected data, so data management and model governance become real roles, especially at larger investment managers. Someone has to own the firm's templates and assumption libraries that AI tools depend on.

What stays human. Setting the assumptions that drive returns, defending a deal in committee, negotiating with sellers and lenders, and maintaining the LP relationships that fund everything.

6. Friction and Risk

Confidentiality. Deal documents usually arrive under a nondisclosure agreement. Uploading them to a general-purpose AI tool may breach it, so firms need approved tools with clear data handling terms.

Model risk. A model populated automatically can look complete while carrying a single wrong input that changes the answer. Firms need review checkpoints for inputs that drive returns, not just a final glance at the output.

Seller data is marketing. A T12 in an OM may be normalized in the seller's favor. AI extracts what's on the page. It doesn't know what was left off.

Investor communications. Fund managers communicating with investors operate under securities rules and fiduciary duties. AI-drafted investor materials need the same compliance review as anything else, and claims about a firm's own AI capabilities need to be accurate.

Data fragmentation. AI can only analyze a portfolio if the portfolio data is connected. For many firms, the first AI project turns out to be a data cleanup project.

7. Residential Callout

Homebuilders underwrite land, not buildings. A production builder's core model is a land or community pro forma: land price, development costs per lot, the number and timing of lots, projected home prices, and absorption. It's almost always built in Excel, and the key judgment is the sales pace the market will support.

Land deals are structured around risk. Builders often control land through option contracts or land banking arrangements, paying for lots as they need them rather than buying everything upfront. Underwriting these structures is its own specialty.

Single-family rental and build-to-rent underwriting happens at scale. Investors buying or building thousands of rental homes underwrite with automated valuation models, rent estimates, and standardized expense assumptions, closer to a data pipeline than a one-off model.

Smaller sponsors use lighter tools. Syndicators raising capital for small multifamily deals often use simpler investor management portals alongside a spreadsheet model, rather than institutional platforms.

8. Signals to Watch

- **ARGUS Assist adoption and the new ARGUS modules** for acquisitions and development, which would put Altus in more direct competition with deal and development management platforms.
- **AI built into Excel itself.** If spreadsheet-native AI gets good enough, it strengthens Excel's position against specialized platforms.
- **Source traceability as a feature.** Tools that link every extracted number to its location in the source document are the direct answer to the verification tax.
- **Platform convergence.** Deal management, investor management, development management, and valuation vendors are all expanding into each other's territory. Expect consolidation.
- **Data infrastructure spending** at investment managers, as the prerequisite for AI that actually saves time.

9. Self-Check Exercise

The OM stress test. Brokers post offering memorandums publicly for many listed properties, and some include a rent roll and operating statement. Find one for a small multifamily or retail property.

1. Use a general-purpose AI assistant to extract the rent roll and T12 into a structured table.
2. Check every extracted number against the OM. Count the errors and record how long verification took compared with extraction.
3. Build a simple net operating income calculation from the verified data and compare it with the NOI the broker presents.
4. Ask the AI to draft a one-page investment memo, then mark every sentence that states something you couldn't verify.

Questions to answer afterward:

1. Did AI save time once verification was counted? What would have to change for it to save more?
2. Where did the broker's NOI differ from yours, and why might a seller present it that way?
3. Which assumptions in your analysis would an investment committee challenge first, and could AI have set them for you?
4. If you were an analyst two years into your career, which skills from this exercise would you invest in, given how the job is changing?

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3.4 - Due Diligence & Acquisition

1. Where This Sits

Chapter 3 ended with a decision to pursue a deal. Once a buyer and seller sign a purchase agreement, a diligence period begins. The buyer now has a limited window to confirm that the property is what the seller says it is before its deposit becomes nonrefundable and the deal closes.

The owner still hasn't started a design or construction project, but decisions made here shape everything that follows. An undiscovered easement can block a building footprint. Contaminated soil can add months and millions to site work. A lease clause can prevent a planned renovation. Part 1 covers what diligence investigates and why. This chapter covers the software and what AI is changing.

Due diligence has a distinctive shape. It's a search for exceptions: the one lease with an unusual termination right, the one title exception that matters, or the one historical record showing a gas station on the site in 1958. Most documents are routine. The cost of missing the one that isn't is enormous. That asymmetry explains most of how AI is being adopted in this phase.

2. Who Does the Work

The buyer's acquisitions team manages the diligence checklist and timeline and decides what findings mean for price or terms.

Real estate attorneys review the purchase agreement, title, survey, leases, and entity documents, then negotiate fixes before closing. Junior associates and paralegals carry much of the document review.

Title companies search the property's recorded history, issue a title commitment listing exceptions to coverage, and ultimately insure the title. In many transactions they also run escrow and move the money.

Surveyors produce an ALTA/NSPS land title survey showing boundaries, improvements, easements, and encroachments.

Environmental professionals perform a Phase I Environmental Site Assessment under the ASTM E1527-21 standard, which also satisfies the EPA's All Appropriate Inquiries rule.

Property condition assessors evaluate the physical state of existing buildings and estimate near-term repair costs.

Lenders and appraisers run their own parallel diligence to protect the loan.

3. The Software Landscape

There is no single diligence platform. The software is split by task, and each task has its own specialists.

Virtual data rooms are where sellers post diligence documents and buyers review them. Datasite and Intralinks are the enterprise leaders for large transactions. Both now offer AI features such as automated redaction and AI question-and-answer across documents. Some cite source documents in their answers, and some data room vendors are connecting their platforms to general-purpose AI assistants.

Environmental records are dominated by LightBox's EDR product line, which environmental professionals have relied on for more than 30 years. Its reports draw on more than 2,000 federal, state, tribal, local, and proprietary databases, along with historical sources like Sanborn fire insurance maps, aerial photos, and city directories. LightBox uses an AI-powered geocoder to match historical addresses to present-day properties. It's an unglamorous but essential job, because street names and numbers change over decades.

Title and closing runs through title companies and the large national title underwriters, supported by title production and closing software. Qualia is a leading digital closing platform, especially in residential transactions. In June 2026 it released Qualia Clear 2.0, describing it as a step toward fully automated title and escrow.

Lease abstraction turns long commercial leases into structured data such as rent schedules, escalations, renewal options, and critical dates. Prophia is a specialist for office, retail, and industrial leases. It extracts more than 200 fields and links every extracted value back to its location in the source document. MRI and Yardi offer abstraction within their property management platforms, and many firms still outsource abstraction to service providers.

E-signature is effectively settled, with DocuSign as the dominant platform for executing transaction documents.

Zoning reports, which lenders often require at closing, come from providers like LightBox and Zoneomics, covered in Chapter 2.

4. Where AI Is Landing

Automated with sign-off: lease abstraction. This is the most mature AI use case in the phase. Specialist platforms deliver abstracts in minutes, and vendors offering human-validated tiers report accuracy around 99%. The critical feature is source linking. When every extracted rent step or termination right links to the exact clause, a reviewer can verify it in one click instead of rereading the lease. Related comparisons are also largely automated, like checking tenant estoppel certificates against the rent roll and leases for discrepancies.

Automated with sign-off: document organization and redaction. Indexing a data room, flagging missing documents, and redacting sensitive information across thousands of files is now mostly machine work.

Augmented: first-pass legal and title review. AI can read a title commitment and flag the Schedule B exceptions most likely to matter, summarize recorded easements, and compare a survey against the title commitment. It can review contracts and leases for unusual clauses. The attorney still decides which issues to object to and how to negotiate them.

Augmented: environmental records review. AI speeds up reading historical records and regulatory database hits and helps draft sections of reports. The report itself still depends on the environmental professional.

Resistant: work that carries legal weight. Some diligence tasks are resistant for a specific legal reason. A Phase I ESA's value to a buyer comes partly from the protection it can provide against federal environmental cleanup liability, and that protection depends on the assessment meeting the All Appropriate Inquiries standard. The standard requires a qualified environmental professional, site reconnaissance, and interviews. A tool can accelerate the research, but it can't be the professional. Similar logic applies to a surveyor's certification and a title insurer's policy. The human signature is what the buyer is actually paying for.

Resistant: judgment on findings. Deciding whether a problem kills the deal, reduces the price, or just needs a closing condition belongs to the buyer and counsel.

AI on offense: wire fraud

This phase also shows AI working against the industry. Closings involve large wire transfers, published timelines, and many parties exchanging email, which makes them a prime target for fraud. Criminals impersonate sellers, attorneys, or title officers to redirect funds, and tools like deepfakes make impersonation more convincing. In Qualia's 2026 survey of more than 800 title and escrow professionals, 80% of companies reported a fraud attempt in the past year. The industry's response is also increasingly automated, with identity checks, bank account verification, and fraud screening built directly into closing workflows. The lesson extends beyond title: **as AI lowers the cost of producing convincing documents and communications, verification of identity and authenticity becomes a core part of the workflow.**

5. Jobs and Process Change

What compresses. Manual lease abstraction, both in-house and outsourced, is shrinking quickly. First-pass document review by paralegals and junior associates compresses, as does routine title examination and data room administration.

What shifts. Diligence teams can review everything instead of sampling. A buyer that once abstracted the largest leases and spot-checked the rest can now abstract every lease. Review work shifts from reading documents to investigating flagged exceptions.

What grows. Fraud prevention and verification roles grow in title and escrow. Firms also need people who manage secure AI access to confidential deal documents.

What stays human. Environmental site reconnaissance, survey certification, legal negotiation of title and contract issues, and the buyer's final judgment about risk.

The training question again. Junior attorneys have traditionally learned real estate law by reviewing hundreds of leases and title documents. If AI does the first pass, law firms and investment firms will need another way to build that expertise.

6. Friction and Risk

The cost of a missed exception. A tool that correctly handles 99 routine leases and misses the one unusual termination right has failed on the only lease that mattered. Review processes should focus human attention on what the AI flags and also test a sample of what it doesn't.

Confidentiality and access. Data rooms exist to control who sees what. Connecting them to AI assistants creates new questions about permissions, logging, and whether documents leave the controlled environment.

Gaps in the record. Older records may be handwritten, poorly scanned, or never digitized. AI can only review what exists in usable form, and missing records don't flag themselves.

Standards and liability. Where a professional standard or legal protection requires human work, as with the Phase I ESA, cutting corners with AI can undermine the very protection the buyer is paying for.

Fraud. Any workflow that relies on emailed instructions or documents is exposed. Verification has to be built in, not added when something looks suspicious.

7. Residential Callout

Homebuyer closings are the high-volume version. Residential closings run through the same title and escrow infrastructure at far greater volume, which is why digital closing platforms and fraud prevention tools have developed fastest there. Homebuyers are also the most exposed to wire fraud, since they are unfamiliar with the process and often wiring most of their savings.

Home inspection reports are a new AI use case. Buyers and agents increasingly use AI to summarize long inspection reports and estimate repair costs. It's helpful for triage, but an estimate generated from a report isn't a contractor's quote.

Builders run land diligence. For homebuilders, diligence focuses on land: soils and geotechnical conditions, wetlands and environmental constraints, utility availability and capacity, access, and any existing entitlements or development agreements. Much of that work depends on civil engineers and consultants in the field.

HOA and community documents. Resale buyers review homeowners association documents, budgets, and rules, a smaller-scale version of commercial lease and document review that AI summarizes well.

8. Signals to Watch

- **Fully automated title.** Watch whether automated title search and examination expand beyond simpler residential transactions into commercial deals.
 - **AI connections to data rooms.** Watch how data room vendors balance opening their platforms to AI assistants with the security their clients expect.
 - **Changes to diligence standards.** Watch whether ASTM or other standards bodies address the use of AI in environmental and property condition assessments.
 - **Fraud escalation.** More convincing impersonation will push the industry toward stronger identity verification at every step.
 - **Full-population review becoming the norm,** and whether lenders and investment committees begin expecting it.
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9. Self-Check Exercise

The lease abstraction test. Public companies file material contracts with the SEC, and many filings on the EDGAR database include full commercial leases as exhibits. Find a commercial office or retail lease there.

1. Before using any tool, read the lease yourself and write a one-page abstract covering the parties, term, base rent and escalations, renewal and termination options, operating expense responsibilities, assignment restrictions, and critical dates. Record how long it took.

2. Ask a general-purpose AI assistant to abstract the same lease, requiring it to cite the section number for every item.
3. Compare the two abstracts. Mark every discrepancy and determine which is correct.
4. Search the lease for one unusual clause the AI didn't mention.

Questions to answer afterward:

1. How long did verifying the AI abstract take compared with writing your own? Did the section citations help?
 2. Did either abstract miss something a buyer or lender would care about? Why do you think it was missed?
 3. Why can't AI produce a Phase I Environmental Site Assessment on its own, even if it can review every historical record faster than a person?
 4. If you ran a title company, which part of your closing process would you change first in response to AI-enabled fraud?
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3.5 - Owner's Program, Budget & Delivery Method

1. Where This Sits

The site is acquired. Before an architect draws anything, the owner has to answer three questions that shape the rest of the project:

1. **What do we need?** This is the program: the spaces, sizes, relationships, performance requirements, and quality level.
2. **What can we spend?** This is the budget, usually set from a conceptual estimate before any real design exists.
3. **How will we buy it?** This is the delivery method, which determines who designs, who builds, when the contractor gets involved, and who carries which risks.

These decisions come before any design contract is signed. The delivery method is the least familiar of the three, so here is a quick summary:

- Design-bid-build: the owner hires a designer, then bids the finished documents to contractors.
- Construction manager at risk: a contractor joins during design and later guarantees a maximum price.
- Design-build: one company is contracted to both design and build.
- Integrated project delivery: the owner, designer, and contractor sign a single agreement and share risk and reward.

This chapter covers the software owners use to plan and govern projects, and what AI is changing.

This is the bridge between the real estate side of Part 3 and the design and construction side. The owner is the one party present for the entire lifecycle, and the decisions made here have more influence over cost than anything that happens later, precisely because they're made when the least is known.

2. Who Does the Work

Owners come in many types: developers, universities, health systems, school districts, government agencies, corporations, and institutional investors. Large owners run capital programs with many projects at once.

Owner's representatives and program managers manage projects on the owner's behalf. They may be in-house staff or consultants hired for their construction expertise.

Facility planners and architects develop the program, often through a formal predesign or programming phase that involves interviewing the people who will use the building.

Cost consultants and estimators produce conceptual estimates. Contractors also provide this during early preconstruction when brought in under collaborative delivery methods.

Public procurement officials manage how public owners select designers and contractors under procurement laws.

Boards, lenders, and funders approve budgets and release money, often at fixed milestones.

3. The Software Landscape

Owner software is different from contractor software in one important way. A general contractor manages one project at a time for the owner. A large owner manages a portfolio: dozens or hundreds of projects, multiple funding sources, board reporting, and governance rules. That's why the category is often called capital program management, or a project management information system (PMIS).

Enterprise owner platforms were built for this from the start. **Oracle Primavera Unifier** handles cost control, schedule management, and capital planning for very large programs. **e-Builder**, owned by Trimble, is widely used by public agencies, education, and healthcare owners. **Kahua** focuses on configurable workflows for owners and has emphasized public-sector requirements, including authorization to use AI on government projects. **InEight** serves large capital projects with integrated project controls.

Contractor platforms are moving into the owner's seat. In June 2026, Procore introduced a suite of owner capabilities covering concept projects, portfolio monitoring, capital planning, funding source management, and asset management, all built on Procore's AI. Several pieces were in beta at announcement, with general availability planned through 2026. Procore's pitch to owners rests on a network advantage: when an owner's contractors already run projects in Procore, their data flows into the owner's portfolio view. **Autodesk Construction Cloud** makes a similar case for owners, built around its design and model data.

Development management platforms such as Northspyre, covered in Chapter 3, serve private developers managing budgets, invoices, and draws.

Conceptual cost data anchors early budgets. **RSMeans**, owned by Gordian, is the industry-standard construction cost database, with location-adjusted costs for labor, materials, and equipment. In March 2026, Gordian launched Flash AI Estimating, which uses RSMeans data to generate conceptual estimates from limited project information. Gordian reported that early

adopters cut the time to produce a conceptual estimate from several days to under an hour. **DESTINI Profiler**, from Beck Technology, is widely used by contractors to build conceptual estimates from early models during preconstruction.

And as everywhere in Part 3, **spreadsheets** still carry a large share of capital planning, especially at mid-size owners.

4. Where AI Is Landing

Conceptual estimates: fast is not the same as accurate

Augmented: conceptual estimating. AI is making early estimates dramatically faster, and that matters because owners can now test many scenarios quickly: a bigger building versus a smaller one, a steel frame versus concrete, or phasing versus building all at once.

But a conceptual estimate's accuracy is limited by how little is defined, not by how long it takes to produce. The industry has long recognized that estimates made at the concept stage carry a wide accuracy range, far wider than estimates made from completed drawings. An AI tool can produce a concept-stage number in minutes, and it will often look more precise than it is. The risk is anchoring. Once a board approves a budget based on an early number, raising it later is politically painful, even if the original number never reflected the real scope.

Cost data also moves. Gordian's 2026 RSMeans data showed construction labor wages rose 4.6% over the previous year. An estimate built on stale, national-average, or poorly location-adjusted data can be wrong before design starts.

Augmented: program development. AI can turn a stack of requirements, such as user interviews, organizational standards, benchmark programs from similar facilities, and code-driven minimums, into a draft space program that lists rooms, sizes, and adjacencies. Facility planners then validate it with the people who will use the building, which is where the real work of programming happens.

Augmented: portfolio monitoring. Across a capital program, AI can flag projects trending over budget or behind schedule and answer questions about portfolio health from connected project data. It works only when that data is actually connected, which is exactly what owner and contractor platforms are competing to provide.

Automated with sign-off: procurement documents. Drafting requests for qualifications and proposals, scoring templates, and standard contract exhibits is largely automatable, with legal and procurement review.

Assisted: choosing a delivery method. AI can lay out the tradeoffs of each delivery method clearly. The decision depends on the owner's risk tolerance, schedule pressure, in-house expertise, local market conditions, and, for public owners, what procurement law allows.

Resistant: alignment and approval. Getting users, executives, boards, funders, and sometimes the community to agree on what to build and how much to spend is human work.

Delivery method decides how data flows

Delivery method matters more for technology than most people realize, and it's shifting. FMI's research for the Design-Build Institute of America projects that design-build will represent more than 47% of construction spending in the segments it studied by 2028, and progressive design-build, where the owner and team develop the design and price together, makes up roughly a third of design-build procurements.

Here's why that matters for AI. In traditional design-bid-build, the design team's models and data are typically reduced to drawings for bidding, and the contractor starts over. In design-build and other collaborative methods, the contractor is involved early and the same team carries information from concept through construction. AI tools depend on connected, continuous data.

Collaborative delivery methods create the conditions where AI can deliver the most value, while fragmented delivery methods reset the data at every handoff. That's an argument, not a settled finding, but it's worth testing against every project you see.

5. Jobs and Process Change

What compresses. Status reporting and portfolio roll-ups, which consume much of an owner's representative's week, shrink when project data flows automatically. Conceptual estimate production compresses for cost consultants and precon teams. Drafting procurement documents gets faster.

What shifts. Cost consultants' value moves from producing estimates to challenging them, meaning local market knowledge, escalation judgment, and knowing which scope an early estimate is missing. Program managers become portfolio analysts who spend more time on decisions and less on assembling reports.

What grows. Owners need people who understand data: which contractor platforms their projects run on, what data the owner is entitled to, and how it connects to their finance systems. Owners with years of their own historical cost data have a growing advantage, and someone has to manage that data.

What stays human. Defining what the organization actually needs, negotiating priorities among stakeholders, choosing a delivery method, and securing funding approval.

6. Friction and Risk

Data ownership. When an owner relies on a contractor's platform for project data, it needs a clear contract answer to who owns that data and whether the owner keeps access after the project ends. This becomes a strategic question as platforms compete to become the owner's system of record.

False precision in early budgets. Fast, polished conceptual estimates can make uncertain numbers feel settled. Owners should present early budgets as ranges, with contingency tied to how much of the design is defined.

Public procurement constraints. Public owners must follow procurement laws that limit how they select teams and what delivery methods they can use. Some also face security authorization requirements before adopting cloud or AI tools.

Integration with finance systems. Capital plans have to reconcile with the owner's accounting and budgeting systems. A beautiful portfolio dashboard that doesn't match the general ledger won't survive a board meeting.

Benchmark bias. Programs and estimates drafted from comparable projects inherit those projects' assumptions, including their mistakes and outdated standards.

7. Residential Callout

Production builders don't program projects; they manage plan libraries. A production homebuilder maintains a set of home plans with structural options and design selections, each priced by community. The core system work is managing plans, options, and pricing, typically in homebuilder-specific enterprise software, and keeping that pricing current as costs change.

Custom builders and remodelers program through conversation. The program is whatever the client describes across several meetings, and the budget often starts as a cost per square foot plus allowances for finishes. Residential project management tools carry the client's selections forward from there.

Residential delivery methods are simpler but still consequential. The key decisions are usually fixed-price versus cost-plus contracts and whether to hire a design-build firm or separate designer and builder. Design-build is common among remodelers because it keeps design and pricing together.

Homeowners are starting their own programs with AI. Houzz's 2026 research found homeowners using AI mostly for renovation ideas and visualization, with far fewer using it for budgeting. That gap is a residential version of the conceptual estimating problem: easy inspiration, harder numbers.

8. Signals to Watch

- **Procore's owner suite reaching general availability**, and whether large owners adopt a contractor-origin platform as their system of record over purpose-built owner platforms.
 - **Calibration of AI conceptual estimates.** Owners with historical data can compare early AI estimates against final costs. Published results would show whether AI actually narrows early budget error or just produces it faster.
 - **Continued growth of collaborative delivery**, especially progressive design-build, and whether it measurably improves data continuity.
 - **Public-sector AI authorization.** More security-authorized platforms would open AI to government capital programs, one of the largest owner segments.
 - **Owner data strategies**, such as contract language requiring project data delivery in usable formats.
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9. Self-Check Exercise

The one-hour project definition. Pick a realistic project in a metro you know, such as a 20,000-square-foot medical office building or a small K–8 school addition.

1. Ask a general-purpose AI assistant to draft a space program based on a short description of the users and services.
2. Ask it for a conceptual cost estimate, with the location, building type, and assumptions stated explicitly.
3. Find an independent reference point for cost per square foot for that building type and location, such as published cost data, a public agency's recent bid results, or a comparable project reported in local news.
4. Ask the AI to recommend a delivery method, then write your own recommendation and explain where you disagree.

Questions to answer afterward:

1. How far apart were the AI estimate and your independent reference? Which assumptions explain the difference?
 2. Which three assumptions in the program or estimate would move the budget the most?
 3. How would you present this early budget to a board so it isn't mistaken for a firm number?
 4. How would your delivery method choice change the way project data flows from design into construction?
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3.6 - Conceptual & Schematic Design

1. Where This Sits

Section A ended with an owner who has a site, a program, a budget, and a delivery method. Now the design team gives the project a shape.

This is the start of design. Design traditionally moves through three phases: schematic design, design development, and construction documents. In schematic design, the team establishes the building's form and massing, its placement on the site, the basic plan organization, a direction for materials and character, and preliminary approaches to structure and systems. The phase ends when the owner approves the direction.

This phase has a particular economic character. Changing a design here is cheap, since it means redrawing or remodeling, but the decisions are highly consequential. The massing chosen now largely determines the building's cost, energy performance, and constructability. That makes it the phase where exploring more options has the most value, and where AI has arrived fastest in architecture.

2. Who Does the Work

Design principals and lead designers set the concept and make the key design decisions.

Junior designers have traditionally produced much of the volume: massing studies, plan options, diagrams, and presentation materials.

Visualization specialists produce renderings and animations, either inside the architecture firm or at dedicated architectural visualization studios.

Computational designers at larger firms build scripts and parametric tools that generate and evaluate options.

Consultants including structural engineers, sustainability advisors, and landscape architects weigh in early on major decisions.

Owners and their stakeholders review options and approve the direction.

3. The Software Landscape

Modeling for concept work is dominated by tools built for speed and flexibility. **SketchUp**, owned by Trimble, is the fast, approachable default for massing and early 3D. **Rhino**, from McNeel, paired with its visual programming environment **Grasshopper**, is the standard for complex geometry and computational design. Many firms also start directly in **Revit**, covered in Chapter 7.

Autodesk Forma, covered in Chapter 2, is Autodesk's cloud tool for early-stage site planning and environmental analysis such as sun, daylight, and wind. Autodesk has been tying it more closely to Revit, and with the Revit 2027 release in April 2026, it's reported to be included in Revit subscriptions.

A group of cloud-native challengers is attacking the space between concept tools and Revit. Industry coverage often calls them "BIM 2.0." **Snaptrude** is a browser-based modeling platform that has focused its AI investment on conceptual and schematic design. A major 2026 release aims to take its models far enough that architects could stay in it through schematic design. **Arcol** offers browser-based, collaborative building design. **Hypar**, co-founded by the creator of Autodesk's Dynamo scripting tool, focuses on generating space plans from building requirements and works alongside Revit. Most of these challengers support Revit's file format, because no challenger can ignore where the industry's work already lives.

Visualization has its own market. **Chaos** owns several of the leading tools: **Enscape** for real-time rendering inside design software, and **V-Ray** and **Corona** for high-end photorealistic rendering. In May 2026, Chaos made **Veras**, its AI visualization tool, part of every version of Enscape, V-Ray, and Corona. Veras turns sketches, simple models, and images into presentation-quality renderings, and in September 2026 Chaos added the ability to turn a still render into a short animation in minutes. Competitors include **Twinmotion** from Epic Games, **Lumion**, and **D5 Render**. General-purpose image generators are also widely used for mood boards and inspiration.

4. Where AI Is Landing

Automated with sign-off: presentation imagery. Producing a compelling rendering used to take hours or days of specialized work. AI tools now generate styled, photorealistic images from a rough model or even a hand sketch in seconds, and animations in minutes. For most early-stage presentation needs, rendering is effectively solved.

Augmented: generating design options. Tools can generate many massing and layout options from a site, program, and set of constraints, and space-planning tools can turn a room list into draft floor plans. Early environmental analysis, such as sun and daylight studies, runs quickly enough to inform choices while they're still being made.

Augmented: exploring variations. Designers can test different materials, facade treatments, and moods on the same massing in minutes, which used to require separate rendering passes.

Assisted: the concept and narrative. AI can help articulate a design idea and draft presentation text, but the idea itself has to come from the designer.

Resistant: design judgment. Deciding which option is right for the client, the site, the budget, and the community is the core of the architect's work. So is the relationship with the client that makes the decision stick.

From producing options to curating them

The shift in this phase can be summarized simply. For most of the profession's history, the constraint was production: how many options a team could draw and render before the presentation. That constraint is disappearing. The new constraint is judgment: knowing which of 50 generated options deserves development, and explaining why to a client. The architect's job moves from producing options to curating them.

The rendering trap

AI imagery brings a subtle risk. A photorealistic image implies decisions that haven't been made yet: specific materials, structural spans, facade details, and a level of finish. A client who sees a polished rendering at the concept stage can reasonably assume the design is further along than it is, and can become attached to details that won't survive budget or code review. There's also a technical issue. Tools that generate images from screenshots can quietly alter geometry, while tools that work from the actual model tend to stay truer to the design. Chaos makes this distinction itself in its guidance on AI rendering. **A rendering is no longer evidence of design progress**, and firms need to manage client expectations accordingly.

Adoption is real, but measured differently

Surveys show architecture adopting AI quickly, particularly for imagery, but they measure very different things. An AIA study published in early 2025 found only 6% of architects routinely using AI tools. A 2026 survey reported by ArchDaily found 64% of architects experimenting with AI in their daily work and about one in five firms integrating it across multiple processes, with concept imagery and rendering among the most common uses. Part of that gap is real change, and part is the difference between routine use and experimentation.

5. Jobs and Process Change

What compresses. The volume work of junior designers, such as producing massing variations, diagrams, and presentation images, shrinks sharply. Architectural visualization studios face pressure on routine rendering work.

What shifts. Junior designers move toward editing, evaluating, and refining generated options. Visualization specialists move toward high-end work: marketing imagery, immersive experiences, and art direction that clients will still pay a premium for.

What grows. Again, mostly shrinking here.

The fee question. When options and renderings take minutes, clients may expect more of them for the same fee. Firms that bill by the hour for production face the most pressure. Firms that price on the value of design decisions are better positioned.

The apprenticeship problem. Junior architects have traditionally developed design judgment by producing options and seeing which ones survive critique. If AI produces the options, firms have to find another way to build that judgment.

6. Friction and Risk

Client expectations. Photorealistic early imagery can lock in client expectations for decisions that haven't been made. Firms need to label images clearly and pace their level of detail to the phase.

Geometry fidelity. AI images that drift from the actual model misrepresent the design. For anything shown to a client or the public, the image needs to match the model.

Intellectual property. Legal questions about how AI image models were trained and who owns AI-generated imagery remain unsettled. Firms should understand the terms of the tools they use, especially for marketing materials.

Confidentiality. Uploading client designs to cloud AI services raises the same data handling questions as any confidential project information.

Homogenization. When many firms use the same tools trained on similar images, designs can drift toward a common look. Distinctive design becomes a competitive advantage rather than a default.

7. Residential Callout

Many homes are designed without an architect. Building designers, plan services, and builders' in-house drafters design most new single-family homes. Concept design is often choosing and adapting an existing plan rather than creating a new one.

Rendering sells homes before they exist. Production builders rely on renderings, virtual tours, and interactive floor plans to sell homes that haven't been built. AI makes these far cheaper to produce for every plan, elevation, and option combination.

Remodelers use visualization to close the sale. Showing a homeowner what a finished kitchen or addition will look like is one of the most effective sales tools in residential construction, and Houzz's 2026 research found pre-project visuals to be the most concentrated AI use among design firms. The rendering trap applies even more strongly here, since homeowners rarely distinguish a concept image from a promise.

Homeowners generate their own concepts. Consumers now arrive with AI-generated images of what they want. That can speed up conversations, but it also sets expectations for finishes and budgets that professionals then have to reset.

8. Signals to Watch

- **Cloud-native challengers moving deeper into design.** Watch whether tools like Snaptrude and Arcol can carry projects past schematic design, which would challenge Revit's position rather than just feed it.
 - **Bundling by incumbents.** Autodesk including Forma with Revit, and Chaos including Veras across its renderers, make it harder for standalone AI tools to compete.
 - **AI animation and video** becoming standard for client presentations and marketing.
 - **Changes in architectural fee structures** as production time falls.
 - **Legal decisions on AI training data and image ownership**, which could change what firms are comfortable using.
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9. Self-Check Exercise

Options versus judgment. Pick a simple project, such as a two-story library branch on a corner lot or a small office building.

1. Build a basic massing model in a free or trial modeling tool, or sketch the massing by hand and photograph it.
2. Use an AI rendering tool to produce three different renderings of the same massing in different styles and materials.

3. List everything each rendering implies that your model or sketch didn't actually decide: materials, structure, window systems, landscaping, and level of finish.
4. Ask an AI assistant to generate 10 different massing concepts in writing for the same site and program. Choose the three you would develop and write one paragraph each explaining why.

Questions to answer afterward:

1. How did the renderings change what a client would believe about the project?
 2. What criteria did you use to choose three options out of 10? Could an AI have applied those criteria for you?
 3. If you were running a small architecture firm, how would you explain to a client why more options don't mean a lower fee?
 4. How would you train a junior designer to develop design judgment when AI produces most of the options?
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3.7 - BIM Authoring & Engineering Analysis

1. Where This Sits

Chapter 6 ended with an approved schematic design. Now the team has to make it real: resolving how the structure works, how mechanical, electrical, and plumbing systems fit, how the site grades and drains, and how every assembly goes together. Then it has to document all of it.

This is design development flowing into construction documents, where the approved concept becomes a fully resolved and documented building. This chapter covers the software that models the building and analyzes its engineering systems.

The key idea is **Building Information Modeling (BIM)**. A BIM model isn't just a 3D picture. It's a database of building elements, such as walls, doors, beams, ducts, and fixtures, each with properties like size, material, fire rating, and cost. Drawings, schedules, and quantities are generated from that database. When the model changes, the drawings update. That's the theory, and it's also why this phase is where AI meets the industry's core production software.

2. Who Does the Work

Project architects lead design development and coordinate the technical resolution of the design.

BIM managers and coordinators set modeling standards, manage model quality, and run coordination between disciplines.

Technical designers and drafters build much of the model and produce the drawing sheets.

Structural engineers design the structural system and verify it through analysis software.

MEP engineers and designers design mechanical, electrical, and plumbing systems.

Civil engineers design grading, drainage, utilities, and site access.

Specialty contractors' VDC teams increasingly build detailed models too. Electrical, mechanical, and steel contractors produce detailed coordination and fabrication models from the design team's intent. VDC stands for virtual design and construction.

3. The Software Landscape

Architectural authoring

Revit, from Autodesk, is the dominant BIM authoring platform in the United States for architecture, and it's widely used by structural and MEP engineers as well. Its dominance comes from years of network effects. Firms use Revit because their consultants, clients, and future hires use Revit.

Autodesk released **Revit 2027** in April 2026, and AI was the headline. It introduced **Autodesk Assistant**, released as a technology preview, which lets users work with the model in plain language. Users can query elements, create and organize sheets, manage schedules and rooms, and edit parameters across many elements at once. It's built on the Model Context Protocol (MCP), an open standard for connecting AI tools to software, and Autodesk has also opened Revit's MCP server so firms can connect other AI tools to their models. That openness is a notable shift for a platform long criticized as closed.

Archicad, from Graphisoft, part of the Nemetschek Group, is the most established alternative to Revit, with loyal users especially among design-focused firms. Graphisoft introduced an AI Assistant in beta with Archicad 29 in late 2025. In May 2026, Nemetschek announced it is bringing its design tools together into a single cloud-native platform, and in June Graphisoft announced a connection between Archicad and Autodesk Forma. **Vectorworks**, also owned by Nemetschek, is popular with landscape architects and smaller design-driven practices. **Allplan**, another Nemetschek brand, is much more common in Europe than in the U.S. **Bentley Systems** is strongest with infrastructure owners and engineering firms.

Structural engineering

Tekla Structures, from Trimble, is the leading tool for detailed steel and concrete modeling used in fabrication. For structural analysis, **ETABS** and **SAP2000** from Computers & Structures, Inc. (not to be confused with the Construction Specifications Institute, also known as CSI) are industry standards, along with **RISA**, now part of Nemetschek. Structural documentation often happens in Revit.

MEP engineering

Revit is the dominant MEP design environment. Contractor-side tools such as **Trimble MEP** handle detailing and fabrication. **Augmenta**, a Toronto company founded by people who worked on generative design at Autodesk, is one of the clearest examples of AI designing building systems rather than just operating software. Its platform automatically routes and coordinates electrical conduit through a building. Augmenta reports customers working about three times faster than manual methods. It released a major update in June 2026 and plans to extend the approach to mechanical and plumbing systems.

Civil engineering

Civil 3D, from Autodesk, and **OpenRoads** and **OpenSite**, from Bentley, dominate civil design. Bentley is particularly strong with state transportation departments and large infrastructure owners.

4. Where AI Is Landing

Two kinds of AI in the model

This chapter depends on one distinction.

AI that runs the software. This includes assistants like Autodesk Assistant and Archicad's AI Assistant. They translate plain-language requests into software operations: find these elements, create these sheets, update these parameters. Language models are well suited to this, because the task is understanding intent and executing known commands.

AI that designs the systems. This is AI that generates actual building geometry, such as conduit routes, duct layouts, or structural framing, that must be physically correct, coordinated, and constructable. Augmenta frames this directly, arguing that building design is a geometry problem rather than a language problem. That work relies on spatial reasoning and optimization, not just language.

Both matter, but they mature at different speeds and change different jobs.

Task ratings

Automated with sign-off: documentation production. Setting up sheets, placing views, numbering, tagging elements, building schedules, and making batch edits are the repetitive tasks that consume enormous drafting hours. They're the first targets for AI assistants. Model audits, such as finding untagged doors or elements with missing fire ratings, are also becoming conversational queries.

Augmented: MEP routing. For electrical raceway today, and likely for other systems soon, AI generates coordinated routing that a VDC team or engineer reviews and adjusts. Demand from very large projects like data centers is accelerating adoption, because those projects contain miles of repetitive systems.

Augmented: structural and energy optimization. Engineering calculations were computerized decades ago. The new capability is exploring alternatives quickly, such as testing many framing schemes or envelope options against cost, weight, carbon, and performance before choosing one.

Assisted: technical detailing. AI can suggest assemblies, surface relevant code requirements, and draft details from firm standards. Choosing the right detail for specific conditions still depends on experience.

Resistant: engineering judgment and responsible charge. The licensed engineer or architect who stamps the drawings is personally responsible for them. Coordination decisions between disciplines, like whose system moves when a duct hits a beam, are negotiations with cost and schedule consequences. Neither can be delegated to software.

Model quality is the foundation

AI assistants operate on whatever model they're given. A well-structured model with consistent families, parameters, and standards produces useful results. A messy model produces messy results faster. Practitioners reviewing Revit 2027 emphasized exactly this point. For many firms, the first real AI project will be cleaning up modeling standards and content libraries.

5. Jobs and Process Change

What compresses. Drafting and documentation hours shrink first: sheet setup, tagging, annotation, and schedule production. Much of the modeling time electrical designers and VDC teams spend routing systems also compresses.

What shifts. BIM managers move from enforcing standards manually to designing automation and governing how AI is used on models. Engineers spend less time modeling and more time checking, adjusting, and deciding. Specialty contractors' VDC teams can take on more projects with the same staff, which matters given how hard those roles are to fill.

What grows. Firms need people who can build automation, whether through scripting tools like Dynamo and Grasshopper, custom AI connections to models, or both. Model data quality becomes a real specialty.

What stays human. Engineering judgment, responsible charge, coordination negotiations, and the experience-based knowledge of how buildings actually go together.

The apprenticeship problem, again. Technical designers have traditionally learned construction by drawing thousands of details and seeing which ones cause problems in the field. As production automates, firms need to deliberately teach what drafting used to teach by repetition.

6. Friction and Risk

Responsible charge. Licensing boards require that a licensed professional exercise control over the work they seal. As AI generates more of the design content, firms need clear review processes that demonstrate that control.

Maturity of the tools. Autodesk Assistant launched as a technology preview. Early AI features inside production software should be tested on non-critical work before firms rely on them for deadlines.

Security of model connections. Opening models to external AI tools creates new questions about where project data goes and who can modify a model.

Platform dependence. Revit's dominance gives Autodesk significant pricing and product power over the industry. Firms have long complained about licensing costs and product pace. AI features embedded in the incumbent platform can deepen that dependence even as open standards like MCP create some new openness.

Interoperability. Projects often involve several authoring tools across disciplines. Exchanges through open formats like IFC still lose information, and AI tools trained on one platform's data may not transfer to another's.

7. Residential Callout

Chief Architect leads residential design. Most residential designers and many builders use Chief Architect, with SoftPlan as a notable competitor. These tools are built around residential construction: framing, roofs, cabinetry, and automatic generation of construction drawings for typical wood-frame houses. Revit use in single-family residential is limited.

Engineering comes through the supply chain. Most wood-frame homes don't have a full engineering team. Roof trusses and floor systems are typically engineered by component manufacturers using specialized software, with MiTek's tools dominant in truss design. Structural engineers stamp specific elements like beams, foundations, or lateral systems where codes or conditions require it.

Production builders manage plan sets, not one-off models. A production builder maintains drawing sets for each plan with its elevations and options, and has to keep every combination accurate as codes and products change. That's a strong fit for automation: generating option variants, updating drawing sets, and producing framing takeoffs.

Offsite construction connects design to fabrication. Wall panelization and modular construction depend on accurate digital models that feed factory production. That's where residential design software, component engineering, and automation are converging.

8. Signals to Watch

- **Autodesk Assistant moving out of technology preview**, and how many firms connect their own AI tools to Revit through its open MCP server.
 - **Nemetschek's cloud-native platform**, and whether a unified alternative can meaningfully challenge Revit's U.S. position.
 - **AI routing expanding beyond electrical** to mechanical and plumbing systems, which would change MEP design and coordination economics.
 - **Cloud-native challengers reaching construction documents**, the step that would let firms leave Revit entirely.
 - **Licensing board guidance** on AI-generated design content and responsible charge.
-

9. Self-Check Exercise

Running software versus designing systems. Find a sample BIM project. Autodesk publishes sample Revit projects, and buildingSMART and others publish sample IFC files viewable in free viewers.

1. Open the model in a free viewer or trial and explore how elements carry properties like type, material, and fire rating.
2. List every step required to produce a set of floor plan sheets with door tags and a door schedule.
3. Sort those steps into two groups: tasks that are really about operating the software, and tasks that involve design or engineering decisions.
4. Pick one building system visible in the model, such as ductwork, conduit, or structural framing, and write down what a person would need to decide before any AI could route or size it.

Questions to answer afterward:

1. Which group of tasks would an AI assistant handle well today, and which would it struggle with?
 2. Why might MEP routing be automated before architectural detailing?
 3. What would a licensed engineer need to see in order to responsibly seal drawings that included AI-generated systems?
 4. If you managed a 30-person architecture firm, what would you fix in your models and standards before rolling out AI assistants?
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3.8 - Coordination, Common Data Environments & Drawing QA

1. Where This Sits

Chapter 7 covered how each discipline models its part of the building. This chapter covers the hard part: making all of those parts agree.

A set of construction documents for a commercial building can run to hundreds of sheets produced by several firms. The architect, structural engineer, MEP engineers, civil engineer, and specialty consultants each draw their own systems. Those systems have to fit in the same physical space, reference each other correctly, and describe one building. When they don't, the conflicts show up as RFIs, change orders, and rework in the field.

This work spans design development and construction documents, and it continues into construction as contractors coordinate their own detailed models before fabricating and installing. This chapter covers three connected functions:

1. **Coordination:** finding and resolving conflicts between disciplines.
 2. **Common data environments (CDEs):** the controlled places where project information lives.
 3. **Drawing quality assurance:** checking a set before it's issued.
-

2. Who Does the Work

BIM and VDC coordinators, at design firms and general contractors, run clash detection, manage federated models (combined models from all disciplines), and lead coordination meetings.

Project architects and senior reviewers perform quality reviews of drawing sets before major issues, often through detailed red-line reviews that consume some of a firm's most experienced hours.

Discipline engineers resolve the conflicts assigned to them.

Document controllers manage versions, transmittals, and who has access to what.

Specialty contractors' detailers, such as mechanical, electrical, plumbing, fire protection, and steel, build detailed models during construction and coordinate them with each other before fabrication.

3. The Software Landscape

Common data environments

The international standard ISO 19650 defines a common data environment as a single controlled source of project information with defined workflows for sharing and approving it. In the United States, formal adoption of that standard is uneven, so in practice a project's CDE is usually whichever platform the owner or lead firm chooses. Large projects often end up with several: the architect's, the contractor's, and sometimes the owner's.

Autodesk Construction Cloud is the leading CDE for teams working in Revit, with document management, model coordination, and issue tracking in one cloud platform. It pairs with **Navisworks Manage**, the long-standing industry standard for combining models from different disciplines and running clash detection. Clash results can become tracked issues that sync between Navisworks and the cloud.

Bluebeam Revu, owned by Nemetschek, is the standard tool for reviewing and marking up PDF drawings, and its Studio Sessions let many reviewers mark up the same set at once. For most of the industry, drawing review still happens on PDFs, which makes Bluebeam one of the most widely used tools in AEC.

Revizto combines clash detection with issue tracking in one environment and automates grouping clashes and assigning them to the responsible party.

Bentley ProjectWise and **iTwin** are strongest with infrastructure owners and engineering firms. **Trimble Connect** serves Trimble's ecosystem. **Oracle Aconex** is used on very large projects where a strict audit trail across many organizations matters. **Newforma** is widely used by design firms to manage project information like email, transmittals, and submittal records from the architect's side.

Speckle takes a different approach. It's an open-source data platform that breaks models from Revit, Rhino, Archicad, and other tools into structured, versioned data objects rather than files. Large firms use it to move data between tools and build dashboards and automation on their own project data. In April 2026, Suffolk Technologies, the venture arm of Suffolk Construction, invested in Speckle following Suffolk's company-wide deployment, a signal that contractors see structured design data as a foundation for AI.

4. Where AI Is Landing

The real problem is noise, not detection

Clash detection software has found geometric conflicts for years. The problem was never finding clashes. It was finding thousands of them, most of them trivial, duplicated, or already known. A pipe running through a hundred ceiling hangers can generate a hundred clashes that are really one issue. Coordinators spend their time sorting noise from problems.

The same pattern applies to drawing review. A senior reviewer's time goes mostly to searching: flipping between sheets to check that a detail callout points to a detail that exists, that a door on the plan matches the door schedule, that a revision didn't break something elsewhere.

AI's main contribution in this phase is triage. It shifts reviewers' time from finding candidate issues to confirming which ones matter.

Task ratings

Automated with sign-off: clash grouping and assignment. Tools now group related clashes into single issues, remove duplicates, and route them to the responsible discipline. A coordinator reviews the groups rather than the raw list.

Automated with sign-off: revision comparison. Identifying exactly what changed between two versions of a drawing set is now largely automated. Bluebeam's **Smart Overlay**, part of its Bluebeam Max plan launched globally in May 2026, compares sets across sheets and disciplines and shows where changes occurred.

Augmented: cross-sheet consistency and drawing QA. Tools scan a whole set for broken references, missing information, and inconsistencies between sheets. Bluebeam's **Smart Review** produces an issues list with locations and priorities. Both Smart Review and Smart Overlay grew out of technology from Firmus AI and were still labeled preview features for a period after launch. Bluebeam Max also adds a natural-language integration with Anthropic's Claude for automating repetitive Revu tasks. Other drawing review tools are emerging from startups and from construction platforms.

Augmented: model QA. Checking models against firm standards, such as naming, required parameters, and correct classification, is becoming a routine automated pass before models are shared.

Assisted: prioritizing issues. AI can rank clashes by apparent severity, but knowing which conflicts will actually cost money or delay installation depends on construction knowledge: sequence, access, fabrication, and trade practice.

Resistant: resolving conflicts and owning the set. Deciding whose system moves when a duct hits a beam is a negotiation with cost and schedule consequences. The professionals who seal the drawings remain responsible for them, whatever tools checked them. And some of the

most expensive problems aren't geometric at all: equipment that can't be maintained, corridors too tight to move materials through, or systems that can't be installed in the planned sequence.

Files versus data

A deeper shift sits underneath these tools. Most coordination still moves whole files, such as model files and PDFs, between firms. AI works far better on structured data it can query directly. Platforms like Speckle, along with Autodesk's and Nemetschek's moves toward cloud-based data models and open AI connections, point toward CDEs that hold queryable data instead of document folders. That transition is a prerequisite for most of the ambitious AI promises in design and construction.

5. Jobs and Process Change

What compresses. Running and sorting clash tests shrinks for junior coordinators. Senior reviewers spend far less time hunting for cross-reference errors and revision changes. Routine document control tasks automate.

What shifts. Coordinators become facilitators of decisions rather than producers of clash reports. Senior reviewers focus on constructability, design intent, and the issues automated tools can't see. Coordination meetings can spend their time on resolution instead of discovery.

What stays human. Conflict resolution, constructability judgment, responsibility for the issued set, and the field experience that recognizes a problem no algorithm flags.

6. Friction and Risk

False positives and alert fatigue. An AI review that flags 400 possible issues, 350 of them irrelevant, can cost more time than it saves and teach teams to ignore flags. The value of these tools depends heavily on precision.

Wrong version, right answer. Automated review is only useful if it's run on the current set. With multiple CDEs on one project, version confusion remains a real risk.

AI review is not professional review. A clean automated check doesn't mean a set is complete or correct. Firms need to be careful not to let tool output substitute for the quality review their standard of care requires.

Data security. Connecting drawing sets and models to AI services raises questions about where confidential project information goes, especially for sensitive facilities.

Platform fragmentation. When the architect, contractor, and owner each run a different CDE, information still moves by export and upload, and each handoff loses context.

7. Residential Callout

Formal coordination is mostly skipped. Most single-family homes are built from PDF or paper plan sets without federated models or clash detection. Conflicts, like a plumbing drain landing on a floor joist, get resolved in the field by framers and trade partners.

Production builders coordinate through repetition. A production builder's first build of a new plan often surfaces its coordination problems. Those fixes then get folded back into the plan set, so later homes go smoother. The coordination cost is real; it's just paid on the first houses instead of in software.

Remodelers coordinate with existing conditions. For renovation work, the biggest conflicts are between the plan and what's actually inside the walls, which no drawing review can catch without investigation.

Simple plan sets are a strong AI fit. Residential plan sets are shorter and more standardized than commercial ones, which makes automated completeness and consistency checks before permit submission a practical near-term use. Chapter 10 covers the permitting side.

8. Signals to Watch

- **Accuracy of AI drawing review.** Watch for published false-positive rates and case studies as tools like Smart Review move from preview to general availability.
 - **Open AI connections to CDEs and markup tools.** Watch whether firms build their own review agents on top of these platforms.
 - **Data-based CDEs** replacing file-based folders, and whether contractors keep investing in design data platforms.
 - **Owner mandates** for information management standards like ISO 19650 on U.S. public and institutional projects.
 - **Issues beyond geometry.** Watch whether AI begins catching maintainability and sequencing problems, not just physical clashes.
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9. Self-Check Exercise

The addendum hunt. Public agencies post bid documents and addenda for construction projects on their procurement websites. Find a project with an original drawing set and at least one addendum or revised set.

1. Pick five sheets that changed. Without any comparison tool, list every change you can find and record how long it took.
2. Run the same sheets through a PDF comparison tool. Many PDF editors and trial versions include one. List what the tool found that you missed, and anything it flagged that wasn't a real change.
3. Choose 20 detail or section callouts in the set and verify that each one points to a detail that exists and matches.
4. For every problem you found, decide whether a geometry check, a text check, or only a human reviewer with construction knowledge could have caught it.

Questions to answer afterward:

1. How did the comparison tool change where you spent your time?
 2. Which of your findings would cause an RFI or change order if missed?
 3. How would you design a review process that uses automated checks without letting them replace professional review?
 4. If a project has three different CDEs, what's the first thing you'd standardize?
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3.9 - Specifications

1. Where This Sits

Drawings show what goes where: quantities, locations, dimensions, and relationships.

Specifications define quality. They state what materials and products are acceptable, what standards they must meet, how they must be installed, how the contractor proves compliance, and the administrative procedures for the project. Together, drawings and specifications make up the construction documents, and specifications are bound into the **project manual** along with bidding and contract requirements.

Specifications are organized using standards from the Construction Specifications Institute (CSI). MasterFormat organizes them into numbered divisions and sections, such as Division 08 for openings and Division 09 for finishes. SectionFormat structures each section into three parts: General, Products, and Execution. PageFormat standardizes page layout. These standards matter more for AI than they might seem, as this chapter will show.

The most important fact about specifications is legal. **Specifications are contract documents.** A requirement in a spec is a term the contractor is bound to deliver and the owner is paying for. That fact shapes everything AI can and can't do in this phase.

This chapter also takes on a common claim directly: that AI is making specification writing obsolete.

2. Who Does the Work

Specification writers, often called specifiers, produce and coordinate the project manual. Many hold CSI's Certified Construction Specifier credential. Some work inside large architecture firms, and many small and mid-size firms hire independent specification consultants.

Project architects and engineers make the product and performance decisions the specs record, and engineers typically write their own discipline sections.

Manufacturer representatives provide guide specifications and technical information for their products. CSI offers a credential for them too, the Certified Construction Product Representative.

Owners with standards, such as universities, hospital systems, and federal agencies, require their own specification requirements on every project.

Contractors and subcontractors read specs to price the work, order materials, and prepare submittals. General contractors' project engineers and the architect review those submittals against the specs.

That last group matters for the market. A handful of people write a project's specifications. Dozens of firms read them.

3. The Software Landscape

Deltek is the central player in U.S. specification software. Its **Specpoint** platform is the home of **AIA MasterSpec**, the leading master guide specification library, with content authored and peer-reviewed by licensed architects and engineers and updated regularly. Deltek also owns **SpecLink**, another established specification system. Specpoint includes an AI assistant, Dela, that answers questions grounded in MasterSpec content, product listings, and supporting documents. Its "Best Practices" feature applies a firm's master specifications and owner standards to new projects through smart tagging. When Deltek introduced it in 2024, the company said it could potentially generate 80% of a project's specifications automatically at creation. Specpoint can also generate a submittal register directly from the specifications.

SpecsIntact is the free specification system used for federal work with agencies like NASA, the Army Corps of Engineers, and the Naval Facilities Engineering Command, built around their unified guide specifications. **NBS Chorus** plays a role in the UK and internationally similar to MasterSpec's in the U.S.

Manufacturer guide specifications are published by product manufacturers in MasterFormat structure, both on their own websites and through free libraries like ARCAT. They're useful, but they're written to favor the manufacturer's product.

Office masters in word processors are still common. Many firms maintain their own master specifications as word processing documents, edited project by project.

Downstream tools read specifications for contractors. This is a growing and important part of the market. **Pype AutoSpecs**, owned by Autodesk, reads a project's spec book and automatically generates the submittal log and closeout requirements. **Procore** offers a Submittal Reviewer Agent that checks submittals against project specifications and flags discrepancies. Specialists like **BuildSync** extract individual technical characteristics from a submittal, such as ratings, dimensions, and certifications, and mark each one compliant, non-compliant, or unclear against the specification.

4. Where AI Is Landing

Automated with sign-off: building logs and registers from specs. Extracting every submittal, test, warranty, and closeout requirement from a spec book used to take a project engineer days. It's now largely automated on both the design and contractor sides.

Automated with sign-off: first-draft project specifications. Starting from a master and project information, software can select applicable sections, remove options that don't apply, and apply firm and owner standards. The result is a draft far closer to finished than a raw master.

Augmented: coordination checks. AI can flag sections referenced but not included, inconsistent terminology, outdated references to industry standards, and some conflicts between specifications and drawings.

Augmented: submittal review. Checking a product data sheet against a specification requirement by requirement is exactly the kind of structured comparison AI handles well. Reviewers get a list of pass, fail, and unclear items instead of reading every page themselves.

Augmented: product research. Finding products that meet a set of performance requirements, and comparing them, is faster with AI across manufacturer data.

Assisted: performance specifications and substitutions. Writing specs that define required performance rather than naming products, and evaluating whether a proposed substitution is truly equal, require technical judgment and an understanding of risk.

Resistant: responsibility for the requirements. Deciding what the project actually requires, standing behind it as a contract term, approving substitutions, and interpreting intent when questions arise during construction all remain professional responsibilities.

Is specification writing becoming obsolete?

The honest answer is that **specification drafting labor is disappearing, but specification responsibility is not.**

The drafting argument is strong. Most of a project manual is adapted from masters, and software is increasingly good at the adaptation. Deltek's own claim about generating most of a project's specs automatically points the same way. The hours a firm spends assembling and formatting a project manual are going to shrink dramatically.

But a specification isn't a document; it's a set of promises. Someone has to decide that the roofing membrane must meet a particular wind uplift rating, that the door hardware must match the owner's key system, and that Division 01 requires a specific submittal and closeout process suited to the delivery method. Someone has to catch that the boilerplate section requires a product that conflicts with the one on the drawings. When the wrong requirement causes a failure, the question in the dispute will be what a reasonably careful professional would have specified, not which software drafted it.

The likely future is **fewer specification writers covering more projects, with the role shifting from author to editor and quality authority**. For a firm, that means the specifier's value lies in judgment, product knowledge, and risk management, not word processing.

The bigger AI market is reading specs

There's a second, less discussed shift. Because far more people read specifications than write them, much of the AI investment is flowing downstream to contractors and subcontractors, into tools that extract requirements and check compliance. As that loop tightens, specifications start to function less like documents and more like structured data that drives automated checks.

That's where CSI's formats become more valuable, not less. MasterFormat numbering and SectionFormat's consistent three-part structure give AI a reliable map of where every requirement lives. **A standard designed for human readers turns out to be excellent scaffolding for machines.**

5. Jobs and Process Change

What compresses. Project manual assembly, formatting, and first-draft editing shrink sharply. So does the work of building submittal logs and first-pass submittal review on the contractor side, covered further in Chapter 14.

What shifts. Specifiers move toward editing, coordination, product strategy, and risk review. Architects' submittal review can focus on the items flagged as non-compliant or unclear. Manufacturer representatives spend less time distributing specs and more time as technical advisors, since AI can surface product information on its own.

What stays human. Deciding what the project requires, evaluating substitutions, standing behind requirements as contract terms, and interpreting intent during construction.

The training gap. Junior architects used to learn construction materials and methods partly by editing specifications. As that work automates, firms lose one of their traditional ways of teaching technical knowledge.

6. Friction and Risk

Boilerplate bloat. AI makes it easy to include everything. Over-inclusive specifications create internal conflicts, unnecessary cost, and requirements no one intended. Easier drafting makes disciplined editing more important.

Conflicts with drawings. Many of the costliest document problems are disagreements between specs and drawings. Automated tools catch some of them, but not reliably across every format.

Outdated references. Specifications cite industry standards by edition. An AI draft that references a superseded standard or discontinued product creates a real contract problem.

Over-reliance on automated submittal review. An automated compliance check that marks a product "compliant" can create false confidence. Reviews still need a professional's judgment, especially for life safety, structural, and building envelope items.

Licensed content. MasterSpec and other master libraries are licensed content. Firms need to understand what their licenses permit when that content is used with outside AI tools.

7. Residential Callout

CSI-format project manuals are rare. Most single-family homes are built without a formal project manual. Residential "specifications" are selections sheets, notes on the plans, standard specifications for production builders, and manufacturer installation instructions.

Manufacturer instructions often carry legal weight. Building codes frequently require products to be installed according to the manufacturer's instructions, and manufacturer warranties depend on it. In residential construction, those instructions often function as the specification.

Selections are the residential spec. For custom homes and remodels, the selections process, choosing every fixture, finish, cabinet, and appliance, is how quality gets defined. AI helps compare products and manage selections, but the homeowner's choices still have to be captured accurately and tied to allowances.

It's a real skill gap for career changers. Residential professionals moving into commercial work often find specifications the least familiar part of the construction documents. Learning to read and navigate a project manual is one of the most valuable crossover skills in this curriculum.

8. Signals to Watch

- **How far automated drafting goes.** Watch whether platforms can reliably produce nearly complete project specifications, and how firms change staffing and fees in response.
- **Adoption of AI submittal review,** and whether architects accept submittals that have been pre-screened by contractors' AI tools.

- **Specifications as structured data**, and whether owners begin requiring requirement-level data alongside the traditional project manual.
 - **Professional organizations' response**, including how CSI and AIA address AI in specification practice and credentials.
 - **Disputes involving AI-drafted specifications**, which will begin defining the standard of care.
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9. Self-Check Exercise

The submittal check. Public agencies post project manuals with their bid documents. Find a project manual and pick one product-heavy section, such as door hardware, roofing, or interior paint. Then find a manufacturer's product data sheet for a product that might be submitted for it.

1. Read the specification section and list every requirement the product must meet.
2. Compare the product data sheet against your list, marking each requirement as compliant, non-compliant, or unclear.
3. Ask an AI assistant to do the same comparison, citing the spec paragraph for every item.
4. Compare the two reviews and resolve every disagreement.
5. Then ask the AI to draft a short specification section for a simple product, like a toilet accessory or interior door. Mark every requirement in the draft that you would personally be accountable for if it were a contract term.

Questions to answer afterward:

1. Where did the AI review disagree with yours, and who was right?
 2. Which requirements in the AI-drafted section would you remove or change, and why?
 3. How would you explain to a skeptical colleague the difference between specification drafting and specification responsibility?
 4. How do MasterFormat and SectionFormat make automated review easier? What would be harder without them?
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3.10 - Code Compliance, Plan Review & Permitting

1. Where This Sits

Before construction can start, a government has to agree that the building is safe to build. That agreement is the building permit, and the process that produces it is one of the most frustrating bottlenecks in American construction.

Permitting comes at the end of construction documents, when the completed set goes to the authority having jurisdiction (AHJ) for review. The rules come in layers:

- Model codes, like the International Building Code and International Residential Code, published by the International Code Council (ICC).
- NFPA codes, like the National Electrical Code.
- Referenced standards, which the codes incorporate by reference.
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- States and local governments adopt specific editions, often with their own amendments. Accessibility adds a separate layer of federal civil rights requirements.

Code compliance has two sides.

The design side. Architects and engineers are responsible for designing a code-compliant building and documenting it through code analyses and life safety plans.

The government side. Plan reviewers check the documents, the building official issues the permit, and inspectors verify the work in the field.

AI is arriving on both sides at once, and this is the phase where public funding and public accountability shape how it's adopted.

2. Who Does the Work

Architects and engineers perform code analysis and design to comply.

Code consultants handle complex occupancies, fire and life safety strategy, and performance-based design.

Permit expeditors navigate the process for applicants, especially in large cities.

Permit technicians at the building department handle intake, checking that applications are complete before they go to review.

Plan reviewers, often ICC-certified plans examiners, check documents against codes across building, fire, structural, mechanical, electrical, plumbing, and energy disciplines.

Building officials hold the legal authority to approve permits and interpret codes.

Inspectors verify construction in the field.

Applicants, from national developers to homeowners, prepare and correct submissions.

3. The Software Landscape

Government permitting systems

Every building department needs a system of record for applications, fees, reviews, and inspections. **Accela** is a long-standing leader in government permitting and licensing software. **Tyler Technologies**, one of the largest public sector software companies, sells EnerGov for community development and permitting. **Clariti** offers cloud-based permitting and has moved directly into AI. It acquired **CivCheck**, one of the most widely piloted AI plan review tools. Other systems serve smaller jurisdictions, and many departments review drawings electronically using PDF markup tools like Bluebeam.

AI plan review for governments

CivCheck, now part of Clariti, checks permit applications against local codes and requirements, flags missing information, and guides applicants through corrections. It told Stateline in 2026 that it was working with more than 20 cities, from pilots to full implementations. **Archistar**, covered in Chapter 2, offers AI prescreening to cities, including a residential zoning pilot in Austin and use in Los Angeles fire rebuilding. **Symbium**, also covered in Chapter 2, encodes rules as deterministic logic and powers instant rooftop solar permitting in Bakersfield, California.

Design-side tools

Tools for architects and engineers are more fragmented. **UpCodes** built a searchable database of adopted building codes and tools that check designs against them. The ICC offers its own digital code access. And a large, fast-growing group of startups sells AI plan-check tools that review drawings for code issues before submission.

The legal question underneath: who owns the code?

Every AI code tool depends on access to code text, and that access has been fought over in court. Model codes are written by private organizations like the ICC, which fund their work partly by selling access, but governments adopt them as law. In 2017, the ICC sued UpCodes for publishing adopted codes. In 2020, a federal district court found that UpCodes' posting of adopted codes was covered by fair use and public domain principles. In April 2026, in a separate case brought by ASTM International, the Third Circuit Court of Appeals found that UpCodes' verbatim republication of standards incorporated into binding law likely constitutes fair use. That ruling came at an early stage of the case, so the question isn't fully settled. Still, the direction of the courts matters for anyone building AI on top of building codes.

4. Where AI Is Landing

Completeness versus compliance

This chapter depends on one distinction.

A completeness check asks whether an application includes everything it's supposed to: the right drawings, forms, calculations, and information. It's a checklist problem.

A compliance check asks whether the design actually meets the code. Some compliance checks are objective, like a minimum corridor width or a maximum travel distance to an exit. Others require interpretation or discretion.

AI's clearest results so far are in completeness checking, and that turns out to matter more than it sounds. Incomplete applications drive repeated rounds of corrections, and each round adds weeks.

What the early results show

Honolulu was one of the first cities to use CivCheck. City and CivCheck officials reported that applications using it averaged 7.7 corrections, compared with 23.5 for applications that didn't. They also reported that the average time applicants spent in the process fell from 73 days to 32.5 days.

Seattle ran a pilot and measured CivCheck at 87% accuracy on completeness checks and 92% on design-compliance checks. The city saw roughly a 50% reduction in average intake review time and 35% fewer correction cycles per review. As of a June 2026 report, however, Seattle was still exploring funding and logistics and wasn't ready to adopt it fully.

The same number means different things

Here's the lesson in those accuracy figures. A tool that's right about 90% of the time is transformative as a **prescreener**. It catches most problems before a human ever looks, and the reviewer still checks everything. The same tool would be unacceptable as an **approver**, because

one in ten wrong answers about fire exits is a public safety failure. Nearly every government deploying AI in permitting has drawn exactly this line. The tool flags and guides; people decide.

Task ratings

Automated with sign-off: intake and completeness screening. Checking for missing documents and information, and steering applicants to the right permit type, is where AI is delivering measurable time savings now.

Augmented: objective code checks. Prescriptive, measurable requirements, such as occupant loads, egress widths, fixture counts, and setbacks, are increasingly checked automatically on both the design and government sides, with a reviewer confirming the results.

Augmented: design-side code research. Finding every applicable requirement across the adopted code edition, local amendments, and referenced standards is much faster with AI.

Assisted: code interpretation. Where code language is ambiguous, or a design uses an alternative method, AI can surface relevant sections and past interpretations. The building official's judgment governs.

Resistant: approval, discretion, and inspection. Issuing a permit is a legal act by a public official. Variances, appeals, discretionary reviews, and field inspections remain human work.

The applicant is half the problem

Officials in Honolulu made a point worth remembering. Applicants account for at least half of the time spent in the permitting process, through incomplete submissions, wrong permit types, and slow responses to corrections. That means AI prescreening on the applicant side can matter as much as AI inside the government. Design-side and government-side tools are converging on the same checks, which could eventually let applicants submit documents that have already passed the government's own automated screening.

Why the rulebook problem is worse here

Chapter 2 showed that local zoning isn't the whole rulebook. Building codes have the same problem at larger scale. Jurisdictions adopt different code editions on different schedules and add their own amendments. A tool checking against the wrong edition, or missing a local amendment, will give a confident wrong answer. **Which code, which edition, and which amendments** is the first question to ask of any AI code tool.

Public money is accelerating adoption

HUD is offering up to \$3 million in grants for local governments to deploy automated building code permitting systems. A federal housing law enacted in July 2026 also created an Innovation Fund with \$200 million annually for local efforts to speed permitting. The American Planning

Association has identified more than 70 state and local cases of AI being used across planning functions, including 13 for zoning and development.

5. Jobs and Process Change

What compresses. Intake work by permit technicians shrinks, as do the repeated correction cycles that consume plan reviewer time. Routine code research for architects gets faster. The routine checklist side of permit expediting compresses.

What shifts. Plan reviewers move toward complex projects, interpretation, and direct conversations with applicants. Honolulu officials described wanting to return to face-to-face meetings and the more complex parts of review. Expeditors' value shifts toward navigating difficult projects and relationships with agencies.

What grows. Building departments need staff who can configure and oversee AI tools, maintain local code data, and evaluate vendor accuracy. Code consultants' performance-based and complex-occupancy work grows as routine compliance automates.

What stays human. Legal approval authority, interpretation, discretionary decisions, field inspection, and accountability to the public.

Capacity, not replacement. Many building departments are chronically understaffed. Officials deploying these tools have generally framed them as a way to handle volume with existing staff rather than to cut positions.

6. Friction and Risk

Accountability for misses. If an AI prescreen passes an application with a life safety problem that a reviewer then misses, responsibility still rests with the people and agencies involved. Governments need clear processes that keep human review meaningful rather than a rubber stamp.

Transparency and fairness. Permitting decisions are public actions. Residents and applicants can reasonably expect to understand why an application was flagged, and agencies need tools whose results can be explained and audited.

Budgets and procurement. Pilots are one thing; permanent funding is another. Seattle's experience shows that promising results don't automatically lead to adoption.

Code data maintenance. Keeping every local amendment and adopted edition current is ongoing work, and outdated code data produces wrong results.

Access to code text. Courts are trending toward allowing adopted codes to be republished, but licensing arrangements and unresolved cases still shape what tools can include.

7. Residential Callout

Residential is where results show up first. Residential permits are reviewed under the IRC and are high-volume, repetitive, and relatively simple. That makes them the natural first target for automated prescreening. Many early government deployments focus on housing.

Some residential permits are becoming instant. Rooftop solar is the clearest example. Because systems are standardized, automated platforms can approve many residential solar permits without waiting for manual review. Symbium's work in Bakersfield is one example, and the National Renewable Energy Laboratory developed SolarAPP+ for the same purpose. ADUs are a similar opportunity, especially where cities offer pre-approved plans.

Homeowners are applicants too. Homeowners pulling their own permits for renovations or additions are the least familiar with the process, which makes AI guidance at intake especially valuable.

Energy code compliance has its own tools. Residential energy code compliance often runs through REScheck or HERS rating software such as Ekotrope and REM/Rate, and it's a frequent source of corrections.

8. Signals to Watch

- **Federal funding flowing to local permitting**, and whether pilots funded by HUD grants and the new Innovation Fund become permanent programs.
 - **Final outcomes in the UpCodes litigation**, and how code publishers adapt their business models to AI access.
 - **Pre-screened submissions.** Watch whether agencies begin accepting applications that have already passed the same automated checks the agency uses.
 - **Published accuracy data** from more jurisdictions, especially for compliance checks beyond completeness.
 - **Private plan review.** Several states, including Florida, allow licensed private providers to perform plan review and inspections. Watch whether AI tools grow fastest where the law already allows outside review.
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9. Self-Check Exercise

The completeness test. Most city building departments publish submission checklists for common permit types. Find the checklist for a commercial tenant improvement or a residential addition in a city you know, and find a publicly available plan set for a similar project. Public agencies' bid documents work well.

1. Check the plan set against the city's submission checklist and list everything missing or unclear.
2. Ask an AI assistant to perform the same completeness check using the checklist, and compare results.
3. Identify which building code edition the city has adopted and whether it has local amendments.
4. Pick five objective code requirements, such as occupant load, number of exits, corridor width, plumbing fixture count, and accessible route, and check the plan set against them.

Questions to answer afterward:

1. What did the AI catch that you missed, and what did it get wrong?
2. Would you trust its compliance checks as a prescreen? As an approval? What's the difference?
3. How would your answers change if the AI had checked against the wrong code edition?
4. If you ran a building department, how would you explain to the public what your AI tool does and doesn't decide?

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3.11 - Estimating & Quantity Takeoff

1. Where This Sits

Chapter 5 covered the owner's conceptual estimate, a budget built before real design exists. This chapter covers the estimates that turn completed construction documents into prices contractors commit to.

This is procurement, the stage where the owner obtains prices and selects a contractor. Under design-bid-build, general contractors and subcontractors price the finished documents competitively. Under construction manager at risk and design-build, estimating happens continuously during design as the price is developed with the owner.

An estimate has two parts that are easy to confuse.

Quantity takeoff is measurement: how many square feet of drywall, linear feet of pipe, cubic yards of concrete, and doors of each type the drawings show.

Estimating is pricing and judgment: what those quantities will cost to buy and install, given labor productivity, crew and equipment choices, subcontractor quotes, general conditions, schedule, risk, and market conditions.

That distinction is the key to understanding what AI is doing here.

2. Who Does the Work

General contractor estimators and preconstruction managers assemble the overall bid, perform takeoffs for self-performed work, solicit and level subcontractor bids, and build general conditions.

Subcontractor estimators price their trade in detail. At small subcontractors, the estimator is often the owner, who also runs projects and sells work.

Heavy civil estimators price roads, utilities, and earthwork, where equipment productivity and production rates dominate.

Suppliers and distributors often perform takeoffs for contractors as a sales service, especially for lumber, electrical materials, and building products.

Owner-side cost consultants check contractor pricing, especially on negotiated work.

3. The Software Landscape

Estimating software is fragmented for good reasons. Every trade measures and prices differently, general contractors and subcontractors need different tools, and heavy civil work runs on different logic than buildings.

Takeoff

Bluebeam Revu, owned by Nemetschek, is widely used for takeoff simply because nearly every estimator already has it for reviewing drawings. Its Bluebeam Max plan adds AI features that automate repetitive markups for takeoffs. **On-Screen Takeoff** and **PlanSwift**, both owned by ConstructConnect, are long-established dedicated takeoff tools. **STACK** is a cloud-based takeoff and estimating platform with AI features that detect doors, windows, rooms, and walls.

Autodesk Takeoff works from both 2D drawings and models within Autodesk Construction Cloud, and Autodesk also owns the estimating platform **ProEst**.

Estimating

Trimble WinEst and **Sage Estimating** are long-standing commercial estimating systems. **DESTINI Estimator**, from Beck Technology, supports model-based estimating. **Trimble Accubid** is a standard for electrical contractors. **HCSS HeavyBid** is the leading estimating system in heavy civil construction, and Nemetschek's agreement to acquire HCSS for about \$2.4 billion was one of the largest AEC software deals of 2026. As everywhere in Part 3, **Excel** remains the default for a large share of estimating, especially at smaller firms.

AI-native takeoff

A wave of startups is built around AI that reads drawings directly.

Togal.AI, founded in Miami in 2019 with seed funding from Coastal Construction, uses computer vision to detect, measure, and label spaces and features on architectural plans. It focuses on takeoff and exports quantities for pricing elsewhere.

Kreo offers cloud takeoff with AI detection at a low price point.

Beam AI takes a hybrid approach, combining AI with human expert review to deliver finished takeoffs as a service.

Bobyard, founded in 2023, raised a \$35 million Series A led by 8VC in December 2025. It builds AI models trained for specific trades and says it automates up to 70% of takeoff while keeping the estimator in a verification loop. It launched a commercial plumbing product in July 2026.

Countfire focuses on automated symbol counting for trades like fire alarm and electrical.

4. Where AI Is Landing

Takeoff is not estimating

The most important thing to understand is that AI is automating the measurement part of estimating much faster than the judgment part. Industry comparisons consistently point out that even the best AI takeoff tools leave estimators supervising the automation, checking results, and building scope and pricing themselves. Takeoff hours can collapse while the total hours spent producing a bid fall far less.

Task ratings

Automated with sign-off: routine takeoff on clean drawings. Areas, lengths, and counts from well-drawn architectural plans, such as rooms, flooring, walls, doors, and fixtures, are now largely automated. An estimator verifies the results rather than tracing every line.

Augmented: trade-specific takeoff on complex drawings. Mechanical, electrical, and plumbing drawings are dense and symbol-heavy. Trade-specific AI models are making progress, but review is essential, and accuracy varies with drawing quality.

Augmented: model-based quantities. When a reliable BIM model exists, quantities come directly from it. The limitation is that contractors often receive models that aren't built accurately enough to price from.

Augmented: scope review. AI can read drawings and specifications together to flag scope that's easy to miss, such as requirements buried in specifications, conflicts between documents, and work assigned ambiguously between trades.

Assisted: pricing and productivity. Historical cost data and AI can suggest unit costs and production rates. But material prices move, labor markets differ by region, and every project has conditions that historical averages don't capture.

Resistant: bid strategy and relationships. Deciding which projects to pursue, how to handle risk, what markup the market will bear, and which subcontractors to trust depends on judgment and relationships. So does defending the number to a project executive or owner.

The capacity problem AI is solving

Estimating has a harsh economic reality. On competitive work, contractors lose most of the bids they price, so most estimating effort produces nothing. Meanwhile, experienced estimators are hard to find. One 2026 vendor survey of U.S. industrial estimators reported that bid invitations had grown about 24% over five years while the number of estimators grew only about 4%. It's vendor-reported, but it reflects what contractors commonly describe. It's no surprise that

ServiceTitan's 2026 contractor survey found cost estimating and bid management to be the most common AI uses.

Faster takeoff gives firms two strategic options. They can **bid more work** with the same team, or they can **bid more selectively**, spending the time saved on scope review, pricing strategy, and risk analysis for the projects they most want. The second path is less discussed and often more valuable.

The winner's curse

There's a specific risk in competitive bidding that makes estimating accuracy unusually important. In a hard-bid market, the lowest price usually wins. An estimate that misses quantities or scope tends to come in low, which makes it more likely to win. Estimating errors don't average out. They select themselves into contracts, where they become the contractor's losses.

That's why verification matters so much in this phase. An AI takeoff that's slightly low on a few items doesn't just produce a slightly wrong number. It can produce a winning bid on a job that loses money.

5. Jobs and Process Change

What compresses. Manual takeoff shrinks dramatically. That work has traditionally been assigned to junior estimators and has consumed much of their time. Suppliers' takeoff services face pressure as contractors produce quantities themselves.

What shifts. Estimators move toward scope analysis, pricing strategy, subcontractor coverage, and risk. Preconstruction teams can spend more time with owners and designers on value engineering and constructability, especially under collaborative delivery methods.

What stays human. Bid strategy, risk judgment, means and methods decisions, subcontractor relationships, and responsibility for the number.

The learning path. Takeoff has traditionally taught junior estimators how buildings go together, one measured assembly at a time. Firms that automate it will need to teach that understanding deliberately.

6. Friction and Risk

Verification discipline. AI takeoff saves time only if verification is efficient, and it's safe only if verification actually happens. The winner's curse punishes skipped checks.

Drawing quality. AI performs best on clean, consistent, vector-based drawings. Scanned sets, inconsistent symbols, and poorly organized documents reduce accuracy.

Takeoff-to-estimate handoffs. When quantities are exported from one tool and re-entered into another, errors creep in. Platforms combining takeoff and estimating are responding to exactly this problem.

Vendor accuracy claims. Nearly all accuracy and time-savings figures in this market are reported by vendors. Firms should test tools on their own completed projects, where the real quantities are known.

Price volatility. Accurate quantities don't protect a bid from swings in material and labor costs. Escalation and procurement risk remain estimating problems AI doesn't solve.

7. Residential Callout

Much residential estimating doesn't start with a plan set. Custom builders work from plans, but many remodelers and residential trades estimate from a site visit, photos, and conversations with the homeowner. Plan-based AI takeoff fits only part of the residential market.

Photo and aerial measurement is the mature residential AI. Hover generates exterior measurements for siding, windows, and painting from smartphone photos, and EagleView produces roof measurements from aerial imagery. Both are established tools for roofing and exterior contractors.

Supplier takeoffs are the norm. Lumberyards and building material suppliers commonly provide free framing and material takeoffs to builders who buy from them.

Builders estimate inside their management platforms. Custom builders and remodelers often build estimates in Buildertrend, JobTread, or Excel, typically as cost per square foot plus allowances refined into line items.

Production builders manage cost differently. A production builder doesn't estimate each home from scratch. It maintains budgets per plan and option, negotiates trade pricing by community, and controls costs through purchasing systems and variance purchase orders.

8. Signals to Watch

- **Nemetschek's combination of Bluebeam and HCSS,** and whether it connects PDF takeoff with heavy civil estimating in one workflow.

- **Trade-specific AI models** expanding across mechanical, electrical, plumbing, and fire protection.
 - **Consolidation of AI takeoff startups** into larger estimating and construction platforms.
 - **Independent accuracy benchmarks** replacing vendor-reported figures.
 - **Automated scope-gap detection**, which would address more of the bid's real risk than faster measurement.
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9. Self-Check Exercise

Measure, then price. Download a public bid set for a small commercial project, such as a library, fire station, or school renovation, from a public agency's procurement website.

1. For one floor, manually take off the flooring area by finish type and count the doors by type. Record your time.
2. Use a trial AI takeoff tool, or an AI assistant with the floor plan images, to produce the same quantities. Compare the results.
3. List scope that the flooring and door work requires but that neither takeoff captured directly, such as floor preparation, transitions, base, hardware, and frames. Check the specifications for requirements you'd otherwise miss.
4. Price the flooring using a published cost reference or supplier pricing. Then reprice it assuming your quantities were 5% low.

Questions to answer afterward:

1. How much time did the AI save on measurement, and how much work remained after the takeoff was done?
 2. What scope did the specifications add that the drawings alone didn't show?
 3. How does the winner's curse change how you'd verify AI takeoffs on a hard-bid job?
 4. If your firm's estimators suddenly had 30% more time, would you bid more projects or bid more carefully? Why?
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3.12 - Bidding, Prequalification & Contracts

1. Where This Sits

Chapter 11 covered how contractors price the work. This chapter covers how the work gets bought: how bids are solicited and compared, how contractors and subcontractors are vetted, and how the contracts that allocate risk get written, signed, and followed.

This is still procurement, and it happens in two layers.

The owner buys from a contractor. Through competitive bidding or negotiation, the owner selects a general contractor, construction manager, or design-builder.

The contractor buys from subcontractors. The general contractor breaks the work into bid packages, solicits subcontractor prices, compares them, and awards subcontracts. This is often called buyout.

Both layers end in contracts. Most U.S. projects use standard forms from a few sources: **AIA Contract Documents** from the American Institute of Architects, **ConsensusDocs** from a coalition of industry associations, **EJCDC** documents common in engineering and infrastructure, and **DBIA** forms for design-build. Those contracts decide who pays when something goes wrong, which makes this phase the place where most of a project's risk is priced, assigned, and accepted.

2. Who Does the Work

Owner procurement and legal teams write bid requirements, evaluate proposals, and negotiate owner-contractor agreements. Public owners follow procurement laws designed to ensure fairness.

General contractor preconstruction teams solicit subcontractor bids, track coverage for every trade, compare proposals, and recommend awards.

Subcontractors find bid opportunities, submit pricing and qualifications, and negotiate subcontract terms.

Risk managers prequalify subcontractors by reviewing financial strength, safety records, insurance, and bonding capacity.

Construction attorneys draft and negotiate contracts and handle disputes when they arise.

Sureties and insurers back performance and payment bonds and insure project risks.

Project managers inherit the contract after signing and live with its obligations for the rest of the project.

3. The Software Landscape

Bid management and subcontractor networks

BuildingConnected, owned by Autodesk, is widely regarded as the largest subcontractor bidding network in North America. General contractors use it to send invitations to bid, track responses, and compare proposals side by side. Subcontractors can join free to receive invitations, which is how the network grows. **Procore Bid Management** serves general contractors already running projects in Procore.

ConstructConnect combines project lead data with bidding tools and owns the SmartBid bid management platform. **Dodge Construction Network** is a long-standing source of project lead intelligence, tracking projects from planning through bidding. **PlanHub** connects general contractors with subcontractors and suppliers. Public owners typically run bids through their own e-procurement portals.

Prequalification and compliance

TradeTapp, part of BuildingConnected, is a leading subcontractor prequalification tool. It collects financial statements, insurance certificates, safety records, and references. **Highwire** and **ISNetworld** are widely used, especially by owners and large contractors in industrial and high-risk work, to track contractor safety performance and compliance requirements.

Contract review

Document Crunch built an AI platform that reads construction contracts and flags risk: payment terms, notice requirements, specification compliance, and problematic provisions. By the time **Trimble** agreed to acquire it in April 2026, it had been used on more than 10,000 projects by more than 500 general contractors and construction managers, including Balfour Beatty and DPR Construction. Trimble plans to make it the risk intelligence layer across its Construction One platform, pushing contract obligations directly into project management and accounting workflows.

AI bid leveling

Bid leveling tools, built into bid management platforms or sold separately, read subcontractor proposals, normalize their scope, and flag exclusions so general contractors can compare bids accurately.

4. Where AI Is Landing

The lowest number isn't always the lowest cost

Subcontractor proposals rarely arrive in a comparable form. One drywall bid includes framing, another excludes it. One mechanical bid excludes controls wiring, another includes it but qualifies its schedule. A bid that looks lowest may simply leave out work that someone else will have to pay for. **Bid leveling** is the process of adjusting every proposal to the same scope so the numbers can be compared honestly. It's tedious, detail-heavy work done under deadline pressure on bid day, which makes it a strong fit for AI.

Task ratings

Automated with sign-off: bid logistics. Sending invitations, tracking which subcontractors have responded, following up, and confirming coverage for every trade are largely automated in modern bid platforms.

Automated with sign-off: proposal extraction. Pulling prices, inclusions, exclusions, and qualifications out of subcontractor PDFs and into a leveling sheet is increasingly handled by AI, with the estimator reviewing the result.

Augmented: scope-gap detection. AI can compare bid packages and proposals against the drawings and specifications to find work that falls between trades, such as blocking, firestopping, or equipment connections that every subcontractor assumes someone else is providing.

Augmented: prequalification analysis. Reviewing a subcontractor's financial statements, safety history, and insurance against the risk of a specific project goes faster with AI summarizing and flagging concerns.

Augmented: contract review. AI flags provisions that shift risk, such as indemnity, pay-when-paid terms, notice requirements, and limits on delay damages, and compares a proposed contract against a firm's preferred positions.

Assisted: selection and negotiation strategy. AI can surface past performance data and suggest negotiating positions. Deciding which subcontractor to trust with a critical scope, and what terms to push back on, stays with experienced people.

Resistant: relationships, negotiation, and award. A subcontractor who takes the call on a problem job, a negotiated compromise on a risky clause, and a public owner's award decision all depend on trust, authority, and accountability.

Contract knowledge on the project manager's desk

The most consequential change in this phase may happen after the contract is signed. Traditionally, an attorney reviewed a contract at signing, and then it went into a file. Project managers worked from memory and habit. Many construction disputes turn on procedure rather than merit: whether a party gave written notice of a delay or change within the contract's deadline. A contractor with a valid claim can lose it by missing a notice window.

AI contract tools change that by turning a contract into a working set of obligations: notice periods, payment conditions, and required documentation, surfaced to the project manager at the moment they matter. That's why Trimble plans to embed contract intelligence in project management and accounting workflows rather than keep it in a legal review tool. **The contract stops being a document someone read once and becomes a set of rules the project runs on.**

Contracts are starting to address AI itself

The contracts themselves are catching up slowly. According to industry analysis in early 2026, the AIA's standard agreements don't yet contain AI-specific provisions, and design firms are adding their own language on AI disclosure, data ownership, restrictions on training AI with client data, and preservation of the standard of care. At the same time, some insurers have introduced broad AI exclusions in professional liability policies, which means a firm's use of AI could affect its coverage. Anyone negotiating contracts in 2026 should read both the contract and the insurance policy with AI in mind. Chapter 20 covers liability in depth.

5. Jobs and Process Change

What compresses. Bid coordination work, such as sending invitations, chasing responses, and assembling leveling sheets, shrinks sharply. Routine first-pass contract review compresses for attorneys and risk managers.

What shifts. Estimators and preconstruction managers spend less time assembling comparisons and more time analyzing scope and subcontractor risk. Attorneys focus on negotiation, unusual provisions, and disputes. Project managers become more accountable for contract compliance because the obligations are now in front of them.

What stays human. Negotiation, subcontractor relationships, risk acceptance, legal judgment, and the decision to award.

6. Friction and Risk

Network lock-in. Bid networks gain power as more subcontractors join. Subcontractors often must maintain profiles on whatever platforms their general contractor customers use, and switching platforms means rebuilding relationships and data.

Missed contract context. Construction contracts are assembled from a base form, modifications, exhibits, riders, and incorporated documents, with rules for which one controls. AI tools that read only the base agreement, or miss a rider, can give a clean and wrong answer.

Confidentiality of pricing. Subcontractor prices are sensitive. Tools and data practices that expose pricing, or make it easier to shop one subcontractor's number to another, raise ethical and relationship concerns.

Fairness in public procurement. Public owners using AI to evaluate proposals face questions about transparency, consistency, and records that can be challenged in a bid protest.

Insurance gaps. Broad AI exclusions in liability policies can leave firms uninsured for problems connected to AI use, even when AI was embedded in a vendor's software rather than chosen deliberately.

7. Residential Callout

Bidding is informal and relationship-driven. Custom builders and remodelers typically get prices from a small group of trusted trade partners by phone, text, and email. Formal bid platforms are rare, and the most valuable asset is a reliable crew that shows up.

Production builders negotiate by plan and community. A production builder negotiates trade pricing for each plan in a community, then issues purchase orders house by house through purchasing and supply systems like Hyphen SupplyPro. Changes are handled through variance purchase orders.

Consumer contracts carry extra requirements. Contracts with homeowners are often governed by state consumer protection laws that can require specific disclosures, cancellation rights, warranty terms, and lien notices. Builder association contract forms help, but requirements vary by state. AI-drafted homeowner contracts that miss a state requirement can be unenforceable or expose the builder to penalties.

Subcontract paperwork is often thin. Many residential builders work with trades under minimal written agreements. The practical risk management is tracking insurance certificates and lien waivers, and even that is often manual.

8. Signals to Watch

- **Document Crunch inside Trimble**, and whether contract intelligence remains available to contractors who don't run Trimble software.
 - **AI-specific provisions** in future editions of AIA, ConsensusDocs, and EJCDC documents.
 - **Insurance market treatment of AI**, including whether broad AI exclusions spread or narrow.
 - **Subcontractor performance data** becoming a standard input to bid leveling and award decisions.
 - **Public procurement rules** for the use of AI in evaluating bids and proposals.
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9. Self-Check Exercise

The notice calendar. The AIA and ConsensusDocs publish sample versions of many standard contract forms. Find a sample owner-contractor agreement and its general conditions, or a sample standard subcontract.

1. Read the document yourself and list every provision that shifts risk to the contractor, such as indemnity, payment conditions, and limits on delay damages.
2. Build a notice calendar listing every obligation that requires written notice within a set time, such as claims, changes, and concealed conditions. Include the deadline and the consequence of missing it.
3. Ask an AI assistant to do both tasks, citing the section number for each item.
4. Compare the results and resolve every difference.

Questions to answer afterward:

1. Which notice requirement would be easiest for a busy project manager to miss, and what would it cost?
 2. Did the AI miss or misstate anything that would matter in a dispute?
 3. If this contract had three riders and a modified general conditions exhibit, how would you make sure an AI review accounted for them?
 4. How would you change a general contractor's buyout process if bid leveling took one hour instead of one day?
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3.13 - Scheduling & Planning

1. Where This Sits

The contracts are signed. Now the contractor has to plan how the building will actually get built: in what sequence, with which crews, at what pace, and with what happens when something goes wrong.

This is where the project moves from paper to the jobsite, and where Part 2 of this series picks up, following a construction site from handoff through closeout. Scheduling starts before the first crew arrives and continues through the entire build. Most contracts require the contractor to submit a schedule and update it regularly. Division 01 of the specifications often spells out exactly what the schedule must include and how often it must be updated.

Most commercial construction scheduling uses the **Critical Path Method (CPM)**. A CPM schedule breaks a project into activities with durations and logical relationships: this wall can't be framed until that slab is poured. The longest chain of dependent activities is the **critical path**. A delay to any activity on it delays the whole project, while activities off the critical path have **float**, meaning slack. Contractors build a baseline schedule, update it as work progresses, and break it into short-term look-ahead schedules for the field.

A schedule plays three roles at once:

1. **A plan** for building the project.
2. **A coordination tool** for dozens of trades.
3. **A legal record**. When a project runs late, the schedule becomes the central evidence in deciding who pays.

AI affects each role differently.

2. Who Does the Work

Schedulers and project controls staff build and update CPM schedules, especially on large projects.

Superintendents run the field and plan the short-term work: what happens this week and next.

Project managers own the schedule commercially, including requests for time extensions.

Subcontractor foremen commit to durations and handoffs, especially in collaborative pull planning sessions.

Owners' schedule reviewers check contractor schedules for quality and realism.

Forensic schedulers and claims consultants reconstruct what happened on delayed projects and analyze who caused the delay.

3. The Software Landscape

CPM scheduling

Oracle Primavera P6 is the standard for large commercial, industrial, and infrastructure projects, and many owner contracts require schedules in P6 format. Oracle also offers **Primavera Cloud** and **Primavera Unifier** for owners' capital programs. In April 2026, Oracle added AI features to Unifier that automatically generate structured chronologies of project workflows, recording who did what, when, and what was decided. That kind of record matters a great deal when a project ends in a dispute.

Microsoft Project is common on mid-size projects and among firms that don't need P6's depth. **Asta Powerproject**, from Elecosoft, has a strong base in the UK and a smaller U.S. presence. **InEight Schedule** serves large capital projects with integrated planning and risk analysis. Construction management platforms like **Procore** and **Autodesk Construction Cloud** import CPM schedules and provide look-aheads and schedule visibility for project teams.

Collaborative planning

Touchplan digitizes pull planning based on the Last Planner System, a lean construction method in which trade foremen plan work backward from milestones and make commitments to each other. It focuses less on calculation and more on getting reliable commitments from the people doing the work.

AI-native scheduling

ALICE Technologies offers generative scheduling. It works from "recipes" for how building elements get constructed, simulates large numbers of possible sequences, and ranks the alternatives by time, cost, and resources. It can work directly from an existing P6 schedule without requiring a full BIM model. In April 2026, ALICE and McKinsey formalized an alliance to deploy generative scheduling on capital projects, with the partners claiming construction timelines can be reduced by up to 20%.

nPlan focuses on schedule risk. It compares a project's schedule against a database of roughly 750,000 historical schedules to forecast which activities are most likely to slip.

SmartPM analyzes schedule data to grade schedule quality, identify problems, and flag gaps between reported progress and likely actual status.

4. Where AI Is Landing

Planning versus monitoring

AI scheduling tools answer two different questions, and it's worth keeping them separate.

Generative scheduling asks, "What's the best way to build this?" It creates and compares alternative sequences and is most valuable before construction starts, when the plan can still change.

Schedule risk and analytics asks, "Where is this plan likely to fail?" It checks schedule quality, forecasts slippage from historical patterns, and monitors progress during construction.

Task ratings

Automated with sign-off: schedule quality checks. Finding open-ended activities, missing logic, unrealistic constraints, and other structural problems is rule-based work. Industry assessments like the DCMA 14-point check have long defined it, and AI tools now run it automatically.

Automated with sign-off: update narratives and reports. Drafting the monthly narrative explaining what changed, what slipped, and why is increasingly generated from schedule data.

Augmented: scenario planning. Testing alternative sequences, crew sizes, shift work, or acceleration strategies used to take days of manual rework. Generative tools produce and compare many options quickly.

Augmented: risk forecasting. Comparing a schedule against how similar activities actually performed on past projects gives a more realistic picture of risk than manually estimated ranges.

Augmented: delay analysis support. Reconstructing what happened on a delayed project means assembling daily logs, RFIs, emails, meeting minutes, and schedule updates into a chronology. AI dramatically speeds up that assembly.

Assisted: durations and logic. Historical data can suggest durations and sequences, but site access, crane locations, weather, labor availability, and a project's particular conditions require people who know the job.

Resistant: commitment and entitlement. A trade foreman's commitment to finish on Thursday so the next trade can start on Friday is a promise between people. The superintendent's daily decisions are made on site. Whether a contractor is entitled to more time

is decided under the contract, and in disputes, by expert testimony and ultimately arbitrators or courts.

A schedule is a promise, not a calculation

CPM math has been computerized for decades. The hard part of scheduling has never been calculating the critical path. It's getting durations that reflect reality and trades that actually commit to them. Many schedules are built to satisfy contract requirements and then drift away from what's happening in the field.

That creates a practical limit on AI. A generative engine can produce a better sequence, and a risk model can forecast slippage, but neither helps if the schedule isn't updated with accurate progress or the field doesn't follow it. Data quality is the constraint. That's also why the connection between scheduling and field data capture, covered in Chapter 16, matters so much. **The most valuable AI scheduling improvement may be a schedule that knows what actually happened today.**

AI changes the economics of delay disputes

When a project finishes late, both sides reconstruct history to argue who caused the delay. That reconstruction has traditionally been slow and expensive, requiring consultants to read thousands of documents. AI can assemble chronologies in a fraction of the time, which lowers the cost of analyzing delay claims. But when both sides have that capability, the advantage shifts to whoever kept better records during the project. Contemporaneous daily logs, timely notices, and updated schedules become even more valuable.

Read performance claims carefully

Nearly all published results for AI scheduling, including double-digit reductions in project duration, come from vendors and their partners. Some are likely real on the projects measured. But a shorter optimized schedule on paper only becomes a shorter project if the plan is executed, and independent comparisons across many projects are still scarce.

5. Jobs and Process Change

What compresses. Mechanical schedule update work, quality checking, and report writing shrink. Assembling documents for delay analysis compresses dramatically.

What shifts. Schedulers move from maintaining schedules to planning strategy: testing scenarios, advising on recovery options, and quantifying risk. Superintendents get better look-aheads and earlier warning of problems. Forensic schedulers spend less time assembling records and more time on analysis and expert judgment.

What grows. Project controls roles that combine scheduling, cost, and data analysis grow, especially on large capital projects.

What stays human. Field leadership, trade commitments, sequencing judgment on site, contract entitlement arguments, and expert testimony.

6. Friction and Risk

Schedule fiction. A schedule that isn't updated with real progress produces confident wrong forecasts. AI amplifies whatever data it's given.

Field acceptance. Superintendents are rightly skeptical of sequences generated by software that has never seen the site. Optimized plans that ignore crane reach, laydown space, or crew availability won't be followed.

Contract format requirements. Owners often require P6 schedules in specific formats with specific update procedures. AI tools have to work within those requirements rather than around them.

Records in disputes. Scenario files, AI-generated forecasts, and analyses created during a project may become evidence if a dispute arises. Firms should consider how those records are created and kept.

Unverified performance claims. Duration reductions reported by vendors are difficult to verify independently. Pilot tools on real projects and compare planned sequences with actual outcomes.

7. Residential Callout

Custom builders schedule in their management platforms. Custom builders and remodelers typically schedule in Buildertrend or JobTread using Gantt views, linked to homeowner communication and trade notifications. Formal CPM analysis is rare.

Production builders run repeatable schedules at scale. A production builder may build the same plan hundreds of times, so it uses standardized schedule templates, often managed in systems like Hyphen BuildPro. The automation targets are trade dispatch and rebalancing: when rain or a failed inspection pushes one house back, dozens of trade appointments across a community have to move.

Inspections and trade availability drive the schedule. Residential schedules are shaped less by complex logic and more by when inspectors and reliable trades are available. AI can optimize a sequence, but it can't create an extra framing crew.

Homeowners expect updates. Keeping homeowners informed about schedule changes is a large part of a residential project manager's workload, and AI-drafted client updates are already shipping in residential platforms.

8. Signals to Watch

- **The ALICE and McKinsey alliance** on capital projects, and whether independent results confirm duration reductions.
 - **AI layered on P6** versus AI built into Oracle's own products, and whether P6's position as the contractual standard holds.
 - **Field data feeding schedules automatically**, connecting progress tracking from reality capture to schedule updates.
 - **Owners requiring schedule quality metrics** as part of contract compliance.
 - **How arbitrators and courts treat AI-assembled delay analyses.**
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9. Self-Check Exercise

Build, break, and explain. Choose a simple project, such as a 10,000-square-foot single-story retail shell building.

1. Build a CPM schedule of about 25 activities from site work through turnover, using a spreadsheet or a free scheduling tool such as ProjectLibre. Include durations and logic.
2. Ask an AI assistant to generate a schedule for the same project, then compare the two. Note differences in sequence, logic, and durations.
3. Check both schedules for open-ended activities, missing logic, and unrealistic constraints.
4. Introduce a delay: structural steel delivery slips three weeks. Determine the impact on completion in both schedules.
5. List every record you would need to prove to an owner that the delay entitled you to a time extension.

Questions to answer afterward:

1. Where did the AI schedule's logic or durations differ from yours, and whose was more realistic?
 2. Did the steel delay hit the critical path in both schedules? Why or why not?
 3. Which records on your list would most likely be missing on a real project, and how could better daily documentation fix that?
 4. How would you get a skeptical superintendent to trust an AI-generated sequence?
-

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3.14 - Project Management & Contract Administration

1. Where This Sits

Construction has started. For the next year or more, the project runs on a steady stream of questions, documents, decisions, and records. A contractor finds a conflict in the drawings and asks the architect about it. A subcontractor submits product data for approval. The owner requests a change. The superintendent records what happened on site today. Every one of those exchanges has a defined process, a deadline, and consequences.

This is contract administration: the formal processes that govern how the project team communicates during construction. Part 2 follows these processes on a working jobsite. The core ones are:

- **Requests for information (RFIs):** formal questions about the documents.
- **Submittals:** shop drawings, product data, and samples the contractor provides to show compliance.
- **Change management:** tracking potential changes from first notice through approved change orders.
- **Meeting minutes and daily logs.**
- **Document control:** making sure everyone builds from the current drawings.

This is the administrative core of construction, and it's the phase where AI agents are arriving fastest. It's also where one of the most important workforce questions in this curriculum comes into focus.

2. Who Does the Work

Project managers at the general contractor own the contract, the budget, the owner relationship, and change management.

Project engineers handle much of the document flow: writing and tracking RFIs, logging and routing submittals, preparing meeting minutes, and maintaining records. It's typically an early-career role.

Superintendents and assistant superintendents run the field and keep daily logs.

Architects' and engineers' construction administration staff answer RFIs, review submittals, and evaluate proposed changes.

Subcontractor project managers prepare submittals, raise questions, and price changes for their scopes.

Owner's representatives track progress, approve changes, and protect the owner's interests.

3. The Software Landscape

Procore is the leading construction project management platform for commercial general contractors in the U.S. It connects owners, designers, contractors, and subcontractors in one system for RFIs, submittals, changes, logs, and documents.

Autodesk Construction Cloud, through Autodesk Build, is Procore's main competitor, with particular strength where teams already work in Autodesk's design tools. **Oracle Aconex** is used on very large projects, often at the owner's requirement, because of its strict document control and audit trail across many organizations. **Trimble** offers **ProjectSight** and Viewpoint's project tools, tied to its construction ERP. **CMiC** combines project management with its enterprise accounting system. **Fieldwire**, owned by Hilti, focuses on field task management and is popular with subcontractors and superintendents.

The agent push

In May 2026, Procore introduced AI agents it describes as "digital coworkers," built on Datagrid, the AI company it acquired in January 2026. The agents include:

- **Deep Search:** searches specifications, drawings, and RFIs and links back to source files.
- **Submittal Reviewer:** checks submittals against specifications and flags discrepancies.
- **RFI Agent:** checks RFIs for completeness and clarity and attaches relevant documents.
- **Daily Log Agent:** drafts daily logs from photos, emails, and voice notes for review.
- **Contract Review:** reviews contract provisions.

Procore also added capabilities that let agents take actions inside the platform and respond automatically to events like a new submittal. The agents rolled out as a limited-availability paid offering beginning in April 2026, priced on credit consumption.

Trunk Tools is the most prominent AI-native company in this space. Its platform lets teams ask natural-language questions across project documents and runs agents for tasks like submittal review. It raised a \$40 million Series B in July 2025. Gilbane Building Company then selected it for an enterprise rollout across more than 200 projects, reporting that submittal cycle times fell by nearly half on early projects.

4. Where AI Is Landing

Automated with sign-off: daily logs and meeting minutes. Drafting a daily log from photos, voice notes, and messages, or turning a meeting recording into minutes and action items, is now routine. A person reviews and finalizes the record.

Automated with sign-off: finding information. Answering "what does the spec say about this?" or "has this already been asked in an RFI?" across thousands of documents used to take a project engineer real time. Search agents now answer in seconds with links to sources.

Augmented: RFI preparation. AI checks whether a question has already been answered, attaches the relevant drawings and specifications, and sharpens the question so it gets a usable answer the first time.

Augmented: submittal pre-review. As Chapter 9 described, AI compares submittals against specification requirements and flags likely non-compliance before the package goes to the design team. One Procore customer reported running 10 submittal reviews in an hour with an agent, saving about 12 hours of work. That's a vendor-published example, but it shows the scale of the change.

Augmented: identifying potential changes. AI can scan RFIs, field notes, and revised drawings for items that likely affect cost or schedule, so potential change events get captured before notice deadlines pass.

Assisted: pricing and negotiating changes. AI can assemble backup and compare pricing to history, but negotiating a change order is a commercial conversation with real stakes.

Resistant: professional responses and decisions. The architect's answer to an RFI is a professional interpretation of the design. Submittal approval carries responsibility. Deciding what counts as a change, and how to resolve it with the owner, depends on judgment and relationships.

AI on both sides of the RFI

A new dynamic is emerging. When contractors use AI to draft RFIs and architects use AI to research and draft responses, parts of the exchange start happening between machines. That can be efficient. It also creates two risks. When writing an RFI costs almost nothing, RFI volume can rise, burying design teams in questions. And an AI-drafted answer that nobody meaningfully reviewed is still the design professional's answer. **Faster documents don't change who is accountable for them.**

The record still has to be true

As Chapter 13 explained, daily logs, meeting minutes, and RFIs become evidence when disputes arise. An AI-drafted daily log is useful only if it accurately reflects what happened on site. Superintendents who approve AI drafts without reading them create records that can undermine their own company's position later.

5. The Apprenticeship Problem

This chapter deserves a longer look at workforce change, because the project engineer role is where the question is sharpest.

The project engineer job has never been only administrative. It's how the construction industry trains its future project managers and executives. Writing RFIs teaches people to read drawings closely enough to find the conflicts. Processing submittals teaches materials, products, and specifications. Keeping logs and running meetings teaches how a job actually runs and how problems unfold. A project manager's judgment is built on years of that work.

Now the agents described in this chapter are designed to do exactly those tasks. The short-term economics are compelling: fewer hours on document handling and more projects per person. The long-term risk is less visible. If early-career staff never do the work, where does the next generation of project managers get its judgment?

"They'll review what the AI produces" is a partial answer at best. Reviewing a submittal summary isn't the same as learning why a product doesn't comply. People who haven't done the work often can't tell when the output is wrong.

Firms that take this seriously are likely to try several approaches:

- **Redesign the project engineer role** around deliberate learning rather than volume.
- **Rotate early-career staff** through field work and cost responsibilities sooner.
- **Occasionally require manual work** on selected tasks for training purposes.
- **Pair new staff with experienced mentors** who explain what the tools are doing.

The firms that solve this will have a talent advantage a decade from now. The firms that simply cut entry-level hiring may find they've automated away their own leadership pipeline.

6. Jobs and Process Change

What compresses. Document handling, logging, searching, routing, and minute-taking shrink for project engineers and assistant superintendents. First-pass submittal review compresses for both contractors and design teams.

What shifts. Project engineers move toward reviewing agent output, managing exceptions, and taking on field and cost responsibilities earlier. Project managers spend more time on commercial management and relationships. Architects' construction administration staff can concentrate on substantive design questions.

What grows. General contractors are building construction technology teams to configure agents, manage data and permissions, and train project teams.

What stays human. Professional interpretation, change negotiation, field leadership, relationships across the project team, and accountability for the record.

7. Friction and Risk

Pricing and access. AI agents are being sold as premium, usage-based add-ons. Smaller contractors and subcontractors may have much less access to them than large general contractors.

Multi-party permissions. Project platforms hold information from many companies. Agents that search and act across that data have to respect who is allowed to see and change what.

Accuracy and over-reliance. An agent that's right most of the time can make it easy to stop checking. Submittal errors discovered after installation are expensive.

Volume inflation. When drafting is effortless, the number of RFIs, notices, and submittal comments can grow faster than anyone's capacity to answer them.

Multiple platforms per project. When the owner, designer, and contractor use different systems, agents may only see part of the picture.

8. Residential Callout

Buildertrend leads custom and remodel. Buildertrend, which absorbed CoConstruct, is the leading project management platform for custom builders and remodelers. JobTread is a fast-growing challenger, and smaller firms use tools like Houzz Pro.

The administration is homeowner-facing. Residential contract administration is less about RFIs and formal submittals and more about selections, change orders, allowances, and keeping the homeowner informed. Client portals and regular updates are central, and AI-drafted homeowner updates are already shipping in residential platforms.

Changes are the biggest risk. Informal verbal changes made on site, without written change orders, are one of the most common sources of residential disputes. AI tools that turn a conversation or text message into a documented change order request address a real problem.

Production builders manage quality at scale. A production builder's version of contract administration includes standardized quality inspections, purchase order management, and handling trade issues across many homes at once.

Early-career training looks different. Residential project managers often learn on site as assistant superintendents or through the trades, so the apprenticeship problem shows up less in document work and more in field supervision.

9. Signals to Watch

- **General availability and pricing of platform agents**, including how many contractors adopt them beyond early customers.
 - **Independent measurement of agent accuracy** on submittals and RFIs.
 - **How architects respond** to AI-generated RFIs and pre-reviewed submittals, including any contract language addressing them.
 - **Changes in project engineer hiring and training** at large general contractors.
 - **Agents that work across platforms**, reaching information held in different companies' systems on the same project.
-

10. Self-Check Exercise

The RFI round trip. Download drawings and specifications for a public project from an agency procurement website.

1. Find a genuine conflict or gap between the drawings and specifications, or between two drawing sheets.
2. Write an RFI yourself: state the question, cite the documents, and propose a solution.
3. Ask an AI assistant to draft an RFI for the same issue, then compare the two.
4. Switch roles. As the architect, write a response to the better RFI and note what you would need to verify before sending it.
5. Write a one-page plan for how you would develop a new project engineer's judgment over their first year if AI agents handled most RFI drafting and submittal review.

Questions to answer afterward:

1. What did writing the RFI yourself teach you about the documents that reviewing the AI's draft wouldn't have?
2. As the architect, what would make you trust or distrust an AI-drafted RFI?
3. How would you prevent RFI volume from overwhelming a design team when drafting becomes effortless?
4. What's the one task you'd insist a new project engineer do manually, and why?

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3.15 - Construction Financials

1. Where This Sits

Every construction project is also a financial machine. Money flows down a chain, from a lender to the owner, the general contractor, subcontractors, and their suppliers. Paperwork flows back up: applications for payment, invoices, change orders, lien waivers, and compliance documents. When the paperwork is late or wrong, the money stops.

The contract defines how money moves. Part 2 covers how these procedures play out on a real project. The core pieces are:

- **The schedule of values**, which breaks the contract sum into billable line items.
- **Monthly applications for payment** based on work completed.
- **Retainage**, the portion of each payment withheld until completion.
- **Lien waivers**, documents in which a party gives up its right to file a lien against the property in exchange for payment.
- **Final payment**.

Contractors also run their own internal accounting. **Job costing** tracks every cost against the budget for each project. The **work-in-progress (WIP) schedule** compares costs, billings, and estimated completion to show whether each job is overbilled or underbilled and whether its profit is holding. Contractors typically recognize revenue using percentage-of-completion methods, so cost-to-complete forecasts affect the financial statements directly.

A personal note: During my time as a software engineer, I built accounting software for the oil and gas industry, and the two fields have more in common than most people assume. Oil and gas tracks costs by well against authorized budgets, the way construction tracks costs by project. Both also deal with complicated money flows. A single well can have thousands of owners with different kinds of interests. Working interest owners share in the costs, while royalty owners receive a share of revenue without paying costs, so every expense and every dollar of revenue has to be split correctly across all of them. Construction's complexity runs in a different direction. Instead of many owners splitting one asset, money moves down a chain of separate companies, from owner to general contractor to subcontractors to suppliers, with retainage held back at each level and lien rights governed by laws that vary by state. In both industries, that complexity is where the paperwork piles up, and where AI has the most to offer.

2. Who Does the Work

Controllers and CFOs at contracting firms own financial reporting, WIP, cash flow, and relationships with sureties and banks.

Project accountants manage billing, job cost coding, and compliance for specific projects.

Accounts payable and receivable staff process invoices, pay applications, and collections.

Payroll staff handle union and certified payroll, including prevailing wage reporting on public projects.

Project managers forecast cost to complete, approve invoices, and prepare billings.

Subcontractors' billing staff, often the owner at small firms, prepare pay applications and chase payment.

Lender and owner draw administrators review each funding request against the budget, progress inspections, and lien waivers before releasing money.

3. The Software Landscape

Construction ERP and accounting

Sage is a long-standing leader in construction accounting. **Sage 300 Construction and Real Estate** is its mature, traditionally on-premises system, and **Sage Intacct Construction** is its cloud platform, often favored for multi-entity financial consolidation.

Trimble Viewpoint offers **Vista**, a deeply configurable ERP for large contractors that's especially strong in heavy civil work, equipment costing, and complex payroll. It also offers **Spectrum**, a cloud ERP for mid-size and large contractors. Vista's native AI so far includes automated invoice processing and an AI help assistant released in late 2025.

CMiC is a single-database enterprise ERP for large general contractors that combines accounting, payroll, and project management. **Foundation Software** is well regarded for mid-size contractors, especially those with union and certified payroll requirements. **Acumatica Construction Edition** serves cloud-first mid-market firms. **Procore Financials** manages budgets, commitments, and change costs on the project management side and integrates with contractors' accounting systems.

Payments and compliance

Oracle Textura is widely used by large general contractors to manage subcontractor payment applications, compliance documents, and lien waivers. **GCPay** and **Autodesk Payapps** serve the same need. **Procore Pay** routes pay applications and payments inside Procore, and **Levelset**, also owned by Procore, focuses on lien rights: preliminary notices, waivers, and deadlines that vary by state. **Siteline** helps subcontractors build and submit pay applications.

Construction lending and draws

Built is a leading platform for construction lenders, owners, and contractors to manage draw requests, inspections, budgets, and lien waivers. **Rabbit** serves developers and lenders managing draws and project budgets.

4. Where AI Is Landing

The money is usually waiting on paperwork

Billd's 2026 National Subcontractor Market Report, based on a survey of more than 600 industry respondents, found that subcontractors reported waiting an average of 51 days for payment. General contractors estimated they paid in 35. Built, which sees transaction data from its own platform, reports an average closer to 57 days and argues that most of that wait is document friction rather than contractors holding cash. Neither source is neutral, but they point to the same conclusion: much of construction's payment delay comes from documents that are missing, wrong, or waiting for someone to process them. That's squarely the kind of problem AI is good at.

Task ratings

Automated with sign-off: invoice processing. Capturing invoice data, assigning job and cost codes, and matching invoices to purchase orders and subcontracts is increasingly automated. Accounting staff handle exceptions rather than keying every invoice.

Automated with sign-off: lien waiver and compliance tracking. AI extracts waiver type, amount, and dates from incoming documents, matches waivers to payments, flags missing waivers from lower-tier subcontractors before a draw closes, and tracks expiring insurance certificates.

Automated with sign-off: pay application assembly. Building a draft pay application from the schedule of values, approved changes, and reported progress is largely automatable, with the project manager confirming the percentages.

Augmented: cost-to-complete forecasting. AI can combine committed costs, pending change events, spending trends, labor productivity, and schedule progress to forecast where a job will finish financially.

Augmented: WIP preparation. Assembling the WIP schedule from job cost actuals, approved changes, and cost-to-complete estimates consumes significant controller time every month. AI speeds up the assembly and highlights jobs whose numbers moved unexpectedly.

Augmented: anomaly and fraud detection. AI flags duplicate invoices, billing that runs ahead of physical progress, unusual pricing, and suspicious changes to a vendor's bank account details. That last one matters more as AI-enabled impersonation fraud grows, as Chapter 4 described.

Assisted: claims and change valuation. Quantifying the cost impact of disruptions and changes requires judgment about causation and contract terms.

Resistant: authority and relationships. Approving payments, deciding whether to file a lien, negotiating with a surety or bank, and signing financial statements remain human responsibilities.

The forecast problem is partly human

Cost-to-complete forecasting has a problem that isn't technical. Project managers are often optimistic about how their jobs will finish, sometimes deliberately. Reporting a margin decline early invites scrutiny, so problems tend to surface late. The industry calls it "profit fade," when a job's expected profit shrinks quarter after quarter until the final number is far below the original.

An AI forecast built from spending patterns and progress data doesn't share those incentives. It may flag a fading job months before the project manager reports it. That's valuable to a CFO, a surety, or a bank. It also changes office politics, because it makes optimistic forecasts harder to sustain. Firms adopting AI forecasting should expect some resistance from exactly the people whose forecasts it checks.

Garbage in, still garbage out

Financial AI depends on project data. A change event that was never entered, a subcontract commitment recorded late, or progress reported inaccurately will produce a confidently wrong forecast. That's why the connection between project management platforms (Chapter 14), contract intelligence (Chapter 12), and accounting systems matters so much. Trimble's plan to push contract obligations from Document Crunch into both project management and ERP workflows is an example of vendors trying to close that gap.

5. Jobs and Process Change

What compresses. Invoice data entry, coding, waiver chasing, compliance tracking, and pay application assembly all shrink. Controllers spend fewer hours assembling WIP schedules.

What shifts. Accounting staff move from processing to exception handling and analysis. Project accountants become financial analysts for their projects. Project managers spend less time on billing paperwork and face more scrutiny on forecasts.

What grows. Contractors need financial planning and analysis capability, meaning people who can interpret forecasts and cash flow across a portfolio. Data integration between field, project, and accounting systems becomes a core finance function.

What stays human. Payment approval, legal decisions about liens and claims, negotiation with owners, sureties, and banks, and responsibility for financial statements.

6. Friction and Risk

Legacy systems. Many contractors run long-established ERP systems that are deeply customized and hard to integrate. AI layered on top can only work with the data those systems expose.

State lien law. Lien rights, notice deadlines, and waiver requirements vary by state, and some states require specific statutory waiver forms. Using the wrong form, or missing a deadline, can void a party's rights entirely.

Controls and audit. Sureties and lenders rely on contractors' WIP schedules and financial statements. Automation has to strengthen financial controls, not bypass them, and every automated adjustment needs an audit trail.

Coding accuracy. An invoice coded to the wrong job or cost code distorts job cost reports and forecasts, and automated coding errors can spread quietly across many transactions.

Payment fraud. Construction's large payments and many parties make it a target. Automated payment workflows need strong verification of bank details and approval authority.

7. Residential Callout

QuickBooks runs most custom and remodel businesses. Custom builders and remodelers typically keep their books in QuickBooks, often integrated with Buildertrend or JobTread for job costing, estimates, and invoicing.

Construction loan draws are a core residential workflow. Custom homes are usually financed with construction loans that release money in draws after an inspector verifies progress. Built and Land Gorilla are reference platforms for managing that process, and photo-based inspections are an early AI entry point.

Deposits and payment schedules are regulated. Some states limit how much a contractor can collect upfront on home improvement contracts. California, for example, limits down payments on home improvement contracts. AI-generated payment schedules have to follow those rules.

Production builders use enterprise systems. Production homebuilders run enterprise ERPs and control costs through plan budgets, trade pricing, and variance purchase orders across thousands of homes.

Cash flow is survival. For small residential contractors, late payments and thin cash reserves are an existential risk. Faster, cleaner invoicing and collections may be the most valuable AI benefit for this segment.

8. Signals to Watch

- **Agentic accounts payable and receivable** built directly into construction ERPs, and how quickly controllers trust it.
 - **Consolidation of payment platforms** among Procore, Autodesk, Oracle, and independent providers.
 - **Contract data flowing into accounting systems**, such as payment terms and notice requirements applied automatically.
 - **Sureties and lenders using AI** to analyze contractor WIP schedules and draw requests, which would change what contractors are expected to report.
 - **Actual reductions in payment times** for subcontractors, measured by transaction data rather than surveys.
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9. Self-Check Exercise

Build a small job, then break it. Set up a spreadsheet for a hypothetical \$1 million subcontract with 10% retainage.

1. Create a schedule of values with eight line items and bill three months of progress, including one approved change order.
2. Track costs against budget and build a simple WIP calculation showing whether the job is overbilled or underbilled each month.
3. Create a lien waiver log for the three payments, including one lower-tier supplier.
4. Plant three problems: a duplicate invoice, a billing percentage that runs ahead of the reported cost progress, and a missing conditional waiver.
5. Give the spreadsheet to an AI assistant and ask it to review the job for errors and risks.

Questions to answer afterward:

1. Did the AI find all three planted problems? Did it flag anything else, and was it right?
2. What does your WIP calculation show about the job's profit, and what would a surety think?
3. Which of the planted problems would delay payment, and which would create financial or legal risk?
4. If an AI forecast showed profit fade that the project manager hadn't reported, how would you handle that conversation?

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3.16 - Reality Capture, Progress & Quality

1. Where This Sits

Chapters 13 through 15 kept returning to the same weakness. Schedules, project records, and financial forecasts all depend on knowing what has actually been built, and that knowledge has traditionally come from someone walking the site and estimating. A superintendent says the third floor is 60% framed. A subcontractor bills for 75% of its ductwork. Everyone's numbers are a little different, and some are optimistic on purpose.

Reality capture turns the physical site into a digital record using photos, 360-degree video, drones, and laser scans. Once the site is captured, AI can compare what was built with what was designed and scheduled.

Reality capture supports the quality responsibilities built into every construction contract: the contractor's quality control, the architect's site observations, and inspections and testing. Part 2 covers how that work happens on site. Reality capture feeds schedule updates (Chapter 13), change documentation (Chapter 14), billing verification (Chapter 15), and eventually as-built records at closeout (Chapter 18). It also runs in the other direction, putting the digital model onto the physical site through automated layout.

2. Who Does the Work

Superintendents and field engineers document progress and conditions, traditionally with phone photos and site walks.

VDC teams manage the models that captured data is compared against.

Quality managers run inspections and track deficiencies.

Surveyors and layout crews establish control points and mark where walls, sleeves, and embeds go.

Drone pilots, who must hold FAA remote pilot certification for commercial work, capture aerial imagery and site measurements.

Scanning specialists produce precise point clouds with laser scanners.

Owners' representatives and lenders' inspectors verify progress before approving payment.

3. The Software Landscape

360-degree capture and progress tracking

OpenSpace is one of the best-funded companies in 360-degree jobsite documentation, having raised about \$199 million. A worker walks the site with a camera, and the software maps the imagery to the floor plan automatically so anyone can review conditions remotely. In late 2025, OpenSpace acquired **Disperse**, a progress-tracking specialist. In June 2026, the company said it had passed 1,000 data center projects.

Drones and ground capture

DroneDeploy, which Procore acquired for about \$845 million in a deal that closed in September 2026, combines aerial drone mapping with ground-level capture. Drones measure earthwork and stockpile volumes, document site progress, and inspect roofs and facades.

Matterport, now owned by CoStar, creates navigable 3D captures of spaces and is especially common for documenting existing buildings.

Laser scanning and survey

Leica Geosystems (part of Hexagon), **Faro**, and **Trimble** make the high-precision scanners and survey instruments used to create point clouds. Point clouds are dense 3D measurements accurate enough to check whether installed work matches the model, or to document existing conditions for renovation design.

Automated layout

Dusty Robotics makes the FieldPrinter, a robot that prints full-scale layout from BIM models directly onto concrete slabs to about 1/16 of an inch, covering 10,000 to 15,000 square feet per day for multiple trades at once. **HP SitePrint** is the main competitor. It's a robot the contractor owns that works with a robotic total station many contractors already have, and it prints on a wider range of surfaces.

4. Where AI Is Landing

Settling arguments about what's built

The founders of Buildots describe their original idea as a camera that could settle arguments about what had been built. That captures the category's core value. Progress reporting in construction has always been partly negotiation: over billing, over schedule delays, over who's

behind. An objective, element-by-element record changes that negotiation. It gives owners and lenders better grounds to question billing, gives contractors evidence when others delay their work, and gives everyone earlier warning that a schedule is slipping.

Task ratings

Automated with sign-off: organizing captured imagery. Mapping thousands of photos and 360-degree frames to their location on the plans, by date, is now automatic. Finding what a wall looked like before drywall takes seconds.

Automated with sign-off: progress detection. Computer vision identifies installed elements and compares them with the model and schedule. Some vendors add human verification to reach high accuracy, which is worth knowing when comparing claims.

Automated with sign-off: site measurements. Drone-based earthwork quantities and stockpile volumes are routine and far faster than traditional survey methods.

Augmented: deviation detection. Comparing scans or imagery to the model flags work installed in the wrong place, missing elements, and conflicts before the next trade covers them up.

Augmented: delay forecasting. Measured installation pace gives an early, objective signal that an activity won't finish on time.

Augmented: layout. Robots execute layout quickly and precisely, but the output is only as good as the coordinated model they print from. Layout automation raises the stakes on model coordination (Chapter 8).

Assisted: quality judgment. AI can flag that something looks wrong. Judging workmanship, deciding whether a deviation matters, and determining the fix require experienced people.

Resistant: inspection and acceptance. Code inspections by building inspectors and special inspectors, testing, and formal acceptance of work remain human responsibilities. Much of what matters for quality, like connections, fastening, and concealed conditions, isn't visible to a camera.

Capture timing is the hidden discipline

Cameras see surfaces. Once insulation, drywall, or concrete covers the work, it's gone from view. The value of reality capture depends on capturing systematically at the right moments, especially before cover-up. That's a field management discipline, not a software feature. The best tools don't help if nobody walks the site with the camera before the drywall crew arrives.

Data centers are accelerating adoption

Data center construction appears repeatedly in this category's recent news. These projects are enormous, repetitive, and extremely schedule-sensitive, with miles of identical systems where small delays carry huge costs. They're an ideal environment for automated progress tracking, and they're pushing the technology forward quickly.

The competitive pressure ahead

Most current progress tracking depends on aligning captures with a detailed BIM model. Industry analysts have noted that newer vision-language models could infer progress directly from imagery without precise model alignment, which could make progress tracking cheaper and usable on projects without detailed models. If that happens, pricing and competition in the category could change quickly. Vendor claims about delay reductions, such as Buildots' claim of up to 50% fewer delays, should be read as company claims until independent results are available.

5. Jobs and Process Change

What compresses. Manual photo documentation, progress walks, and percent-complete estimation shrink. Traditional layout with tape measures and chalk lines shrinks where robots are adopted. Drone surveys replace much traditional topographic survey work on active sites.

What shifts. Superintendents move from estimating progress to acting on measured progress. Layout crews and surveyors shift toward preparing models, setting control, and running robots. Quality managers focus on the deviations the tools flag rather than finding problems by walking.

What grows. VDC work grows, because progress tracking and robotic layout both depend on accurate, coordinated models. Capture technicians and reality capture service providers become a real category of work.

What stays human. Inspection, acceptance, workmanship judgment, and decisions about how to fix problems.

6. Friction and Risk

Capture discipline and cost. Someone has to walk the site regularly, and some vendors' accuracy depends on human verification labor built into their pricing. Costs per project can be significant.

Model dependence. Model-based progress tracking and robotic layout require accurate models. Projects with poor or outdated models get less value.

Worker privacy. Cameras capture people as well as work. Firms need clear policies about what imagery is used for, especially if it could be used to monitor individual workers.

Data ownership and access. Owners, lenders, and contractors may all want the progress data, and it can become evidence in disputes. Contracts should address who owns and can access it.

Drone rules and site limits. Federal drone regulations, airspace restrictions near airports, and site safety rules limit where and how drones can fly.

7. Residential Callout

Phone photos are the reality capture. Most residential documentation is smartphone photos. **CompanyCam** is the leading photo documentation app for residential builders and trades, organizing photos by project and location and sharing them with homeowners.

Aerial measurement is mature. Roofing and solar contractors use aerial imagery from EagleView and drone captures to measure roofs without climbing them.

Pre-drywall documentation is valuable. Photographing or scanning walls before drywall creates a permanent record of plumbing, wiring, and framing that helps with warranty service and future renovations.

Lender draws depend on progress photos. Construction loan draws for custom homes increasingly rely on photo-based inspections, as Chapter 15 described.

Remodelers capture existing conditions. Scans and 3D captures of existing rooms help remodelers design and estimate accurately without repeated site visits.

8. Signals to Watch

- **Procore's integration of DroneDeploy**, and whether reality capture becomes a standard part of the project management platform.
 - **Buildots' expansion into workforce data** after acquiring Genda, and OpenSpace's integration of Disperse.
 - **Progress tracking without detailed BIM models**, enabled by newer vision AI.
 - **Owners and lenders requiring objective progress data** for payment approval.
 - **Robotic layout adoption** spreading from large general contractors to specialty trades.
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9. Self-Check Exercise

The percent-complete test. Find a space under construction or renovation you can legally access, or use a set of progress photos from a public project update.

1. Create a simple list of the elements that should be installed in one area: walls, doors, electrical boxes, ductwork, fixtures.
2. Estimate the area's percent complete by looking.
3. Capture photos or a video walk of the area and ask an AI vision tool to identify which elements from your list appear installed, missing, or unclear.
4. Compare the AI's element-level assessment with your estimate.
5. For a three-story office building, write a capture schedule identifying the moments when imagery must be captured before work is covered.

Questions to answer afterward:

1. How different were your estimate and the element-level assessment? Which would you trust for billing?
 2. What important quality issues would never be visible in photos?
 3. If an owner used this data to challenge your pay application, how would your negotiation change?
 4. What policy would you set so workers trust that site cameras are used to track work, not to watch them?
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3.17 - Safety, Workforce & the Connected Jobsite

1. Where This Sits

Every chapter so far has been about information: documents, models, schedules, and money. This chapter is about people doing physical work in dangerous conditions, and about the sensors, equipment, and data systems increasingly surrounding them.

Standard contracts assign responsibility for jobsite safety to the contractor. In AIA's general conditions, for example, it's covered in the article on protection of persons and property. Federal and state OSHA regulations set the legal requirements, and Part 2 covers how safety is managed on a working site.

According to the Bureau of Labor Statistics, 1,034 construction workers died on the job in 2024, about one in five U.S. workplace deaths. Construction's fatality rate fell to 9.2 per 100,000 full-time workers, the lowest since 2011, but it has hovered between 9 and 10 for more than a decade. Falls, slips, and trips caused 389 of those deaths, roughly 38%. At the same time, the industry can't find enough workers. Associated Builders and Contractors has estimated the industry needs about 349,000 net new workers in 2026 alone.

Safety and workforce shortages are connected problems, and the technology in this chapter is aimed at both.

2. Who Does the Work

Safety directors and site safety managers run safety programs, inspections, training, and incident investigations.

Superintendents and foremen are responsible for safety on site every day and plan the work of their crews.

Craft workers do the work and are the people the technology is meant to protect.

Owners' environmental, health, and safety teams set requirements on large industrial and institutional projects.

Insurers and risk control specialists price workers' compensation and liability coverage and advise on reducing risk.

Operations leaders and resource planners decide which staff and crews go to which projects.

Payroll and timekeeping staff turn hours worked into paychecks and job costs.

Equipment managers track fleet location, use, and maintenance.

3. The Software Landscape

Safety and quality management

Procore Quality & Safety handles inspections, observations, and incident reporting inside Procore. **SafetyCulture**, formerly known for its iAuditor app, is widely used across industries for mobile inspection checklists. **HammerTech** focuses on construction safety management, including worker orientation, permits to work, and subcontractor safety compliance.

Workforce and labor data

Rhumbix captures field labor data for timekeeping, production tracking, and time-and-materials work. Autodesk completed its acquisition of Rhumbix in spring 2026. Autodesk's stated reasoning was that financial systems look backward, so by the time labor costs reach a report, the chance to adjust has passed. Rhumbix's customers include Turner, Suffolk, and DPR Construction.

Bridgit helps general contractors plan which project managers, superintendents, and engineers go to which projects. **Raken** provides daily reports and time tracking for field teams. And as Chapter 16 noted, **Buildots** acquired **Genda**, a workforce and safety management company, to combine labor activity with progress data.

The connected jobsite

A growing set of devices collects data directly from the site:

- **Concrete sensors**, such as those from **Giatec**, embed in fresh concrete to measure temperature and estimate strength, helping teams decide when to strip formwork or apply loads.
- **Wearables and access systems**, from providers such as **Triax**, track who is on site, detect falls, and support evacuations.
- **Environmental monitors** track dust, noise, heat, and water leaks.
- **Equipment telematics** from machine manufacturers report location, hours, fuel use, and fault codes. An industry standard, ISO 15143-3, lets mixed fleets from different manufacturers share that data.
- **Tool and asset tracking** helps contractors keep track of expensive equipment across sites.

4. Where AI Is Landing

Automated with sign-off: safety documentation. Drafting pre-task plans and job hazard analyses from the day's scheduled activities, preparing toolbox talks, and drafting incident reports are increasingly automated. A competent person still has to review them against actual site conditions.

Automated with sign-off: timekeeping and certification tracking. Capturing crew hours, coding them to jobs and cost codes, and tracking expiring training certifications are largely automatable.

Augmented: predictive safety risk. Models combining safety observations, schedule activities, weather, subcontractor history, and past incidents can flag which projects, crews, or weeks carry elevated risk, so safety managers can focus their time.

Augmented: computer vision hazard detection. Site cameras analyzed by AI can detect missing hard hats or vests, unprotected edges, people in equipment swing zones, and similar hazards.

Augmented: labor productivity analytics. Comparing installed quantities with labor hours reveals productivity problems early, feeding cost-to-complete forecasts (Chapter 15).

Augmented: resource planning. Matching staff and crews to upcoming work across a portfolio is faster with AI, which matters more as skilled people get scarcer.

Assisted: safety program design and root cause analysis. AI can find patterns in incidents, but understanding why they happen requires people who know the work.

Resistant: safety leadership. Stopping unsafe work, building a culture where workers speak up, installing guardrails, and training people face to face all depend on people.

Leading indicators depend on trust

Injuries are **lagging indicators**. By the time they're counted, the harm has happened. Safety observations, near-miss reports, and hazard corrections are **leading indicators**, and they're what predictive safety models run on. Those reports come from workers and supervisors choosing to report. If workers believe the data will be used to punish them, reporting dries up and the models go blind. **Predictive safety is only as good as the reporting culture feeding it.**

The surveillance line

The same camera that detects a missing hard hat can identify an individual worker, time their breaks, and measure their pace. That's where safety technology meets worker privacy and trust.

Some states regulate biometric data. Illinois' Biometric Information Privacy Act is the best-known example. Unions and workers have legitimate concerns about monitoring. The technical choices matter: systems can be designed to detect hazards without identifying individuals, and firms can set clear, written limits on what data is used for. Contractors that deploy monitoring without that clarity risk losing the worker trust that safety programs depend on.

A note on hardware reality

I worked at an IoT hardware and SaaS startup, and one lesson carries straight into construction: **hardware is much harder to deploy than software.** Batteries die. Devices get damaged, lost, or buried in concrete. Steel, concrete, and changing site layouts disrupt wireless signals. Someone has to install, maintain, and move every sensor as the site changes. A software pilot can start in a day, while a sensor deployment requires logistics. And a sensor that sends alerts nobody acts on is just an expensive way to generate noise. When evaluating connected jobsite technology, ask who installs it, who maintains it, what happens when connectivity drops, and who responds when it alerts.

Field data connects to cost

Autodesk's reasoning for buying Rhumbix points to one of this chapter's most important links. Labor is one of the largest and most variable costs on a project. When labor hours and production quantities are captured accurately and immediately, cost forecasts (Chapter 15) and schedule forecasts (Chapter 13) get better. **The jobsite is the source of the data every other system depends on.**

AI doesn't lay brick

The workforce shortage won't be solved by software. AI helps indirectly by reducing administrative work for field leaders, reducing rework through better coordination and layout (Chapter 16), and allocating scarce people more effectively. Robotics addresses some physical tasks directly, such as robotic layout and rebar-tying robots like TyBot, but most construction work still requires skilled hands.

5. Jobs and Process Change

What compresses. Safety paperwork, timecard entry, certification tracking, and routine reporting shrink. Safety managers can spend more of their time in the field.

What shifts. Safety professionals move toward analysis, coaching, and responding to risk signals. Resource planners make decisions with better data about who's available and what's coming. Superintendents get earlier warning on productivity and safety trends.

What stays human. The craft itself, safety leadership, crew supervision, and the trust between workers and management.

6. Friction and Risk

Privacy and biometric law. Monitoring systems that identify individuals can trigger state biometric privacy laws and labor relations issues. Legal review should come before deployment, not after a complaint.

The liability of knowing. A system that detects a hazard creates a record. If the hazard wasn't corrected and someone was hurt, that record could become evidence. Detection systems need response procedures, not just alerts.

Alert fatigue. Too many false alarms train people to ignore warnings, including the real ones.

Multi-employer sites. Construction sites involve many employers. OSHA's multi-employer citation policy can hold a controlling contractor responsible for hazards affecting other companies' workers, which complicates who owns safety data and who must act on it.

Connectivity and cost. Sites with poor connectivity limit real-time systems, and the cost of devices is hard for small subcontractors to absorb.

7. Residential Callout

Safety tech adoption is low. Residential construction is dominated by small crews and subcontracted labor, often working informally. Formal safety management software is uncommon outside production builders and larger firms.

Falls are a major residential hazard. Roofing and framing work involves constant fall exposure, and fall protection compliance is a persistent challenge on residential sites.

Production builders run structured programs. Large production builders run safety inspections and trade partner compliance programs across communities, often through mobile inspection apps.

Labor shortages hit residential trades hard. Framing, roofing, and finish trades face persistent shortages, which directly affect schedules and costs.

Offsite construction is the bigger workforce story. Wall panelization, truss plants, and modular construction move work into factories where conditions are safer and labor is more productive. The industry's high-profile failures are useful cautionary cases. Kattera collapsed in

2021 after raising billions of dollars, and Veev shut down in 2023. Their stories show that offsite construction's challenges are as much about business models and demand as technology.

8. Signals to Watch

- **Autodesk's integration of Rhumbix** and Buildots' integration of Genda, and whether field labor data becomes a standard input to cost and schedule forecasting.
 - **Privacy regulation** affecting jobsite monitoring and biometric data.
 - **Insurers rewarding technology adoption** with lower premiums for proven safety programs.
 - **Computer vision safety systems** moving from pilots to standard practice on large projects.
 - **Heat illness rules** at the federal and state levels, which could drive adoption of environmental monitoring and wearables.
 - **Robotics for dangerous or repetitive physical tasks.**
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9. Self-Check Exercise

The pre-task plan and the policy. Choose a specific activity with serious hazards, such as setting roof trusses on a two-story wood-frame building.

1. Ask an AI assistant to draft a job hazard analysis identifying steps, hazards, and controls.
2. Review the draft against OSHA's construction fall protection requirements in 29 CFR 1926 Subpart M, and note anything missing or wrong.
3. Add the site-specific conditions the AI couldn't know about, such as weather, crane access, crew experience, and adjacent work.
4. Write a one-page policy for a camera-based safety monitoring system that workers would accept. Define what it detects, who sees the data, how long it's kept, and what it will never be used for.

Questions to answer afterward:

1. What did the AI's hazard analysis miss, and how serious would those gaps be on a real site?
2. What would make a foreman trust an AI-generated pre-task plan, and what would make them ignore it?
3. How would your monitoring policy handle a case where the system caught a worker repeatedly removing fall protection?
4. If you had budget for one connected jobsite technology on a mid-size project, which would you choose and who would maintain it?

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3.18 - Closeout, Commissioning & Turnover

1. Where This Sits

The building looks finished. But between "looks finished" and "handed over" lies one of the most frustrating phases in construction. Punch lists get worked and reworked. Systems get tested. Warranties, manuals, and record drawings get assembled. The final inspection gets scheduled. Retainage waits to be released.

Substantial completion is when the work is complete enough for the owner to use it for its intended purpose. It's typically documented with a certificate, starts warranty periods, and triggers a punch list of remaining items.

Final completion follows once the punch list and all closeout requirements are done, leading to final payment and release of retainage.

Closeout requirements are usually spelled out in Division 01 of the specifications. They include:

- Operation and maintenance (O&M) manuals.
- Warranties.
- **Record drawings** documenting what was actually built.
- Spare parts and extra materials.
- Owner training.
- Lien releases.
- Commissioning documentation.

Commissioning is the quality process that verifies building systems, especially HVAC, controls, and lighting, perform according to the owner's requirements and the design intent. Industry guidance from ASHRAE defines the process, and energy codes and green building programs increasingly require it.

Closeout matters more than its reputation suggests. It determines whether the owner can actually operate the building on day one, and it's the point where most of the information created during design and construction is either handed over usefully or lost.

2. Who Does the Work

General contractor project managers and project engineers track closeout requirements and chase documents from subcontractors.

Superintendents manage punch list completion in the field.

Subcontractors complete punch items and provide warranties, O&M data, and training for their systems.

Architects and engineers inspect for substantial completion, review closeout submittals, and prepare record documents.

Commissioning providers develop test procedures, witness functional testing, document issues, and verify fixes.

Owners' facility staff receive training and accept the building.

Building officials perform final inspections and issue certificates of occupancy.

Owners and lenders release final payment once closeout conditions are met.

3. The Software Landscape

Punch lists and closeout tracking mostly live inside the project management platforms from Chapter 14. **Procore** and **Autodesk Build** both handle punch lists and closeout document tracking. **Fieldwire**, owned by Hilti, is popular with superintendents and subcontractors for field punch work.

Commissioning software is a specialized category. **Facility Grid** manages commissioning checklists, functional testing, and asset readiness, and compiles results into a structured, asset-based turnover package that can export into maintenance systems like IBM Maximo. **CxAlloy** is a long-established commissioning platform owned by Trinity Consultants, which acquired CxAlloy's parent company in 2019. **BlueRithm** and **CxPlanner** are other commissioning tools.

COBie, the Construction Operations Building information exchange, is a standard format for handing over structured facility data: equipment, locations, warranties, spare parts, and maintenance requirements. It's part of the U.S. National BIM Standard and is required by some federal agencies and large owners, though adoption across the broader market is limited.

Reality capture tools from Chapter 16, especially laser scans and 360-degree captures taken before walls were closed, increasingly support record documentation.

4. Where AI Is Landing

Closeout is document archaeology

Most of the information needed at closeout was created months or years earlier. The product data for a rooftop unit was in its submittal. The warranty terms were in the specifications and the manufacturer's documents. The maintenance requirements were in the manufacturer's literature attached to the submittal. At closeout, someone has to dig all of it back up, confirm it matches what was actually installed, and organize it into manuals and registers. Project engineers do that digging while also finishing the punch list, and it's a big reason closeout drags on.

It's also where information disappears. An FMI report found that 95.5% of engineering and construction data goes unused, and closeout is where most of that loss becomes permanent. This is precisely the kind of document retrieval and assembly work AI does well.

Task ratings

Automated with sign-off: O&M manual and warranty assembly. AI can extract product data, maintenance instructions, and warranty terms from approved submittals and organize them into O&M manuals and warranty registers, with warranty start dates tied to substantial completion.

Automated with sign-off: closeout tracking. Comparing Division 01 requirements against documents received, flagging what's missing by subcontractor, and generating follow-ups is largely automatable.

Automated with sign-off: punch list capture. A photo and a voice note can become a punch item with a location, description, and responsible trade.

Augmented: asset data collection. AI can populate asset registers and COBie data from models, submittals, and photos of equipment nameplates, reducing one of the most tedious turnover tasks.

Augmented: as-built verification. Comparing scans and captures against the model helps identify where record documents need updating.

Augmented: commissioning documentation. AI can organize test results, spot recurring issues across systems, and draft commissioning reports.

Assisted: judging completion. Deciding whether a project has reached substantial completion, or who is responsible for a warranty issue, involves contract interpretation and negotiation.

Resistant: physical verification and acceptance. Functional performance testing of equipment, the building official's final inspection, the architect's certification, hands-on owner training, and the owner's acceptance of the building remain human work.

Submitted isn't always installed

There's a trap in automating O&M manuals from submittals. Products get substituted during construction. Equipment models change when a supplier runs out of stock. A manual assembled perfectly from approved submittals can still describe equipment that isn't in the building.

Closeout data needs to be verified against what was actually installed, which is where equipment nameplate photos and field verification matter.

Faster closeout means faster cash

Closeout delays hold up retainage and final payment, often a significant share of a subcontractor's profit on a job. As Chapter 15 described, construction payments already move slowly. Every week cut from closeout returns cash to the contractor and subcontractors sooner.

5. Jobs and Process Change

What compresses. Chasing and assembling closeout documents, formatting O&M manuals, building warranty registers, and entering asset data shrink sharply for project engineers.

What shifts. Project teams spend more closeout time on verification: confirming installed equipment, resolving punch items, and making sure the owner is ready to operate. Commissioning providers spend less time on documentation and more on testing and problem solving.

What stays human. Functional testing, inspections, certifications, training, and acceptance.

6. Friction and Risk

Submittal and installation mismatches. Automated assembly from submittals can carry forward superseded or substituted products. Field verification is essential.

Owner format requirements. Owners may require COBie, specific maintenance system import formats, or their own templates. Turnover data that doesn't match the owner's systems ends up re-keyed or ignored.

Record document responsibility. Contracts assign different responsibilities for record drawings and as-built models, and they're often left incomplete. AI can help identify differences, but someone has to own updating them.

Platform access after the project. Much closeout data lives in the contractor's project platform. When the project ends, owners need the data in their own systems, or they risk losing access to it.

Commissioning shortcuts. Commissioning documentation is easy to generate. Actual testing is not. A well-formatted commissioning report doesn't prove the systems work.

7. Residential Callout

Closeout is the homeowner orientation. For new homes, closeout centers on a walkthrough where the builder shows the homeowner how the home works and identifies final punch items before closing.

Warranty is the long tail. New homes commonly come with a structured warranty, often one year for workmanship and materials, two years for systems, and ten years for major structural defects. Warranty service is a significant cost and a major driver of builder reputation, and residential platforms include warranty modules to manage it.

AI triage of warranty claims is emerging. Homeowners submitting photos and descriptions of issues can be routed automatically to the right trade, with AI distinguishing likely warranty items from normal maintenance.

Home documentation is often thin. Homeowners typically receive appliance manuals and a few warranties rather than a structured record of what's in their home. Builders that deliver organized digital home records, including pre-drywall photos, give buyers something genuinely useful.

Production builders run warranty departments. Large builders manage warranty service across thousands of homes, making claim data a valuable source of feedback about which products and trades cause problems.

8. Signals to Watch

- **AI-assembled turnover packages** becoming standard deliverables, and whether owners trust them.
- **Owners writing structured data requirements into contracts**, specifying exactly what asset information they need and in what format.

- **Commissioning platforms connecting directly to maintenance systems**, so turnover data flows straight into operations.
 - **Reality capture replacing traditional record drawings** for documenting what was built.
 - **Construction platforms extending into operations**, covered in Chapter 19.
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9. Callout: Beyond Closeout, Into Operations

Closeout is where Part 3's lifecycle ends, but it's where the building's life begins. Design and construction take a few years. Operations last decades and account for most of a building's lifetime cost. The major software platforms are extending past handover to capture that phase, and Autodesk's roughly \$3.6 billion acquisition of MaintainX, which closed in August 2026, is the clearest signal.

That shift raises the stakes on everything in this chapter. Owners increasingly expect turnover data they can load directly into their operations systems. Chapter 19 is a short bridge chapter on that handover. Operations and facility management get full treatment in a later part of this series.

10. Self-Check Exercise

The turnover test. Download a project manual for a public building project. Find the closeout sections in Division 01, typically Closeout Procedures (01 77 00) and Closeout Submittals (01 78 00), plus the specification section for one major piece of equipment, such as a rooftop HVAC unit.

1. Build a closeout checklist from Division 01 listing every required document, who provides it, and when.
2. Find the manufacturer's installation and maintenance manual for a real product that meets the equipment specification.
3. Ask an AI assistant to create an O&M manual entry and warranty register entry from the specification and the manufacturer's documents.
4. Verify every item. Mark what's correct, what's wrong, and what could only be confirmed by checking the installed equipment.
5. List the information a facility technician would need on day one of operations that isn't in any closeout document.

Questions to answer afterward:

1. What did the AI get wrong, and would a facility manager have noticed?

2. What would you need in the field to confirm the installed equipment matches the documents?
 3. How would you structure the handover so the owner's maintenance system can use the data immediately?
 4. What gap did you find between what the contract requires at closeout and what operators actually need?
-

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3.19 - Handover to Operations & Digital Twins

This is a short bridge chapter covering the handover from construction to operations. Facility management and operations get full treatment in a later part of this series.

1. Where This Sits

Chapter 18 ended with the keys changing hands. This chapter looks across that line, at what the owner needs to run the building and why the software industry is suddenly so interested in providing it.

This is the transition from construction to operations, the longest phase of a building's life. A building might take two to four years to design and build and then operate for 50 years or more. An estimate widely cited by Autodesk and others puts about 80% of a built asset's lifetime cost in its operation.

That imbalance explains the market's direction. As Chapter 18 noted, an FMI report found that 95.5% of engineering and construction data goes unused. The design and construction industry produces enormous amounts of information, and the people who operate the building for decades receive very little of it in a usable form.

2. Who Does the Work

Facility managers and maintenance technicians keep buildings running day to day.

Owners' engineering and operations teams manage building systems across campuses and portfolios.

Energy managers track and reduce energy use.

Asset and property managers, covered in Chapter 3, look after the building's financial performance.

Controls contractors and IT teams manage building automation systems and the networks they run on.

Design and construction teams produce the handover data, and some increasingly offer to maintain it for owners after the project ends.

3. The Software Landscape

Operations software is its own large market, with several overlapping categories.

Maintenance management systems (CMMS) handle work orders, preventive maintenance, and equipment records. **MaintainX** is a fast-growing platform that Autodesk agreed to acquire for about \$3.6 billion in cash, the largest acquisition in Autodesk's history. The deal closed in August 2026. **Brightly**, a Siemens company, is strong in education, public sector, and municipal facilities.

Enterprise asset management (EAM) systems serve asset-intensive organizations like utilities, infrastructure operators, and large campuses. **IBM Maximo** is the long-standing leader.

Integrated workplace management systems (IWMS) combine space, lease, maintenance, and capital planning for large real estate portfolios. **Planon** serves large enterprise real estate, and **Eptura**, which includes Archibus, serves campuses and office portfolios.

Digital twin platforms connect building models and asset data with live operational data. **Autodesk Tandem** builds digital twins from design and construction data and is now part of Autodesk's operations offering alongside MaintainX. **Bentley iTwin** focuses on infrastructure, and **Willow** is among the specialists building twins for buildings and infrastructure.

Building automation systems from Johnson Controls, Siemens, Schneider Electric, Honeywell, and Tridium control HVAC, lighting, and other systems. They're the main source of live operational data.

Why a design software company bought a maintenance company

Autodesk's MaintainX deal is the clearest statement of strategy in this part of the market. Autodesk built its business on the few years of design and construction. By acquiring a maintenance platform, it's betting it can connect design data, construction data, and decades of operational data in one system. If that works, the platform that holds a building's data from concept through operations would have an advantage no single-phase tool can match. Who owns that data is one of the central questions for the rest of this series.

4. Where AI Is Landing

Data that survives handover

The owner doesn't need everything the project team produced. An operator needs a specific, relatively small set of structured information:

- What equipment exists and where it is.
- What it is (manufacturer, model, specifications).
- When it was installed and what warranties apply.
- What maintenance it requires.
- Its history since installation.

The international information management standard ISO 19650 calls this the owner's asset information requirements. The cheapest way to get that data is to define those requirements at the start of the project, as Chapter 5 described, so the team captures information as it goes rather than reconstructing it at the end.

AI offers a second path: rescuing the data afterward by extracting it from closeout documents, models, and photos. That's genuinely useful for the many existing buildings that were handed over with a pile of PDFs. But it's slower and less reliable than capturing the data correctly in the first place.

Task ratings

Automated with sign-off: building the asset register. AI can extract equipment lists, specifications, and warranty information from closeout documents, models, and nameplate photos.

Automated with sign-off: preventive maintenance schedules. AI can read manufacturer maintenance recommendations and turn them into scheduled maintenance tasks in a maintenance system.

Automated with sign-off: work order intake. Occupant requests can be categorized, prioritized, and routed automatically.

Augmented: predictive maintenance and fault detection. Analyzing building automation and sensor data to detect failing equipment, drifting controls, and wasted energy is an established and growing capability.

Augmented: technician support. Technicians can ask questions about equipment and get answers drawn from the building's own manuals, history, and drawings on a mobile device.

Assisted: capital planning. AI can model repair-versus-replace decisions and forecast capital needs, but funding priorities are organizational decisions.

Resistant: the physical work. Repairs, emergency response, and occupant relationships remain human.

A reality check on digital twins

"Digital twin" means very different things depending on who's selling it. At one end, it's a 3D model linked to an asset list. At the other, it's a live simulation continuously synchronized with

sensor data. For most owners, most of the value today comes from the simple end: knowing what equipment you have, where it is, and what's happened to it. A digital twin also decays like an unupdated schedule. If renovations, replacements, and repairs aren't recorded, the twin stops matching the building.

The feedback loop that rarely closes

Operations data could make design and construction better. It could show which products fail early, which systems use more energy than their models predicted, and which details cause leaks. That feedback almost never reaches designers and contractors today, because the phases run on separate systems owned by separate companies. Closing that loop is a big part of the lifecycle platform pitch. Whether platforms can actually deliver it, and whether owners will share the data, is one of the most important open questions in AEC technology.

5. Jobs and Process Change

What compresses. Manual asset data entry, maintenance schedule setup, and work order triage shrink.

What shifts. Technicians get AI support in the field, which matters in a trade where experienced people are hard to replace. Facility managers spend more time on analysis and planning. Some design and construction firms are positioning themselves to provide ongoing data services to owners after handover.

What grows. Owners need building data specialists who can manage digital twins, integrate systems, and keep asset information current.

What stays human. Hands-on maintenance, emergency response, vendor management, and occupant relationships.

6. Friction and Risk

Cybersecurity. Connecting building systems to cloud platforms and AI tools expands the attack surface. Building automation networks were often not designed with modern security in mind.

Who pays for handover data. Structured handover data costs the project team time. If the contract doesn't require it and pay for it, it usually doesn't get delivered well.

Fragmented operations systems. Many owners run separate maintenance, workplace, automation, and property management systems that don't share data easily.

Lifecycle lock-in. A platform that holds a building's data from design through operations creates real benefits and real dependence. Owners should secure data portability in their contracts.

Skills gaps. Many facility teams lack the staff or training to use advanced data platforms, and the tools only deliver value if someone uses them.

7. Residential Callout

The homeowner is the operator. For most homes, operations means a homeowner with a folder of appliance manuals, or more than likely nothing at all.

Builders are experimenting with home records. Some builders provide digital homeowner portals with manuals, warranties, maintenance reminders, and photos, a residential version of structured handover.

Smart home devices generate operational data. Connected thermostats, water leak sensors, and security systems produce real operating data, though it rarely connects to a record of how the home was built.

Build-to-rent operators run operations at scale. Companies owning thousands of single-family rental homes use maintenance platforms much like commercial owners, and for them, structured handover data from builders has clear financial value.

8. Signals to Watch

- **Autodesk's integration of MaintainX with Tandem and its construction platform,** and whether data truly flows from design through operations.
 - **Owners specifying asset information requirements** in design and construction contracts.
 - **AI-built asset registers for existing buildings** that were never handed over digitally.
 - **Growth in predictive maintenance and fault detection,** especially in energy-intensive facilities like data centers and hospitals.
 - **Building cybersecurity requirements** for connected systems.
 - **Operational feedback reaching design and construction teams.**
-

9. Self-Check Exercise

The day-one test. Choose a building you can access and ask about, such as your office, a school, or a public library. If you can talk to its facility manager, even better.

1. List 10 major pieces of equipment in the building, such as air handlers, boilers, pumps, generators, and elevators.
2. For each, try to find the location, manufacturer and model, installation date, warranty status, and maintenance requirements. Note where each piece of information lives, or whether it exists at all.
3. Find a manufacturer's maintenance manual for one item and ask an AI assistant to turn it into a preventive maintenance schedule. Check it against the manual.
4. Write a one-page list of the minimum information an owner should require at handover for a new building of that type.

Questions to answer afterward:

1. How much of the information you looked for was easy to find, hard to find, or missing?
 2. Where would that information have been easiest to capture during design and construction?
 3. Would you trust the AI-generated maintenance schedule? What would you check first?
 4. What would convince an owner to pay for better handover data on their next project?
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Part 3, Closing

What the Technology Tells Us

Part 3 followed a project from the first market question to the day the keys change hands, stopping at each phase to ask what software runs the work and what AI is changing. The vendors will look different in a year. Several category leaders changed owners in 2026 alone. The patterns underneath them are more durable, and they're worth carrying forward.

AI moves fastest where work is documents. Across every phase, the same kind of task moved first: reading, extracting, comparing, and drafting. That includes market reports, lease abstraction, takeoff, submittal review, daily logs, and O&M manuals. The tasks that barely moved involve authority, physical work, or relationships. Nobody automated a public hearing, a building inspection, or a foreman's commitment to finish on Thursday.

"Automated" almost always means "automated with sign-off." The 95%-accurate extraction tool still gets a few fields wrong on every deal, and it doesn't tell you which ones. A plan review tool that's right about 90% of the time is transformative as a prescreener and unacceptable as an approver. In AEC, someone stays accountable for nearly every output. That's why AI saves time only when checking its work is faster than doing the work. In hard-bid estimating, the winner's curse means an unchecked error doesn't average out. It wins the job.

When analysis gets cheap, data gets valuable. CoStar's rent benchmark is valuable because of the 4 million leases behind it, not the model on top. Autodesk bought Rhumbix for field labor data and MaintainX for decades of operational data. When you evaluate any AI tool in this industry, ask where its data comes from and whether a competitor could get the same data.

Every handoff is where data dies. The feasibility model doesn't feed the design model. The design model gets flattened into PDFs for bidding. Construction data rarely survives into operations. AI works best on connected, structured data, so the firms and delivery methods that keep information continuous from phase to phase will get the most from it. The industry's platform wars are, at bottom, a fight over who holds that continuous record.

Tasks compress; roles shift. No chapter found a job that simply disappears. Every chapter found jobs whose shape changes: the analyst who audits instead of assembles, the specifier who edits instead of drafts, the estimator who spends saved hours on scope and risk. The hardest open question is the apprenticeship problem. The junior work AI absorbs is the same work that has always trained the industry's future leaders.

Accountability doesn't move. Licensure, the building official's authority, contract risk allocation, and the standard of care all slow AI down, and none of them are going away. They also explain where durable careers are: in the judgment and responsibility that software can inform but not carry.

Lifecycle Recap: One Project from Site to Turnover

This module compresses Parts 1 through 3 into one sequence. Each step is a point in a project's life where a group of people answers one question, produces a set of documents, and hands something to the next group. Use it as a review after finishing the course, or as a map before starting it.

Every step uses the same fields:

- **Who:** the people doing the work and making the decision
- **Question:** what this step has to answer
- **Inputs:** what the team needs to answer it
- **Artifacts:** the documents this step produces
- **Tools:** the software most teams use today (Part 3 covers vendors and competitors in depth)
- **AI today:** where AI helps as of 2026, and where it stops
- **Passes forward:** what the next step receives
- **Often lost:** what gets dropped in the handoff
- **Where it's covered:** the modules that teach it

Three threads run through every step. The first is the document chain: a requirement becomes a decision, then a drawing or contract clause, then built work, then a record. The second is money over time: each step either protects the project's return or spends it.

The third is information loss. Each handoff compresses what one group knows into a smaller package for the next group, and the "Often lost" line tracks what falls out. That third thread is where most of the opportunity in AEC technology sits.

Development: Finding, Testing, and Buying the Site

Part 1 is organized around decision gates. At each gate the money at risk goes up and walking away gets more expensive. Gate 0 asks whether the site is worth a look, Gate 1 whether to tie it up, Gate 2 whether to make the earnest money non-refundable, Gate 3 whether the entitlements allow what was underwritten, and Gate 4 whether to close and start.

Step 1: Market and Site Selection (Gate 0)

Before any money is at risk, the developer decides whether a market and a site deserve a closer look.

- **Who:** Developer, land broker, market study firm
- **Question:** Is there demand for this product here, and is this the right site to serve it?
- **Inputs:** Employment, population and migration, household formation, supply pipeline, absorption, rent and sales comps, traffic counts, and parcel data from the county assessor and GIS
- **Artifacts:** Screening memo, site screening matrix, market study, broker opinion of value
- **Tools:** CoStar, Placer.ai, Regrid, Yardi Matrix, county GIS, Census and BLS data
- **AI today:** Strong at screening and ranking sites, pulling comps, and answering market questions in plain language. Weak at knowing which data sources are stale or which broker numbers are optimistic.
- **Passes forward:** A shortlist of sites, each with a highest-and-best-use hypothesis
- **Often lost:** Why the other sites were rejected. That reasoning becomes useful again when the first choice falls through.
- **Where it's covered:** Part 1 Modules 1 and 2; Part 3 Module 3.1

Step 2: Screening Feasibility and Conceptual Budget (Gate 0 to Gate 1)

A quick test of whether a building the site can plausibly hold earns enough to justify tying up the land.

- **Who:** Developer, an architect or planner for the test fit, a GC or estimator for rough cost
- **Question:** Does the deal pencil at a land price the seller might accept?
- **Inputs:** Buildable area assumptions, target unit mix, rents, conceptual cost per square foot, site development cost, cap rate and debt assumptions
- **Artifacts:** Test fit, rough order of magnitude (ROM) budget, preliminary pro forma, conceptual schedule
- **Tools:** Excel, ARGUS, TestFit, Autodesk Forma, SketchUp, RSMeans cost data
- **AI today:** Test fits and yield studies that took weeks now take hours, and AI extracts offering memos, rent rolls, and T12s into a model. The assumptions that drive the answer (exit cap rate, rent growth, contingency) still need a person who will defend them to an investment committee.
- **Passes forward:** A decision on whether to make an offer, a maximum land price (residual land value), and a target return
- **Often lost:** The sensitivity ranges. The pro forma travels as a single number, and the range around it disappears.
- **Where it's covered:** Part 1 Module 5; Part 3 Modules 3.3 and 3.5

Step 3: Site Control (Gate 1)

The developer ties up the land on terms that leave room to investigate before committing.

- **Who:** Developer, seller, attorneys, title company
- **Question:** On what terms do we control the land, and what does it cost to walk away?
- **Inputs:** Residual land value from the pro forma, the seller's motivation, and how much diligence and entitlement time the deal needs
- **Artifacts:** Letter of intent (LOI); purchase and sale agreement (PSA) covering earnest money, the due diligence period, extensions, and the entitlement contingency; title commitment
- **Tools:** Word and DocuSign, Qualia for title, Dealpath for pipeline tracking
- **AI today:** Useful for drafting, comparing contract terms, and flagging missing clauses. Negotiating leverage and the seller relationship remain human work.
- **Passes forward:** A signed PSA, a diligence calendar with hard dates, and a pursuit budget
- **Often lost:** Side agreements and verbal commitments made to get the deal signed
- **Where it's covered:** Part 1 Module 7; Part 3 Module 3.4

Step 4: Zoning and Entitlement (Analysis at Gate 1, Approval at Gate 3)

The step where most deals die, and the one most AEC people underestimate.

- **Who:** Developer, land use attorney, entitlement consultant, architect, civil engineer, planning staff, and elected officials
- **Question:** What can legally be built here, and how certain is the approval?
- **Inputs:** Zoning code (use, density or FAR, height, setbacks, lot coverage, parking), overlays, PUDs and development agreements, CC&Rs and easements from title, impact fees
- **Artifacts:** Zoning report, entitlement schedule, site plan submission, conditions of approval
- **Tools:** Municipal code libraries, parcel and zoning data (Regrid, Zoneomics, Gridics), Autodesk Forma and TestFit for envelopes, municipal permitting portals
- **AI today:** Good at reading zoning text and turning it into a buildable envelope. It cannot predict a planning commission vote or a neighborhood group's opposition, which is where discretionary approvals are decided.
- **Passes forward:** The approved density and envelope, plus every condition of approval the design must now meet
- **Often lost:** The reasoning behind a code interpretation, and conditions of approval recorded in hearing minutes rather than in the drawings
- **Where it's covered:** Part 1 Module 3; Part 3 Module 3.2

Zoning governs what can go on the site. The building code governs how it is built, is reviewed by a different office, and causes problems at a different time (Step 11).

Step 5: Physical Due Diligence (Gate 2)

Everything that could break the budget has to surface before the earnest money goes hard.

- **Who:** Surveyor, geotechnical engineer, environmental consultant, civil engineer, utility providers; for existing buildings, a property condition assessor and code consultant
- **Question:** Is there anything in the ground, on the title, or in the existing building that breaks the budget?
- **Inputs:** ALTA/NSPS survey, geotechnical borings, Phase I ESA (ASTM E1527), wetlands delineation, utility will-serve letters and capacity, traffic study; for existing buildings, a property condition assessment (ASTM E2018), hazardous materials surveys, and an IEBC compliance path
- **Artifacts:** The reports above, a list of budget-changing findings, and a decision to go hard or terminate
- **Tools:** Data rooms (Datasite, Intralinks), LightBox for environmental records, Bluebeam for PDF markup
- **AI today:** Document review and lease abstraction are close to automated. Reading a boring log, or judging whether an environmental finding kills the deal, is still the consultant's professional call.
- **Passes forward:** Non-refundable earnest money, a revised budget (foundation type, sitework, off-site utilities), and a known list of site constraints

- **Often lost:** How the findings relate to each other. The survey, geotech, and utility reports arrive as separate PDFs and are seldom reconciled until design.
- **Where it's covered:** Part 1 Modules 4A and 4B; Part 3 Module 3.4

Step 6: Capital Stack and Closing (Gate 4)

The project gets funded, and the people funding it set the rules the design and construction teams will live under.

- **Who:** Developer, equity partners (LP and GP), construction lender, appraiser, title company
- **Question:** Who funds the project, on what terms, and what do they require before the first draw?
- **Inputs:** Final budget, entitlements in hand, appraisal, a GMP or a clear path to one, completion guaranty
- **Artifacts:** Sources and uses, term sheet and loan agreement, operating agreement with the distribution waterfall, appraisal, closing checklist, title policy
- **Tools:** Excel, ARGUS, Juniper Square for investor management, Built and Rabbet for lender draws
- **AI today:** Drafts investment committee memos and runs scenarios on request. Terms are negotiated between people, and the lender's requirements are fixed in documents AI can check but not change.
- **Passes forward:** Closed land, a funded budget, and the lender's control rights: budget lock, GMP requirement, retainage, and approval of change orders
- **Often lost:** Why each contingency line exists. The design and construction teams inherit the numbers without the reasoning.
- **Where it's covered:** Part 1 Modules 6 and 7; Part 3 Modules 3.3 and 3.15

Design and Construction

Step 7: The Handoff to Design

The deal's assumptions become the design team's requirements.

- **Who:** Owner, owner's rep, architect, and the CM (hired early under CM at-risk)
- **Question:** What is the design team being asked to deliver, for how much, under which delivery method?
- **Inputs:** Pro forma assumptions, entitlement conditions, site constraints package, budget, schedule, lender requirements
- **Artifacts:** Owner's program, design agreement (for example AIA B101), delivery method decision, CM preconstruction agreement
- **Tools:** Excel; owner capital planning tools (Procore, Autodesk, Northspyre, Oracle Unifier); DESTINI Profiler and RSMeans for conceptual cost
- **AI today:** Turns a list of requirements into a draft program and checks it against the pro forma. Choosing a delivery method is a decision about how risk is split among the parties, and AI adds little to it.
- **Passes forward:** A program the architect can design to and a budget the CM can estimate against
- **Often lost:** The financial reason behind each requirement. The architect is told "200 units" and not that the deal fails at 190.
- **Where it's covered:** Part 1 Module 8; Part 2 Module 2.1; Part 3 Module 3.5

Step 8: Design as Progressive Commitment

Each design phase fixes more decisions, and each fixed decision costs more to reverse.

- **Who:** Architect; structural, MEP, and civil engineers; landscape architect; owner; CM
- **Question:** Which decisions are now fixed, and what would it cost to change them?
- **Inputs:** Program, site constraints, code analysis, owner comments at each review
- **Artifacts:** Schematic design (SD) and design development (DD) drawings, outline specifications, code analysis, BIM model
- **Tools:** Rhino, SketchUp, and Autodesk Forma for early massing; Revit for BIM; Navisworks and Autodesk Construction Cloud for coordination; Enscape and Twinmotion for visualization
- **AI today:** Renderings are close to instant and generative massing produces many options quickly, so the architect's work shifts from producing options to choosing among them. Model-to-sheet automation and clash grouping are reducing junior drafting hours.
- **Passes forward:** A design with its major systems fixed by the end of DD
- **Often lost:** The rejected options and the reasons for rejecting them, which matter again when value engineering reopens them
- **Where it's covered:** Part 2 Module 2.2; Part 3 Modules 3.6, 3.7, and 3.8

Step 9: Cost Through Design

The first real estimate arrives, and the building usually doesn't fit the budget.

- **Who:** CM or estimator, architect, owner
- **Question:** Does the design still fit the budget, and if not, what can be cut without cutting the value the budget protects?
- **Inputs:** SD and DD documents, historical cost data, subcontractor input, escalation
- **Artifacts:** SD estimate, DD estimate, value engineering (VE) log of accepted and rejected items
- **Tools:** Bluebeam, On-Screen Takeoff, STACK, and Autodesk Takeoff for quantities; Sage Estimating, Trimble WinEst, and DESTINI for pricing
- **AI today:** AI takeoff is assisted, not hands-off. It counts and measures quickly, and a person still verifies every quantity. Pricing strategy, scope gaps, and deciding what to cut remain the estimator's and owner's judgment.
- **Passes forward:** An estimate reconciled to the budget, and a VE log the design team must carry into the documents
- **Often lost:** VE decisions that reach one document and not another, such as a drawing change with no matching spec change
- **Where it's covered:** Part 2 Module 2.3; Part 3 Modules 3.5 and 3.11

Step 10: Construction Documents

The design becomes a contract: drawings and a project manual that must agree with each other.

- **Who:** Architect, engineers, specifier
- **Question:** Do the drawings and specifications describe the building completely and consistently enough to price, permit, and build?
- **Inputs:** DD set, VE log, code analysis, product selections
- **Artifacts:** Drawings; the project manual (bidding requirements, contract forms, conditions of the contract, Division 01, and technical specifications organized by MasterFormat); addenda during bidding
- **Tools:** Revit and AutoCAD for drawings; Deltek Specpoint and SpecLink with MasterSpec content for specifications; Bluebeam for QA/QC markups
- **AI today:** Spec drafting and cross-sheet consistency checks are becoming commodity work. Specs are contract documents, though, so someone still owns the performance criteria, substitution decisions, Division 01, and the liability for what the spec says. The specifier's role is moving toward editing and quality control, with fewer people covering more projects.
- **Passes forward:** A permit set and a bid set
- **Often lost:** Coordination between drawings and specs. When they conflict, the contract's order-of-precedence clause decides, and a dispute usually follows.
- **Where it's covered:** Part 2 Module 2.4; Part 3 Modules 3.8 and 3.9

Step 11: Permitting and Plan Review

The authority having jurisdiction checks the design against the adopted codes.

- **Who:** Architect and engineers, plans examiners, fire marshal, the authority having jurisdiction (AHJ)
- **Question:** Does the design comply with the adopted building, fire, energy, and accessibility codes, including local amendments?
- **Inputs:** Permit set, code analysis, energy compliance documentation, local amendments
- **Artifacts:** Permit applications, review comments, responses and resubmittals, issued permits (often phased, with grading and site permits ahead of building permits)
- **Tools:** AHJ portals (Accela, Tyler EnerGov, Clariti); Bluebeam for review markups
- **AI today:** Jurisdictions are piloting completeness checks and objective code checks (dimensions, counts, ratios), and design firms use similar tools to prescreen before submitting. Accuracy is good enough to prescreen and not good enough to approve. Interpretive and discretionary review stays with the plans examiner.
- **Passes forward:** Issued permits, plus review comments that change the documents
- **Often lost:** Plan review changes that never make it back into the model or the bid set
- **Where it's covered:** Part 2 Modules 2.4 and 2.6; Part 3 Module 3.10

Step 12: Buying the Work / Procurement

The CM converts the documents into a GMP and a set of subcontracts that together cover every scope.

- **Who:** CM, subcontractors and suppliers, owner, lender
- **Question:** Can the work be bought for the GMP, from subcontractors who can perform?
- **Inputs:** Bid set, scope sheets, prequalification data, bids
- **Artifacts:** Bid leveling sheets; GMP amendment (for example to AIA A133) with assumptions, clarifications, and exclusions; subcontracts; buyout log
- **Tools:** BuildingConnected and TradeApp, Procore Bid Management, AIA Contract Documents, Document Crunch for contract review
- **AI today:** Bid leveling, scope gap detection, and clause-level contract risk flags are in use. Deciding which subcontractor to trust with the framing package still rests on relationships and track record.
- **Passes forward:** A signed GMP and subcontracts that cover every scope
- **Often lost:** Exclusions nobody owns, where one subcontractor excludes a scope and no other picks it up
- **Where it's covered:** Part 2 Module 2.5; Part 3 Modules 3.11 and 3.12

Step 13: Preconstruction Planning

Between the signed contract and the first truck on site, the team decides how long the building will take.

- **Who:** CM project manager and superintendent, key subcontractors, owner, utility providers
- **Question:** In what order is the building built, what must be ordered when, and which risks are being managed?
- **Inputs:** GMP, subcontractor durations, long-lead items, permit path, utility dates, site conditions
- **Artifacts:** Baseline CPM schedule, pull plan, submittal schedule, procurement log, site logistics plan, risk register, notice to proceed
- **Tools:** Primavera P6 and Microsoft Project for CPM scheduling; Touchplan for pull planning; ALICE and nPlan for scenario and risk analysis
- **AI today:** Generates schedule scenarios and flags schedule risk from historical project data. Sequencing still depends on crew availability and site conditions the superintendent knows firsthand.
- **Passes forward:** A baseline schedule that delays will be measured against
- **Often lost:** The assumptions behind each duration. When the schedule slips, nobody can tell which assumption failed.
- **Where it's covered:** Part 2 Module 2.6; Part 3 Module 3.13

Step 14: Construction and Contract Administration

The work is built, documented, paid for, and inspected, one pay period at a time.

- **Who:** CM, subcontractors, architect (construction administration), owner's rep, inspectors, the lender's inspecting consultant
- **Question:** Is the work being built to the documents, on schedule, within the GMP, and safely?
- **Inputs:** Contract documents, schedule, approved submittals, field conditions
- **Artifacts:** Submittals and shop drawings, RFIs, change orders and construction change directives, pay applications (AIA G702/G703) with lien waivers, daily logs, inspection reports, schedule updates
- **Tools:** Procore and Autodesk Build for project management; Sage and Trimble Viewpoint for job cost; Oracle Textura and Procore Pay for payments; OpenSpace, DroneDeploy, and Matterport for reality capture; SafetyCulture for safety
- **AI today:** Agents draft RFIs, check submittals against specs, and write meeting minutes and daily logs, and computer vision compares site progress to the schedule. That work has traditionally been how project engineers learn construction, which raises the question of how future PMs will be trained.
- **Passes forward:** Built work, an as-built record, and a financial history of every change
- **Often lost:** The reason for each field change. The change order records the cost, while the RFI thread that explains it sits in a separate system.
- **Where it's covered:** Part 2 Module 2.7; Part 3 Modules 3.14 through 3.17

Step 15: Closeout, Turnover, and Handback

The building is accepted, documented, and handed to the people who will run it.

- **Who:** CM, architect, commissioning agent, owner, property manager, AHJ
- **Question:** Is the building complete, accepted, and documented well enough for someone else to operate it?
- **Inputs:** Punch lists, test and commissioning results, warranties, submittal records
- **Artifacts:** Certificate of substantial completion (AIA G704), certificate of occupancy, punch list, commissioning report, O&M manuals, warranties, record drawings, attic stock, final pay application and retainage release
- **Tools:** Procore and Autodesk Build for closeout; CxAlloy and Facility Grid for commissioning; COBie for structured handover data
- **AI today:** Assembles O&M manuals and warranty logs from submittals and checks as-builts against scans. Accepting the building remains a contractual act signed by the owner and architect.
- **Passes forward:** A turnover package, keys, and warranty start dates
- **Often lost:** Structured data. The model and submittals hold what the operator needs, but it usually arrives as a stack of PDFs.
- **Where it's covered:** Part 2 Module 2.8; Part 3 Module 3.18

Step 16: Operations, and Back to Step 1, Facility Management

The building starts earning, and its performance becomes the input to the next deal.

- **Who:** Property manager, facility manager, asset manager, owner, investors
- **Question:** Is the building performing as the pro forma assumed, and is it time to hold, refinance, or sell?
- **Inputs:** Leasing results, operating costs, maintenance history, market conditions
- **Artifacts:** Rent roll, T12 operating statement, asset register, maintenance records, refinance or sale package
- **Tools:** Yardi, RealPage, and AppFolio for property management; IBM Maximo, Brightly, and MaintainX for maintenance; Autodesk Tandem and Bentley iTwin for digital twins; building automation systems and IoT sensors
- **AI today:** Predictive maintenance, energy optimization, and tenant service automation are in production use. The unsolved problem is getting design and construction data to survive handover, which is the logic behind Autodesk's 2026 acquisition of MaintainX.
- **Passes forward:** A rent roll and T12 that the next buyer's analyst loads into a model, which is Step 1 for someone else
- **Often lost:** Design intent. The operator knows what equipment was installed but not why it was selected.
- **Where it's covered:** Part 3 Module 3.19

The Pattern Across All Sixteen Steps

Every handoff compresses. A market study becomes a site, a pro forma becomes one number, a design becomes a document set, and a finished building becomes a folder of PDFs. What gets lost is rarely the answer. It is the reasoning behind the answer and the range of uncertainty around it.

AI is strongest where a step involves a high volume of documents and objective checks: comps, zoning text, takeoff, submittal review, O&M assembly. It is weakest where a decision is negotiated, discretionary, or carries a professional stamp: entitlement votes, delivery method, subcontractor selection, acceptance of the work.

The largest opportunity in AEC technology is at the handoffs rather than inside any single step. Classification systems such as MasterFormat, UniFormat, and OmniClass, and exchange standards such as IFC and COBie, are how reasoning could travel with the data from one group to the next. Part 3 Modules 3.20 through 3.23 cover those standards, along with liability, the workforce, and how firms evaluate and adopt new tools.

OUTRO

Hope that helped.

I'll update this as I can.

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